

Comparing the above equation to eqn (4.4.8), we find that

$$e^{i f_2(\theta_1+2\pi, \theta_2)} = e^{i f_2(\theta_1, \theta_2)}$$

We see that $|\theta\rangle$ is quasi-periodic in θ_1 and θ_2 with period 2π :

$$\begin{aligned} |\theta_1 + 2\pi, \theta_2\rangle &= W_1 |\theta_1, \theta_2\rangle \\ |\theta_1, \theta_2 + 2\pi\rangle &= e^{i f_2(\theta_1, \theta_2)} W_2 |\theta_1, \theta_2\rangle \end{aligned} \quad (4.4.9)$$

where $W_{1,2}$ are unitary operators which are independent of θ .

Using $\langle\theta|\partial_\theta|\theta\rangle = \langle\theta|W_{1,2}^\dagger\partial_\theta W_{1,2}|\theta\rangle$ and eqn (4.4.9), we can reduce the Berry phase (4.4.3) to

$$2\pi K = \int d\theta_1 \partial_{\theta_1} f_2(\theta_1, 0) = f_2(2\pi, 0) - f_2(0, 0)$$

As $e^{i f_2(\theta_1, \theta_2)}$ is periodic in θ_1 , we find that $f_2(2\pi, 0) - f_2(0, 0) = 2\pi \times \text{integer}$ and K is quantized as an integer.

INTERACTING FERMION SYSTEMS

Electron systems in nature have quite strong Coulomb interactions. Many interesting properties of materials, such as magnetism, superconductivity, etc. are due to interactions. In this chapter, we will study the effect of electron–electron interactions. In particular, we will discuss Landau’s Fermi liquid theory for metals and symmetry-breaking transitions induced by the interaction.

5.1 Orthogonality catastrophe and X-ray spectrum

- Orthogonality catastrophe is a phenomenon that illustrates how many-body effects affect fermion Green’s functions and the associated experimental results.
- Orthogonality catastrophe is a general phenomenon that can appear in many systems, such as an X-ray spectrum, an I – V curve of tunneling into quantum dots, etc.

5.1.1 Physical model

The X-ray spectrum is caused by a transition between a core level and a conduction band of a metal (see Fig. 5.1(a)). An absorption of X-rays removes an electron from a core level and adds it into an unoccupied band state. An emission removes an electron from an occupied band state to an unoccupied core state. The system can be modeled by the following Hamiltonian:

$$H = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} c_{\mathbf{k}}^{\dagger} c_{\mathbf{k}} - E_C C^{\dagger} C + \Gamma (e^{-i\omega t} c(0) C^{\dagger} + h.c.) \quad (5.1.1)$$

where C describes a core electron at $x = 0$, and $\Gamma e^{-i\omega t}$ is the time-dependent coupling induced by the X-rays. For simplicity, we will assume spinless electrons in this section.

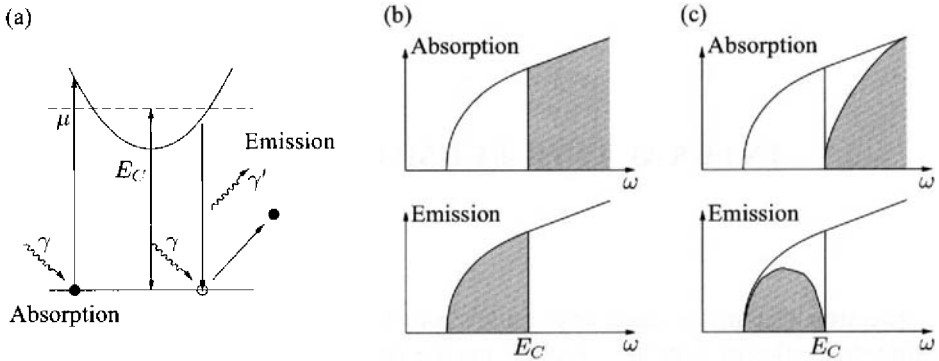


FIG. 5.1. (a) X-ray absorption/emission are caused by transitions between a core level and a conduction band of a metal. Also shown are the X-ray spectra of the transitions (b) with no interaction, and (c) with interaction.

Note that the emission rate is equal to the current from the conduction band to the core state. According to linear response theory, the emission rate is given by

$$\begin{aligned}
 A_{em}(\omega) &= \langle 0 | i\Gamma (e^{-i\omega t} I(t) - h.c.) | 0 \rangle \\
 &= \Gamma^2 \int_{-\infty}^t dt' \left(e^{-i\omega(t-t')} \langle 0 | [I(t), I^\dagger(t')] | 0 \rangle + h.c. \right)
 \end{aligned}$$

where $I(t) = c(0, t)C^\dagger(t)$. We can express the emission rate A_{em} in terms of the spectral function of the core electron and the conduction electron (see eqn (4.2.17)) as follows:

$$A_{em}(\omega) = -2\pi\Gamma^2 \int d\nu \left(A_{c-}^0(\nu) A_{c+}^0(\nu - \omega) - A_{c+}^0(\nu) A_{c-}^0(\nu - \omega) \right)$$

For free fermions, the spectral function is given by the density of states, as in eqn (4.2.18). If we choose our energy to be zero at the Fermi surface, then

$$\begin{aligned}
 A_{c-}^0(\omega) &= n_F(\omega) N(\omega + E_F), & A_{c+}^0(\omega) &= (1 - n_F(\omega)) N(\omega + E_F) \\
 A_{c-}^0(\omega) &= 0, & A_{c+}^0(\omega) &= \delta(\omega + E_C)
 \end{aligned}$$

We note that, before emission, the core state is completely empty. This gives rise to the above non-equilibrium core-electron spectral function. We find that

$$A_{em}(\omega) = 2\pi\Gamma^2 n_F(\omega - E_C) N(E_F - E_C + \omega)$$

and the emission rate directly measures the density of states for free fermions (see Fig. 5.1(b); note that $E_C > 0$). At zero temperature, $A_{em}(\omega)$ has a step-like

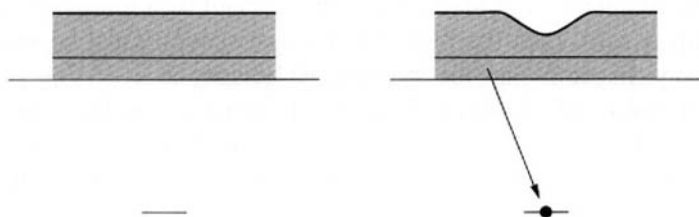


FIG. 5.2. The density of the conduction electrons has to be changed after the tunneling if there is an interaction between the core electron and the conduction electrons.

jump at $\omega = E_C$, indicating the sharpness of the Fermi surface. The above result also applies to interacting electrons if we use the spectral function of interacting electrons.

Problem 5.1.1.

Calculate the absorption rate spectrum $A_{obs}(\omega)$.

5.1.2 The physics of orthogonality catastrophe

- Orthogonality catastrophe is caused by the vanishing overlap between two *many-body* wave functions, which describe the deformed and undeformed Fermi seas.

Now we would like to include an interaction between the core electron and the conduction electrons. We will see that such an interaction can modify the edge singularity of the emission/absorption rate at $\omega = E_C$ in a significant way.

The physics is illustrated in Fig. 5.2. In the absence of the interaction between the core electron and the conduction electrons, the matrix element for the emission at a given frequency involves the matrix element of single-particle wave functions of the conduction electron at a certain energy and the core electron. To generalize to interacting cases, we should view the above matrix element as a matrix element between two many-body eigenstates: one eigenstate has a filled Fermi sea and an empty core state, and the other has a hole in the Fermi sea and a filled core state. In the presence of the interaction, the core electron induces a change in the density of the conduction electrons, and hence a change in the many-body electron wave function. Thus, the matrix element for the emission involves the matrix element between the following two eigenstates. One has a filled but deformed Fermi sea and an empty core state. The other has a hole in the uniform Fermi sea and a filled core state. We can approximate the above matrix element as the product of the matrix element between the single-particle states and the overlap between two many-body states of a uniform Fermi sea and a deformed Fermi sea. If the overlap is finite, then the discussions in the last section remain valid for ω near E_C ,

except that the tunneling amplitude Γ may be reduced by a finite factor. However, if the overlap is zero, then the results in the last section must be modified in a significant way. In particular, the step-like singularity at $\omega = E_C$ is destroyed. This phenomenon is called infra-red catastrophe or orthogonality catastrophe.

We know that a many-body wave function for free fermions can be constructed from single-body wave functions through the Slater determinant as follows:

$$\Psi(\mathbf{x}_1, \dots, \mathbf{x}_N) = \text{Det}[\psi_i(\mathbf{x}_j)]$$

where $\psi_i(\mathbf{x})$ is a single-body wave function and $[\psi_i(\mathbf{x}_j)]$ is the following matrix:

$$[\psi_i(\mathbf{x}_j)] \equiv \begin{pmatrix} \psi_1(\mathbf{x}_1) & \psi_1(\mathbf{x}_2) & \cdots \\ \psi_2(\mathbf{x}_1) & \psi_2(\mathbf{x}_2) & \cdots \\ \cdots & \cdots & \ddots \end{pmatrix}$$

The many-body wave function for the uniform Fermi sea, Ψ_0 , is constructed from the plane waves $\psi_i(\mathbf{x}) = e^{i\mathbf{k}_i \cdot \mathbf{x}}$, where $\mathbf{k}_i$ runs through all momenta whose energies are less than the Fermi energy E_F . The many-body wave function for the deformed Fermi sea, Ψ , is constructed from the eigenfunctions in the presence of the potential of the core electron. We can then calculate the overlap $\langle \Psi_0 | \Psi \rangle$ to determine if there is an orthogonality catastrophe or not.

To illustrate our point in a simple calculation, we will calculate $\langle \Psi_0 | \Psi \rangle$ by treating the fermions as a fluid described by the density $\rho(\mathbf{x})$. This treatment gives us a correct result in one dimension, but an incorrect result beyond one dimension. Such an approach is called the hydrodynamical approach, or the bosonization of fermion systems. Let us first develop the hydrodynamical approach.

5.1.3 Hydrodynamical approach (bosonization)

- The collective density fluctuations in a metal can be described by a bosonic field theory.
- In one dimension, the boson theory provides a complete description of the particle-hole excitations across the Fermi surface.

The potential energy of a compressible fluid is given by

$$V = \int d^d \mathbf{x} \frac{1}{2\chi} \rho^2(\mathbf{x})$$

where χ is the compressibility. Here we assume that $\rho(\mathbf{x})$ is a fluctuation around the uniform density, ρ_0 , of the ground state. The kinetic energy is

$$K = \int d^d \mathbf{x} \frac{m \mathbf{v}^2(\mathbf{x})}{2} \rho_0 = \int d^d \mathbf{x} \frac{m \mathbf{j}^2(\mathbf{x})}{2\rho_0}$$

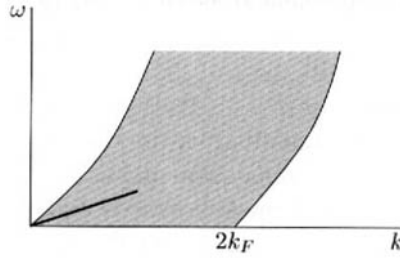


FIG. 5.3. The dispersion of the linear mode always lies within the particle-hole continuum.

where $\mathbf{v}(\mathbf{x})$ is the velocity and $\mathbf{j}(\mathbf{x}) = \mathbf{v}(\mathbf{x})\rho_0$ is the current of the fluid at $\mathbf{x}$. Thus, the Lagrangian that describes the dynamics of the fluid is given by

$$L = \int d^d \mathbf{x} \left[\frac{m \mathbf{j}^2(\mathbf{x})}{2\rho_0} - \frac{1}{2\chi} \rho^2(\mathbf{x}) \right] \quad (5.1.2)$$

From current conservation $\partial_t \rho + \partial_{\mathbf{x}} \cdot \mathbf{j} = 0$, we find that $\mathbf{j} = -\frac{1}{\partial_{\mathbf{x}}} \partial_t \rho$ (assuming that $\partial_{\mathbf{x}} \times \mathbf{j} = 0$), and the Lagrangian can be written in terms of density only as follows:

$$L = \int d^d \mathbf{x} \left(\frac{m}{2\rho_0} \dot{\rho} - \frac{1}{\partial_{\mathbf{x}}^2} \dot{\rho} - \frac{1}{2\chi} \rho^2 \right) \quad (5.1.3)$$

$$= \mathcal{V}^{-1} \sum_{\mathbf{k}} \left(\frac{m}{2\rho_0} \dot{\rho}_{-\mathbf{k}} \frac{1}{\mathbf{k}^2} \dot{\rho}_{\mathbf{k}} - \frac{1}{2\chi} \rho_{-\mathbf{k}} \rho_{\mathbf{k}} \right) \quad (5.1.4)$$

where $\rho_{\mathbf{k}} = \int d^d \mathbf{x} \rho(\mathbf{x}) e^{-i\mathbf{k} \cdot \mathbf{x}}$ and $\mathcal{V}$ is the volume of the system.

Such a Lagrangian is identical to the phonon Lagrangian (3.3.15) studied previously, with $A_{\mathbf{k}} = m/\mathcal{V}\rho_0 \mathbf{k}^2$ and $B_{\mathbf{k}} = \frac{1}{\mathcal{V}\chi}$. The phonons are described by the following quantum Hamiltonian (see eqn (3.3.18)):

$$H = \sum_{\mathbf{k} \neq 0} \Omega_{\mathbf{k}} \alpha_{\mathbf{k}}^{\dagger} \alpha_{\mathbf{k}}, \quad \Omega_{\mathbf{k}} = \sqrt{\frac{B_{\mathbf{k}}}{A_{\mathbf{k}}}} = \sqrt{\frac{\rho_0}{m\chi}} |\mathbf{k}| \quad (5.1.5)$$

For free fermions, the compressibility is equal to the density of states: $\chi = N(E_F)$. We find the velocity of the hydrodynamical mode to be

$$v = v_F \quad \text{for } d = 1$$

$$v = \frac{1}{\sqrt{2}} v_F \quad \text{for } d = 2$$

$$v = \frac{1}{\sqrt{3}} v_F \quad \text{for } d = 3$$

The dispersion of the hydrodynamical mode always lies within the particle-hole continuum (see Fig. 5.3). One can show that the specific heat of the single linear mode is less than the specific heat of the corresponding free fermion system in $d > 1$ dimensions. Thus, for $d > 1$, the hydrodynamical treatment does not reproduce the full low-lying excitations; it only describes some average density fluctuations. However, in $d = 1$ dimension and at low temperatures, the specific heat of the single hydrodynamical mode is exactly equal to the specific heat of the corresponding free fermion system, suggesting that the single boson mode reproduces all low-lying excitations of the free fermion system. This indicates that the single boson mode is a faithful description of the fermion system at low energies, and we can use a boson theory to describe the low-energy properties of a fermion system. The method of using bosons to describe a one-dimensional fermion system is called bosonization (Tomonaga, 1950; Luttinger, 1963; Coleman, 1975).

To compare the hydrodynamical model and the fermion model in more detail, let us consider the many-body low-energy excitations in the quantum Hamiltonian (5.1.5). In one dimension, the hydrodynamical Hamiltonian (5.1.5) reproduces the spectrum of all of the low-energy excitations in the corresponding fermion system. To see this, let us put the system on a circle of size L and assume that the ground state has zero energy. The hydrodynamical model has two eigenstates with energy $E = 2k_0v$ and momentum $K = 2k_0$, namely $(\alpha_{k_0}^\dagger)^2|0\rangle$ and $\alpha_{2k_0}^\dagger|0\rangle$, where $k_0 = 2\pi/L$. The fermion system also has two particle-hole excited states at momentum $2k_0$ which have exactly the same energy $E = 2k_0v_F = 2k_0v$. The two particle-hole states are given by $c_{k_F+2k_0}^\dagger c_{k_F}|\psi_0\rangle$ and $c_{k_F+k_0}^\dagger c_{k_F-k_0}|\psi_0\rangle$. Here $|\psi_0\rangle$ is the ground state of the fermion system where the single particle states $|k_F\rangle$, $|k_F - k_0\rangle$, ..., $|-k_F\rangle$ are filled by one fermion. The hydrodynamical model has one eigenstate with energy $E = 2k_0v$ and momentum $K = 0$, namely $a_{-k_0}^\dagger a_{k_0}^\dagger|0\rangle$. The fermion system also has one particle-hole state with the same momentum and energy, namely $c_{-k_F-k_0}^\dagger c_{-k_F} c_{k_F+k_0}^\dagger c_{k_F}|\psi_0\rangle$. In fact, the excitation spectrum of the hydrodynamical model is identical to the particle-hole excitation spectrum of the fermion model (see Problem 5.1.3). The hydrodynamical model is a faithful description of the fermion system at low energies and small momenta.

Problem 5.1.2.

Write α_k in terms of ρ_k and $\dot{\rho}_k$. Show that α_k carries a definite momentum.

Problem 5.1.3.

Show that, in one-dimension, both the hydrodynamical model (with a fixed number of particles) and the fermion system (with a fixed number of fermions) have the following properties.

- There are p_n excited states with energy nk_0v_F and momentum nk_0 , where $(p_0, p_1, p_2, \dots) = (1, 1, 2, 3, 5, 7, \dots)$ are the partition numbers. (If you cannot find the general proof, then you may check the result up to p_5 .)
- There are p_n excited states with energy nk_0v_F and momentum $-nk_0$.

(c) There are $p_{n-m}p_m$ states with energy nk_0v_F and momentum $(n-2m)k_0$.

Thus, the hydrodynamical model and the fermion system have an identical excitation spectrum at low energies and small momenta.

Problem 5.1.4.

(a) From eqn (4.3.2), we see that the compressibility $\chi(\omega, \mathbf{k}) = -\Pi^{00}(\omega, \mathbf{k})$ contains an imaginary part. Using such a compressibility and the equation of motion of the Lagrangian (5.1.2), find the decay rate of the density mode (the phonons) in $d = 1$, $d = 2$, and $d = 3$ dimensions. Compare the decay rate with the frequency of the mode. Does the density mode give rise to a well-defined phonon?

(b) In fact, part (a) is not quite right. One should calculate the density response function of the hydrodynamical model using eqn (5.1.2) and then obtain $\chi(\omega, \mathbf{k})$ by choosing it to fit the fermion results of eqn (4.3.2). Find $\chi(\omega, \mathbf{k})$ using this method. Are the main results in (a) changed by the new $\chi(\omega, \mathbf{k})$?

5.1.4 Orthogonality catastrophe from the hydrodynamical approach

- In the hydrodynamical approach, the overlap of many-body wave functions can be calculated easily from the shifted harmonic oscillators.

In the presence of the potential caused by the core electron, the Hamiltonian (5.1.5) contains an additional term (see eqn (3.3.17)):

$$\delta H = \int d^d x \rho(x) V(x) = \mathcal{V}^{-1} \sum_{\mathbf{k}} \rho_{\mathbf{k}} V_{-\mathbf{k}} = \mathcal{V}^{-1} \sum_{\mathbf{k}} V_{-\mathbf{k}} \frac{\alpha_{\mathbf{k}} + \alpha_{\mathbf{k}}^\dagger}{2u_{\mathbf{k}}}$$

$$u_{\mathbf{k}} = \frac{1}{\sqrt{2}} (A_{\mathbf{k}} B_{\mathbf{k}})^{1/4} = \left(\frac{m}{4\mathcal{V}^2 \rho_0 \chi k^2} \right)^{1/4}$$

Here $H + \delta H$ describes a collection of shifted oscillators. The ground state of the system in the presence of the potential is given by

$$|\Psi\rangle = \frac{e^{-\mathcal{V}^{-1} \sum_{\mathbf{k}} \frac{V_{-\mathbf{k}}}{2\Omega_{\mathbf{k}} u_{\mathbf{k}}} \alpha_{\mathbf{k}}^\dagger} |\Psi_0\rangle}{\sqrt{\langle \Psi_0 | e^{-\mathcal{V}^{-1} \sum_{\mathbf{k}} \frac{V_{-\mathbf{k}}}{2\Omega_{\mathbf{k}} u_{\mathbf{k}}} \alpha_{\mathbf{k}}^\dagger} e^{-\mathcal{V}^{-1} \sum_{\mathbf{k}} \frac{V_{-\mathbf{k}}}{2\Omega_{\mathbf{k}} u_{\mathbf{k}}} \alpha_{\mathbf{k}}} |\Psi_0\rangle}}$$

where $|\Psi_0\rangle$ is the ground state in the absence of the potential. Here $|\Psi\rangle$ corresponds to the deformed state. The overlap between $|\Psi\rangle$ and $|\Psi_0\rangle$ is

$$\langle \Psi_0 | \Psi \rangle = e^{-\mathcal{V}^{-2} \sum_{\mathbf{k}} \frac{|V_{\mathbf{k}}|^2}{4\Omega_{\mathbf{k}}^2 u_{\mathbf{k}}^2}} = e^{-\int \frac{d^d k}{(2\pi)^d} \frac{\rho_0 |V_{\mathbf{k}}|^2}{2m v_F^2 |k|}}$$

For a short-range potential, $V_{\mathbf{k}}$ is nonzero for small $\mathbf{k}$. In $d = 1$ dimension, the integral diverges for small $\mathbf{k}$ and $\langle \Psi_0 | \Psi \rangle = 0$. We have an orthogonality catastrophe. In $d > 1$ dimensions, the integral is finite for small $\mathbf{k}$. For large $\mathbf{k}$, the integral

is cut off by a short distance scale (say $l = 1/k_F$). Therefore, the overlap is finite within the hydrodynamical approach. (This result turns out to be incorrect.)

In the following, we will calculate the correlation of the tunneling operator $I(t)$ to obtain the ω dependence of the emission rate. We approximate the time-ordered imaginary-time correlation as

$$\langle T(I(\tau)I(0)) \rangle = -G_C(\tau)G_C(-\tau) \quad (5.1.6)$$

The above is exact only when there are no interactions between the core and the conduction electrons.

It turns out that the influence of the core electron on the conduction electron Green's function is not important and we can approximate $G_C(\tau)$ by the free fermion Green's function. To calculate $G_C(\tau)$, we note that the path of the core electron in space-time is a straight line in the time direction. Thus,

$$G_C(\tau) = \left\langle e^{-\int_0^\tau d\tau' V_C \rho(\tau', 0)} \right\rangle e^{E_C \tau}$$

where $\rho(\tau)$ is the density operator of the conduction electrons, and $-E_C$ is the energy of the core electron. Within the hydrodynamical approach, we may use the path integral to evaluate the above expression. In Problem 5.1.5 it will be shown that the Lagrangian for the hydrodynamical approach (5.1.2) is equivalent to the standard Lagrangian of the XY-model, namely

$$L = \frac{\chi}{2} ((\partial_t \theta)^2 - v^2 (\partial_x \theta)^2), \quad \rho = -\chi \partial_t \theta$$

In imaginary time ($t \rightarrow -i\tau$), we have

$$L = \frac{\chi}{2} ((\partial_\tau \theta)^2 + v^2 (\partial_x \theta)^2), \quad \rho = -i\chi \partial_\tau \theta$$

Therefore

$$\begin{aligned} G_C(\tau) e^{-E_C \tau} &= \frac{\int \mathcal{D}\theta \, e^{i \int_0^\tau d\tau' V_C \chi \partial_\tau \theta(\tau', 0) - \int d\tau' d^d \mathbf{x} L(\theta)}}{\int \mathcal{D}\theta \, e^{-\int d\tau' d^d \mathbf{x} L(\theta)}} \\ &= \left\langle e^{i V_C \chi \theta(\tau, 0)} e^{-i V_C \chi \theta(0, 0)} \right\rangle \end{aligned}$$

The long-time behavior of $G_C(\tau)$ depends on the spatial dimension. In $d = 1$ dimension (see eqn (3.3.25)), we have

$$G_C(\tau) e^{-E_C \tau} = \left(\frac{l^2}{v^2 \tau^2} \right)^{\chi V_C^2 / 4\pi v}$$

where we have assumed that the temperature $T = 0$. In real time, we have

$$G_C(t) = e^{-i \frac{\pi \chi V_C^2}{4\pi v}} \left(\frac{l}{v|t|} \right)^{\chi V_C^2 / 2\pi v} e^{i E_C t}$$

We see that the Green's function is very different from the one for the free core electron, namely $G_C(t) = e^{iE_C t}$. The spectral function for the dressed core electron is

$$A_{C+}(\omega) \sim \Theta(\omega + E_C)(\omega + E_C)^{\frac{\chi V_C^2}{2\pi v} - 1}$$

and the emission rate behaves like (see Fig. 5.1(c))

$$A_{em} \sim \Theta(E_C - \omega)(E_C - \omega)^{\chi V_C^2 / 2\pi v}$$

In one dimension, the exponent is given by

$$\frac{\chi V_C^2}{2\pi v} = \frac{V_C^2 N^2(E_F)}{2}. \quad (5.1.7)$$

In $d > 1$ dimensions, $\langle e^{iV_C \chi \theta(\tau, 0)} e^{-iV_C \chi \theta(0, 0)} \rangle$ approaches a constant for large τ and $G_C(t) \rightarrow e^{iE_C t}$, which is the same as the free core-electron Green's function. Thus, according to the hydrodynamical model, the emission rate should be similar to the one for the non-interacting case. This result is consistent with the above analysis of wave function overlap. However, as we will see in the next section, such a result is incorrect for fermion systems.

Problem 5.1.5.

In the hydrodynamical approach, the energy (Hamiltonian) of the conduction electrons is given by

$$H = \int d^d \mathbf{x} \left[\frac{m \mathbf{j}^2(\mathbf{x})}{2\rho_0} + \frac{1}{2\chi} \rho^2(\mathbf{x}) \right]$$

We may replace $\mathbf{j}$ by $\partial_{\mathbf{x}} \phi$ if we assume that $\partial_{\mathbf{x}} \times \mathbf{j} = 0$.

1. If we view $\rho(\mathbf{x})$ as the canonical coordinates, $q_{\mathbf{x}}$, then $C\phi(\mathbf{x})$ can be viewed as the canonical momenta, $p_{\mathbf{x}}$. (Here $\mathbf{x}$ can be regarded as an index which labels different canonical coordinates and momenta.) Show that the current conservation $\dot{\rho} + \partial_{\mathbf{x}} \mathbf{j} = 0$ can be reproduced by one of the Hamiltonian equations

$$\dot{q}_{\mathbf{x}} = \frac{\delta H}{\delta p_{\mathbf{x}}}$$

if we choose the value of C properly. Find the value of C .

2. Show that the Hamiltonian equations

$$\dot{p}_{\mathbf{x}} = -\frac{\delta H}{\delta q_{\mathbf{x}}}, \quad \dot{q}_{\mathbf{x}} = +\frac{\delta H}{\delta p_{\mathbf{x}}}$$

reproduce the equation of motion of the Lagrangian (5.1.2). Show that the Hamiltonian and the canonical coordinate-momentum pairs $(q, p) = (\rho(\mathbf{x}), C\phi(\mathbf{x}))$ reproduce the Lagrangian (5.1.2).

3. Now let us view $C\phi(\mathbf{x})$ as coordinates and $\rho(\mathbf{x})$ as momenta. Show that the Hamiltonian and the canonical coordinate–momentum pairs $(q, p) = (C\phi(\mathbf{x}), \rho(\mathbf{x}))$ reproduce the Lagrangian of the XY-model.
4. We know that, in the standard XY-model, the θ field is periodic, i.e. θ and $\theta + 2\pi$ describe the same point. Find the periodic condition of the ϕ field. (Hint: Consider the $\mathbf{k} = 0$ mode and find the relationship between the quantization of the total particle number and the periodic condition of the ϕ field.) We see that, after a proper scaling, $D\phi$ can be identified as the θ field and our hydrodynamical model can be identified as the XY-model.
5. Compare the density correlation function (say, time-ordered) calculated from eqn (5.1.2) and the XY-model. Comment on your results.

Problem 5.1.6.

Use the relation between the hydrodynamical model and the XY-model obtained in the previous problem, to show that the one-dimensional hydrodynamical model can reproduce the low-energy excitations near $\pm 2k_F, \pm 4k_F, \dots$ in the one-dimensional fermion system (see Section 3.6).

5.1.5 Direct calculation for fermion systems

- The hydrodynamical approach can only produce the corrected overlap of many-body wave functions in one dimension.
- In higher dimensions, the single boson mode is not sufficient for describing particle–hole excitations across the Fermi surface. The particle–hole excitations correspond to many-boson modes, which leads to a smaller overlap.

Let us directly calculate

$$G_C(\tau) = \left\langle e^{-\int_0^\tau d\tau' V_C \rho(\tau', 0)} \right\rangle e^{E_C \tau}$$

for $\tau > 0$ in a fermion system. Here we view ρ as the fluctuations around the constant density ρ_0 . (The constant part can be absorbed into E_C .) We can remove the constant part from ρ through a normal ordering with respect to the filled Fermi sea, i.e. $\rho(\mathbf{x}) =: c^\dagger(\mathbf{x})c(\mathbf{x}) :.$ Here $: c_{\mathbf{k}}^\dagger c_{\mathbf{k}} := c_{\mathbf{k}}^\dagger c_{\mathbf{k}}$ if $\xi_{\mathbf{k}} > 0$, and $: c_{\mathbf{k}}^\dagger c_{\mathbf{k}} := -c_{\mathbf{k}} c_{\mathbf{k}}^\dagger$ if $\xi_{\mathbf{k}} < 0$.

To calculate $G_C(\tau)$, we first expand $e^{-\int_0^\tau d\tau' V_C \rho(\tau', 0)}$ as follows:

$$G_C(\tau) = \left\langle 1 + \left(-\int_0^\tau d\tau' V_C \rho(\tau', 0) \right) + \frac{1}{2!} \left(-\int_0^\tau d\tau' V_C \rho(\tau', 0) \right)^2 + \dots \right\rangle$$

Using the linked-cluster theorem, we find that

$$G_C(\tau) = e^{\left\langle -\int_0^\tau d\tau' V_C \rho(\tau', 0) \right\rangle_c + \frac{1}{2!} \left\langle \left(-\int_0^\tau d\tau' V_C \rho(\tau', 0) \right)^2 \right\rangle_c + \dots}$$

where $\langle \dots \rangle_c$ contains only the connected graphs. The first term $\langle -\int_0^\tau d\tau' V_C \rho(\tau', 0) \rangle_c$ vanishes, and the second term is the leading term. If we assume that the density fluctuations are small, then we may ignore the higher-order terms. Thus,

$$G_C(\tau) = e^{\frac{V_C^2}{2} \int_0^\tau d\tau_1 d\tau_2 \langle \rho(\tau_1, 0) \rho(\tau_2, 0) \rangle}$$

Using $\mathcal{G}(\tau, 0) = N(E_F)/\tau$, we find that

$$\langle \rho(\tau, 0) \rho(0, 0) \rangle = \mathcal{G}(\tau, 0)(-) \mathcal{G}(-\tau, 0) = \frac{N^2(E_F)}{\tau^2}$$

and

$$G_C(\tau) = e^{V_C^2 N^2(E_F) (\frac{\tau}{\tau_l} - \ln \frac{\tau}{\tau_l})} = \left(\frac{\tau_l}{\tau} \right)^\alpha e^{V_C^2 N^2(E_F) \tau / \tau_l}$$

where $\tau_l \sim 1/E_F$ is a short-time cut-off and

$$\alpha = V_C^2 N^2(E_F)$$

The emission rate behaves like

$$A_{em} \sim \Theta(E_C - \omega)(E_C - \omega)^\alpha$$

We see that the edge singularity is modified as long as the fermions have a finite density of states, regardless of the number of spatial dimensions. The result from the hydrodynamical approach is incorrect for $d > 1$.

In one dimension, the result (5.1.7) from the hydrodynamical approach is similar to the one obtained here. However, if we compare the value of the exponent, then the exponent obtained here is twice as large. So, one may wonder, what can go wrong in our one-dimensional hydrodynamical approach? After all, the hydrodynamical modes reproduce the specific heat of the particle-hole excitations near the Fermi surface in one dimension. It appears that the hydrodynamical modes capture all of the low-lying excitations of the fermion systems in one dimension. Thus, one expects that the one-dimensional hydrodynamical approach should give us the correct result. As we will see below, the one-dimensional hydrodynamical approach indeed gives us a correct result in a certain sense.

To understand the discrepancy, we note that the fermion Green's function at the equal-space point, $\mathcal{G}(\tau, 0) = N(E_F)/\tau$, contains two contributions from the two Fermi points. For a small but finite x , we have

$$\mathcal{G}(\tau, x) = \frac{N(E_F)}{2} \left(\frac{v e^{i k_F x}}{v\tau + ix} + \frac{v e^{-i k_F x}}{v\tau - ix} \right)$$

Therefore, the density correlation also contains two parts with momenta $k \sim 0$ and $k \sim \pm 2k_F$:

$$\begin{aligned} \langle \rho(\tau, x) \rho(0, 0) \rangle &= \mathcal{G}(\tau, x)(-) \mathcal{G}(-\tau, -x) \\ &= \frac{N^2(E_F)}{4} \left(\frac{v^2}{(v\tau + ix)^2} + \frac{v^2}{(v\tau - ix)^2} \right) + \frac{N^2(E_F) \cos(2k_F x)}{2} \frac{v^2}{v^2 \tau^2 + x^2} \end{aligned}$$

In the hydrodynamical approach, only the small-momentum part $\frac{v^2}{(v\tau + ix)^2} + \frac{v^2}{(v\tau - ix)^2}$ is included, which leads to a smaller exponent.

We know that a free fermion system contains low-lying excitations with momenta near $k = 0, \pm 2k_F, \pm 4k_F, \dots$. From the above discussion, it is clear that all of these low-lying excitations could contribute to the suppression of the edge singularity. In general, if the core electron induces a potential of finite range, then the coupling between the core and the conduction electrons will have the form $\int dx V(x)\rho(x)$. The correlation of such an operator is given by

$$\left\langle \int dx V(x)\rho(\tau, x) \int dx V(x)\rho(0, x) \right\rangle = V_0^2 \frac{N^2(E_F)}{2\tau^2} + |V_{2k_F}|^2 \frac{N^2(E_F)}{2\tau^2}$$

where $V_k = \int dx V(x)e^{-ikx}$. In this case, the exponent is given by

$$\alpha = \frac{1}{2}(V_0^2 + |V_{2k_F}|^2)N^2(E_F)$$

For free fermions, the density correlation contains only $k = 0$ and $k = 2k_F$ singularities. Thus, α only contains contributions from V_0 and V_{2k_F} . In the hydrodynamical approach, the contribution from V_{2k_F} is dropped. This is the source of the discrepancy.

Problem 5.1.7.

Calculate the Green's function of the core electron, $\mathcal{G}_C(\tau)$, in the long-time limit, assuming that the interaction between the core and the *one-dimensional* conduction electrons is given by $\int dx V(x)\rho(x)$, with $V(x) = V/|x|^\gamma$ and $0 < \gamma < 1$. What is the resulting emission rate $A_{em}(\omega)$?

Problem 5.1.8.

Calculate the Green's function of the core electron, $\mathcal{G}_C(\tau)$, in the long-time limit, assuming that the core electron interacts with a one-dimensional interacting boson system through a δ -potential, i.e. $H_I = V_C\rho(0)$, where $\rho(x)$ is the density of the bosons. We may describe the interacting boson system by its effective XY-model (3.3.23). Note that the boson density correlation contains singularities at momenta $k = K_n$ (see eqn (3.6.2)). For a given value of (χ, v) , determine which singularity controls the long-time behavior of $\mathcal{G}_C(\tau)$ and calculate this long-time behavior.

5.2 Hartree–Fock approximation

- The Hartree–Fock approximation can be viewed as a variational approach.

When we encounter a condensed matter system, we would like to answer two questions. What are the properties of the ground state and what are the properties of the excitations above the ground state? In this section, we are going to use

the Hartree-Fock approximation to answer these two questions for an interacting electron system.

5.2.1 Ground-state energy and ferromagnetic transition

- The Hartree term represents the interaction between the total densities, while the Fock term represents an effective attraction between the like spins.
- Just like the Hund rule in atoms, a strong repulsion between electrons in a metal can lead to a ferromagnetic state.

Let us assume that the electrons are on a lattice and carry spin-1/2. The electrons interact through $\frac{1}{2} \sum_{i,j} c_{\alpha}^{\dagger}(i) c_{\alpha}(i) V(i-j) c_{\beta}^{\dagger}(j) c_{\beta}(j)$. The Hamiltonian is given by (in momentum space)

$$H = \sum_{\mathbf{k}} \xi_{\mathbf{k}} c_{\alpha\mathbf{k}}^{\dagger} c_{\alpha\mathbf{k}} + \frac{1}{2N_{\text{site}}} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} c_{\alpha\mathbf{k}}^{\dagger} c_{\alpha, \mathbf{k}-\mathbf{q}} V_{\mathbf{q}} c_{\beta\mathbf{k}'}^{\dagger} c_{\beta, \mathbf{k}'+\mathbf{q}} \quad (5.2.1)$$

where N_{site} is the number of sites.

To understand the ground-state properties of the above interacting system, let us use the ground state $|\Psi_0\rangle$ of a non-interacting system as a trial wave function. More precisely, $|\Psi_0\rangle$ is simply a state described by a set of occupation numbers $n_{\mathbf{k}\alpha}$. Here $n_{\mathbf{k}\alpha} = 1$ if the state $\mathbf{k}$ is occupied by an α -spin, and $n_{\mathbf{k}\alpha} = 0$ otherwise. We determine the occupation number $n_{\mathbf{k}\alpha}$ by minimizing the average energy of the trial state.

Using Wick's theorem, we find that $\langle \Psi_0 | H | \Psi_0 \rangle$ contains three terms as follows:

$$\begin{aligned} \langle \Psi_0 | H | \Psi_0 \rangle &= \langle \Psi_0 | H_0 | \Psi_0 \rangle + \frac{1}{2} \sum_{i,j} \langle c_{\alpha}^{\dagger}(i) c_{\alpha}(i) \rangle V(i-j) \langle c_{\beta}^{\dagger}(j) c_{\beta}(j) \rangle \\ &\quad + \frac{1}{2} \sum_{i,j} \langle c_{\alpha}^{\dagger}(i) c_{\beta}(j) \rangle V(i-j) \langle c_{\alpha}(i) c_{\beta}^{\dagger}(j) \rangle \end{aligned}$$

The first term $\langle \Psi_0 | H_0 | \Psi_0 \rangle$ is the kinetic energy, which is minimized when $N_{\uparrow} = N_{\downarrow}$, where $N_{\uparrow}$ is the number of spin-up electrons and $N_{\downarrow}$ is the number of spin-down electrons. The second term is called the Hartree term:

$$\frac{1}{2} \sum_{i,j} \rho(i) V(i-j) \rho(j)$$

which is independent of $N_{\uparrow}/N_{\downarrow}$ if N is fixed. It is clear that the Hartree term is simply the classical potential energy. If we only include the first two terms, then the average energy of the trial state will be minimized at $N_{\uparrow} = N_{\downarrow}$. The (trial) ground state is then a spin singlet and does not break spin-rotation symmetry.

The third term in the above equation is the Fock term, which can be rewritten as

$$\begin{aligned} & \frac{1}{2N_{\text{site}}} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} V_{\mathbf{q}} \langle c_{\alpha\mathbf{k}}^{\dagger} c_{\beta, \mathbf{k}'+\mathbf{q}} \rangle \langle c_{\alpha, \mathbf{k}-\mathbf{q}} c_{\beta\mathbf{k}'}^{\dagger} \rangle \\ &= -\frac{1}{2N_{\text{site}}} \sum_{\mathbf{k}, \mathbf{q}, \alpha} V_{\mathbf{q}} n_{\mathbf{k}\alpha} n_{\mathbf{k}-\mathbf{q}, \alpha} + \frac{1}{2} V(0)N \end{aligned}$$

where N is the total number of electrons. We see that, if we ignore the constant term $\frac{1}{2}V(0)N$, then the Fock term represents an effective attractive interaction between parallel spins, but no interactions between opposite spins. The contribution from the Fock term becomes more negative as $|N_{\uparrow} - N_{\downarrow}|$ increases.

To understand this effective attraction between parallel spins, we note that the average potential energy can be written in terms of density correlation functions as follows:

$$\frac{1}{2} \sum_{\mathbf{i}, \mathbf{j}} V(\mathbf{i} - \mathbf{j}) (\langle \rho_{\uparrow}(\mathbf{i}) \rho_{\uparrow}(\mathbf{j}) \rangle + 2 \langle \rho_{\uparrow}(\mathbf{i}) \rho_{\downarrow}(\mathbf{j}) \rangle + \langle \rho_{\downarrow}(\mathbf{i}) \rho_{\downarrow}(\mathbf{j}) \rangle)$$

where $\rho_{\uparrow}(\mathbf{i})$ and $\rho_{\downarrow}(\mathbf{i})$ are the densities of the spin-up and the spin-down electrons, respectively. Comparing this with the Hartree term

$$\frac{1}{2} \sum_{\mathbf{i}, \mathbf{j}} V(\mathbf{i} - \mathbf{j}) (\langle \rho_{\uparrow}(\mathbf{i}) \rangle \langle \rho_{\uparrow}(\mathbf{j}) \rangle + 2 \langle \rho_{\uparrow}(\mathbf{i}) \rangle \langle \rho_{\downarrow}(\mathbf{j}) \rangle + \langle \rho_{\downarrow}(\mathbf{i}) \rangle \langle \rho_{\downarrow}(\mathbf{j}) \rangle)$$

we find that the Hartree term correctly reproduces the cross-term because $\rho_{\uparrow}(\mathbf{i})$ and $\rho_{\downarrow}(\mathbf{j})$ are independent and $\langle \rho_{\uparrow}(\mathbf{i}) \rho_{\downarrow}(\mathbf{j}) \rangle = \langle \rho_{\uparrow}(\mathbf{i}) \rangle \langle \rho_{\downarrow}(\mathbf{j}) \rangle$. However, the Hartree term overestimates the interaction between the parallel spins. This is because when site $\mathbf{i}$ has an electron, it is less likely to find another electron with the same spin on a nearby site $\mathbf{j}$. In fact the probability will be zero if $\mathbf{j} = \mathbf{i}$ due to the Pauli principle. We find that $\langle \rho_{\uparrow}(\mathbf{i}) \rangle \langle \rho_{\uparrow}(\mathbf{j}) \rangle > \langle \rho_{\uparrow}(\mathbf{i}) \rho_{\uparrow}(\mathbf{j}) \rangle$. The Fock term simply corrects this error and hence represents an effective attraction between parallel spins (assuming that $V(\mathbf{i}) > 0$). As a consequence of the attraction between parallel spins, the Fock term may make a ferromagnetic state (with $N_{\uparrow} \neq N_{\downarrow}$) have a lower energy than the paramagnetic state (with $N_{\uparrow} = N_{\downarrow}$), if the Fock term is larger than the kinetic energy term. This is an example in which an interaction can cause symmetry breaking.

As the trial wave function describes a state of uniform density, the Hartree term is given by

$$\frac{1}{2N_{\text{site}}} V_0 \left(\sum_{\mathbf{k}, \alpha} n_{\mathbf{k}\alpha} \right)^2 = \frac{1}{2N_{\text{site}}} V_0 N^2.$$

Thus, the total energy of the trial state is given by

$$\begin{aligned} & \langle \Psi_{\{n_{\mathbf{k}\alpha}\}} | H | \Psi_{\{n_{\mathbf{k}\alpha}\}} \rangle \\ &= \sum_{\mathbf{k}, \alpha} n_{\mathbf{k}\alpha} (\epsilon_{\mathbf{k}} - \mu + \frac{V_0}{2}) + \frac{V_0}{2N_{\text{site}}} (\sum_{\mathbf{k}, \alpha} n_{\mathbf{k}\alpha})^2 - \frac{1}{N_{\text{site}}} \sum_{\mathbf{k}, \mathbf{q}, \alpha} \frac{V_{\mathbf{q}}}{2} n_{\mathbf{k}\alpha} n_{\mathbf{k}-\mathbf{q}, \alpha} \end{aligned} \quad (5.2.2)$$

If we change $n_{\mathbf{k}\alpha}$ to $n_{\mathbf{k}\alpha} + \delta n_{\mathbf{k}\alpha}$, then the change in the energy is given by

$$\delta E = \sum_{\mathbf{k}, \alpha} \delta n_{\mathbf{k}\alpha} [\epsilon_{\mathbf{k}} - \mu + \rho_0 V_0 + \frac{1}{2} V(0) + \Sigma_{\mathbf{k}\alpha}] \quad (5.2.3)$$

where

$$\Sigma_{\mathbf{k}\alpha} = -\frac{1}{N_{\text{site}}} \sum_{\mathbf{q}, \alpha} V_{\mathbf{q}} n_{\mathbf{k}-\mathbf{q}, \alpha}, \quad (5.2.4)$$

$\rho_0 = N/N_{\text{site}}$ is the number of electrons per site, and we have only kept the terms which are linear in $\delta n_{\mathbf{k}\alpha}$.

The occupation $n_{\mathbf{k}\alpha}$ that minimizes the energy has the property that δE is positive for any $\delta n_{\mathbf{k}\alpha}$. Such a set of occupation numbers satisfies

$$\begin{aligned} n_{\mathbf{k}\alpha} &= 1, & \text{if } \epsilon_{\mathbf{k}} - \mu' + \Sigma_{\mathbf{k}\alpha} < 0 \\ n_{\mathbf{k}\alpha} &= 0, & \text{if } \epsilon_{\mathbf{k}} - \mu' + \Sigma_{\mathbf{k}\alpha} > 0 \end{aligned} \quad (5.2.5)$$

where $\mu' = \mu - \rho_0 V_0 - \frac{1}{2} V(0)$. We see that $\epsilon_{\mathbf{k}} - \mu' + \Sigma_{\mathbf{k}\alpha} = 0$ at the Fermi surface, where $n_{\mathbf{k}\alpha}$ change from 0 to 1. The ground-state occupation numbers $n_{\mathbf{k}\alpha}$ are obtained by solving the coupled eqns (5.2.5) and (5.2.4).

5.2.2 Spectrum of excitations in the Hartree-Fock approximation

After obtaining the ground-state occupation numbers $n_{\mathbf{k}\alpha}$, we can consider the excitations above the ground state. Let us assume that the trial wave function for an excitation is described by a different set of occupation numbers $n_{\mathbf{k}\alpha} + \delta n_{\mathbf{k}\alpha}$. The (average) energy for such an excitation has been calculated above:

$$\delta E = \sum_{\mathbf{k}, \alpha} \delta n_{\mathbf{k}\alpha} [\epsilon_{\mathbf{k}} - \mu' + \Sigma_{\mathbf{k}\alpha}] \quad (5.2.6)$$

We see that, under the Hartree-Fock approximation, the low-lying excitations of the interacting system are described by a free fermion system. In particular, the following statements hold.

1. The many-body eigenstates (the ground state and the low-lying excitations) are labeled by the occupation numbers $n_{\mathbf{k}\alpha} = 0, 1$. The number of low-lying excitations is the same as in free fermion theory.

2. The energy of a many-body eigenstate is given by the sum of the energies of the occupied states. To be more precise, we can assign an energy $\xi_{\mathbf{k}\alpha}^*$ to each momentum state $(\mathbf{k}, \alpha)$ such that the total energy of an eigenstate can be expressed as $\sum_{\mathbf{k}, \alpha} n_{\mathbf{k}\alpha} \xi_{\mathbf{k}}^* + \text{constant}$.
3. The energy for each momentum state is modified by the interaction $\xi_{\mathbf{k}\alpha}^* = \epsilon_{\mathbf{k}} - \mu' + \Sigma_{\mathbf{k}\alpha}$. The correction $\Sigma_{\mathbf{k}\alpha}$ is called the self-energy of the electron.

The above statements can be summarized by the low-energy effective Hamiltonian

$$H_{\text{eff}} = \sum_{\mathbf{k}, \alpha} (\epsilon_{\mathbf{k}} - \mu + \Sigma_{\mathbf{k}\alpha}) c_{\alpha\mathbf{k}}^\dagger c_{\alpha\mathbf{k}} \quad (5.2.7)$$

One can directly derive the above effective Hamiltonian using mean-field theory (see Problem 5.2.2).

For a Coulomb interaction in three dimensions, $V(\mathbf{x}) = e^2/|\mathbf{x}|$, we find that³²

$$\Sigma_{\mathbf{k}\alpha} = -\frac{e^2 k_{F\alpha}}{\pi} \left(1 + \frac{1-y^2}{2y} \ln \left| \frac{1+y}{1-y} \right| \right), \quad y = \frac{k}{k_{F\alpha}} \quad (5.2.8)$$

We note that the renormalized Fermi velocity

$$v_{F\alpha}^*(k) = v_{F\alpha}(k) + \frac{\partial \Sigma_{\mathbf{k}\alpha}}{\partial k}$$

diverges as $-\ln|k - k_{F\alpha}|$ when $k \rightarrow k_{F\alpha}$ (or $y \rightarrow 1$). However, this divergence has not been observed experimentally, and this particular result from the Hartree–Fock approximation turns out to be incorrect.

Problem 5.2.1.

Consider a one-dimensional electron system on a lattice with an on-site potential interaction, so that

$$H = -t \sum_i (c_{\alpha,i}^\dagger c_{\alpha,i+1} + h.c.) + V \sum_i \left(\sum_\alpha c_{\alpha,i}^\dagger c_{\alpha,i} \right)^2$$

1. Find the energy ϵ_k of a single-particle state with momentum k for a non-interacting system with $V = 0$.
2. Use the Hartree–Fock approximation to find the ground-state energy $E(N_\uparrow, N_\downarrow)$, assuming that the numbers of spin-up and spin-down electrons are given by $N_\uparrow$ and

³² As $V_q = 4\pi e^2/q^2$, we have

$$\begin{aligned} \Sigma_{\mathbf{k}\alpha} &= - \int \frac{d^3 \mathbf{q}}{(2\pi)^3} \frac{4\pi e^2}{|\mathbf{k} - \mathbf{q}|^2} n_{\mathbf{q}\alpha} = -\frac{e^2}{\pi} \int_0^{k_{F\alpha}} q^2 dq \int_{-1}^{+1} \frac{dt}{k^2 + q^2 - 2kqt} \\ &= -\frac{e^2}{\pi k} \int_0^{k_{F\alpha}} q dq \ln \left| \frac{k+q}{k-q} \right| = -\frac{e^2 k_{F\alpha}}{\pi} \left(1 + \frac{1-y^2}{2y} \ln \left| \frac{1+y}{1-y} \right| \right) \end{aligned}$$

$N_{\downarrow}$, respectively. You may assume that the total number of electrons N is less than the number of lattice sites N_{site} .

3. Consider the energy $E(N_{\uparrow}, N_{\downarrow})$ in the neighborhood of $N_{\uparrow} - N_{\downarrow} = 0$ and for fixed N . Show that $E(N_{\uparrow}, N_{\downarrow})$ has a local minimum at $N_{\uparrow} - N_{\downarrow} = 0$ if $V < V_0$, and a local maximum at $N_{\uparrow} - N_{\downarrow} = 0$ if $V > V_0$. Find the critical value V_0 .
4. Calculate the spin susceptibility χ of the spin-unpolarized state for $V < V_0$. What is the behavior of χ as $V \rightarrow V_0$?
5. Minimize the ground-state energy obtained above with fixed N . Determine the critical value V_c of V beyond which the system becomes a ferromagnet. Determine the value of the total spin S_{z0} in the minimized ground state, assuming that the spin of the ferromagnet is pointing in the z direction.
6. Calculate $\Sigma_{k\alpha}$ for both $V < V_c$ and $V > V_c$, and determine the energy spectrum of the excitations above the minimized ground state.
7. Find the minimum energy cost to create an excitation of flipping one spin (i.e. change the total spin S_z by 1), for both $V < V_c$ and $V > V_c$. Comment on your result (do you believe it?).

Problem 5.2.2.

Derive eqn (5.2.7) by replacing a pair $c_{\alpha k}^{\dagger} c_{\beta k'}$ by its average $\langle c_{\alpha k}^{\dagger} c_{\beta k'} \rangle$. (This is a mean-field approximation.) There are several contributions from different replacements. Note that $\langle H_{\text{eff}} \rangle$ is different from the ground-state energy $\langle H \rangle$ calculated in the Hartree–Fock approximation. Include a constant term (which can be written in terms of $\langle c_{\alpha k}^{\dagger} c_{\beta k'} \rangle$), so that $\langle H_{\text{eff}} \rangle = \langle H \rangle$.

5.3 Landau Fermi liquid theory

- Low-energy excitations of an interacting fermion system are described by free quasiparticles. In other words, interacting fermions $\approx$ free fermions.
- Perturbation theory works even when the interaction is much much larger than the level spacing.

Landau Fermi liquid theory (Landau, 1956, 1959) is one of the two cornerstones of traditional many-body theory. It essentially states that a metal formed by interacting electrons behaves almost like a free fermion system. Landau Fermi liquid theory is very useful because it describes (almost) all known metals. It also forms the foundation of our understanding of many non-metallic states, such as superconductors, anti-ferromagnetic states, etc. These non-metallic states are realized as certain instabilities of a Fermi liquid. On the other hand, Landau Fermi liquid theory is very mysterious because the Coulomb interaction between electrons in ordinary metals is as large as the Fermi energy. It is much larger than

the energy-level spacing near the Fermi energy. Perturbation theory (in the usual sense) breaks down for such a strong interaction. It is hard to see why a gapless system with such a strong interaction resembles a non-interacting system.

To appreciate the brilliance of Landau Fermi liquid theory, let us look at the many-body Hamiltonian of interacting electrons, namely

$$H = \sum_i \left(\frac{\hbar^2}{2m} \partial_{\mathbf{x}_i}^2 + U(\mathbf{x}_i) \right) + \sum_{i < j} \frac{e^2}{|\mathbf{x}_i - \mathbf{x}_j|}$$

It is hopeless for a theorist to solve such a ‘nasty’ system, not to mention to guess that such a system behaves almost like a free electron system. Certainly, condensed matter physicists did not provide such a bold guess. It is nature itself who hints to us over and over again that metals behave just like a free electron system, despite the strong Coulomb interaction. Even now, I am amazed that so many metals can be described by Landau Fermi liquid theory, and puzzled by the difficulty to find a metal that cannot be described by Landau Fermi liquid theory.

On a technical level, Landau Fermi liquid theory implies that a perturbative expansion in interaction works, despite the interaction being much larger than the level spacing. Due to this, to understand the physical properties of a metal, we can start with a free electron system and use perturbation theory to calculate various quantities. This approach was so successful that Landau Fermi liquid theory became the ‘standard model’ for interacting fermion systems. The dominant position of Landau Fermi liquid theory was only challenged not so long ago by fractional quantum Hall states in 1982, and by high- T_c superconductors in 1987.

5.3.1 Basic assumptions and their consequences

- The concept of a quasiparticle.

Landau Fermi liquid theory has only one really basic assumption.

The low-energy eigenstates (including the ground state and the low-lying excitations) are labeled by a set of quantum numbers $n_{\mathbf{k}\alpha} = 0, 1$, which are called the ‘occupation numbers’.

As the excitations are in one-to-one correspondence with those in the free fermion system, we can use the language of the free fermion systems to describe the excitations in the Landau Fermi liquid, and we call these excitations quasiparticle excitations. The energy of an eigenstate is a function of $n_{\mathbf{k}\alpha}$. Expanding around the ground-state occupation numbers $n_{0,\mathbf{k}\alpha}$, we may write

$$E_{\{n_{\mathbf{k}\alpha}\}} = E_g + \sum_{\mathbf{k},\alpha} \delta n_{\mathbf{k}\alpha} \xi_{\mathbf{k}\alpha}^* + \frac{1}{2V} \sum_{\mathbf{k}\alpha, \mathbf{k}'\beta} f(\mathbf{k}, \alpha; \mathbf{k}', \beta) \delta n_{\mathbf{k}\alpha} \delta n_{\mathbf{k}'\beta} \quad (5.3.1)$$

where $\xi_{\mathbf{k}\alpha}^*$ is the energy of the quasiparticles, and $f(\mathbf{k}, \alpha; \mathbf{k}', \beta)$ represents the interaction between quasiparticles. Here $f(\mathbf{k}, \alpha; \mathbf{k}', \beta)$ is called the Fermi liquid function, which is also very important in determining the low-energy properties of the system. We note that the energy obtained in the Hartree-Fock approximation has the form (5.3.1) (see eqn (5.2.2)). The quadratic term in $\delta n_{\mathbf{k}\alpha}$ which we ignored in the last section has the form

$$\frac{1}{2V} V_0 \left(\sum_{\mathbf{k}, \alpha} \delta n_{\mathbf{k}\alpha} \right)^2 - \frac{1}{2V} \sum_{\mathbf{k}, \mathbf{q}, \alpha} V_{\mathbf{q}} \delta n_{\mathbf{k}\alpha} \delta n_{\mathbf{k}-\mathbf{q}, \alpha}$$

Thus, we have

$$\begin{aligned} \xi_{\mathbf{k}\alpha}^* &= \epsilon_{\mathbf{k}} - \mu + \Sigma_{\mathbf{k}\alpha} \\ f(\mathbf{k}, \alpha; \mathbf{k}', \beta) &= V_0 - V_{\mathbf{k}-\mathbf{k}'} \delta_{\alpha\beta} \end{aligned} \quad (5.3.2)$$

within the Hartree-Fock approximation.

Equation (5.3.1) determines the energies of all low-lying excitations, and many low-energy properties of the interacting fermion system can be expressed in terms of the quasiparticle energy ξ^* and the Fermi liquid function f . Creating an excitation above the ground state by changing an occupation number $n_{\mathbf{k}\alpha}$ from 0 to 1 (or from 1 to 0) costs an energy $\xi_{\mathbf{k}\alpha}^*$. However, in the presence of other excitations (say, due to finite temperatures) the energy cost is different from $\xi_{\mathbf{k}\alpha}^*$ due to the interaction between quasiparticles. From eqn (5.3.1), we find the new energy cost to be

$$\epsilon_{\mathbf{k}\alpha}^T = \xi_{\mathbf{k}\alpha}^* + \frac{1}{V} \sum_{\mathbf{k}'\beta} f(\mathbf{k}, \alpha; \mathbf{k}', \beta) \delta n_{\mathbf{k}'\beta}$$

The momentum change due to the creation of such a quasiparticle is the same as the free fermion system, i.e. $\Delta \mathbf{K} = \mathbf{k}$, and the total momentum of a state described by $n_{\mathbf{k}\alpha}$ is

$$\mathbf{K} = \sum_{\mathbf{k}\alpha} \mathbf{k} n_{\mathbf{k}\alpha}$$

This result can be obtained from the second assumption of Landau Fermi liquid theory, which is stated as follows.

As we turn off the interaction, the low-energy eigenstates adiabatically change into the corresponding eigenstates of the free fermion system labeled by the same occupation numbers $n_{\mathbf{k}\alpha}$.

As the momenta are quantized, they cannot change during the adiabatic turning on of the interaction.

From the momentum and the energy, we find the velocity of the quasiparticle to be

$$\mathbf{v}^T(\mathbf{k}) = \partial_{\mathbf{k}} \epsilon_{\mathbf{k}\alpha}^T$$

At zero temperature, it changes to

$$\mathbf{v}^*(\mathbf{k}) = \partial_{\mathbf{k}} \xi_{\mathbf{k}\alpha}^*$$

We note that, near the Fermi surface, the quasiparticle behaves like a particle with mass

$$m_{\alpha}^* = \frac{k_{F\alpha}}{|\mathbf{v}^*(\mathbf{k}_{F\alpha})|}$$

Here m^* is called the effective mass of the quasiparticle, which may not be equal to the mass of the original electrons.

We see that the energy of the quasiparticle ϵ^T depends on the presence of other quasiparticles and is not a constant. However, because the sum $\frac{1}{V} \sum_{\mathbf{k}\alpha, \mathbf{k}'\beta} f(\mathbf{k}, \alpha; \mathbf{k}', \beta) \delta n_{\mathbf{k}'\beta}$ effectively averages over many quasiparticles (assuming that $f(\mathbf{k}, \alpha; \mathbf{k}', \beta)$ is not too singular), its thermal fluctuations are small and ϵ^T can be treated as a constant:

$$\epsilon_{\mathbf{k}\alpha}^T = \xi_{\mathbf{k}\alpha}^* + \frac{1}{V} \sum_{\mathbf{k}\alpha, \mathbf{k}'\beta} f(\mathbf{k}, \alpha; \mathbf{k}', \beta) \langle \langle \delta n_{\mathbf{k}'\beta} \rangle \rangle$$

where $\langle \langle \dots \rangle \rangle$ represents a thermal average. From this, we obtain the average occupation number

$$\langle \langle n_{\mathbf{k}\alpha} \rangle \rangle = n_F(\epsilon_{\mathbf{k}\alpha}^T) = \frac{1}{e^{\epsilon_{\mathbf{k}\alpha}^T/T} + 1}$$

Certainly, $\epsilon_{\mathbf{k}\alpha}^T$ depends on $\langle \langle n_{\mathbf{k}\alpha} \rangle \rangle$, and we have to solve the above equation to obtain $\langle \langle n_{\mathbf{k}\alpha} \rangle \rangle$. Given $\langle \langle n_{\mathbf{k}\alpha} \rangle \rangle$, all of the thermodynamical properties of the Fermi liquids can be determined.

To calculate the specific heat at low temperatures, we note that the average occupation numbers can be approximated by

$$\langle \langle n_{\mathbf{k}\alpha} \rangle \rangle = n_F(\epsilon_{\mathbf{k}\alpha}^T) = \frac{1}{e^{\epsilon_{\mathbf{k}\alpha}^T/T} + 1} \approx \frac{1}{e^{\xi_{\mathbf{k}\alpha}^*/T} + 1}$$

because $(\epsilon_{\mathbf{k}\alpha}^T - \xi_{\mathbf{k}\alpha}^*) \sim \langle \langle \delta n_{\mathbf{k}\alpha} \rangle \rangle$ vanishes as T^2 when $T \rightarrow 0$ (note that $\delta n_{\mathbf{k}\alpha}$ changes its sign across the Fermi surface). Therefore, $\langle \langle n_{\mathbf{k}\alpha} \rangle \rangle$ are the same as the average occupation numbers of a mass- m^* free electron system. The specific heat C_V is given by the same formula. In terms of the density of states (see eqn (4.2.12) but with m replaced by m^*), we have $C_V = 2 \frac{2\pi^2}{3} N(E_F) T$. The interaction contributes to a term of order T^3/E_F^3 . We see that the specific heat depends only on

the effective mass m^* . This gives us a way to experimentally measure the effective mass m^* of a Landau Fermi liquid. In Problem 5.3.1, we will see that other low-energy properties, such as the compressibility χ , depend on both the effective mass m^* and the Fermi liquid function f .

We have mentioned that, in Fermi liquid theory, a quasiparticle above the ground state is created by changing $n_{\mathbf{k}\alpha}$ from 0 to 1. Can we create such a quasiparticle state $|\Psi_{\mathbf{k}}\rangle$ by applying $c_{\mathbf{k}\alpha}^\dagger$ to the ground-state wave function $|\Psi_0\rangle$? The answer is no. Although $c_{\mathbf{k}\alpha}^\dagger|\Psi_0\rangle$ carries the same momentum as the quasiparticle state, it may not be an energy eigenstate. The state $c_{\mathbf{k}\alpha}^\dagger|\Psi_0\rangle$ may be a superposition of a state with one quasiparticle, and a state with two quasiparticles and one quasihole, etc. All of these states can carry the same momentum $\mathbf{k}$ but different energies. Therefore, the overlap between $c_{\mathbf{k}\alpha}^\dagger|\Psi_0\rangle$ and $|\Psi_{\mathbf{k}}\rangle$ given by

$$Z = |\langle\Psi_{\mathbf{k}}|c_{\mathbf{k}\alpha}^\dagger|\Psi_0\rangle|^2$$

may not be 1. The third assumption of Landau Fermi liquid theory is stated as follows.

The overlap $Z = |\langle\Psi_{\mathbf{k}}|c_{\mathbf{k}\alpha}^\dagger|\Psi_0\rangle|^2$ is not zero in the thermodynamic limit (i.e. the $\mathcal{V} \rightarrow \infty$ limit) for $\mathbf{k}$ near the Fermi surface.

This implies that the electron spectral function $\pi^{-1}\text{Im}G(\mathbf{k}, \omega)$ at $T = 0$ contains a δ -function: $Z\delta(\omega - \xi_{\mathbf{k}\alpha}^*)$. The energies of the states with two quasiparticles and one quasihole can spread over a finite range. These states contribute a finite background to the spectral function.

Problem 5.3.1.

Calculate the compressibility χ and the spin susceptibility χ_s of a Landau Fermi liquid. Calculate the ratios between χ , χ_s , and C_V , and compare them with the corresponding ratios for a free electron system.

5.3.2 Boltzmann equation of a Fermi liquid at $T = 0$

In Section 5.1.3, we have tried to use density fluctuations to describe Fermi liquids. The hydrodynamical approach works in one dimension but fails badly beyond one dimension. The reason is that beyond one dimension there are a lot more particle-hole excitations across the Fermi surface than a single density mode can possibly describe. On the other hand, Landau Fermi liquid theory suggests that a uniform Fermi liquid can be completely described by the occupation number $n_{\mathbf{k}}$. The density $\sum_{\mathbf{k}} n_{\mathbf{k}}$ is just one combination of $n_{\mathbf{k}}$. This suggests that, if we can generalize Landau theory to non-uniform and time-dependent $n_{\mathbf{k}}$, then we can obtain the hydrodynamical theory that provides a complete description of Fermi

liquids beyond one dimension. In this section, we are going to develop such a theory. The hydrodynamical theory can also be regarded as bosonization beyond one dimension (Luther, 1979; Haldane, 1992; Houghton and Marston, 1993; Neto and Fradkin, 1994).

Let us consider a spin-1/2 interacting electron system. The ground state is described by the occupation number $n_{0,\mathbf{k}\alpha}$. A collective excited state is described by $n_{\mathbf{k}\alpha}(\mathbf{x}, t)$. Let $\delta n_{\mathbf{k}\alpha}(\mathbf{x}, t) = n_{\mathbf{k}\alpha}(\mathbf{x}, t) - n_{0,\mathbf{k}\alpha}$. Here we assume that $\delta n_{\mathbf{k}\alpha}(\mathbf{x}, t)$ is a smooth function of $\mathbf{x}$ and t , and treat it as a local constant. The quasiparticle energy on the background of a collective excited state is given by

$$\tilde{\epsilon}_{\mathbf{k}\alpha}(\mathbf{x}, t) = \xi_{\mathbf{k}\alpha}^* + \frac{1}{\mathcal{V}} \sum_{\mathbf{k}'\beta} f(\mathbf{k}, \alpha; \mathbf{k}', \beta) \delta n_{\mathbf{k}'\beta}(\mathbf{x}, t) \quad (5.3.3)$$

Thus, the changes in the position and the momentum of the quasiparticle are given by

$$\dot{\mathbf{x}} = \partial_{\mathbf{k}} \tilde{\epsilon}_{\mathbf{k}\alpha}(\mathbf{x}, t), \quad \dot{\mathbf{k}} = -\partial_{\mathbf{x}} \tilde{\epsilon}_{\mathbf{k}\alpha}(\mathbf{x}, t). \quad (5.3.4)$$

The motion of the quasiparticle causes the following change in $n(\mathbf{x}, t)$: $\partial n_{\mathbf{k}\alpha} / \partial t = -(\partial n_{\mathbf{k}\alpha} / \partial \mathbf{x}) \dot{\mathbf{x}} - (\partial n_{\mathbf{k}\alpha} / \partial \mathbf{k}) \dot{\mathbf{k}} = -(\partial n_{\mathbf{k}\alpha} / \partial \mathbf{x}) \cdot (\partial \tilde{\epsilon} / \partial \mathbf{k}) + (\partial n_{\mathbf{k}\alpha} / \partial \mathbf{k}) \cdot (\partial \tilde{\epsilon} / \partial \mathbf{x})$. Such a hydrodynamical equation is also written as follows: $dn/dt = (\partial n_{\mathbf{k}\alpha} / \partial t) + (\partial n_{\mathbf{k}\alpha} / \partial \mathbf{x}) \cdot (\partial \tilde{\epsilon} / \partial \mathbf{k}) - (\partial n_{\mathbf{k}\alpha} / \partial \mathbf{k}) \cdot (\partial \tilde{\epsilon} / \partial \mathbf{x}) = 0$. The Boltzmann equation is obtained by letting dn/dt be equal to the redistribution of n due to the additional scattering caused by electron-electron interactions:

$$\frac{\partial n_{\mathbf{k}\alpha}}{\partial t} + \frac{\partial n_{\mathbf{k}\alpha}}{\partial \mathbf{x}} \cdot \frac{\partial \tilde{\epsilon}}{\partial \mathbf{k}} - \frac{\partial n_{\mathbf{k}\alpha}}{\partial \mathbf{k}} \cdot \frac{\partial \tilde{\epsilon}}{\partial \mathbf{x}} = I[n_{\mathbf{k}\alpha}]$$

The Boltzmann equation describes the dynamics of collective excitations.

As an application of the Boltzmann equation, let us calculate the current carried by a quasiparticle. We know that the velocity of the quasiparticle is $\mathbf{v}^* = \partial \xi_{\mathbf{k}}^* / \partial \mathbf{k}$. So, naively, one expects that the current should be $\mathcal{V}^{-1} \sum_{\mathbf{k}} \delta n_{\mathbf{k}\alpha} \mathbf{v}^*(\mathbf{k})$. It turns out that the quasiparticle interaction has a non-trivial correction to the current. The electron number conservation implies that $\sum_{\mathbf{k}} I[n] = 0$. Thus, $\sum_{\mathbf{k}} dn_{\mathbf{k}\alpha} / dt = 0$. This allows us to show that

$$\partial_t \mathcal{V}^{-1} \sum_{\mathbf{k}} n_{\mathbf{k}\alpha}(\mathbf{x}, t) + \partial_{\mathbf{x}} \mathbf{J}(\mathbf{x}, t) = 0$$

where

$$\mathbf{J}(\mathbf{x}, t) = \mathcal{V}^{-1} \sum_{\mathbf{k}} n_{\mathbf{k}\alpha}(\mathbf{x}, t) \frac{\partial \tilde{\epsilon}}{\partial \mathbf{k}} \quad (5.3.5)$$

As

$$\rho(\mathbf{x}, t) = \mathcal{V}^{-1} \sum_{\mathbf{k}} n_{\mathbf{k}\alpha}(\mathbf{x}, t)$$

is the electron number density, $\mathbf{J}$ can be interpreted as the number current density. If we only keep the linear δn term, then $\mathbf{J}$ becomes

$$\mathbf{J}(\mathbf{x}, t) = \mathcal{V}^{-1} \sum_{\mathbf{k}} \delta n_{\mathbf{k}\alpha}(\mathbf{x}, t) \tilde{\mathbf{v}} \\ \tilde{\mathbf{v}}(\mathbf{k}) = \mathbf{v}^*(\mathbf{k}) + \int \frac{d^d \mathbf{k}'}{(2\pi)^d} \sum_{\beta} f(\mathbf{k}, \alpha; \mathbf{k}', \beta) \mathbf{v}^*(\mathbf{k}') \delta(\xi_{\mathbf{k}'}^*) \quad (5.3.6)$$

The second term in the expression of $\tilde{\mathbf{v}}$ is a correction to the free fermion result. It is called the drag term or the back-flow term. To understand the correction, we note that a quasiparticle at $\mathbf{k}$ also causes a slight change in the velocity of the quasiparticles in the filled Fermi sea, as a result of interaction between the quasiparticles. Thus, the electrons in the Fermi sea also contribute to the number current density. This is the source of the drag/back-flow term.

In the relaxation-time approximation, we assume that $I[n] = -\tau^{-1} \delta n$. To include an external force $\mathbf{F}$ on the quasiparticle, we need to replace $-\partial_{\mathbf{x}} \tilde{\epsilon}$ in eqn (5.3.4) by $\mathbf{F} - \partial_{\mathbf{x}} \tilde{\epsilon}$. The resulting Boltzmann equation becomes

$$\frac{\partial n_{\mathbf{k}\alpha}}{\partial t} + \frac{\partial n_{\mathbf{k}\alpha}}{\partial \mathbf{x}} \cdot \frac{\partial \tilde{\epsilon}}{\partial \mathbf{k}} + \frac{\partial n_{\mathbf{k}\alpha}}{\partial \mathbf{k}} \cdot \left(\mathbf{F} - \frac{\partial \tilde{\epsilon}}{\partial \mathbf{x}} \right) = -\tau^{-1} \delta n_{\mathbf{k}\alpha} \quad (5.3.7)$$

We can use the above equation to calculate many different transport properties.

Within Landau Fermi liquid theory, the quasiparticle is assumed to have infinite lifetime and does not decay into any other states. For a uniform $\delta n_{\mathbf{k}\alpha}$ and vanishing $\mathbf{F} = 0$, eqn (5.3.7) is reduced to $\frac{\partial n_{\mathbf{k}\alpha}}{\partial t} = -\tau^{-1} \delta n_{\mathbf{k}\alpha}$. We see that τ is the quasiparticle lifetime. Thus, the collision term $I[n]$ is assumed to be zero in Landau Fermi liquid theory. For a real interacting electron system, $I[n]$ behaves like $|k_F - k|^2 \delta n$ (see Section 5.4.4) and can be ignored at low energies.

Problem 5.3.2.

Prove eqns (5.3.5) and (5.3.6). (Hint: $\frac{\partial n}{\partial \mathbf{x}} \cdot \frac{\partial \tilde{\epsilon}}{\partial \mathbf{k}} - \frac{\partial n}{\partial \mathbf{k}} \cdot \frac{\partial \tilde{\epsilon}}{\partial \mathbf{x}} = \frac{\partial}{\partial \mathbf{x}} (n \frac{\partial \tilde{\epsilon}}{\partial \mathbf{k}}) - \frac{\partial}{\partial \mathbf{k}} (n \frac{\partial \tilde{\epsilon}}{\partial \mathbf{x}})$. Also, $\partial_{\mathbf{k}} n_{0,\mathbf{k}} = -\mathbf{v}^*(\mathbf{k}) \delta(\xi_{\mathbf{k}}^*)$.)

5.3.3 Hydrodynamical theory of a Fermi liquid

In this section, we will obtain a reduced Boltzmann equation, which serves as the classical equation of motion for the hydrodynamical description of a Fermi liquid (Kim *et al.*, 1995). Such a reduced Boltzmann equation generalizes the equation of motion of the density mode discussed in Section 5.1.3.

The key to obtaining the reduced Boltzmann equation is to use the Fermi surface displacement h to describe the collective fluctuations (see Fig. 5.4(a)). One may wonder why we do not use the occupation number $n_{\mathbf{k}\alpha}$ to describe the collective fluctuations. One important step in developing a hydrodynamical theory of a

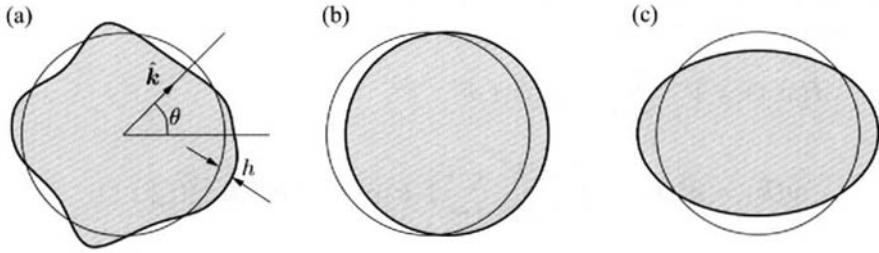


FIG. 5.4. (a) Collective fluctuations in a Fermi liquid can be described by the displacement of the Fermi surface. (b) The $l = 1$ mode corresponds to a dipole fluctuation, and (c) the $l = 2$ mode corresponds to a quadrupole fluctuation.

Fermi liquid is to identify the classical field that we can use to formulate the classical hydrodynamical theory. The uniform occupation numbers $n_{\mathbf{k}\alpha}$ can already describe all excited states. After including the spatial dependence, $n_{\mathbf{k}\alpha}(\mathbf{x}, t)$ will be an over complete set of variables. It turns out that h , with its spatial dependence, is a proper choice.

For simplicity, we limit our discussion to two-dimensional systems. We will also suppress the spin index. The displacement h and the occupation number $n_{\mathbf{k}}$ are related as follows:

$$\tilde{\rho}(\theta) = \int \frac{k dk}{(2\pi)^2} \delta n_{\mathbf{k}=\hat{k}}, \quad h(\theta) = (2\pi)^2 \tilde{\rho}(\theta) / k_F$$

where $\hat{k} = \cos(\theta)\mathbf{x} + \sin(\theta)\mathbf{y}$ is the unit vector in the θ direction (see Fig. 5.4(a)). It turns out that $\tilde{\rho}(\theta)$ is more convenient. So we will use $\tilde{\rho}$ instead of h . From the definition, we see that $\int d\theta \tilde{\rho}(\theta)$ is the fluctuation of the total density of the electrons, and it corresponds to the ρ in eqn (5.1.2).

For small fluctuations, h is close to zero and the quasiparticle energy (5.3.3) can be simplified to give

$$\tilde{\epsilon}_{\mathbf{k}}(\mathbf{x}, t) = \xi_{\mathbf{k}}^* + \int d\theta' f(\theta, \theta') \tilde{\rho}(\theta', \mathbf{x}, t) \quad (5.3.8)$$

where θ is the angle of $\mathbf{k}$ and $f(\theta, \theta') = f(k_F \hat{k}, k_F \hat{k}')$ is the Fermi function between two points on the Fermi surface. The reduced Boltzmann equation can be obtained from eqn (5.3.7) by performing the integration $\int \frac{dk}{(2\pi)^2}$ on both sides of

the equation. Assuming rotational symmetry, we find that³³

$$\frac{\partial \tilde{\rho}}{\partial t} + \frac{\partial \tilde{\rho}}{\partial \mathbf{x}} \cdot \frac{\partial \tilde{\epsilon}}{\partial \mathbf{k}} + k_F^{-1} \left(\hat{\mathbf{k}}_{\perp} \frac{\partial \tilde{\rho}}{\partial \theta} - \hat{\mathbf{k}} \tilde{\rho} - \frac{\hat{\mathbf{k}} k_F^2}{(2\pi)^2} \right) \cdot \left(\mathbf{F} - \frac{\partial \tilde{\epsilon}}{\partial \mathbf{x}} \right) = -\tau^{-1} \tilde{\rho} \quad (5.3.9)$$

where $\hat{\mathbf{k}}_{\perp}$ is the unit vector which is perpendicular to $\hat{\mathbf{k}}$ and $\tilde{\rho}$ is a function $\tilde{\rho}(\theta, \mathbf{x}, t)$. The above can be viewed as a classical equation of motion for a field in $(1+2)$ -dimensional *space* parametrized by $(\theta, \mathbf{x})$. The linearized equation has the form

$$\frac{\partial \tilde{\rho}}{\partial t} + v_F^* \hat{\mathbf{k}} \cdot \frac{\partial \tilde{\rho}}{\partial \mathbf{x}} - \frac{\hat{\mathbf{k}} k_F}{(2\pi)^2} \cdot \left(\mathbf{F} - \int d\theta' f(\theta, \theta') \frac{\partial \tilde{\rho}(\theta', \mathbf{x}, t)}{\partial \mathbf{x}} \right) = -\tau^{-1} \tilde{\rho} \quad (5.3.10)$$

When $\mathbf{F} = \tau^{-1} = 0$, eqn (5.3.10) can be viewed as an equation of motion for a Fermi liquid which has a form:

$$\frac{\partial \tilde{\rho}}{\partial t} + v_F^* \hat{\mathbf{k}} \cdot \frac{\partial \tilde{\rho}}{\partial \mathbf{x}} + \frac{k_F}{(2\pi)^2} \int d\theta' f(\theta, \theta') \hat{\mathbf{k}} \cdot \frac{\partial \tilde{\rho}(\theta', \mathbf{x}, t)}{\partial \mathbf{x}} = 0 \quad (5.3.11)$$

The corresponding Fermi liquid energy (or Hamiltonian) (5.3.1) can also be written in terms of $\tilde{\rho}$:

$$\begin{aligned} H &= E_g + \int d^2 \mathbf{x} d\theta \tilde{\rho} \frac{h v_F^*}{2} + \frac{1}{2} \int d^2 \mathbf{x} d\mathbf{k} d\mathbf{k}' f(\mathbf{k}, \mathbf{k}') \delta n_{\mathbf{k}} \delta n_{\mathbf{k}'} \\ &= E_g + \int d^2 \mathbf{x} d\theta \frac{2\pi^2 v_F^*}{k_F} \tilde{\rho}^2 + \frac{1}{2} \int d^2 \mathbf{x} d\theta d\theta' f(\theta, \theta') \tilde{\rho}(\theta) \tilde{\rho}(\theta') \end{aligned} \quad (5.3.12)$$

The Lagrangian that reproduces the equation of motion and the Hamiltonian is given by

$$\begin{aligned} L &= \int d^2 \mathbf{x} d\theta \left(\frac{1}{2k_F} \hat{\mathbf{k}} \cdot \partial_{\mathbf{x}} \tilde{\varphi} \partial_t \tilde{\varphi} - \frac{v_F^*}{2k_F} (\hat{\mathbf{k}} \cdot \partial_{\mathbf{x}} \tilde{\varphi})^2 \right) \\ &\quad - \frac{1}{8\pi^2} \int d^2 \mathbf{x} d\theta d\theta' f(\theta, \theta') \hat{\mathbf{k}} \cdot \partial_{\mathbf{x}} \tilde{\varphi}(\theta, \mathbf{x}, t) \hat{\mathbf{k}}' \cdot \partial_{\mathbf{x}} \tilde{\varphi}(\theta', \mathbf{x}, t) \end{aligned} \quad (5.3.13)$$

where $\tilde{\varphi}$ is related to $\tilde{\rho}$ according to $\tilde{\rho}(\theta, \mathbf{x}, t) = (2\pi)^{-1} \hat{\mathbf{k}} \cdot \frac{\partial \tilde{\varphi}(\theta, \mathbf{x}, t)}{\partial \mathbf{x}}$. Equation (5.3.13) is a hydrodynamical description of a two-dimensional Fermi liquid, or,

³³ Note that $\frac{\partial n}{\partial \mathbf{k}} = -\hat{\mathbf{k}} \delta(|\mathbf{k}| - k_F) + \frac{\partial \delta n}{\partial \mathbf{k}}$. We find that $\int \frac{k dk}{(2\pi)^2} \hat{\mathbf{k}} \delta(|\mathbf{k}| - k_F) = \frac{k_F \hat{\mathbf{k}}}{(2\pi)^2}$ and

$$\int \frac{k dk}{(2\pi)^2} \left(\hat{\mathbf{k}}_{\perp} k_F^{-1} \frac{\partial \delta n}{\partial \theta} + \hat{\mathbf{k}} \frac{\partial \delta n}{\partial k} \right) = \hat{\mathbf{k}}_{\perp} k_F^{-1} \frac{\partial \tilde{\rho}}{\partial \theta} - \int \frac{k dk}{(2\pi)^2} k^{-1} \hat{\mathbf{k}} \delta n = k_F^{-1} (\hat{\mathbf{k}}_{\perp} \frac{\partial \tilde{\rho}}{\partial \theta} - \hat{\mathbf{k}} \tilde{\rho})$$

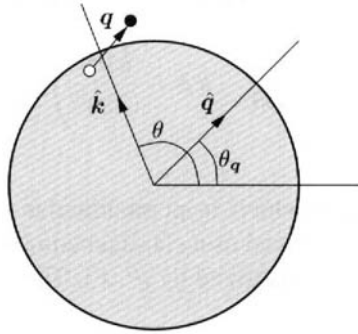


FIG. 5.5. A particle-hole excitation of momentum q at the angle θ .

in other words, a bosonization of a two-dimensional Fermi liquid. Comparing this with the hydrodynamical description discussed in Section 5.1.3, we see that eqn (5.3.13) contains more than just the density mode.

Problem 5.3.3.

In this section, we have derived the reduced Boltzmann equation and the associated Hamiltonian for a two-dimensional rotationally-invariant system. Generalize these results (eqns (5.3.11) and (5.3.12)) to a two-dimensional system with no rotational symmetry.

Problem 5.3.4.

- Derive the bosonized Lagrangian (the one similar to eqn (5.3.13)) for a one-dimensional Fermi liquid.
- Consider eqn (5.1.3) in one dimension. Introduce ϕ through $\rho = (2\pi)^{-1} \partial_x \phi$. Write eqn (5.1.3) in terms of ϕ . Find the canonical momentum of ϕ (denoted by $\tilde{\phi}$), and the Hamiltonian H in terms of ϕ and $\tilde{\phi}$. Compare the action $S = \int dt (\tilde{\phi} \dot{\phi} - H)$ with the result in (a).

5.3.4 Application of the hydrodynamical description of a Fermi liquid

As the first application of the bosonized description of a Fermi liquid, let us calculate the spectrum of the collective modes. In q space, the Fermi liquid equation of motion (5.3.11) becomes

$$i \partial_t \tilde{\rho} = q \cos(\theta_q - \theta) \int d\theta' \left(v_F^* \delta(\theta - \theta') + \frac{k_F}{(2\pi)^2} f(\theta, \theta') \right) \tilde{\rho}(\theta', q, t) \quad (5.3.14)$$

where θ_q is the angle of q (see Fig. 5.5).

Equation (5.3.14) can be solved easily if $f(\theta, \theta') = 0$. The eigenmode has the form $\tilde{\rho}(\theta, q, t) = \delta(\theta - \theta_0) \delta(q - q_0) e^{-i\omega(q_0, \theta_0)t}$, with a frequency $\omega(q_0, \theta_0) = v_F^* q_0 \cos(\theta_{q_0} - \theta_0)$. Such an eigenmode describes an excitation with momentum q_0 . Depending on θ , the energy of such an excitation ranges for 0 to $v_F^* q_0$, which

agrees with the energy range of a particle-hole excitation of the same momentum (see Fig. 4.11). In fact, the eigenmode corresponds to the particle-hole excitation in Fig. 5.5.

To find the energies of the collective mode for nonzero $f(\theta, \theta')$, we note that eqn (5.3.14) can be rewritten as

$$iq^{-1}\partial_t\tilde{\rho}(\theta, \mathbf{q}, t) = KM\tilde{\rho}(\theta, \mathbf{q}, t) \quad (5.3.15)$$

where the two real symmetric operators K and M are given by

$$\begin{aligned} K(\theta, \theta') &= \cos(\theta_{\mathbf{q}} - \theta)\delta(\theta - \theta') \\ M(\theta, \theta') &= v_F^*\delta(\theta - \theta') + \frac{k_F}{(2\pi)^2}f(\theta, \theta') \end{aligned} \quad (5.3.16)$$

From the energy of the Fermi liquid, see eqn (5.3.12), we find that the stability of the Fermi liquid requires M to be positive definite. Thus, we can write M as $M = \tilde{M}\tilde{M}^\top$. Letting $u = \tilde{M}^\top\tilde{\rho}$, then eqn (5.3.15) becomes $i\partial_t u = q\tilde{M}^\top K\tilde{M}u$. The energies of the collective excitations with a fixed momentum $\mathbf{q}$ are given by the eigenvalues of the real symmetric matrix $q\tilde{M}^\top K\tilde{M}$.

The rotational symmetry requires that $f(\theta, \theta') = f(\theta - \theta')$. Because the energy spectrum does not depend on the direction of $\mathbf{q}$, we can choose $\mathbf{q}$ to be in the x direction, i.e. $\mathbf{q} = q\mathbf{x}$ and $\theta_{\mathbf{q}} = 0$. Introducing (see Fig. 5.4)

$$\tilde{\rho}_l(\mathbf{q}, t) = \int d\theta \frac{e^{-il\theta}}{\sqrt{2\pi}} \tilde{\rho}(\theta, \mathbf{q}, t),$$

eqn (5.3.15) can be written as

$$\begin{aligned} i\partial_t\tilde{\rho}_l(q\mathbf{x}, t) &= qK_{lm}M_{m'l'}\tilde{\rho}_{l'}(q\mathbf{x}, t), \quad f(\theta - \theta') = \sum_l f_l e^{il(\theta - \theta')}, \\ K_{ll'} &= \frac{1}{2}(\delta_{l, l'+1} + \delta_{l, l'-1}), \quad M_{ll'} = \left(v_F^* + \frac{k_F f_l}{2\pi}\right)\delta_{ll'}. \end{aligned}$$

From M , we see that the Fermi liquid is unstable if one of the f_l is less than $-2\pi v_F^*/k_F$. We also see that $\tilde{M}_{ll'} = \sqrt{v_F^* + \frac{k_F f_l}{2\pi}}\delta_{ll'}$ and

$$q(\tilde{M}^\top K\tilde{M})_{ll'} \equiv \tilde{H}_{ll'} = \frac{q}{2}(\delta_{l, l'+1} + \delta_{l, l'-1})\sqrt{v_F^* + \frac{k_F f_l}{2\pi}}\sqrt{v_F^* + \frac{k_F f_{l'}}{2\pi}}$$

The eigenvalues of $\tilde{H}$ give us the frequencies of the collective modes.

Here $\tilde{H}$ describes a particle hopping on a one-dimensional lattice. The nearest-neighbor hopping amplitude between site l and site $l + 1$ is given by

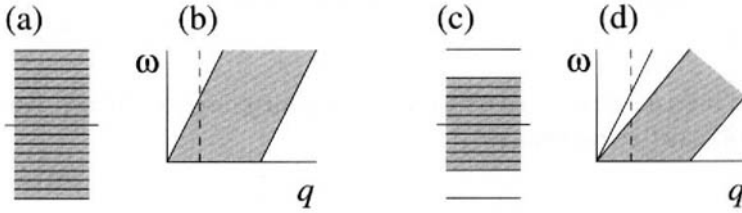


FIG. 5.6. (a) The spectrum of $\tilde{H}$ when $f_l = 0$, and (b) the corresponding particle-hole spectrum. (c) The spectrum of $\tilde{H}$ when $f_l > 0$, and (d) the corresponding particle-hole spectrum.

$\frac{q}{2} \sqrt{\frac{v_F^*}{2\pi} + \frac{k_F f_l}{(2\pi)^2}} \sqrt{\frac{v_F^*}{2\pi} + \frac{k_F f_{l+1}}{(2\pi)^2}}$. When $f_l = 0$, $\tilde{H}$ has a continuous spectrum ranging from $-v_F^* q$ to $v_F^* q$ (see Fig. 5.6(a)). It reproduces the particle-hole continuum of a free fermion system (see Fig. 5.6(b)). However, if $f_l > f_\infty$, then the particle described by H_{lv} is attracted to the small- l region. Thus, the spectrum of $\tilde{H}$ has isolated eigenvalues corresponding to the bound states in the attractive region, in addition to the continuous spectrum (see Fig. 5.6(c)). The corresponding particle-hole spectrum contains well-defined collective modes (see Fig. 5.6(d)).³⁴

In the second application, we would like to use the linearized hydrodynamical equation of motion (5.3.10) to calculate the d.c. conductance of the Fermi liquid. In the presence of a uniform static electric field $\mathbf{E}$, the force is given by $\mathbf{F} = e\mathbf{E}$. The force $\mathbf{F}$ will induce a uniform and static $\tilde{\rho}$, i.e. $\partial_t \tilde{\rho} = \partial_x \tilde{\rho} = 0$. Equation (5.3.10) becomes

$$\tilde{\rho} = e\tau \frac{k_F}{(2\pi)^2} \hat{\mathbf{k}} \cdot \mathbf{E}$$

From eqn (5.3.6), we see that the above $\tilde{\rho}$ induces an electric current:

$$e\mathbf{J} = e \int d\theta \tilde{\rho} \tilde{\mathbf{v}}$$

$$\tilde{\mathbf{v}}(\mathbf{k}) = \mathbf{v}^*(\mathbf{k}) + \int \frac{d^2 \mathbf{k}'}{(2\pi)^2} f(\mathbf{k}, \mathbf{k}') \mathbf{v}^*(\mathbf{k}') \delta(\xi_{\mathbf{k}}^*)$$

³⁴ A bosonic mode must have a positive energy. A negative energy implies an instability. However, $\tilde{H}$ has both positive and negative eigenvalues, corresponding to positive and negative frequencies. If the positive eigenvalues of $\tilde{H}$ correspond to the positive energies of the bosonic collective modes, then do the negative eigenvalues of $\tilde{H}$ also correspond to the negative energies of the collective modes? Here we would like to point out that so far we have treated eqn (5.3.13) as a classical Lagrangian and only discussed the resulting classical equation of motion. After quantization, the equation of motion becomes an equation for an operator $\hat{\tilde{\rho}}$. It turns out that the modes with positive frequencies correspond to creation operators of the bosonic mode and the modes with negative frequencies correspond to annihilation operators of the bosonic mode. The quantization of a one-dimensional version of eqn (5.3.13) will be discussed in Section 7.4.5, see eqns (7.4.35) and (7.4.36).

Due to the rotational symmetry, $\tilde{\mathbf{v}} = |\tilde{\mathbf{v}}|\hat{\mathbf{k}}$. We find that

$$e\mathbf{J} = e^2\tau \frac{|\tilde{\mathbf{v}}|k_F}{(2\pi)^2} \int d\theta \hat{\mathbf{k}}(\hat{\mathbf{k}} \cdot \mathbf{E}) = \frac{e^2\tau|\tilde{\mathbf{v}}|\rho}{k_F} \mathbf{E}$$

The conductance is $\sigma = \frac{e^2\tau|\tilde{\mathbf{v}}|\rho}{k_F}$. The above reduces to the Drude result $\sigma = \frac{e^2\tau\rho}{m^*}$ if the Fermi liquid function $f(\mathbf{k}, \mathbf{k}')$ vanishes.

Problem 5.3.5.

In the presence of a uniform magnetic field $\mathbf{B} = B\mathbf{z}$, the force $\mathbf{F}$ in the two-dimensional reduced Boltzmann equation (5.3.9) has the form $\mathbf{F}(\mathbf{k}, \mathbf{x}) = \frac{e}{c}\mathbf{v}^*(\mathbf{k}) \times \mathbf{B}$.

(a) Show that, after linearization, the reduced Boltzmann equation in the magnetic field has the following form (Kim *et al.*, 1995):

$$(i\partial_t + i\omega_c\partial_\theta - v_F^*q \cos(\theta - \theta_q))\tilde{\rho} - \frac{qk_F \cos(\theta_q - \theta)}{(2\pi)^2} \int d\theta' f(\theta - \theta')\tilde{\rho}(\theta', \mathbf{q}, t) = 0$$

if we set $\tau^{-1} = 0$. Find ω_c .

(b) Show that the energy spectrum of the collective mode with momentum $\mathbf{q}$ is determined by the eigenvalues of

$$\tilde{H}_{ll'} = \omega_c l \delta_{ll'} + \frac{q}{2}(\delta_{l,l'+1} + \delta_{l,l'-1}) \sqrt{v_F^* + \frac{k_F f_l}{2\pi}} \sqrt{v_F^* + \frac{k_F f_{l'}}{2\pi}}$$

Show that the spectrum of $\tilde{H}$ is symmetric around zero. (Hint: $f_l = f_{-l} = \text{real}$.)

(c) Show that, when $f_l = 0$, the eigenvalues of $\tilde{H}$ are given by $\omega_c \times \text{integer}$. (Hint: Consider $\tilde{\rho} = \exp(i l \theta - i \frac{v_F^* q}{\omega_c} \sin(\theta - \theta_q))$.) So we can recover the free fermion result using hydrodynamical theory.

Problem 5.3.6.

The a.c. transport of a two-dimensional Fermi liquid

(a) Assume that the force $\mathbf{F}$ in eqn (5.3.9) has the form $\mathbf{F} = e\mathbf{E}e^{i\omega t - \mathbf{q} \cdot \mathbf{x}}$. Find $\tilde{\rho}$, assuming that $f(\theta - \theta') = 0$.

(b) Find $\tilde{\rho}$ to the first order in $f(\theta - \theta')$.

(c) Find the induced electric current $e\mathbf{J} = \sigma(\omega, \mathbf{q})\mathbf{E}$, and the optical conductance $\sigma(\omega, \mathbf{q})$ to the first order in $f(\theta - \theta')$. Here $\mathbf{J}$ is the quasiparticle current given in eqn (5.3.6).

5.3.5 The essence of Fermi liquid theory

It is commonly believed that the essence of Fermi liquid theory is the concept and the existence of well-defined quasiparticles. We adopted this point of view in Section 5.3.1. The existence of well-defined quasiparticles implies that, not only is the total number of quasiparticles conserved, but the number of quasiparticles in a given momentum direction is also conserved. Thus, a Fermi liquid has an infinite number of conserved quantities (Luther, 1979; Haldane, 1992). This allows us to develop a hydrodynamical theory for a Fermi liquid which contains many bosonic

modes, one mode for each conserved quantity. Thus, the existence of an infinite number of conserved quantities is also an essence of Fermi liquid theory. In Section 5.3.3, we developed a Fermi liquid theory based on this point of view.

We would like to point out that the two points of view are not equivalent, and the second point of view is more general. The existence of well-defined quasiparticles implies that the number of quasiparticles at each momentum is conserved. In the second point of view, we only assume that the quasiparticles in each momentum direction are conserved. In one dimension, the hydrodynamical theory in Section 5.3.3 actually describes one-dimensional Tomonaga–Luttinger liquids.³⁵ In higher dimensions, the hydrodynamical theory describes Fermi liquids.

5.4 Perturbation theory and the validity of Fermi liquid theory

In this section, we will test the correctness of Fermi liquid theory. We will develop a perturbation theory to calculate the electron Green's function for an interacting system. We will check if $\text{Im}G(\mathbf{k}, \omega)$ contains a δ -function. The presence of the δ -function indicates the presence of well-defined Landau quasiparticles and the validity of Landau Fermi liquid theory.

5.4.1 Path integrals and perturbation theory for fermions

- The path integral for fermions is a bookkeeping formalism that allows us to formally express the correlation functions of a fermion system. Unlike the path integral for bosons, the physical meaning of the fermion path integral is unclear.
- The path integral for fermions allows us to systematically develop a perturbation theory for an interacting fermion system.
- The structure of perturbation theory is captured by Feynman diagrams and Feynman rules.

To develop a path integral for fermions, we first note that the time-ordered correlation of fermion operators satisfies $\langle T[c(t)c^\dagger(t')] \rangle = -\langle T[c^\dagger(t')c(t)] \rangle$. Thus, we will use (complex) Grassmann numbers ξ_α to represent the fermion operators. The reader may ask, what do you mean by 'use Grassmann numbers to represent the fermion operators'? Well, I have to say I do not know. I do not know the physical meaning of a Grassmann number. I will treat the following fermion path integral as some kind of bookkeeping device, which allows us to formally pack various fermion correlation functions into a compact formula. The Grassmann numbers are anti-commuting numbers which

³⁵ See Section 7.4.4 for a discussion of a Tomonaga–Luttinger liquid.

satisfy

$$\xi_\alpha \xi_\beta = -\xi_\beta \xi_\alpha, \quad \xi_\alpha \xi_\beta^* = -\xi_\beta^* \xi_\alpha, \quad \xi_\alpha^2 = 0,$$

and

$$(\xi_{\alpha_1} \dots \xi_{\alpha_n})^* = \xi_{\alpha_n}^* \dots \xi_{\alpha_1}^*$$

Functions of the Grassmann numbers are defined by the following expansion:

$$f(\xi) = f_0 + f_1 \xi, \quad A(\xi, \xi^*) = a_0 + a_1 \xi + \bar{a}_1 \xi^* + a_{12} \xi^* \xi$$

The derivative is identical to the ordinary derivative, except that, in order for the derivative operator $\frac{\partial}{\partial \xi}$ to act on ξ , the variable ξ has to be anti-commuted through until it is adjacent to $\frac{\partial}{\partial \xi}$. For example, $\frac{\partial}{\partial \xi}(\xi^* \xi) = \frac{\partial}{\partial \xi}(-\xi \xi^*) = -\xi^*$. With these definitions, we have

$$\begin{aligned} \frac{\partial}{\partial \xi} A(\xi, \xi^*) &= a_1 - a_{12} \xi^* \\ \frac{\partial}{\partial \xi^*} A(\xi, \xi^*) &= \bar{a}_1 + a_{12} \xi \\ \frac{\partial}{\partial \xi^*} \frac{\partial}{\partial \xi} A(\xi, \xi^*) &= -a_{12} = -\frac{\partial}{\partial \xi} \frac{\partial}{\partial \xi^*} A(\xi, \xi^*) \end{aligned}$$

We see that

$$\frac{\partial}{\partial \xi^*} \frac{\partial}{\partial \xi} = -\frac{\partial}{\partial \xi} \frac{\partial}{\partial \xi^*}$$

The Grassmann integration, as a linear mapping, is defined as the Grassmann derivative:

$$\int d\xi = \frac{\partial}{\partial \xi}$$

We have that

$$\int d\xi \, 1 = 0, \quad \int d\xi \, \xi = 1$$

The Grassmann integration has the following properties of standard integration:

$$\int d\xi \, \frac{\partial}{\partial \xi} f(\xi) = 0, \quad \int d\xi \, f(\xi + \eta) = \int d\xi \, f(\xi)$$

From the definition of the determinant, namely

$$\text{Det}(H_0) = \sum_P (-)^P (H_0)_{1P_1} \dots (H_0)_{nP_n}$$

where P are permutations $(1, \dots, n) \rightarrow (P_1, \dots, P_n)$, we find the Grassmann Gaussian integral:

$$\int \prod_{i=1}^n d\eta_i^* d\eta_i \, e^{-\eta_i^* (H_0)_{ij} \eta_j} = \text{Det}(H_0)$$

and

$$\int \prod_{i=1}^n d\eta_i^* d\eta_i e^{-\eta_i^* (H_0)_{ij} d\eta_j + \zeta_i^* \eta_i + \zeta_i \eta_i^*} = \text{Det}(H_0) e^{\zeta_i^* (H_0^{-1})_{ij} \zeta_j}$$

Note that, compared to the boson Gaussian integral, the determinant appears in the numerator instead of the denominator.

The fermion Gaussian integral allows us to obtain the Wick theorem for fermion correlations (defined by the path integral):

$$\begin{aligned} & \langle \psi_1 \dots \psi_n \psi_n^* \dots \psi_1^* \rangle \\ &= \frac{\int \mathcal{D}\psi^* \mathcal{D}\psi \psi_1 \dots \psi_n \psi_n^* \dots \psi_1^* e^{-\sum_{ij} \psi_i^* (H_0)_{ij} \psi_j}}{\int \mathcal{D}\psi^* \mathcal{D}\psi e^{-\sum_{ij} \psi_i^* (H_0)_{ij} \psi_j}} \\ &= \sum_P \zeta^P (H_0^{-1})_{P_n, n} \dots (H_0^{-1})_{P_1, 1} \end{aligned} \quad (5.4.1)$$

where $\zeta = -1$. (If we choose $\zeta = +1$, then the above will be the Wick theorem for boson operators.)

Let us apply the above formalism to a free fermion system at zero temperature. We consider the following path integral:

$$\begin{aligned} Z_0 &= \int \mathcal{D}\psi^* \mathcal{D}\psi e^{i \int_0^\beta dt L_0} \\ L_0 &= \sum_{\mathbf{k}} i \psi_{\mathbf{k}}(t)^* \partial_t \psi_{\mathbf{k}}(t) - \psi_{\mathbf{k}}(t)^* \xi_{\mathbf{k}} \psi_{\mathbf{k}}(t) \end{aligned}$$

We can calculate the path integral average

$$i G_{\mathbf{k}}(t, t') = \langle \psi_{\mathbf{k}}(t) \psi_{\mathbf{k}}^*(t') \rangle = \frac{1}{(-i)(i\partial_t - \xi_{\mathbf{k}})}$$

In frequency space, we have

$$G_{\mathbf{k}, \omega} = \frac{1}{\omega - \xi_{\mathbf{k}}}$$

which agrees with the time-ordered average³⁶ $\langle 0 | T(\psi_{\mathbf{k}}(t) \psi_{\mathbf{k}}^\dagger(t')) | 0 \rangle$ in eqn (4.2.9). Using the Wick theorem, we can derive the following more general relationship between the path integral average and the time-ordered average:

$$\langle \psi_{i_1} \dots \psi_{i_n} \psi_{j_n}^* \dots \psi_{j_1}^* \rangle = \frac{\text{Tr} \left(T(\psi_{i_1} \dots \psi_{i_n} \psi_{j_n}^\dagger \dots \psi_{j_1}^\dagger e^{-\int_0^\beta dt H_0}) \right)}{\text{Tr} \left(T(e^{-\int_0^\beta dt H_0}) \right)}$$

Thus, we can use path integrals to calculate arbitrary time-ordered correlations.

³⁶ The calculation presented here is a formal calculation, which fails to reproduce the important 0^+ term. Section 2.2.2 discussed how to produce the 0^+ term.

The above results are for free fermion systems. Let us apply the path integral to an interacting system with

$$H = H_0 + H_I(t)$$

where H_0 describes a free fermion system. Let us assume that $H_I(t)$ is a slowly varying function of t , that $H_I(t) = H_I$ for finite t , and $H_I(\pm\infty) = 0$. Let

$$Z = \langle 0|T(e^{-i \int dt (H_0 + H_I(t))})|0\rangle$$

be the partition function, where $|0\rangle$ is the ground state for H_0 . The partition function Z has the following perturbative expansion:

$$Z = Z_0 \left(1 + Z_0^{-1} \langle 0|T \left((-i \int dt H_I) e^{-i \int dt H_0} \right) |0\rangle + \dots \right)$$

$$Z_0 = \langle 0|T(e^{-i \int dt H_0})|0\rangle.$$

As $Z_0^{-1} \langle 0|T(O_1(t_1) \dots O_n(t_n) e^{-i \int dt H_0})|0\rangle$ in the Schrödinger picture is equal to $\langle 0|T(O_1(t_1) \dots O_n(t_n))|0\rangle$ in the Heisenberg picture, both of these quantities represent the time-ordered average. We see that the partition function Z for an interacting system can be expressed in terms of the time-ordered averages in the free fermion system H_0 . This allows us to express Z in terms of a path integral as follows:

$$Z = \int \mathcal{D}\psi^* \mathcal{D}\psi e^{i \int_0^\beta dt (L_0 + L_I)}, \quad L_I = -H_I$$

Similarly, using a perturbative expansion, we can show that the time-ordered correlation for an interacting system (in the Schrödinger picture) can be expressed in terms of path integrals as follows:

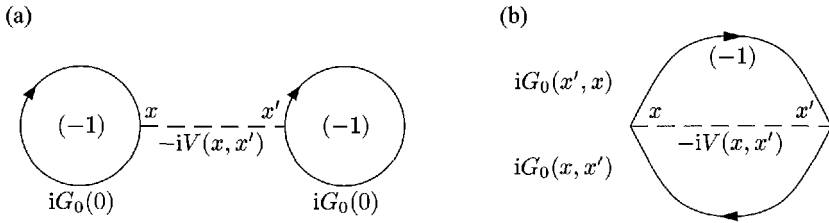
$$\frac{\langle 0|T(O_1(t_1) \dots O_n(t_n) e^{-i \int dt H})|0\rangle}{\langle 0|T(e^{-i \int dt H})|0\rangle} = \frac{\int \mathcal{D}\psi^* \mathcal{D}\psi O_1(t_1) \dots O_n(t_n) e^{i \int_0^\beta dt (L_0 + L_I)}}{\int \mathcal{D}\psi^* \mathcal{D}\psi e^{i \int_0^\beta dt (L_0 + L_I)}}$$

Now, let us calculate the partition function Z perturbatively using path integrals:

$$\begin{aligned} Z &= Z_0 \frac{\int \mathcal{D}\psi^* \mathcal{D}\psi e^{i \int dt (L_0 + L_I)}}{\int \mathcal{D}\psi^* \mathcal{D}\psi e^{i \int dt L_0}} \\ &= Z_0 \left(1 + \left\langle \left(i \int dt L_I \right) \right\rangle_0 + \frac{1}{2!} \left\langle \left(i \int dt L_I \right)^2 \right\rangle_0 + \dots \right) \end{aligned}$$

If we assume that the interaction has the form

$$\int dt L_I = - \int dx dx' \psi^*(x^\mu) \psi(x^\mu) V(x^\mu, x'^\mu) \psi^*(x'^\nu) \psi(x'^\nu)$$

FIG. 5.7. Diagrams that contribute to Z/Z_0 at first order in V .

where $x^\mu = (t, x^1, \dots, x^d)$ and $dx = dt d^d x$, then we find that

$$\begin{aligned} \frac{Z}{Z_0} &= 1 + \left\langle \int dx dx' \psi^*(x^\mu) \psi(x^\mu) [-iV(x^\mu, x'^\nu)] \psi^*(x'^\nu) \psi(x'^\nu) \right\rangle_0 + \dots \\ &= 1 + \int dx dx' (-) iG_0(0) [-iV(x^\mu, x'^\nu)] (-) iG_0(0) \\ &\quad + \int dx dx' (-) iG_0(x^\mu - x'^\nu) iG_0(x'^\nu - x^\mu) [-iV(x^\mu, x'^\nu)] + \dots \end{aligned}$$

The two terms of order V can be represented by the Feynman diagrams shown in Fig. 5.7. The following Feynman rules for fermions are almost identical to those of bosons, except that each closed loop contributes an extra factor of -1 .

(a) Each line in the Feynman diagrams represents a propagator $G_0(x, x')$, where x and x' are the coordinates of the two vertices at the ends of the line.

(b) Each vertex represents an integration over the location of the vertex $\int dx$.

(c) Each dashed line represents the interaction potential $-iV(x, x')$.

(d) Each closed fermion loop contributes a factor of -1 .

(e) The value of a connected diagram is given by rules (a)–(d) and the value of a disconnected diagram is given by the product of connected sub-diagrams.

Feynman diagrams capture the structure of a perturbative expansion. They not only produce the order- V terms, but they also produce order- V^2 terms and all other higher-order terms.

In the perturbative expansion of Z/Z_0 , the path integral averages $\langle \dots \rangle_0$ contain both connected and disconnected diagrams. We can use the linked-cluster expansion to rewrite the expansion of Z/Z_0 as

$$\ln \frac{Z}{Z_0} = \sum_{n=0}^{\infty} \frac{1}{n!} \left\langle \left(i \int dt L_I \right)^n \right\rangle_{0c} = \left\langle e^{i \int dt L_I} \right\rangle_{0c} \quad (5.4.2)$$

where the path integral average $\langle \dots \rangle_{0c}$ includes only the connected graphs.

Problem 5.4.1.

- (a) Calculate the order- V^2 terms in the perturbative expansion of Z/Z_0 . Draw the corresponding Feynman diagrams.
- (b) Calculate the order- V^2 terms in the perturbative expansion of $\ln(Z/Z_0)$. Draw the corresponding Feynman diagrams.

5.4.2 Self-energy and two-body interactions

Now we are ready to calculate the electron Green's function perturbatively for the following interacting electron system:

$$H = \sum_{\mathbf{k}} \xi_{\mathbf{k}} c_{\alpha\mathbf{k}}^\dagger c_{\alpha\mathbf{k}} + \frac{1}{2V} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} c_{\alpha\mathbf{k}}^\dagger c_{\alpha, \mathbf{k}-\mathbf{q}} V_{\mathbf{q}} c_{\beta\mathbf{k}'}^\dagger c_{\beta, \mathbf{k}'+\mathbf{q}}$$

The time-ordered Green's function is given by the following path integral average:

$$iG(\mathbf{x}, t) \delta_{\alpha\beta} = \frac{\int \mathcal{D}c \mathcal{D}c^* c_{\alpha}(\mathbf{x}, t) c_{\beta}^*(0, 0) e^{i \int dt L}}{\int \mathcal{D}c \mathcal{D}c^* e^{i \int dt L}} \equiv \langle c_{\alpha}(\mathbf{x}, t) c_{\beta}^*(0, 0) \rangle$$

for interacting electrons, where

$$L = i c_{\alpha}^* \partial_t c_{\alpha} - H$$

is the Lagrangian of the interacting electrons. Note that here we have assumed that $\langle c_{\alpha}(\mathbf{x}, t) c_{\beta}^*(0, 0) \rangle$ is spin-rotation invariant. As a result, it has the form $iG(\mathbf{x}, t) \delta_{\alpha\beta}$.

To calculate iG , it is convenient to introduce

$$Z[\eta_{\alpha}, \eta_{\alpha}^*] = \int \mathcal{D}c \mathcal{D}c^* e^{i \int dt L + \int dt d^d \mathbf{x} (\eta_{\alpha} c_{\alpha}^* + \eta_{\alpha}^* c_{\alpha})}$$

where $\eta(\mathbf{x}, t)$ and $\eta^*(\mathbf{x}, t)$, just like $c(\mathbf{x}, t)$ and $c^*(\mathbf{x}, t)$, are Grassmann numbers. We see that

$$iG(\mathbf{x}, t) \delta_{\alpha\beta} = \frac{\partial}{\partial \eta_{\alpha}^*(\mathbf{x}, t)} \frac{\partial}{\partial \eta_{\beta}(0, 0)} \ln Z[\eta_{\alpha}, \eta_{\alpha}^*]$$

Using eqn (5.4.2), we obtain

$$\begin{aligned} iG(\mathbf{x}, t) \delta_{\alpha\beta} &= \frac{\partial}{\partial \eta_{\alpha}^*(\mathbf{x}, t)} \frac{\partial}{\partial \eta_{\beta}(0, 0)} \left\langle e^{i \int dt L_I + \int dt d^d \mathbf{x} (\eta_{\alpha} c_{\alpha}^* + \eta_{\alpha}^* c_{\alpha})} \right\rangle_{0c} \\ &= \left\langle c_{\alpha}(\mathbf{x}, t) c_{\beta}^*(0, 0) e^{i \int dt L_I} \right\rangle_{0c} \end{aligned} \quad (5.4.3)$$

This formula lays the foundation for using Feynman diagrams to calculate the time-ordered Green's function of interacting electrons.

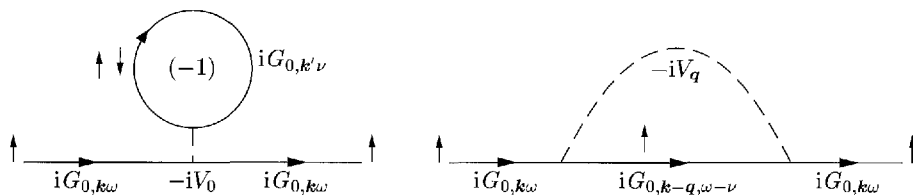


FIG. 5.8. The two Feynman diagrams that contribute to $iG_{k\omega}$. The dashed line represents the potential $-iV_q$.

To first order in V_q , we find that

$$iG(\mathbf{x}, t)\delta_{\alpha\beta} = \left\langle c_{\alpha}(\mathbf{x}, t)c_{\beta}^{\dagger}(0, 0) \right\rangle_0 + \left\langle c_{\alpha}(\mathbf{x}, t)c_{\beta}^{\dagger}(0, 0)(-i) \int dt' H_I(t') \right\rangle_{0c}$$

Using the Wick theorem, we can express the above equation in terms of the free electron Green's function G_0 . In $\mathbf{k}-\omega$ space, we find that (see Problem 5.4.2)

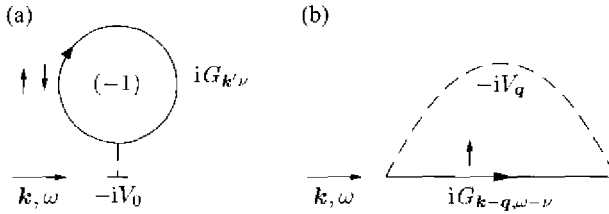
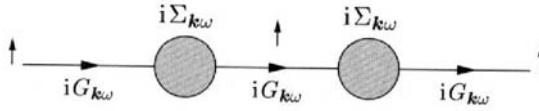
$$\begin{aligned} iG_{k\omega} = & iG_{0,k\omega} + (-1)(iG_{0,k\omega})^2(-iV_0)\frac{1}{\mathcal{V}} \sum_{\mathbf{k}', \alpha'} \int \frac{d\nu}{2\pi} iG_{0,\mathbf{k}'\nu} e^{i0^+\nu} \\ & + (iG_{0,k\omega})^2 \frac{1}{\mathcal{V}} \sum_{\mathbf{q}} \int \frac{d\nu}{2\pi} iG_{0,\mathbf{k}-\mathbf{q},\nu} (-iV_{\mathbf{q}}) e^{-i0^+\nu} \end{aligned} \quad (5.4.4)$$

where the free electron Green's function $G_{0,k\omega}$ is given by eqn (4.2.4). The two order- V_q terms are summarized by the Feynman diagrams shown in Fig. 5.8. The Feynman rules in momentum space are different from those in real space and are as follows.

- (a) Each line in the Feynman diagrams carries an energy-momentum vector, say, $(\mathbf{q}, \nu)$. It represents a propagator $iG_{q\nu} = i/(\nu - \xi_{\mathbf{q}})$.
- (b) The energy-momentum flowing into a node is conserved.
- (c) Each loop represents the integral $(\beta\mathcal{V})^{-1} \int \frac{d\mathbf{t}}{2\pi} \sum_{\mathbf{q}}$.
- (d) The dashed line represents the interaction $-iV_{\mathbf{q}}$, where $\mathbf{q}$ is the momentum flowing through the dashed line.
- (e) Each closed loop contributes an extra factor of -1 due to the anti-commuting property of the fermion operators. (The momentum space Feynman rules for boson systems do not have such a -1 factor.)

Equation (5.4.4) can be written as

$$iG_{k\omega} = iG_{0,k\omega} + (iG_{0,k\omega})^2(-i\Sigma_{k\omega})$$


 FIG. 5.9. Two diagrams that contribute to the self-energy $-i\Sigma_{\mathbf{k}\omega}$.

 FIG. 5.10. Higher-order diagrams that contribute to $iG_{\mathbf{k}\omega}$.

where

$$\begin{aligned}
 & -i\Sigma_{\mathbf{k}\omega} \\
 &= (-1) \frac{-iV_0}{\mathcal{V}} \sum_{\mathbf{k}', \alpha'} \int \frac{d\nu}{2\pi} iG_{0, \mathbf{k}'\nu} e^{i0^+ \nu} + \frac{1}{\mathcal{V}} \sum_{\mathbf{q}} \int \frac{d\nu}{2\pi} iG_{0, \mathbf{k}-\mathbf{q}, \nu} (-iV_{\mathbf{q}}) e^{-i0^+ \nu}.
 \end{aligned}$$

The above can be represented by the diagrams in Fig. 5.9. If we include some of the higher-order terms represented by the diagrams in Fig. 5.10, then the Green's function can be written as

$$iG_{\mathbf{k}\omega} = iG_{0, \mathbf{k}\omega} \sum_{n=0}^{\infty} [(iG_{0, \mathbf{k}\omega})(-i\Sigma_{\mathbf{k}\omega})]^n$$

We find that

$$G_{\mathbf{k}\omega} = \frac{1}{\omega - \xi_{\mathbf{k}} - \Sigma_{\mathbf{k}\omega} + i\text{sgn}(\omega)0^+}$$

The dispersion relation can be obtained from the position of the pole in $G_{\mathbf{k}\omega}$, or from the position of the zero in $\omega - \xi_{\mathbf{k}} - \Sigma_{\mathbf{k}\omega}$. We see that $\Sigma_{\mathbf{k}\omega}$ corrects the energy of the electron (or quasiparticle) and is called the self-energy.

Note that

$$\int \frac{d\nu}{2\pi} iG_{0, \mathbf{k}'\nu} e^{i0^+ \nu} = i \int \frac{d\nu}{2\pi} \frac{1}{\nu - \xi_{\mathbf{k}'} + i0^+ \text{sgn}(\nu)} e^{i0^+ \nu} = -\Theta(-\xi_{\mathbf{k}'}) = -n_{\mathbf{k}'}$$

As we must close the contour in the upper-half complex ν plane, only the poles with negative ν contribute. As $\sum_{\mathbf{k}', \alpha} \Theta(-\xi_{\mathbf{k}'})$ is simply the total number of

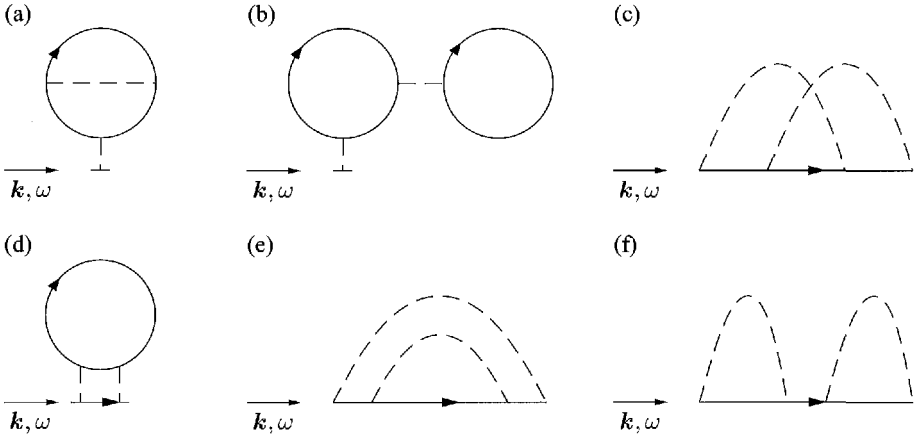


FIG. 5.11. The diagrams (a)–(e) contribute to the self-energy at order V_q^2 . The diagram (f) does not contribute to the self-energy because it is already included in Fig. 5.10.

electrons, the first term in the self-energy is simply the Hartree contribution (Fig. 5.9(a)):

$$\Sigma_{\mathbf{k}\omega} = V_0 \rho_0 + \dots$$

Similarly, the second term in the self-energy (Fig. 5.9(b)) has the following form:

$$\Sigma_{\mathbf{k}\omega} = \dots + \frac{1}{V} \sum_{\mathbf{q}} (-n_{\mathbf{k}-\mathbf{q}}) V_{\mathbf{q}}$$

which is just the Fock contribution to the self-energy.

In the above, we have seen how to calculate the self-energy $\Sigma_{\mathbf{k}\omega}$ and the quasiparticle dispersion $\xi_{\mathbf{k}}^* = \xi_{\mathbf{k}} + \Sigma_{\mathbf{k}\omega}|_{\omega=\xi_{\mathbf{k}}^*}$ from the diagrams. This approach reproduces the Hartree–Fock result to first order in V . However, the diagram approach allows us to calculate higher-order contributions to the self-energy in a systematic way (see Fig. 5.11).

We can also use diagrams to calculate the Fermi liquid function in Fermi liquid theory. We know that the interaction $V_{\mathbf{q}}$ can scatter two electrons with momenta $\mathbf{k}_1$ and $\mathbf{k}_2$ into two electrons with momenta $\mathbf{k}'_1$ and $\mathbf{k}'_2$. The interaction term $n_{\mathbf{k}_1} n_{\mathbf{k}_2} V(\mathbf{k}_1 - \mathbf{k}_2)$ in Fermi liquid theory corresponds to terms that scatter two electrons with momenta $\mathbf{k}_1$ and $\mathbf{k}_2$ into two electrons with the same momenta. Thus, the Fermi liquid function f receives two contributions, which are illustrated by the two diagrams in Fig. 5.12:

$$-i f(\mathbf{k}_1, \alpha; \mathbf{k}_2, \beta) = (-i V_0) + (-)(-i V_{\mathbf{q}}) \delta_{\alpha\beta}$$

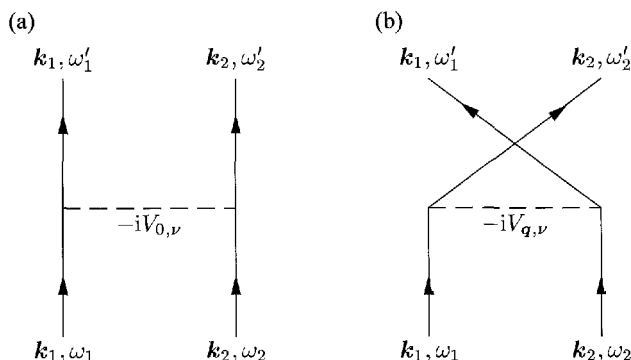


FIG. 5.12. Diagrams that contribute to the Fermi function $f(\mathbf{k}_1, \alpha; \mathbf{k}_2, \beta)$. (a) Here $\mathbf{q} = \mathbf{k}_1 - \mathbf{k}_1 = 0$ is the momentum transfer and $\nu = \omega_1 - \omega_1'$ is the energy transfer. (b) Here $\mathbf{q} = \mathbf{k}_1 - \mathbf{k}_2$ is the momentum transfer and $\nu = \omega_1 - \omega_2'$ is the energy transfer.

The $(-)$ sign arises from the exchange and $\delta_{\alpha\beta}$ arises from the requirement that the two electrons must have the same spins before and after the scattering. As the interaction considered here is instantaneous, $V_{\mathbf{q}}$ does not depend on the energy transfer ν (see Fig. 5.12). The resulting Fermi liquid function f is also instantaneous and independent of the energy transfer ν :

$$f(\mathbf{k}_1, \alpha; \mathbf{k}_2, \beta) = V_0 - V_{\mathbf{k}_1 - \mathbf{k}_2} \delta_{\alpha\beta}$$

The above is just the Hartree–Fock result (5.3.2) that we obtained before.

Problem 5.4.2.

Prove eqn (5.4.4) using perturbation theory. We note that $\int \frac{d\nu}{2\pi} iG_{0,\mathbf{k}'\nu} e^{i0^+\nu}$ represents $\langle c^\dagger c \rangle$ at equal time, while $\int \frac{d\nu}{2\pi} iG_{0,\mathbf{k}'\nu} e^{-i0^+\nu}$ represents $\langle cc^\dagger \rangle$ at equal time.

Problem 5.4.3.

Find the expressions for the order- V_q^2 terms in the perturbative expansion of the self-energy $\Sigma_{\mathbf{k}\omega}$, as described in Fig. 5.11(a)–(e).

5.4.3 Random phase approximation and the effective potential

- The screening of the interaction potential by particle–hole excitations can be included in the random phase approximation.

We have seen that, in three dimensions and for a Coulomb interaction, the Hartree–Fock approximation results in a self-energy that is singular near the Fermi surface. The singularity gives rise to an infinite Fermi velocity. The singularity arises from the long-range potential of the Coulomb interaction. A short-ranged interaction cannot produce any singular self-energy. However, we know that two

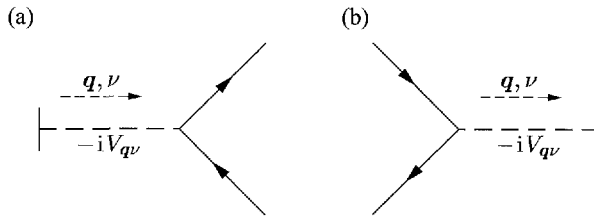


FIG. 5.13. (a) The interaction generates particle-hole excitations, and (b) the particle-hole excitations modify the interaction.

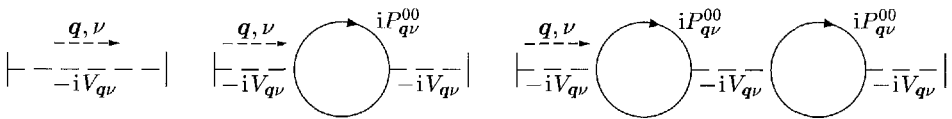


FIG. 5.14. The effective potential in the RPA approximation.

charges in a metal do not experience a long-range interaction due to the screening from other electrons. Therefore, one may doubt the correctness of the Hartree-Fock result about the diverging Fermi velocity; in particular, after realizing that the Hartree-Fock approximation includes only direct interaction and no screening effects.

To go beyond the Hartree-Fock approximation and include the screening effects, we must include the effect that the interaction generates particle-hole excitations (Fig. 5.13(a)), and the particle-hole excitations in turn modifying the potential (Fig. 5.13(b)). In the random phase approximation (RPA), we include the diagrams in Fig. 5.14 to calculate the screened effective potential as follows:

$$-iV_{q\nu}^{\text{eff}} = -iV_{q\nu} \left(1 + (-iV_{q\nu})(iP_{q\nu}^{00}) + (-iV_{q\nu})^2(iP_{q\nu}^{00})^2 + \dots \right)$$

We find that

$$V_{q\nu}^{\text{eff}} = \frac{V_{q\nu}}{1 - V_{q\nu}P_{q\nu}^{00}}$$

In the $\nu \ll v_F q$ limit, $P_{q\nu}^{00}$ is equal to the negative of the compressibility, namely $P_{q\nu}^{00} = -N(E_F)$. We have

$$V_{q\nu}^{\text{eff}} = \frac{V_{q\nu}}{1 + V_{q\nu}N(E_F)}$$

and for the Coulomb interaction we have

$$V_q^{\text{eff}} = \frac{4\pi e^2}{q^2 + 4\pi e^2 N(E_F)}$$

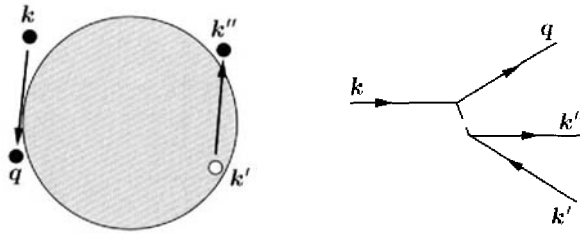


FIG. 5.15. The decay of a quasiparticle.

It is clear that, in the $\nu \ll v_F q$ limit, the effective potential is not singular as $q \rightarrow 0$. Such an effective potential represents a screened short-range interaction. The same result can also be obtained through the Thomas–Fermi model.

The relationship between the RPA calculation and the screening can be seen more clearly in the following way. The first term in the RPA calculation (see Fig. 5.14) represents the direct (i.e. unscreened) interaction between two densities (represented by two short vertical bars in Fig. 5.14) at two different places. The second term in the RPA calculation represents the effect that the density at one place induces a particle–hole excitation via the interaction $V_{q\nu}$. The particle–hole excitation, representing a deformed density, interacts in turn with the density at another place. In this way, the second term (and other higher-order terms) modify or screen the interaction between the densities at the two places.

In standard practice, we replace the bare potential $V_{q\nu}$ with the screened potential $V_{q\nu}^{\text{eff}} = V_{q\nu} / (1 + V_{q\nu} N(E_F))$, and then use this effective potential to calculate the self-energy and the Fermi liquid function. In this approximation, we ignore the frequency dependence of P^{00} . An instantaneous bare interaction (a frequency-independent interaction) is approximated by an instantaneous effective interaction. If we include the frequency dependence of P^{00} , then an instantaneous bare interaction can result in a frequency-dependent effective interaction. Such a frequency dependence represents the retardation effects.

5.4.4 Justification of Landau Fermi liquid theory

- The interaction causes the quasiparticle to decay, but the decay rate is much less than the quasiparticle energy. Landau Fermi liquid theory remains to be validated.

An electron–electron interaction can cause an electron with momentum k to decay into an electron with momentum q and a particle–hole excitation with momentum $k - q$ (see Fig. 5.15). The particle and the hole in the particle–hole excitation can independently have energies in the range between 0 and ξ_k . The

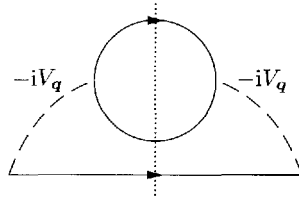


FIG. 5.16. A contribution to the self-energy at order V_q^2 . The diagram is formed by two halves, each representing the decay process in Fig. 5.15. The above diagram produces the decay rate, while the one in Fig. 5.15 leads to the decay amplitude.

energy of the particle at q is adjusted accordingly to maintain the energy conservation. Thus, the number of available decay channels is proportional to the square of the initial electron energy ξ_k . As a result, the decay rate is also proportional to ξ_k^2 . As the decay amplitude is proportional to V_{k-q} , we find the decay rate to be

$$\Gamma_k \propto |V_k|^2 \xi_k^2$$

The decay of a quasiparticle can be described by the imaginary part of the self-energy. When $\Sigma_{k\omega}$ has an imaginary part, the pole of the electron Green's function $1/(\omega - \xi_k - \Sigma_{k\omega})$ is off the real- ω axis. This leads to a decay in the real-time Green's function:

$$G(t, k) = \int \frac{d\omega}{2\pi} \frac{1}{\omega - \xi_k - \Sigma_{k\omega}} e^{-i\omega t} \sim e^{-i\xi_k^* t} e^{-|\text{Im}\Sigma_{k\xi_k^*}|t}$$

where $\xi_k^* = \xi_k + \text{Re}\Sigma_{k\xi_k^*}$. We see that the imaginary part of the self-energy is given by

$$|\text{Im}\Sigma_{k\xi_k^*}| = \Gamma_k$$

We would like to mention that such a self-energy can be obtained from the Feynman diagram in Fig. 5.16.

As the decay rate $\Gamma = \text{Im}\Sigma_{k\xi_k^*} \propto (\xi_k^*)^2$ is much less than the quasiparticle energy ξ_k^* in the low-energy limit, we can ignore the decay of the quasiparticle. In this limit, the quasiparticle excitations can be regarded as eigenstates. This is the foundation of Landau Fermi liquid theory. We see that $\text{Im}G_{k\omega}$ is almost a δ -function at ξ_k^* . The width of the peak is only proportional to $(\xi_k^*)^2$.

Near the Fermi surface we may use the expansion

$$\text{Re}\Sigma_{k\omega} = (1 - \frac{1}{Z})\omega + (\frac{v_F^*}{Z} - v_F)(k - k_F)$$

and rewrite the Green's function as follows:

$$G_{k\omega} = \frac{1}{\omega - v_F(k - k_F) - \text{Re}\Sigma_{k\omega}} = \frac{Z}{\omega - v_F^*(k - k_F)}$$

Here Z is the overlap factor introduced near the end of Section 5.3.1. In this way we justify Landau Fermi liquid theory within a perturbative calculation.

5.5 Symmetry-breaking phase and the spin-density-wave state

- Fermi liquids can have many instabilities. Interactions can lead to symmetry-breaking phases.

Interacting fermion systems can have many different states, which lead to materials with so many different properties, such as superconductors, magnets, charge-density-wave states, etc. All of these different states of materials can be understood as certain symmetry-breaking states induced by interactions. In this section, we are going to discuss only one of the symmetry-breaking states—the spin-density-wave state. The picture, the methods, and the results discussed here can be easily generalized to other symmetry-breaking states, such as the superconducting state.

5.5.1 Linear responses and instabilities

- A free fermion system, with so many gapless excitations, has many potential instabilities. Interactions between fermions can turn the potential instabilities into real instabilities. Different interactions induce different instabilities.

Consider an interacting electron system on a lattice with an on-site interaction:

$$\begin{aligned} H &= - \sum_{\langle ij \rangle, \alpha} t (c_{\alpha j}^\dagger c_{\alpha i} + h.c.) - \mu N + U \sum_i n_i^2 \\ &= - \sum_{\langle ij \rangle, \alpha} t (c_{\alpha j}^\dagger c_{\alpha i} + h.c.) - \mu' N + U \sum_i c_{\beta i}^\dagger c_{\alpha i}^\dagger c_{\alpha i} c_{\beta i} \end{aligned} \quad (5.5.1)$$

where $n_i = \sum_\alpha c_{\alpha i}^\dagger c_{\alpha i}$ is the number of electrons at site i and $N = \sum_i n_i$ is the total number of electrons. The above model is called the Hubbard model.

In the following, we will concentrate on the Hubbard model in two dimensions. In momentum space, the two-dimensional Hubbard model can be rewritten as

$$H = \sum_{\mathbf{k}} \xi_{\mathbf{k}} c_{\alpha \mathbf{k}}^\dagger c_{\alpha \mathbf{k}} + U \frac{1}{N_{\text{site}}} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} c_{\alpha \mathbf{k}}^\dagger c_{\beta \mathbf{k}'}^\dagger c_{\beta \mathbf{k}'+\mathbf{q}} c_{\alpha \mathbf{k}-\mathbf{q}}$$

where

$$\xi_{\mathbf{k}} = -2t (\cos(k_x) + \cos(k_y)) - \mu'$$

and N_{site} is the number of lattice sites. We note that, when $\mu' = 0$, the Fermi sea is half-filled ($\langle n_i \rangle = 1$) and the Fermi surface is a square (see Fig. 5.17) which

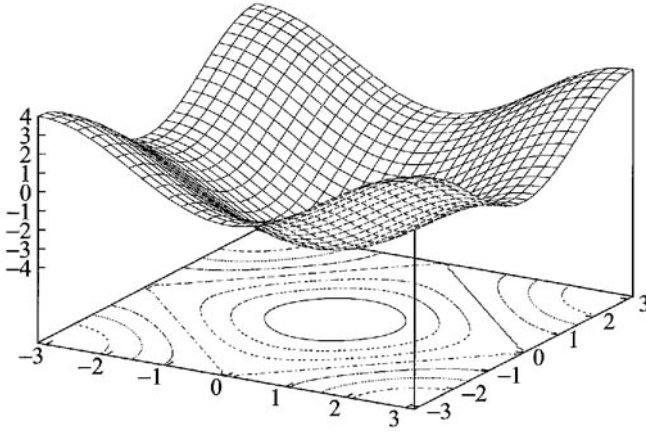
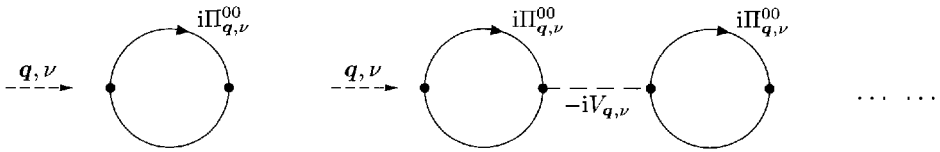
FIG. 5.17. The electron dispersion ξ_k in the Hubbard model.

FIG. 5.18. The compressibility and the spin susceptibility in the RPA.

satisfies a nesting condition, with nesting vector $\mathbf{Q} = (\pi, \pi)$. From eqn (4.3.3), we see that the free fermion compressibility and the spin susceptibility at the nesting vector $\mathbf{Q}$ diverge at low temperatures. As the density of states $N(E_F)$ (now $N(E_F)$ means the number of states per site per unit energy) also has a logarithmic divergence, we have

$$\chi_0(\mathbf{Q}) \sim \frac{1}{E_F} (\ln \frac{E_F}{T})^2 \quad (5.5.2)$$

The divergent $\chi_0(\mathbf{Q})$ at $T = 0$ implies that the system is on the verge of developing a density order or a spin order. In the following, we will show that, for an interacting system, the susceptibility $\chi(\mathbf{Q})$ diverges as T approaches a finite critical temperature T_c . Below T_c , the system spontaneously generates a charge-density-wave (CDW) or a spin-density-wave (SDW) state.

For interacting electrons, the compressibility and the spin susceptibility receive corrections due to the interaction. Within the RPA, we only consider corrections described by the graphs in Fig. 5.18. The one-loop bubble is given by the correlation

$$\langle \rho_{\alpha; q\nu} \rho_{\beta; -q, -\nu} \rangle = i\Pi_{\alpha\beta; q\nu}^{00} = i\Pi_{q\nu}^{00} \delta_{\alpha\beta}$$

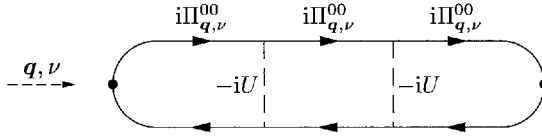


FIG. 5.19. The ladder graphs that contribute to the spin and density correlations.

and the interaction line is given by

$$-iV_{\alpha\beta;q\nu} = -iUC_{\alpha\beta}$$

where ρ_α is the density of the spin- α electron and $C = (C_{\alpha\beta})$ is a 2×2 matrix whose elements are all equal to one. We find that, within the RPA,

$$\begin{aligned} i\Pi_{\alpha\beta;q\nu}^{\text{RPA}} &= i\Pi_{\alpha\beta;q\nu}^{00} + \sum_{\gamma,\lambda} i\Pi_{\alpha\gamma;q\nu}^{00} (-iV_{\gamma\lambda;q\nu}) i\Pi_{\lambda\beta;q\nu}^{00} + \dots \\ &= \sum_{\gamma} i\Pi_{q\nu}^{00} (1 - U\Pi_{q\nu}^{00}C)^{-1}_{\alpha\beta} = i\Pi_{q\nu}^{00} \left(\delta_{\alpha\beta} + \frac{U\Pi_{q\nu}^{00}}{1 - 2U\Pi_{q\nu}^{00}} C_{\alpha\beta} \right) \end{aligned}$$

where we have used $(a + bC)^{-1} = a^{-1} - b(a^2 + 2ab)^{-1}C$. Therefore, the density response function is given by

$$\begin{aligned} i\Pi_{\text{den};q\nu}^{\text{RPA}} &= \langle (\rho_{\uparrow;q\nu} + \rho_{\downarrow;q\nu})(\rho_{\uparrow;-q,-\nu} + \rho_{\downarrow;-q,-\nu}) \rangle \\ &= i\Pi_{\uparrow\uparrow;q\nu}^{\text{RPA}} + i\Pi_{\downarrow\downarrow;q\nu}^{\text{RPA}} + 2i\Pi_{\uparrow\downarrow;q\nu}^{\text{RPA}} = 2i\Pi_{q\nu}^{00} \frac{1}{1 - 2U\Pi_{q\nu}^{00}} \end{aligned}$$

and the spin response function is given by

$$\begin{aligned} i\Pi_{\text{spin};q\nu}^{\text{RPA}} &= \left\langle \frac{1}{2}(\rho_{\uparrow;q\nu} - \rho_{\downarrow;q\nu}) \frac{1}{2}(\rho_{\uparrow;-q,-\nu} - \rho_{\downarrow;-q,-\nu}) \right\rangle \\ &= \frac{1}{4}i\Pi_{\uparrow\uparrow;q\nu}^{\text{RPA}} + \frac{1}{4}i\Pi_{\downarrow\downarrow;q\nu}^{\text{RPA}} - \frac{1}{2}i\Pi_{\uparrow\downarrow;q\nu}^{\text{RPA}} = \frac{1}{2}i\Pi_{q\nu}^{00} \end{aligned}$$

However, the above results are not quite complete. For an on-site interaction, the ladder graphs in Fig. 5.19 have similar contributions to the graphs in Fig. 5.18. Including the mixed ladder and bubble graphs, we have

$$\begin{aligned} i\Pi_{\alpha\beta;q\nu}^{\text{RPA}+} &= i\Pi_{q\nu}^{00}\delta_{\alpha\beta} + \sum_{\gamma\lambda} (i\Pi_{q\nu}^{00}\delta_{\alpha\gamma}(-iUC_{\gamma\lambda})i\Pi_{q\nu}^{00}\delta_{\lambda\beta}) \\ &\quad + \sum_{\gamma\lambda} ((-i)\Pi_{q\nu}^{00}\delta_{\alpha\gamma}(-iU\delta_{\gamma\lambda})i\Pi_{q\nu}^{00}\delta_{\lambda\beta}) + \dots \\ &= \sum_{\gamma} i\Pi_{q\nu}^{00} (1 - U\Pi_{q\nu}^{00}C + U\Pi_{q\nu}^{00})^{-1}_{\alpha\beta} \\ &= i\Pi_{q\nu}^{00} \left(\frac{1}{1 + U\Pi_{q\nu}^{00}} \delta_{\alpha\beta} + \frac{U\Pi_{q\nu}^{00}}{1 - (U\Pi_{q\nu}^{00})^2} C_{\alpha\beta} \right) \end{aligned}$$

The new density response function is given by

$$i\Pi_{\text{den};q\nu}^{\text{RPA}+} = i\Pi_{q\nu}^{00} \left(2 \frac{1}{1 + U\Pi_{q\nu}^{00}} + 2 \frac{U\Pi_{q\nu}^{00}}{1 - (U\Pi_{q\nu}^{00})^2} + 2 \frac{U\Pi_{q\nu}^{00}}{1 - (U\Pi_{q\nu}^{00})^2} \right) = 2 \frac{i\Pi_{q\nu}^{00}}{1 - U\Pi_{q\nu}^{00}}$$

and the new spin response function is given by

$$i\Pi_{\text{spin};q\nu}^{\text{RPA}+} = i\Pi_{q\nu}^{00} \left(2 \frac{1}{1 + U\Pi_{q\nu}^{00}} + 2 \frac{U\Pi_{q\nu}^{00}}{1 - (U\Pi_{q\nu}^{00})^2} - 2 \frac{U\Pi_{q\nu}^{00}}{1 - (U\Pi_{q\nu}^{00})^2} \right) = \frac{1}{2} \frac{i\Pi_{q\nu}^{00}}{1 + U\Pi_{q\nu}^{00}}$$

We find that the compressibility χ_c and the spin susceptibility χ_s within the RPA are given by

$$\chi_c^{\text{RPA}+}(\mathbf{q}) = \frac{\chi_c^{(0)}(\mathbf{q})}{1 + \frac{1}{2}U\chi_c^{(0)}(\mathbf{q})}$$

$$\chi_s^{\text{RPA}+}(\mathbf{q}) = \frac{\chi_s^{(0)}(\mathbf{q})}{1 - 2U\chi_s^{(0)}(\mathbf{q})}$$

where $\chi_c^{(0)}(\mathbf{q}) = -2\Pi_{q,0}^{00}$ and $\chi_s^{(0)}(\mathbf{q}) = -\Pi_{q,0}^{00}/2$ are the compressibility and the spin susceptibility for free electrons.

We see that a (short-range) repulsive interaction enhances the spin susceptibility and suppresses the charge compressibility, while an (short-range) attractive interaction suppresses the spin susceptibility and enhances the charge compressibility. For a nested Fermi surface and a repulsive interaction, $\chi_s^{\text{RPA}+}(\mathbf{Q})$ diverges as T approaches a finite critical temperature T_c determined by $1 - U\chi_s^{(0)}(\mathbf{Q}) = 0$. Below T_c , the system spontaneously generates an SDW and spontaneously breaks the spin-rotation symmetry and translational symmetry of the lattice. The transition temperature can be estimated from the low-temperature behavior of $\chi_s^{(0)}(\mathbf{Q})$ (see eqn (5.5.2)):

$$T_c \sim E_F e^{-C_1 \sqrt{\frac{E_F}{\mathcal{U}}}}$$

We see that, for a nested Fermi surface, a repulsive interaction always generates an SDW, no matter how weak the interaction. Similarly, for a nested Fermi surface, an attractive interaction always generates a CDW, no matter how weak the interaction. The transition temperature is

$$T_c \sim E_F e^{-C_2 \sqrt{\frac{E_F}{\mathcal{U}}}}$$

5.5.2 Mean-field approach for the spin-density-wave state

- In the mean-field approach, we replace pairs of fermion operators by their ground-state average to convert an interacting Hamiltonian to a non-interacting Hamiltonian (the mean-field Hamiltonian).

We have seen that a repulsive interaction in a system with a nested Fermi surface may cause an SDW instability. In this and the following sections, we are going study the ground state for such a system and show that the ground state indeed

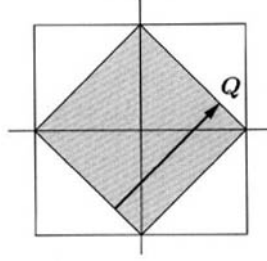


FIG. 5.20. The shaded area is the reduced Brillouin zone. It is also the filled Fermi sea for a half-filled band.

has an SDW order. We are going to use two approaches, namely the mean-field approach and the variational approach. The variational approach is conceptually simpler and more correct. It can also calculate the interaction between quasiparticles. The mean-field approach is mathematically simpler. We first consider the mean-field approach.

To use the mean-field approach to study the SDW state at $T = 0$, we first rewrite the Hamiltonian in a more convenient form. Using the identity

$$Un_i^2 = -U(c_i^\dagger \sigma^z c_i)^2 + 2Un_i$$

and absorbing the term $2Un_i$ into the chemical potential term, we can rewrite the Hubbard model as

$$H = \sum_{\mathbf{k}, \alpha} \xi_{\mathbf{k}} c_{\alpha \mathbf{k}}^\dagger c_{\alpha \mathbf{k}} - U \sum_i (c_i^\dagger \sigma^z c_i)^2$$

Now we can formally obtain a mean-field Hamiltonian by replacing one of the spin $c_i^\dagger \sigma^z c_i$ by its average $(-)^i M$, and replacing $(c_i^\dagger \sigma^z c_i)^2$ by $2M(-)^i (c_i^\dagger \sigma^z c_i) - M^2$. We obtain

$$H_{\text{mean}}^{\text{sub}} = \sum_{\mathbf{k}, \alpha} \xi_{\mathbf{k}} c_{\alpha \mathbf{k}}^\dagger c_{\alpha \mathbf{k}} - 2U \sum_i (-)^i M c_i^\dagger \sigma^z c_i + N_{\text{site}} U M^2$$

Within the mean-field approximation, we say that the physical properties of our system are described by $H_{\text{mean}}^{\text{sub}}$ instead of the original interacting Hamiltonian (5.5.1).

The mean-field Hamiltonian can be solved exactly. In the momentum space, we have

$$\begin{aligned} H_{\text{mean}}^{\text{sub}} &= \sum_{\mathbf{k}, \alpha} 2t(\cos k_x + \cos k_y) c_{\mathbf{k}}^\dagger c_{\mathbf{k}} - B \sum_{\mathbf{k}} (-)^i c_{\mathbf{k}+\mathbf{Q}}^\dagger \sigma^z c_{\mathbf{k}} + \text{constant} \\ &= \sum_{\mathbf{k}}' \psi_{\mathbf{k}}^\dagger M_{\mathbf{k}} \psi_{\mathbf{k}} + \text{constant} \end{aligned}$$

where $B = 2UM$, $\sum'_{\mathbf{k}}$ is a summation over a reduced Brillouin zone (see Fig. 5.20), $(-)^i = (-)^{i_x + i_y}$,

$$M_{\mathbf{k}} = \begin{pmatrix} \xi_{\mathbf{k}} & -B\sigma^z \\ -B\sigma^z & -\xi_{\mathbf{k}} \end{pmatrix}, \quad \xi_{\mathbf{k}} = 2t(\cos k_x + \cos k_y) \quad (5.5.3)$$

and

$$\psi_{\alpha\mathbf{k}} = \begin{pmatrix} c_{\alpha\mathbf{k}} \\ c_{\alpha,\mathbf{k}+\mathbf{Q}} \end{pmatrix}$$

We can diagonalize the above mean-field Hamiltonian $H_{\text{mean}}^{\text{sub}}$ by introducing

$$\eta_{\alpha\mathbf{k}} = -v_{\mathbf{k}}\sigma_{\alpha\beta}^z c_{\beta\mathbf{k}} + u_{\mathbf{k}}c_{\alpha,\mathbf{k}+\mathbf{Q}}$$

$$\lambda_{\alpha\mathbf{k}} = u_{\mathbf{k}}c_{\alpha\mathbf{k}} + v_{\mathbf{k}}\sigma_{\alpha\beta}^z c_{\beta,\mathbf{k}+\mathbf{Q}}$$

where

$$u_{\mathbf{k}} = \frac{1}{\sqrt{2}} \sqrt{1 - \frac{\xi_{\mathbf{k}}}{\sqrt{\xi_{\mathbf{k}}^2 + B^2}}} \quad v_{\mathbf{k}} = \text{sgn}(B) \frac{1}{\sqrt{2}} \sqrt{1 + \frac{\xi_{\mathbf{k}}}{\sqrt{\xi_{\mathbf{k}}^2 + B^2}}}$$

As $u_{\mathbf{k}}^2 + v_{\mathbf{k}}^2 = 1$, $\eta_{\alpha\mathbf{k}}$ and $\lambda_{\alpha\mathbf{k}}$ satisfy the following commutation relations for the fermion operators:

$$\{\eta_{\alpha\mathbf{k}}, \eta_{\alpha'\mathbf{k}}^\dagger\} = \delta_{\mathbf{k}\mathbf{k}'} \delta_{\alpha\alpha'}$$

$$\{\lambda_{\alpha\mathbf{k}}, \lambda_{\alpha'\mathbf{k}}^\dagger\} = \delta_{\mathbf{k}\mathbf{k}'} \delta_{\alpha\alpha'}$$

$$\{\eta_{\alpha\mathbf{k}}, \lambda_{\alpha'\mathbf{k}}^\dagger\} = 0$$

In terms of $\eta_{\alpha\mathbf{k}}$ and $\lambda_{\alpha\mathbf{k}}$, we have

$$H_{\text{mean}}^{\text{sub}} = \sum'_{\alpha,\mathbf{k}} E_{\mathbf{k}} \eta_{\alpha\mathbf{k}}^\dagger \eta_{\alpha\mathbf{k}} + \sum'_{\alpha,\mathbf{k}} -E_{\mathbf{k}} \lambda_{\alpha\mathbf{k}}^\dagger \lambda_{\alpha\mathbf{k}} + N_{\text{site}} U M^2$$

and

$$E_{\mathbf{k}} = \sqrt{\xi_{\mathbf{k}}^2 + B^2}$$

We can see that the mean-field ground-state energy is given by

$$E_{\text{mean}} = -2 \sum'_{\mathbf{k}} E_{\mathbf{k}} + N_{\text{site}} \frac{B^2}{4U}$$

The spectrum of quasiparticle excitations is given by $\pm E_{\mathbf{k}}$ using mean-field theory.

However, the above mean-field results contain an unknown parameter M —the average of the staggered spins. To determine M , we minimize the mean-field ground-state energy E_{mean} . We obtain

$$\frac{1}{4} - U N_{\text{site}}^{-1} \sum_{\mathbf{k}}' \frac{1}{E_{\mathbf{k}}} = 0. \quad (5.5.4)$$

This equation is called the gap equation. We can also obtain the same gap equation by imposing the self-consistency relation

$$M = (-)^i \langle \Phi_{\text{mean}} | c_i^\dagger \sigma^z c_i | \Phi_{\text{mean}} \rangle$$

where $|\Phi_{\text{mean}}\rangle$ is the ground state of $H_{\text{mean}}^{\text{sub}}$. That is, the average spin obtained from the mean-field ground state should be equal to the assumed average spin that we used to obtain the mean-field Hamiltonian.

If the gap equation has a solution with $M \neq 0$, then the ground state of our interacting system spontaneously breaks the spin-rotation symmetry and develops an anti-ferromagnetic order. As $\int \frac{d^2 \mathbf{k}}{(2\pi)^2} \frac{1}{E_{\mathbf{k}}} \rightarrow \infty$ as $B \rightarrow 0$, we find that eqn (5.5.4) always has $M \neq 0$ solutions as long as $U > 0$. Thus, the positive- U Hubbard model always develops an SDW, regardless of how small U is. This agrees with the result obtained in the last section.

We would like to remark that the above result remains valid as long as $\xi_{\mathbf{k}} = -\xi_{\mathbf{k}+\mathbf{Q}}$. In this case, different points on the Fermi surface (where $\xi_{\mathbf{k}} = 0$) are connected by $\mathbf{Q}$, or, in other words, the Fermi surface satisfies the nesting condition. For more general $\xi_{\mathbf{k}}$, where the Fermi surface does not satisfy the nesting condition, the quasiparticle dispersion is given by

$$\pm \sqrt{\frac{1}{4}(\xi_{\mathbf{k}} - \xi_{\mathbf{k}+\mathbf{Q}})^2 + B^2} + \frac{1}{2}(\xi_{\mathbf{k}} + \xi_{\mathbf{k}+\mathbf{Q}})$$

In this case, the gap equation (5.5.4) no longer applies. It turns out that the SDW is not developed for small U .

Although the mean-field approach (and the variational approach discussed below) correctly reproduces the anti-ferromagnetic ground state of the repulsive Hubbard model, the resulting excitation spectrum is incorrect. This is because the anti-ferromagnetic ground state spontaneously breaks the spin-rotation symmetry and there should be gapless spin wave excitations above the ground state. The spin wave excitations can be probed through the spin correlation $iS^{ab}(\mathbf{i} - \mathbf{j}) = \langle (c_i^\dagger \sigma^a c_i)(c_j^\dagger \sigma^b c_j) \rangle$. Within mean-field theory, we have $S^{ab} \sim \delta_{ab} G_{\text{mean}}^2$, where G_{mean} is the electron Green's function determined from H_{mean} . Thus, $\text{Im} S_{\mathbf{k}\omega}^{ab}$ is nonzero only when $|\omega| > 2\Delta$. However, if we go beyond mean-field theory and calculate S^{ab} using the RPA and the ladder graphs in Figs 5.18 and 5.19, then

$\text{Im}S_{\mathbf{k}\omega}^{ab}$ will be nonzero down to $\omega = 0$, indicating the presence of a gapless excitation. Also, $S_{\mathbf{k}\omega}^{ab}$ has an isolated pole from which we can obtain the dispersion of the spin wave excitations.

5.5.3 Variational approach for the spin-density-wave state—the hard way

- In the variational approach, we use the ground state of a non-interacting Hamiltonian as the trial wave function to approximate the ground state of an interacting Hamiltonian.

In the variational approach, we concentrate on the ground-state wave function, instead of the Hamiltonian. We have seen that the Hubbard model with an on-site repulsive interaction has an SDW instability. To understand the ground-state properties using the variational approach, we just guess a trial wave function for the ground state. One possible choice is to use the ground state $|\Psi_0\rangle$ of $H_0 = \sum_{\mathbf{k},\alpha} \xi_{\mathbf{k}} c_{\alpha\mathbf{k}}^\dagger c_{\alpha\mathbf{k}}$ as the trial wave function, as we did in the Hartree–Fock approach. However, $|\Psi_0\rangle$ does not break spin and translational symmetries. The presence of an SDW instability suggests that we should be able to find a different trial wave function which has a lower energy. From symmetry considerations, we would like to use the ground state $|\Psi_B\rangle$ of

$$H_B = \sum_{\mathbf{k},\alpha} 2\tilde{t}(\cos k_x + \cos k_y) c_{\alpha\mathbf{k}}^\dagger c_{\alpha\mathbf{k}} - B \sum_i (-)^i c_i^\dagger \sigma^z c_i$$

as the trial wave function. Instead of writing down the many-body trial wave function explicitly, we first diagonalize the ‘variational’ Hamiltonian H_B , as we did in the mean-field approach. The ground state of H_B satisfies $\lambda_{\alpha\mathbf{k}}^\dagger |\Psi_B\rangle = \eta_{\alpha\mathbf{k}} |\Psi_B\rangle = 0$. Such an algebraic relation determines our trial wave function.

Note that the Hamiltonian H_B is merely an operator that is used to obtain the trial wave function. Scaling H_B by a constant will not change the trial wave function. Thus, we can set $\tilde{t} = t$, and hence $\tilde{\xi}_{\mathbf{k}} = \xi_{\mathbf{k}}$. In this case, H_B has the same form as the mean-field Hamiltonian $H_{\text{mean}}^{\text{sub}}$. Here B is a variational parameter that changes the shape of the wave function. We will later vary B to minimize the energy.

To calculate the average energy of our trial state, we note that, for our trial wave function $|\Psi_B\rangle$, we have $\langle \eta_{\alpha\mathbf{k}}^\dagger \eta_{\alpha\mathbf{k}} \rangle = 0$ and $\langle \lambda_{\alpha\mathbf{k}}^\dagger \lambda_{\alpha\mathbf{k}} \rangle = 1$. From

$$c_{\alpha\mathbf{k}} = u_{\mathbf{k}} \lambda_{\alpha\mathbf{k}} - v_{\mathbf{k}} \sigma_{\alpha\beta}^z \eta_{\beta\mathbf{k}} \quad c_{\alpha,\mathbf{k}+\mathbf{Q}} = v_{\mathbf{k}} \sigma_{\alpha\beta}^z \lambda_{\beta\mathbf{k}} + u_{\mathbf{k}} \eta_{\alpha\mathbf{k}}$$

we find that

$$\begin{aligned} \langle c_{\alpha\mathbf{k}}^\dagger c_{\beta\mathbf{k}} \rangle &= u_{\mathbf{k}}^2 \delta_{\alpha\beta} & \langle c_{\alpha,\mathbf{k}+\mathbf{Q}}^\dagger c_{\beta,\mathbf{k}+\mathbf{Q}} \rangle &= v_{\mathbf{k}}^2 \delta_{\alpha\beta} \\ \langle c_{\alpha\mathbf{k}}^\dagger c_{\beta,\mathbf{k}+\mathbf{Q}} \rangle &= u_{\mathbf{k}} v_{\mathbf{k}} \sigma_{\alpha\beta}^z & \langle c_{\alpha,\mathbf{k}+\mathbf{Q}}^\dagger c_{\beta\mathbf{k}} \rangle &= u_{\mathbf{k}} v_{\mathbf{k}} \sigma_{\alpha\beta}^z \end{aligned}$$

In real space, we have

$$\langle c_{\alpha i}^\dagger c_{\beta i} \rangle = \frac{1}{2} \delta_{\alpha\beta} + 2(-)^i \sigma_{\alpha\beta}^z N_{\text{site}}^{-1} \sum_{\mathbf{k}}' u_{\mathbf{k}} v_{\mathbf{k}} = \frac{1}{2} \delta_{\alpha\beta} + (-)^i \sigma_{\alpha\beta}^z N_{\text{site}}^{-1} \sum_{\mathbf{k}}' \frac{B}{E_{\mathbf{k}}}$$

As expected, our trial state $|\Psi_B\rangle$ has an anti-ferromagnetic order:

$$\langle S_{z,i} \rangle = \frac{1}{2} \langle c_i^\dagger \sigma^z c_i \rangle = (-)^i M, \quad M = N_{\text{site}}^{-1} \sum_{\mathbf{k}}' \frac{B}{E_{\mathbf{k}}}$$

The average energy of our trial state is given by (see Section 5.5.4.1)

$$\langle \Psi_B | H | \Psi_B \rangle = N_{\text{site}} \left(-2 N_{\text{site}}^{-1} \sum_{\mathbf{k}}' \frac{\xi_{\mathbf{k}}^2}{E_{\mathbf{k}}} - U \left(N_{\text{site}}^{-1} \sum_{\mathbf{k}}' \frac{B}{E_{\mathbf{k}}} \right)^2 + \text{constant} \right)$$

The value of B is obtained by minimizing $\langle \Psi_B | H | \Psi_B \rangle$. We find that such a B satisfies the gap equation (see Section 5.5.4.2):

$$1 - U N_{\text{site}}^{-1} \sum_{\mathbf{k}}' \frac{1}{\sqrt{\xi_{\mathbf{k}}^2 + B^2}} = 0 \quad (5.5.5)$$

We see that eqns (5.5.4) and (5.5.5) do not agree. This illustrates the point that mean-field approaches may lead to different results at the leading order. This is because different mean-field approaches correspond to different expansions, which in general have different leading terms.

The variational approach also allows us study the properties of excitations. The trial wave functions for excitations are created by $\eta_{\alpha\mathbf{k}}^\dagger$ and $\lambda_{\alpha\mathbf{k}}$ from the trial ground state $|\Psi_B\rangle$. For the excited states, we have $\langle \eta_{\alpha\mathbf{k}}^\dagger \eta_{\alpha\mathbf{k}} \rangle = n_{\alpha\mathbf{k}}^+$ and $\langle \lambda_{\alpha\mathbf{k}}^\dagger \lambda_{\alpha\mathbf{k}} \rangle = n_{\alpha\mathbf{k}}^-$ with $n_{\alpha\mathbf{k}}^+ \neq 0$ and $n_{\alpha\mathbf{k}}^- \neq 1$. After relating $\langle c^\dagger c \rangle$ to $n_{\alpha\mathbf{k}}^+$ and $n_{\alpha\mathbf{k}}^-$, we obtain the average energy of our trial excited state (see Section 5.5.4.3) as follows:

$$\langle \Psi_B^{\text{exc}} | H | \Psi_B^{\text{exc}} \rangle = E_{\text{ground}} + \sum_{\mathbf{k}, \alpha}' E_{\mathbf{k}} (\delta n_{\alpha\mathbf{k}}^+ - \delta n_{\alpha\mathbf{k}}^-) + U \delta N + O(\delta n^2)$$

where $n_{\alpha\mathbf{k}}^- = 1 + \delta n_{\alpha\mathbf{k}}^-$ and $n_{\alpha\mathbf{k}}^+ = \delta n_{\alpha\mathbf{k}}^+$ describe small fluctuations around the ground state. We see that the quasiparticle excitations have an energy spectrum $\pm E_{\mathbf{k}}$. Note that there is a finite energy gap $\Delta = \min E_{\mathbf{k}} = |B|$ for quasiparticle excitations. The interaction between the quasiparticles can be easily included through the $(\delta n_{\alpha\mathbf{k}}^\pm)^2$ terms.

5.5.4 Remarks: some calculation details

5.5.4.1 Calculation of the average energy

We calculate the average energy of the trial state $|\Psi_B\rangle$ as follows:

$$\begin{aligned}
 & \langle \Psi_B | H | \Psi_B \rangle \\
 &= 2 \sum_k' (u_k^2 - v_k^2) \xi_k + \frac{U}{2} \sum_i \left(\langle c_{\alpha i}^\dagger c_{\alpha i} \rangle \langle c_{\beta i}^\dagger c_{\beta i} \rangle + \langle c_{\alpha i}^\dagger c_{\beta i} \rangle \langle c_{\alpha i} c_{\beta i}^\dagger \rangle \right) \\
 &= 2 \sum_k' (u_k^2 - v_k^2) \xi_k + \frac{U}{2} \sum_i \left(1 + \text{Tr} \left(\frac{1}{2} + (-)^i M \sigma^z \right) \left(\frac{1}{2} - (-)^i M \sigma^z \right) \right) \\
 &= N_{\text{site}} \left(-UM^2 + \frac{3U}{4} + 2N_{\text{site}}^{-1} \sum_k' - \frac{\xi_k}{E_k} \xi_k \right)
 \end{aligned}$$

5.5.4.2 Calculation of the gap equation

To obtain the value of B that minimizes the average energy, we calculate $\partial \langle \Psi_B | H | \Psi_B \rangle / \partial B$ and obtain

$$\begin{aligned}
 & 2N_{\text{site}}^{-1} \sum_k' - \xi_k^2 \delta E_k^{-1} - 2U \left(N_{\text{site}}^{-1} \sum_k' \frac{B}{E_k} \right) \left(N_{\text{site}}^{-1} \sum_k' \frac{\delta B}{E_k} - \frac{B^2 \delta B}{E_k^3} \right) \\
 &= 2N_{\text{site}}^{-1} \sum_k' - \xi_k^2 \delta E_k^{-1} - 2U \left(N_{\text{site}}^{-1} \sum_k' \frac{B}{E_k} \right) \left(N_{\text{site}}^{-1} \sum_k' \frac{\xi_k^2 \delta B}{E_k^3} \right) \\
 &= \left(2N_{\text{site}}^{-1} \sum_k' - \xi_k^2 \delta E_k^{-1} \right) - 2U \left(N_{\text{site}}^{-1} \sum_k' \frac{B}{E_k} \right) \left(N_{\text{site}}^{-1} \sum_k' - \xi_k^2 B^{-1} \delta E_k^{-1} \right)
 \end{aligned}$$

The requirement that the above vanishes leads to the gap equation (5.5.5).

5.5.4.3 Calculation of the average energy for excited states

From the relations

$$\begin{aligned}
 \langle c_{\alpha k}^\dagger c_{\beta k} \rangle &= u_k^2 n_{\alpha k}^- \delta_{\alpha\beta} + v_k^2 n_{\alpha k}^+ \delta_{\alpha\beta}, & \langle c_{\alpha, k+Q}^\dagger c_{\beta, k+Q} \rangle &= v_k^2 n_{\alpha k}^- \delta_{\alpha\beta} + u_k^2 n_{\alpha k}^+ \delta_{\alpha\beta}, \\
 \langle c_{\alpha k}^\dagger c_{\beta, k+Q} \rangle &= u_k v_k (n_{\alpha k}^- - n_{\alpha k}^+) \sigma_{\alpha\beta}^z, & \langle c_{\alpha, k+Q}^\dagger c_{\beta k} \rangle &= u_k v_k (n_{\alpha k}^- - n_{\alpha k}^+) \sigma_{\alpha\beta}^z,
 \end{aligned}$$

we find that, in real space,

$$\langle c_{\alpha i}^\dagger c_{\beta i} \rangle = \delta_{\alpha\beta} N_{\text{site}}^{-1} \sum_k' (n_{\alpha k}^- + n_{\alpha k}^+) + 2(-)^i \sigma_{\alpha\beta}^z N_{\text{site}}^{-1} \sum_k' u_k v_k (n_{\alpha k}^- - n_{\alpha k}^+)$$

This allows us to calculate

$$\begin{aligned}
& \sum_i \left(\langle c_{\alpha i}^\dagger c_{\alpha i} \rangle \langle c_{\beta i}^\dagger c_{\beta i} \rangle + \langle c_{\alpha i}^\dagger c_{\beta i} \rangle \langle c_{\alpha i} c_{\beta i}^\dagger \rangle \right) \\
&= \sum_i (n_{\uparrow i} + n_{\downarrow i})^2 + n_{\uparrow i}(1 - n_{\uparrow i}) + n_{\downarrow i}(1 - n_{\downarrow i}) \\
&= \sum_i n_{\uparrow i} + n_{\downarrow i} + 2n_{\uparrow i}n_{\downarrow i} \\
&= N + 2N_{\text{site}}^{-2} \sum_i \sum_{\mathbf{k}, \mathbf{k}'}' \left(n_{\uparrow \mathbf{k}}^- + n_{\uparrow \mathbf{k}}^+ + 2(-)^i u_{\mathbf{k}} v_{\mathbf{k}} (n_{\uparrow \mathbf{k}}^- - n_{\uparrow \mathbf{k}}^+) \right) \times \\
&\quad \left(n_{\downarrow \mathbf{k}'}^- + n_{\downarrow \mathbf{k}'}^+ - 2(-)^i u_{\mathbf{k}'} v_{\mathbf{k}'} (n_{\downarrow \mathbf{k}'}^- - n_{\downarrow \mathbf{k}'}^+) \right) \\
&= N + 2N_{\text{site}}^{-1} \sum_{\mathbf{k}, \mathbf{k}'}' \left((n_{\uparrow \mathbf{k}}^- + n_{\uparrow \mathbf{k}}^+)(n_{\downarrow \mathbf{k}'}^- + n_{\downarrow \mathbf{k}'}^+) - \frac{B^2}{E_{\mathbf{k}} E_{\mathbf{k}'}} (n_{\uparrow \mathbf{k}}^- - n_{\uparrow \mathbf{k}}^+)(n_{\downarrow \mathbf{k}'}^- - n_{\downarrow \mathbf{k}'}^+) \right)
\end{aligned}$$

Using this result, we find that

$$\begin{aligned}
\langle \Psi_B^{\text{exc}} | H | \Psi_B^{\text{exc}} \rangle &= \sum_{\mathbf{k}, \alpha}' (u_{\mathbf{k}}^2 - v_{\mathbf{k}}^2) (n_{\alpha \mathbf{k}}^- - n_{\alpha \mathbf{k}}^+) \xi_{\mathbf{k}} \quad (5.5.6) \\
&\quad + \frac{U}{2} \sum_i \left(\langle c_{\alpha i}^\dagger c_{\alpha i} \rangle \langle c_{\beta i}^\dagger c_{\beta i} \rangle + \langle c_{\alpha i}^\dagger c_{\beta i} \rangle \langle c_{\alpha i} c_{\beta i}^\dagger \rangle \right) \\
&= - \sum_{\mathbf{k}, \alpha}' \frac{\xi_{\mathbf{k}}^2}{E_{\mathbf{k}}} (n_{\alpha \mathbf{k}}^- - n_{\alpha \mathbf{k}}^+) + \frac{U}{N_{\text{site}}} \sum_{\mathbf{k}, \mathbf{k}'}' (n_{\uparrow \mathbf{k}}^- + n_{\uparrow \mathbf{k}}^+) (n_{\downarrow \mathbf{k}'}^- + n_{\downarrow \mathbf{k}'}^+) \\
&\quad + \frac{U}{N_{\text{site}}} \sum_{\mathbf{k}, \mathbf{k}'}' \frac{B^2}{E_{\mathbf{k}} E_{\mathbf{k}'}} (n_{\uparrow \mathbf{k}}^- - n_{\uparrow \mathbf{k}}^+) (n_{\downarrow \mathbf{k}'}^- - n_{\downarrow \mathbf{k}'}^+) + \frac{U}{2} N
\end{aligned}$$

where $N = \sum_i n_{\uparrow i} + n_{\downarrow i}$ is the total number of electrons and $n_{\alpha i}$ is the number of spin- α electron at site i .

Expanding $\langle \Psi_B^{\text{exc}} | H | \Psi_B^{\text{exc}} \rangle$ to linear order in $\delta n_{\alpha \mathbf{k}}^\pm$, we obtain

$$\begin{aligned}
\langle \Psi_B^{\text{exc}} | H | \Psi_B^{\text{exc}} \rangle &= E_{\text{ground}} + \frac{U}{2} \delta N - \sum_{\mathbf{k}, \alpha}' \frac{\xi_{\mathbf{k}}^2}{E_{\mathbf{k}}} (\delta n_{\alpha \mathbf{k}}^- - \delta n_{\alpha \mathbf{k}}^+) \\
&\quad + \frac{U}{N_{\text{site}}} \sum_{\mathbf{k}, \mathbf{k}', \alpha}' \left((\delta n_{\alpha \mathbf{k}}^- + \delta n_{\alpha \mathbf{k}}^+) - \frac{B^2}{E_{\mathbf{k}} E_{\mathbf{k}'}} (\delta n_{\alpha \mathbf{k}}^- - \delta n_{\alpha \mathbf{k}}^+) \right) \\
&= E_{\text{ground}} + \sum_{\mathbf{k}, \alpha}' E_{\mathbf{k}} (\delta n_{\alpha \mathbf{k}}^+ - \delta n_{\alpha \mathbf{k}}^-) + U \delta N + O(\delta n^2)
\end{aligned}$$

where we have used $N = \sum_{\mathbf{k}, \alpha}' (n_{\alpha \mathbf{k}}^- + n_{\alpha \mathbf{k}}^+)$, $\sum_{\mathbf{k}}' 1 = N_{\text{site}}/2$, and $1 = \frac{U}{N_{\text{site}}} \sum_{\mathbf{k}'}' E_{\mathbf{k}'}^{-1}$.

5.6 Nonlinear σ -model

5.6.1 Nonlinear σ -model for the spin-density-wave state

- The nonlinear σ -model describes the dynamics of collective fluctuations around a symmetry-breaking ground state.
- The Hubbard–Stratonovich transformation.

The mean-field approach used in the last section fails to reproduce the gapless spin wave excitations in the SDW state. To obtain the spin wave excitations, we have to go beyond mean-field theory and use, for example, the RPA. However, here we will use the semiclassical approach and the related nonlinear σ -model to study the spin wave excitations. In the semiclassical approach, we need to first derive a low-energy effective Lagrangian for the spin waves.

In our variational approach, we have to include a B field in the ‘variational’ Hamiltonian to generate a trial wave function with a uniform anti-ferromagnetic spin polarization M . A natural way to include the spin waves is to replace B by a spatially-dependent B_i field in the ‘variational’ Hamiltonian, which generates a spatially-dependent anti-ferromagnetic spin polarization M_i in the new trial wave function. The energy for the new trial wave function is a functional of M_i , i.e. $E[M_i]$, and can be regarded as the energy for the spin waves. However, $E[M_i]$ is the energy for the time-independent spin wave fluctuations and is only the potential energy of the spin wave. To obtain the effective Lagrangian, we also need the kinetic energy for time-dependent spin wave fluctuations. The easiest way to obtain the kinetic energy (as well as the potential energy) of the spin waves is to use the path integral approach developed in Section 5.4.1.

To obtain the low-energy effective theory for spin waves, we note that the Hubbard term $\frac{U}{2} c_{\alpha i}^\dagger c_{\alpha i} c_{\beta i}^\dagger c_{\beta i}$ is equal to 0 for $n_i = 0$, equal to $U/2$ for $n_i = 1$, and equal to $2U$ for $n_i = 2$. Also note that S_i^2 is equal to 0 for $n_i = 0$, equal to $3/4$ for $n_i = 1$, and equal to 0 for $n_i = 2$, where

$$S_i = \frac{1}{2} c_i^\dagger \sigma c_i$$

is the total spin operator at site i . Therefore

$$\frac{U}{2} c_{\alpha i}^\dagger c_{\alpha i} c_{\beta i}^\dagger c_{\beta i} = U n_i - \frac{2U}{3} S_i^2$$

We will drop the $U n_i$ term by absorbing it into the chemical potential. Thus, the Hubbard model is described by the following path integral:

$$Z = \int \mathcal{D}c^* \mathcal{D}c \, e^{i \int dt \, (L_0 + L_I)}$$

where

$$L_0 = i \sum_i c_{\alpha i}^* \partial_t c_{\alpha i} - \sum_{\langle ij \rangle} -t(c_{\alpha i}^* c_{\alpha j} + c_{\alpha j}^* c_{\alpha i})$$

$$L_I = \frac{U}{6} \sum_i c_i^\dagger \sigma c_i c_i^\dagger \sigma c_i$$

The trick here is to write the interaction term as

$$e^{i \int dt \frac{U}{6} c_i^\dagger \sigma c_i c_i^\dagger \sigma c_i} = \text{constant} \times \int \mathcal{D}[\mathbf{B}_i(t)] e^{i \int dt (-)^i \mathbf{B}_i(t) c_i^\dagger \sigma c_i - \frac{3}{2U} \mathbf{B}_i^2(t)}$$

which converts the fermion quartic term to a fermion quadratic term. Such a transformation is called a Hubbard–Stratonovich transformation. Now the partition function can be rewritten as follows:

$$Z = \int \mathcal{D}c^* \mathcal{D}c \mathcal{D}\mathbf{B} e^{i \int dt L_B - i \int dt \frac{3}{2U} \sum_i \mathbf{B}_i^2} \quad (5.6.1)$$

$$L_B = i \sum_i c_{\alpha i}^* \partial_t c_{\alpha i} - \sum_{\langle ij \rangle} -t(c_{\alpha i}^* c_{\alpha j} + c_{\alpha j}^* c_{\alpha i}) + \sum_i (-)^i \mathbf{B}_i(t) c_i^\dagger \sigma c_i$$

where the fermion Lagrangian is quadratic.

If we integrate out the fermion first, then we obtain

$$Z = \int \mathcal{D}\mathbf{B} e^{i \int dt L_{\text{eff}}(\mathbf{B})}$$

where $L_{\text{eff}}(\mathbf{B})$ is an effective Lagrangian. As $\mathbf{B}$ couples to the electron spin, it describes the spin wave fluctuations.

Let us first study some basic properties of $L_{\text{eff}}(\mathbf{B})$. Assuming that $\mathbf{B}$ is a constant in space–time, then

$$L_{\text{eff}}(\mathbf{B}) = -2 \sum_k' (-E_k) - \frac{3}{2U} N_{\text{site}} \mathbf{B}^2 = -E_{\text{mean}}(\mathbf{B})$$

is the negative of the mean-field energy, where $E_k = \sqrt{\xi_k^2 + |\mathbf{B}|^2}$. Minimizing $E_{\text{mean}}(\mathbf{B})$, we obtain the gap equation

$$\frac{3}{2} - U N_{\text{site}}^{-1} \sum_k' \frac{1}{E_k} = 0 \quad (5.6.2)$$

which determines the mean-field value of $\mathbf{B}_0$. If $\mathbf{B}_0 \neq 0$, then the mean-field ground state is an SDW state. Let us assume that this is the case.

It is clear that the ground states associated with different spin orientations are degenerate. If we have a long-wavelength spin orientation fluctuation, then locally the system is just in one of the many possible ground states. This corresponds to a low-energy fluctuation. To study this kind of low-energy fluctuation, we introduce

$$\mathbf{B}_i(t) = |\mathbf{B}_i(t)|\mathbf{n}_i(t), \quad |\mathbf{n}_i(t)| = 1$$

and replace $|\mathbf{B}_i(t)|$ by $|\mathbf{B}_0|$. This gives us a low-energy effective Lagrangian $L_\sigma(\mathbf{n}) = L_{\text{eff}}(\mathbf{n}\mathbf{B}_0)$. In the long-wavelength limit, the effective Lagrangian can be written as

$$L_\sigma(\mathbf{n}) = \int d^d\mathbf{x} \frac{1}{2g} ((\partial_t \mathbf{n})^2 - v^2 (\partial_x \mathbf{n})^2) + \dots \quad (5.6.3)$$

The terms indicated by ‘...’ are higher-derivative terms. The spin-rotation symmetry forbids the possible potential term (or mass term) $V(\mathbf{n})$ and guarantees gapless spin fluctuations. Such an effective Lagrangian is called an $O(3)$ nonlinear σ -model.

We note that the effective Lagrangian is an expansion in powers of $\partial_\mu \mathbf{n}$. However, when $\mathbf{B} = \mathbf{n}B_0$, L_B in eqn (5.6.1) directly depends on $\mathbf{n}$. It is not obvious how to obtain an expansion in powers of $\partial_\mu \mathbf{n}$. In the following, we discuss a trick to do so. The trick allows us to directly calculate g and v in the nonlinear σ -model.

We introduce $\psi_i = U_i c_i$, where

$$U_i = \begin{pmatrix} z_{1i}^* & z_{2i}^* \\ -z_{2i} & z_{1i} \end{pmatrix}, \quad \psi_i = \begin{pmatrix} \psi_{1i} \\ \psi_{2i} \end{pmatrix}$$

$$z_i = \begin{pmatrix} z_{1i} \\ z_{2i} \end{pmatrix}, \quad z_i^\dagger z_i = 1, \quad z_i^\dagger \sigma z_i = \mathbf{n}_i$$

Here ψ_{1i} corresponds to the spin in the $\mathbf{n}_i$ direction and ψ_{2i} to the spin in the $-\mathbf{n}_i$ direction. Now L_B can be rewritten as

$$L_B = i \sum_i \psi_i^* (\partial_t + U_i \partial_t U_i^\dagger) \psi_i - \sum_{\langle ij \rangle} -t (\psi_i^* u_{ij} \psi_j + h.c.) + B_0 \sum_i (-)^i \psi_i^\dagger \sigma^z \psi_i$$

where

$$u_{ij} = 1 + \begin{pmatrix} z_i^\dagger z_j - 1 & \frac{1}{2}(z_i - z_j)^\dagger i\sigma^y (z_i^* + z_j^*) \\ -\frac{1}{2}(z_i - z_j)^\top i\sigma^y (z_i + z_j) & z_j^\dagger z_i - 1 \end{pmatrix}$$

We see that the B_0 term is independent of $\mathbf{n}$ in the new basis. In the new basis, the L_B dependence on $\mathbf{n}$ is via the derivative of $\mathbf{n}$.

Consider small fluctuations around $\mathbf{n} = \hat{\mathbf{z}}$ as follows:

$$\mathbf{n}_i = \hat{\mathbf{z}} + \text{Re}\phi_i \hat{\mathbf{x}} + \text{Im}\phi_i \hat{\mathbf{y}} \quad (5.6.4)$$

and the corresponding spinor field

$$z_i = \left(\frac{1 - |\phi_i|^2/8}{\phi_i/2} \right) + O(\phi_i^3)$$

Assuming further that ϕ is real, u_{ij} can be simplified as follows:

$$u_{ij} = 1 + \begin{pmatrix} -\frac{1}{8}(\phi_i - \phi_j)^2 & \frac{1}{2}(\phi_j - \phi_i) \\ -\frac{1}{2}(\phi_j - \phi_i) & -\frac{1}{8}(\phi_i - \phi_j)^2 \end{pmatrix} = e^{i \frac{\phi_i - \phi_j}{2} \sigma^y}$$

Also, $U_i \partial_t U_i^\dagger$ can be simplified to

$$U_i \partial_t U_i^\dagger = \begin{pmatrix} 0 & -\frac{1}{2} \partial_t \phi_i \\ \frac{1}{2} \partial_t \phi_i & 0 \end{pmatrix}$$

We note that the coupling of the fermions to the spin fluctuations takes the form of a coupling to an $SU(2)$ gauge field. For small spin fluctuations, L_B can be written as

$$L_B = i \sum_i \psi_i^* \partial_t \psi_i - H_0 - H_I$$

where

$$H_0 + H_I = -t \sum_{\langle ij \rangle} (\psi_i^* e^{i a_{ij}} \psi_j + h.c.) + B_0 \sum_i (-)^i \psi_i^* \sigma^z \psi_i + \sum_i \psi_i^* a_0(i) \psi_i,$$

$$a_0(i) = -\frac{1}{2} \partial_t \phi_i \sigma^y, \text{ and } a_{ij} = \frac{1}{2} (\phi_i - \phi_j) \sigma^y.$$

If $a_0(i)$ and a_{ij} are constant in space, then the Hamiltonian $H_0 + H_I$ can be solved in momentum space. Let $a_0(i) = -\frac{1}{2} B_y \sigma^y$ and $a_{ij} = \frac{1}{2} \mathbf{a} \cdot (\mathbf{i} - \mathbf{j}) \sigma^y$. Note that $B_y = \partial_t \phi$ and $\mathbf{a} = \partial_i \phi_i$. For small B_y and $\mathbf{a}$, the ground-state energy has the form

$$E_0(B_y, \mathbf{a}) = -C_1 B_y^2(i) + C_2 \mathbf{a}^2 \quad (5.6.5)$$

The $a_0^2(i)$ term is the $(\partial_t \mathbf{n})^2$ term in the effective Lagrangian eqn (5.6.3), i.e. $C_1 a_0^2(i) = \int d^d x (\partial_t \phi)^2 / 2g$, and the a_{ij}^2 term is the $(\partial_x \mathbf{n})^2$ term, i.e. $C_2 \mathbf{a}^2 =$

$\int d^d x v^2 (\partial_x \phi)^2 / 2g$. This allows us to obtain g and v in L_σ . We find that

$$g^{-1} = -a^{-d} N_{\text{site}}^{-1} \left. \frac{\partial^2 E_0(B_y)}{\partial B_y^2} \right|_{B_y=0},$$

$$E_0(B_y) = N_{\text{site}}^{-1} \sum_{\mathbf{k}}' (-E_{+, \mathbf{k}}^g - E_{-, \mathbf{k}}^g), \quad (5.6.6)$$

$$E_{\pm, \mathbf{k}}^g = \sqrt{\left(\xi_{\mathbf{k}} \pm \frac{B_y}{2}\right)^2 + B_0^2}$$

and

$$\frac{v^2}{g} = a^{-d} \left. \frac{\partial^2 E_0(\mathbf{a})}{\partial \mathbf{a}^2} \right|_{\mathbf{a}=0},$$

$$E_0(\mathbf{a}) = N_{\text{site}}^{-1} \sum_{\mathbf{k}}' (-E_{+, \mathbf{k}}^v - E_{-, \mathbf{k}}^v), \quad (5.6.7)$$

$$E_{\pm, \mathbf{k}}^v = \left| \sqrt{\frac{1}{4}(\xi_{\mathbf{k}+\mathbf{a}/2} + \xi_{\mathbf{k}-\mathbf{a}/2})^2 + B_0^2} \pm \frac{1}{2}(\xi_{\mathbf{k}+\mathbf{a}/2} - \xi_{\mathbf{k}-\mathbf{a}/2}) \right|,$$

where a is the lattice constant and $\sum_{\mathbf{k}}'$ is the sum over the reduced Brillouin zone (see Fig. 5.20).

Problem 5.6.1.

Prove eqns (5.6.6) and (5.6.7).

5.6.2 Stability of long-range orders

- The nonlinear σ -model allows us to study quantum fluctuations and the stability of the SDW state.
- Stability and the topological term.

Classically, one of the ground states of the nonlinear σ -model (5.6.3) is $\mathbf{n}(\mathbf{x}) = \hat{z}$, which has a long-range spin order. The small-spin wave fluctuations are described by eqn (5.6.4) with a Lagrangian

$$\frac{1}{2g} (|\partial_t \phi|^2 - v^2 |\partial_x \phi|^2)$$

which has a linear dispersion $\epsilon_{\mathbf{k}} = v|\mathbf{k}|$, just like the fluctuations in the XY-model. The quantity $\langle |\phi(\mathbf{x}, t) - \phi(0)|^2 \rangle$ measures the strength of the long-range transverse spin fluctuations. If $\langle |\phi(\mathbf{x}, t) - \phi(0)|^2 \rangle \gg 1$, then the long-range order cannot exist. This problem is identical to the one in the XY-model. The strength of

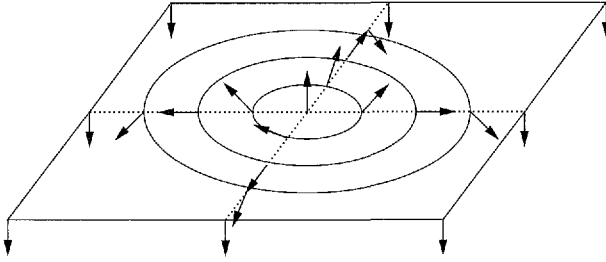


FIG. 5.21. The soliton/instanton solution in the $(1+1)$ -dimensional $O(3)$ nonlinear σ -model.

the fluctuations is controlled by g . A small value of g leads to weak fluctuations. When g is too large, the long-range spin order will be destroyed by spin wave fluctuations. However, when $d \leq 1$ we find that $\langle |\phi(\mathbf{x}, t) - \phi(0)|^2 \rangle$ diverges as $(\mathbf{x}, t) \rightarrow \infty$, no matter how small g is. Therefore, there cannot be long-range order in $d \leq 1$ dimensions for any values of g . For the XY-model, we can still have algebraic long-range order when $d = 1$ under certain conditions. The question now is whether the $O(3)$ nonlinear σ -model has algebraic long-range order for $d = 1$.

To answer this question, let us consider imaginary time and investigate the following $(1+1)$ -dimensional system:

$$\frac{1}{2g} (|\partial_\tau \mathbf{n}|^2 + v^2 |\partial_x \mathbf{n}|^2)$$

The above action has a non-trivial stationary 'path', which is an instanton solution (see Fig. 5.21). The action is independent of the size of the instanton and is equal to $S_0 = 4\pi v/g$. This is because $S(\mathbf{n}(x, \tau)) = S(\mathbf{n}(bx, b\tau))$ for any $\mathbf{n}(x, \tau)$ and any scaling factor b .

As the action of the instanton is finite, we always have a finite (space-time) density of the instantons. The mean separation of the instantons is of order $l e^{S_0/2}$, where l is a short-distance cut-off. The size of the instantons is also of order $l e^{S_0/2}$ (because that action of the instantons is independent of their size). Thus, the long-range spin order is destroyed by the instantons. The spin-spin correlation is short-ranged, with a correlation length

$$\xi = l e^{S_0/2} = l e^{2\pi v/g} \quad (5.6.8)$$

It is interesting to see that ξ or $1/\xi$ is non-perturbative in g .

We see that the $O(3)$ nonlinear σ -model has a short-ranged correlation in $1+1$ dimensions, and all excitations are gapped. However, this result is only half correct. The $O(3)$ nonlinear σ -model can contain a topological term that changes the low-energy properties of the model.

To understand the origin of the topological term, let us consider the large- U limit of the Hubbard model (5.5.1). When $U \gg t$, the low-energy properties of the

Hubbard model (at half-filling) are described by the Heisenberg model:

$$H = \sum_{\langle ij \rangle} J \mathbf{S}_i \cdot \mathbf{S}_j \quad (5.6.9)$$

where $\mathbf{S}_i = c_i^\dagger \frac{\boldsymbol{\sigma}}{2} c_i$ is the spin operator and $J = 4t^2/U$. The effective Lagrangian of the Heisenberg model is

$$L = i \sum_i \dot{z}_i^\dagger \dot{z}_i - \frac{J}{4} \sum_{\langle ij \rangle} \mathbf{n}_i \cdot \mathbf{n}_j \quad (5.6.10)$$

where $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$ and $\mathbf{n} = z^\dagger \boldsymbol{\sigma} z$. Now let us assume that the ground state of the Heisenberg model is the anti-ferromagnetic (AF) state $\mathbf{n}_i = (-)^i \hat{z}$, and we want to study the small fluctuations around the AF ground state.

The gapless spin wave fluctuations are described by

$$\mathbf{n}_i(t) = (-)^i \mathbf{n}(x^i, t)$$

It is very tempting to substitute the above into the Lagrangian (5.6.10) so as to obtain a continuum effective theory. However, this simple-minded approach does not lead to a correct result. For one thing, the resulting Lagrangian contains only a first-order time derivative term and no second-order derivative term. One may say that, at low energy, the first-order time derivative term is more important than the second-order time derivative term. It should be alright to drop the second-order time derivative term. However, it turns out that the first-order time derivative term is a total derivative, which has no effect on the equation of motion. We need to include the second-order time derivative term so that the equation of motion has non-trivial dynamics. The situation here is very similar to what we encountered in Section 3.3.3, where we derived the XY-model from an interacting boson model.

To obtain the second-order time derivative term, we consider the following more general fluctuations:

$$\mathbf{n}_i(t) = (-)^i \mathbf{n}(x^i, t) + \mathbf{m}(x^i, t) \quad (5.6.11)$$

where $\mathbf{n}(x, t)$ is a smooth function of (x, t) with $|\mathbf{n}(x, t)| = 1$, and $\mathbf{m}(x, t)$ is also a smooth function of (x, t) but with $|\mathbf{m}(x, t)| \ll 1$ and $\mathbf{m}(x, t) \cdot \mathbf{n}(x, t) = 0$. Here $\mathbf{n}$ describes the low-energy AF spin wave fluctuations. Also, $\mathbf{m}$ describes the ferromagnetic fluctuations which have high energies and which will be integrated out. As $\int dt \, 2i \dot{z}_i^\dagger \dot{z}_i$ is the solid angle spanned by $\mathbf{n}_i$, we have

$$\int dt \, 2i \dot{z}_i^\dagger \dot{z}_i = \int dt \, 2i (-)^i \dot{z}(x^i, t)^\dagger \dot{z}(x^i, t) + \int dt \, \mathbf{n} \cdot (\dot{\mathbf{n}} \times \mathbf{m})$$

where $\mathbf{n}(x, t) = z^\dagger(x, t)\boldsymbol{\sigma}z(x, t)$, or

$$i \sum_i z_i^\dagger \dot{z}_i = \frac{1}{2} \int dx \frac{1}{2} \mathbf{n} \cdot (\partial_t \mathbf{n} \times \partial_x \mathbf{n}) + \int dx \frac{1}{2a} \mathbf{n} \cdot (\dot{\mathbf{n}} \times \mathbf{m})$$

where a is the lattice spacing. The $\frac{J}{4} \sum_{\langle ij \rangle} \mathbf{n}_i \cdot \mathbf{n}_j$ term induces a term $\mathbf{m}^2$. Thus, after integrating out $\mathbf{m}$, we obtain

$$S = \int dt dx \frac{1}{2g} [(\partial_t \mathbf{n})^2 - v^2 (\partial_x \mathbf{n})^2] + \theta W \quad (5.6.12)$$

with $\theta = \pi$. Comparing this with the effective action that we calculated before, see eqn (5.6.3), the above contains an extra term θW with

$$W = \int dt dx \frac{1}{4\pi} \mathbf{n} \cdot (\partial_t \mathbf{n} \times \partial_x \mathbf{n}) \quad (5.6.13)$$

Such a term is a topological term, because $W(\mathbf{n} + \delta \mathbf{n}) - W(\mathbf{n}) = 0$ for any small variation $\delta \mathbf{n}$.³⁷ Therefore, $W(\mathbf{n})$ takes the same value for different $\mathbf{n}$ s that can continuously deform into each other. Note that $\mathbf{n}$ is a mapping from a two-dimensional plane $\mathcal{R}^2$ to a sphere S^2 , i.e. $\mathcal{R}^2 \rightarrow S^2$. The mapping does have different classes that cannot continuously deform into each other. These different classes are characterized by a winding number—the number of times the $\mathcal{R}^2$ wrap around S^2 . Here $W(\mathbf{n})$ in eqn (5.6.13) taking only integer values is a mathematical expression of the winding number.

In contrast to g and v , the value of $\theta = \pi$ in (5.6.12) is exact and quantized. It does not receive corrections from high-order terms in the perturbative expansion. To see this, we note that the translation by one lattice space induces a transformation $\mathbf{n} \rightarrow -\mathbf{n}$ in our effective theory (5.6.12). Under such a transformation,

³⁷ We note that

$$W(\mathbf{n} + \delta \mathbf{n}) - W(\mathbf{n}) = \int dt dx \left[\frac{1}{4\pi} \mathbf{n} \cdot (\partial_t \delta \mathbf{n} \times \partial_x \mathbf{n}) + \frac{1}{4\pi} \mathbf{n} \cdot (\partial_t \mathbf{n} \times \partial_x \delta \mathbf{n}) \right]$$

because $\mathbf{n} \cdot \delta \mathbf{n} = 0$ and $\partial_t \mathbf{n} \times \partial_x \mathbf{n}$ is parallel to $\mathbf{n}$. Also,

$$\begin{aligned} \int dt dx \mathbf{n} \cdot (\partial_t \delta \mathbf{n} \times \partial_x \mathbf{n}) &= \int dt dx \partial_t \delta \mathbf{n} \cdot (\partial_x \mathbf{n} \times \mathbf{n}) \\ &= - \int dt dx [\delta \mathbf{n} \cdot (\partial_t \partial_x \mathbf{n} \times \mathbf{n}) + \delta \mathbf{n} \cdot (\partial_x \mathbf{n} \times \partial_t \mathbf{n})] \\ &= - \int dt dx \delta \mathbf{n} \cdot (\partial_t \partial_x \mathbf{n} \times \mathbf{n}), \end{aligned}$$

and similarly

$$\int dt dx \mathbf{n} \cdot (\partial_t \mathbf{n} \times \partial_x \delta \mathbf{n}) = + \int dt dx \delta \mathbf{n} \cdot (\partial_t \partial_x \mathbf{n} \times \mathbf{n}).$$

S should be invariant. As $W \rightarrow -W$ under the transformation, only $\theta = 0$ and π are consistent with the symmetry of translation by one lattice space. Thus, as long as the underlying lattice Hamiltonian has the symmetry of translation by one lattice space, then θ can only take the two values 0 and π . Due to the quantization of θ , eqn (5.6.12) not only applies to the Heisenberg model (5.6.9), but it also applies to the AF state of the Hubbard model (5.5.1). We also note that, for the spin- S Heisenberg model, the AF state is also described by the $O(3)$ nonlinear σ -model, but now with $\theta = 2\pi S$. (If we explicitly break the symmetry, say by letting $J_{i,i+1} \neq J_{i+1,i+2}$, then the value of θ is no longer quantized.)

The above discussion about an instanton changing the algebraic long-range order to a short-range one is valid only when $\theta = 0 \bmod 2\pi$. Therefore, the Heisenberg model with integer spin has short-range correlations and a finite energy gap. The ground state is a spin singlet (for a chain with even sites). For the Heisenberg model with half-integer spin, $\theta = \pi$ and the above instanton discussion is not valid. In fact, the spin-spin correlation in this case has algebraic long-range order, i.e. $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle \sim 1/|i-j|$. The ground state is still a spin singlet (for a chain with even sites), but now there are gapless excitations.

The winding number also provides a lower bound for the instanton action S_0 , namely $S_0 \geq 4\pi v|W|/g$.³⁸ In fact, the instanton solutions saturate the bound and $S_0 = 4\pi v/g$.

5.6.3 Quantum numbers and low-energy excitations

- Calculation of the crystal momenta of low-energy excitations from the continuum nonlinear σ -model.

Let us assume that $d > 1$ and the system has a true long-range AF order. The low-energy excitations above the AF ground state are described by the $O(3)$ nonlinear σ -model

$$S = \int dt dx \frac{1}{2g} [(\partial_t \mathbf{n})^2 - v^2 (\partial_x \mathbf{n})^2] \quad (5.6.14)$$

Here we want to use S to calculate the low-lying excitations for a finite system.

³⁸ Assume that $v = 1$. For a small square $\delta\tau \times \delta x$ with $\delta\tau = \delta x$, we have

$$\begin{aligned} \int_{\delta\tau \times \delta x} d\tau dx [(\partial_\tau \mathbf{n})^2 + (\partial_x \mathbf{n})^2] &= (\delta_\tau \mathbf{n})^2 + (\delta_x \mathbf{n})^2 \\ &\geq 2|\mathbf{n} \cdot (\delta_\tau \mathbf{n} \times \delta_x \mathbf{n})| = 2 \left| \int_{\delta\tau \times \delta x} d\tau dx \mathbf{n} \cdot (\partial_\tau \mathbf{n} \times \partial_x \mathbf{n}) \right|. \end{aligned}$$

From the winding number $(4\pi)^{-1} \int d\tau dx \mathbf{n} \cdot (\partial_\tau \mathbf{n} \times \partial_x \mathbf{n}) = \pm 1$, we find that $S_0 \geq 4\pi v|W|/g$ (we have put v back).

First, let us consider the uniform fluctuations, namely

$$\mathbf{n}(\mathbf{x}, t) = \mathbf{n}_0(t) \quad (5.6.15)$$

We have assumed that the lattice has an even number of sites in each direction, so that the above ansatz (5.6.15) can be consistent with the AF order. The effective action for $\mathbf{n}_0$ is

$$S_0 = \int dt \frac{M}{2} \dot{\mathbf{n}}_0^2$$

with $M = \mathcal{V}/g$ and $\mathcal{V}$ being the volume of space. Here S_0 describes a particle of mass M moving on a unit sphere. The energy levels are labeled by the ‘angular momentum’, which is the total spin (S, S_z) in our system: $E_{S, S_z} = \frac{S(S+1)}{2M}$, $S_z = -S, -S+1, \dots, S$. As the lattice has an even number of sites, S is always an integer. We note that E_{S, S_z} scales like $\mathcal{V}^{-1}$, which is much less than the energy of the lowest spin wave excitation, which scales like $\mathcal{V}^{-1/d}$ if $d > 1$.

Now let us consider the momentum for the excited state $|S, S_z\rangle$. Naively, one expects all $|S, S_z\rangle$ states to carry zero momentum because they come from a constant $\mathbf{n}_0$. However, $\mathbf{n}_0$ is invariant only under the translations that translate the even (odd) lattice points to even (odd) lattice points. Thus, $|S, S_z\rangle$ is only invariant under these translations. This restricts the momenta of $|S, S_z\rangle$ to 0 or $Q = (\pi, \pi, \dots, \pi)$. Under the translation by one lattice space, $\mathbf{n}_0 \rightarrow -\mathbf{n}_0$. Thus, the states that are even under $\mathbf{n}_0 \rightarrow -\mathbf{n}_0$ carry momentum 0, and those that are odd under $\mathbf{n}_0 \rightarrow -\mathbf{n}_0$ carry momentum Q . We find that even- S states carry zero momentum and odd- S states carry momentum Q .

Problem 5.6.2.

Calculate the action S in eqn (5.6.12) from eqn (5.6.10) (i.e. determine the value of g and v from J and the lattice spacing a).

6

QUANTUM GAUGE THEORIES

Quantum field theories can be loosely divided into three classes, namely bosonic theory, fermionic theory, and gauge theory. We have discussed quantum field theories of bosons and fermions in the last few chapters. In this chapter, we will discuss gauge theories. Here we will introduce gauge theory formally. A more physical discussion will be given in Chapter 10.

One may wonder why there are three classes of quantum field theories. With which quantum field theories does nature choose to describe itself? According to the $U(1) \times SU(2) \times SU(3)$ standard model, nature chooses fermionic theory and gauge theory. So, gauge theory is important for high energy particle physics. But why does nature choose the more complicated fermionic theory and gauge theory, and skip the simpler bosonic theory? Chapter 10 provides an answer to the above questions. It turns out that we do not have to introduce fermionic theories and gauge theories. They can emerge as effective theories of a local bosonic system. As gauge theory is emergent, it is not surprising that gauge theories also appear in some condensed matter systems. In fact, more varieties of gauge theories could emerge from condensed matter systems than those offered by our vacuum.³⁹

6.1 Simple gauge theories

6.1.1 Gauge ‘symmetry’ and gauge ‘symmetry’ breaking

- Gauge theory is a theory where we use more than one label to label the same quantum state.⁴⁰
- Gauge ‘symmetry’ is not a symmetry and can never be broken.

When two different quantum states $|a\rangle$ and $|b\rangle$ (i.e. $\langle a|b\rangle = 0$) have the same properties, we say that there is a symmetry between $|a\rangle$ and $|b\rangle$. If we use two different labels ‘ a ’ and ‘ b ’ to label the same state, $|a\rangle = |b\rangle$, then $|a\rangle$ and $|b\rangle$ obviously have (or has) the same properties. In this case, we say that there is a

³⁹ This emergent picture also raises an interesting question: can new classes of quantum field theories, other than the fermionic theories and the gauge theories, emerge from bosonic models?

⁴⁰ This notion of gauge theory is quite unconventional, but true. See Section 10.7.4 for a discussion of the historic development of gauge theory.

gauge ‘symmetry’ between $|a\rangle$ and $|b\rangle$, and the theory about $|a\rangle$ and $|b\rangle$ is a gauge theory (at least formally). As $|a\rangle$ and $|b\rangle$, being the same state, always have (or has) the same properties, the gauge ‘symmetry’, by definition, can never be broken.

Usually, when the same ‘thing’ has the same properties, we do not say that there is a symmetry. Thus, the terms ‘gauge symmetry’ and ‘gauge symmetry breaking’ are two of the most misleading terms in theoretical physics. In the following, we will not use the above two confusing terms. We will say that there is a gauge structure (instead of a gauge ‘symmetry’) when we use many labels to label the same state. When we change our labeling scheme, we will say that there is a change of gauge structure (instead of gauge ‘symmetry’ breaking).

6.1.2 Gauge theory without a gauge field

- The concepts of gauge transformation, gauge-invariant states, and gauge-invariant operators.

The above simple example of gauge theory (containing only one state) is too simple. So, in this section, we will study a more complicated example. Consider a particle moving in a one-dimensional periodic potential, with

$$\begin{aligned} H &= \frac{1}{2m}p^2 + V \cos(2\pi x/a) \\ \mathcal{H} &= \{|x\rangle\} \end{aligned} \quad (6.1.1)$$

where $\mathcal{H}$ is the Hilbert space for the possible states in our system. Here we stress that the Hamiltonian alone does not specify the system. It is the Hamiltonian *and* the Hilbert space, $(H, \mathcal{H})$, that specify the system.

The above system has a translational symmetry:

$$T_a^\dagger H T_a = H, \quad T_a |x\rangle = |x + a\rangle$$

We can define a gauge theory by turning the symmetry into a gauge structure. The resulting gauge theory is given by

$$\begin{aligned} H &= \frac{1}{2m}p^2 + V \cos(2\pi x/a) \\ \mathcal{H}_a &= \{|\psi\rangle, T_a |\psi\rangle = |\psi\rangle\} \end{aligned} \quad (6.1.2)$$

That is, we define our gauge theory by modifying the Hilbert space while keeping the ‘same’ Hamiltonian. The new Hilbert space is the invariant subspace under the symmetry transformation. As the Hamiltonian has the symmetry, it acts within the invariant subspace. We call the transformation T_a the *gauge transformation* and the states in the new Hilbert space the *gauge-invariant* states. Our gauge theory simply describes a particle on a circle.

A lot of concepts of gauge theory can be discussed using our ‘more complicated’ gauge theory. For a particle on a line, the states $|x\rangle$ and $|x+a\rangle$ are different states with the same properties. Thus, there is a translational symmetry. For a particle on a circle, $|x\rangle$ and $|x+a\rangle$ are the same state. Here x and $x+a$ are just two labels which label the same quantum state. Gauge transformations are just transformations between those labels which label the same state. Thus, a gauge transformation, by definition, is a ‘do nothing’ transformation. All of the physical states are, by definition, gauge invariant. If a state is not gauge invariant (i.e. in our example, a wave function $\psi(x)$ which is not periodic), then such a state does not belong to the physical Hilbert space (i.e. it cannot be a wave function for a particle on a circle). All of the physical operators must also be gauge invariant, i.e. $O = T_a^\dagger O T_a$. Otherwise, the operator will map a physical state into an unphysical state which is outside the physical Hilbert space. We see that it is very important to maintain gauge invariance in a gauge theory. Only gauge-invariant states and gauge-invariant operators are meaningful.

Another familiar gauge theory is the identical-particle system, where the exchange symmetry is changed into a gauge structure. The following two-particle system:

$$H = H(x^1) + H(x^2)$$

$$\mathcal{H} = \{|x^1, x^2\rangle\}$$

has an exchange symmetry $(x^1, x^2) \rightarrow (x^2, x^1)$. However, it is not an identical-particle system. If we turn the exchange symmetry into a gauge structure (i.e. by reducing the Hilbert space), then the new system is described by

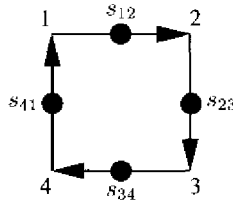
$$H = H(x^1) + H(x^2)$$

$$\mathcal{H}_b = \{|x^1, x^2\rangle \mid |x^1, x^2\rangle = |x^2, x^1\rangle\}.$$

Such a system is an identical-particle system. Note that the identical-particle system does not have exchange *symmetry* because $|x^1, x^2\rangle$ and $|x^2, x^1\rangle$ are simply the same state. All of the special interference phenomena associated with identical particles come from the gauge structure.

6.2 Z_2 lattice gauge theory

In this section, we will study the simplest gauge theory with a gauge field— Z_2 gauge theory. The Z_2 gauge theory is the simplest topological field theory. It appears as an effective theory of several topologically-ordered quantum liquid states which we will discuss in Chapters 9 and 10. So, Z_2 gauge theory is very important for understanding the physical properties of these quantum liquid states.


FIG. 6.1. A Z_2 gauge theory on a square.

Just like any quantum theory, to define a Z_2 gauge theory, we need to define the Hilbert space and the Hamiltonian. To understand the basic properties of the Z_2 gauge theory, we need to find the excitations spectrum. This is what we are going to do in the next few sections.

6.2.1 The Hilbert space

- The states in Z_2 gauge theory have a one-to-one correspondence with the gauge-equivalent classes of configurations.

To define a quantum Z_2 lattice gauge theory (Wegner, 1971), we need to first define its Hilbert space. To be concrete, let us consider a two-dimensional square lattice labeled by i . On every nearest-neighbor link, we assign a link variable $s_{ij} = s_{ji}$ which can take the two values ± 1 . The states in the Hilbert space are labeled by configurations of s_{ij} , i.e. $|\{s_{ij}\}\rangle$. If the labeling was one-to-one, then the Hilbert space would be a Hilbert space of a quantum Ising model (with spins on the links). To obtain the Hilbert space of the Z_2 gauge theory, the labeling is not one-to-one: two gauge-equivalent configurations label the same quantum state.

By definition, two configurations s_{ij} and $\tilde{s}_{ij}$ are gauge equivalent if they are related by a Z_2 gauge transformation:

$$\tilde{s}_{ij} = W_i s_{ij} W_j^{-1} \quad (6.2.1)$$

where W_i is an arbitrary function with values ± 1 . The gauge transformation defines an equivalence relation. If we group all of the gauge-equivalent configurations together to form a class, then such a class will be called a gauge-equivalent class. We see that the states in the Hilbert space have a one-to-one correspondence with the gauge-equivalent classes.

As an exercise, let us consider a Z_2 gauge theory on a square (see Fig. 6.1). We would like to find out the dimension of its Hilbert space. First, there are 4 links and hence $2^4 = 16$ different s_{ij} configurations. Second, there are 4 sites and hence $2^4 = 16$ different Z_2 gauge transformations. If the number of configurations in a gauge-equivalent class is equal to the number of different gauge transformations,

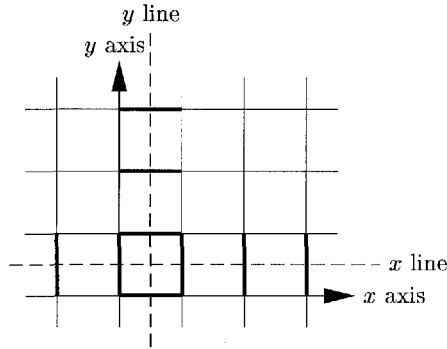


FIG. 6.2. The links cross the x line and/or the y line may receive an additional minus sign.

then there will be only one gauge-equivalent class. However, we note that the two gauge transformations

$$W_i = 1, \quad W_i = -1$$

do not change the configurations s_{ij} . The two gauge transformations form a group which will be called the invariant gauge group (IGG) (see also Section 9.4.2). As a result, the 16 gauge transformations only induce $16/2 = 8$ gauge-equivalent configurations. The factor of 2 is the number of elements in the IGG. As each gauge-equivalent class contains 8 configurations, there are $16/8 = 2$ gauge-equivalent classes, and hence two states in the Hilbert space.

To find a way to explicitly label the states in the Hilbert space, it is important to find gauge-invariant quantities that do not change under gauge transformations. One of the gauge-invariant quantities is the Wegner–Wilson loop variable (Wegner, 1971; Wilson, 1974), defined for a loop C as

$$U(C) = s_{ij}s_{jk}\dots s_{li}$$

where $i, j, k, \dots$, and l are sites on the loop. We will call $U(C)$ the Z_2 flux through the loop. The reader can easily check that $U(C)$ only takes the two values ± 1 and is invariant under the Z_2 gauge transformation. The two states in the Z_2 gauge theory on the square are labeled by $s_{12}s_{23}s_{34}s_{41} = \pm 1$.

Now let us count the number of states in the Z_2 gauge theory on a finite square lattice. We assume that the lattice has a periodic boundary condition in both directions (i.e. the lattice forms a torus). If the lattice has N_{site} sites, then it has $2N_{\text{site}}$ nearest-neighbor links. Thus, there are $2^{2N_{\text{site}}}$ different s_{ij} configurations. There are $2^{N_{\text{site}}}$ different gauge transformations and the IGG still has 2 elements. Thus, there are $\frac{2^{2N_{\text{site}}}}{2^{N_{\text{site}}}/2} = 2 \times 2^{N_{\text{site}}}$ different gauge-equivalent classes which correspond to $2 \times 2^{N_{\text{site}}}$ different states.

To find a way to label the $2 \times 2^{N_{\text{site}}}$ states, we consider the Z_2 flux through a plaquette:

$$F_i = s_{i,i+x} s_{i+x,i+x+y} s_{i+x+y,i+y} s_{i+y,i}$$

As there are N_{site} plaquettes and $F_i = \pm 1$, one naively expects that different $\{F_i\}$ s can provide $2^{N_{\text{site}}}$ different labels. However, the F_i are not independent. They satisfy

$$\prod_i F_i = 1 \quad (6.2.2)$$

Thus, the $\{F_i\}$ only provide $2^{N_{\text{site}}}/2$ different labels. Obviously, we cannot use $\{F_i\}$ to label all of the $2 \times 2^{N_{\text{site}}}$ states. In fact, each flux configuration $\{F_i\}$ corresponds to four different states.

To see this, we consider the following four configurations obtained from one configuration s_{ij}^0 :

$$s_{ij}^{(m,n)} = f_x^m(ij) f_y^n(ij) s_{ij}^0, \quad m, n = 0, 1 \quad (6.2.3)$$

The functions $f_{x,y}(ij)$ take the values -1 or 1 , with $f_x(ij) = -1$ if the link ij crosses the x line (see Fig. 6.2) and $f_x(ij) = 1$ otherwise. Similarly, $f_y(ij) = -1$ if the link ij crosses the y line and $f_y(ij) = 1$ otherwise. The four configurations give rise to the same Z_2 flux F_i through every plaquette. Despite this, the four configurations are not gauge equivalent (see Problem 6.2.2) because they correspond to inserting a different Z_2 flux through the two holes in the torus (see Fig. 9.6). Therefore, $2^{N_{\text{site}}}/2$ different flux configurations $\{F_i\}$ plus the four-fold degeneracy allow us to recover $2 \times 2^{N_{\text{site}}}$ states.

Problem 6.2.1.

Prove eqn (6.2.2) for the Z_2 gauge theory on a torus.

Problem 6.2.2.

Show that the four configurations in eqn (6.2.3) are not gauge equivalent. (Hint: Consider $U(C)$ with C going all the way around the torus.)

Problem 6.2.3.

The Hilbert space $\mathcal{H}_{Z_2}$ of the Z_2 gauge theory can also be constructed from the Hilbert space $\mathcal{H}_{\text{spin}}$ of a spin-1/2 model which has one spin on each link of the square lattice. Here $\mathcal{H}_{\text{spin}}$ has a subspace $\mathcal{H}_{\text{sub}}$ which is formed by states that are invariant under all of the unitary transformations of the form $\prod_{\langle ij \rangle} (\sigma_{ij}^1)^{G_i - G_j}$. Also, G_i is an arbitrary function on the sites taking the values $G_i = 0, 1$, and σ_{ij}^1 is the spin operator acting on the spin on the link $\langle ij \rangle$. Show that $\mathcal{H}_{\text{sub}} = \mathcal{H}_{Z_2}$ and $\prod_{\langle ij \rangle} (\sigma_{ij}^1)^{G_i - G_j}$ generates the Z_2 gauge transformation.

6.2.2 The Hamiltonian

- The Hamiltonian of a Z_2 gauge theory must be gauge invariant.

So far we have only discussed the Hilbert space of the Z_2 lattice gauge theory. What about the Hamiltonian? We cannot just pick any operator in the Z_2 Hilbert space as our Hamiltonian. The Hamiltonian should have some ‘local’ properties. One way to construct a local Hamiltonian is to write it as a local function of s_{ij} . However, because s_{ij} is a many-to-one labeling of the physical states, we have to make sure that the equivalent labels give rise to the same energy. That is, the Hamiltonian must be gauge invariant. A simple way to construct a gauge-invariant Hamiltonian is to write the Hamiltonian as a function of gauge-invariant operators, such as $U(C)$ and F_i . In addition to $U(C)$, the σ_{ij}^1 operator that flips the sign of s_{ij} on a link σ_{ij}^1 , i.e. $s_{ij} \rightarrow -s_{ij}$, is also a gauge-invariant operator (see Problem 6.2.4). Thus, a gauge-invariant local Hamiltonian for the Z_2 gauge theory can be written as

$$H_{Z_2} = -g \sum_i F_i - t \sum_{\langle ij \rangle} \sigma_{ij}^1 \quad (6.2.4)$$

Problem 6.2.4.

Let $\hat{W}$ generate a gauge transformation $\hat{W}|\{s_{ij}\}\rangle = |\{\tilde{s}_{ij}\}\rangle$, where $\tilde{s}_{ij}$ and s_{ij} are related through eqn (6.2.1). An operator $\hat{O}$ is gauge invariant if $\hat{W}\hat{O}\hat{W}^{-1} = \hat{O}$. Show that σ_{ij}^1 is a gauge-invariant operator.

6.2.3 The physical properties

- Despite the finite energy gap, a Z_2 gauge theory at low energies is not trivial. Its low-energy properties on a torus are characterized by the four-fold topologically-degenerate ground states.

When $t = 0$ and $g > 0$, H_{Z_2} has four degenerate ground states on a torus which are characterized by $F_i = +1$. The excitations are created by flipping the signs of F_i (see Fig. 6.3). The excitations behave like local particles and will be called the Z_2 vortices. Due to the constraint (6.2.2), the Z_2 vortices can only be created in pairs on a torus. The energy gap of these excitations is of order g . The t term in H_{Z_2} induces a hopping of the Z_2 vortices. As the σ_{ij}^1 operators commute with each other, using the statistical algebra of the hopping operators, see eqn (4.1.10), we find that the Z_2 vortices are bosons.

We would like to point out that the four-fold degeneracy of the ground states is a so-called topological degeneracy. The topological degeneracy, by definition, is a degeneracy that cannot be lifted by any local perturbations of the Hamiltonian. The topological degeneracy is very important. It allows us to define topological orders in Chapter 8.

To understand the robustness of the four-fold degeneracy in the Z_2 gauge theory, let us treat the t term as a perturbation and see how the t term can lift the four-fold-degenerate ground states of the $t = 0$ Hamiltonian. The four degenerate ground states are given by eqn (6.2.3) with $s_{ij}^0 = 1$. We can see that the only way

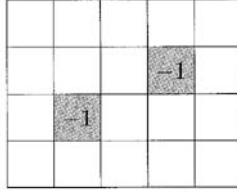


FIG. 6.3. A Z_2 vortex is created by changing F_i from $+1$ to -1 .

to change one state $s_{ij}^{(m,n)}$ into a different state $s_{ij}^{(m',n')}$ is to flip the signs of s_{ij}^0 on a line of links all the way around the torus. If the torus is formed by an $L \times L$ lattice, then we need at least L of the σ_{ij}^1 operators to connect different degenerate states. This can happen only beyond the L th order in perturbation theory. Thus, the t term can lift the four-fold degeneracy. However, the energy splitting ΔE is only of order t^L/g^{L-1} . In the thermodynamical limit, $L \rightarrow \infty$ and $\Delta E \rightarrow 0$. The above discussion does not depend on any symmetry. It is valid even when t is not uniform. The four-fold degeneracy (in the thermodynamical limit) is a robust property of a phase. The four-fold degeneracy reflects the low-energy dynamics of the Z_2 flux and is the most important characteristic of the Z_2 gauge theory.

When $g = 0$ and $t > 0$, the ground state of H_{Z_2} is given by

$$|\Psi_0\rangle = \sum_{\{s_{ij}\}} |\{s_{ij}\}\rangle \quad (6.2.5)$$

where the summation $\sum_{\{s_{ij}\}}$ is over all configurations $\{s_{ij}\}$. The ground state is non-degenerate and has an energy $-tN_l$, where N_l is the number of links. This result is easy to see if we treat H_{Z_2} as an Ising model with spins on the links. Here s_{ij} can be identified as the eigenvalues of the spin operator σ_{ij}^3 in the z direction, and σ_{ij}^1 is the spin operator in the x direction. In the Ising model picture, the $|\Psi_0\rangle$ state is simply a state with all spins pointing in the x direction.

From the above discussion, we see that H_{Z_2} has two phases. When $g \gg t$, the ground states have four-fold degeneracy and the excitations are Z_2 vortices. We will call this phase the Z_2 deconfined phase. The low-energy properties of this phase are described by a Z_2 gauge theory. When $g \ll t$, the ground state is not degenerate. We will call this phase the Z_2 confined phase. The low-energy properties have no characteristics of a Z_2 gauge theory.

If you feel that the definition of the Z_2 gauge theory is formal and the resulting Z_2 gauge theory is strange, then you get the point. The Z_2 gauge theory is actually a non-local theory, in the sense that its total Hilbert space cannot be expressed as a direct product of local Hilbert spaces. In Section 10.3, we will give a more physical description of Z_2 gauge theory. We will see that Z_2 gauge theory is actually a theory of closed strings.

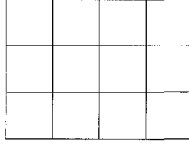


FIG. 6.4. An open two-dimensional square lattice.

Problem 6.2.5.

- (a) A Z_2 gauge theory H_{Z_2} (see eqn (6.2.4)) is defined on an open square lattice of $L_x \times L_y$ sites (see Fig. 6.4). Assuming that $t = 0$, find the ground-state energy and degeneracy.
- (b) Assume that the lattice is periodic in the y direction and forms a cylinder. Assuming that $t = 0$, find the ground-state energy and degeneracy.
- (c) Assume that the lattice is periodic in both the x and y directions. Assuming that $t = 0$ and $g < 0$, find the ground-state energy and degeneracy for the following three cases: (i) $L_x = \text{even}$ and $L_y = \text{even}$; (ii) $L_x = \text{even}$ and $L_y = \text{odd}$; and (iii) $L_x = \text{odd}$ and $L_y = \text{odd}$.

Problem 6.2.6.

Show that eqn (6.2.5) is the ground state of H_{Z_2} when $g = 0$ and $t > 0$.

Problem 6.2.7.

Consider the two configurations s_{ij}^1 and s_{ij}^2 on a torus with $L \times L$ sites. The two configurations generate the same Z_2 flux F_i . Show that, if the two configurations are not gauge equivalent, then one needs to flip the signs of s_{ij} for at least L links to change s_{ij}^1 to s_{ij}^2 (or $s_{ij}^1 s_{ij}^2 = -1$ on more than L links).

Problem 6.2.8.

Note that the hopping of the Z_2 vortices is generated by the hopping operators $P\sigma_{ij}^1 P$, $P\sigma_{ij}^1 P\sigma_{jk}^1 P$, etc., where P is a projection to a subspace with a fixed number of Z_2 vortices. Using the statistical algebra (4.1.10), show that the Z_2 vortices are bosons.

6.3 $U(1)$ gauge theory and the XY-model in $1 + 2$ dimensions

We constructed the Lagrangian of the $U(1)$ gauge theory in Section 3.7.1 and studied its classical equation of motion in Section 3.7.5. In this section, we are going to study the quantum $U(1)$ gauge theory. Instead of studying the $U(1)$ gauge theory by itself, we will study it together with the XY-model in $1 + 2$ dimensions. We will show that there is a duality relation between the XY-model and the $U(1)$ gauge theory. We will use this duality relation to study the instanton effect and confinement in $(1 + 2)$ -dimensional $U(1)$ gauge theory.

6.3.1 Duality between $U(1)$ gauge theory and the XY-model in $1+2$ dimensions

- Gauge fixing and the Coulomb gauge.
- Quantization of the $U(1)$ gauge theory.
- Quantized charge, large gauge transformations, and compact $U(1)$ gauge theory.
- The quantized charge in the $U(1)$ gauge theory is the quantized vortex in the dual XY-model.

The XY-model is described by the path integral

$$Z_{XY} = \int \mathcal{D}\theta \, e^{i \int dt d^2x \, \mathcal{L}_{XY}}$$

$$\mathcal{L}_{XY} = \frac{\chi}{2} \left(\dot{\theta}^2 - (\partial_x \theta)^2 \right) \quad (6.3.1)$$

The $U(1)$ gauge theory can also be described by the path integral

$$Z_{U(1)} = \int \mathcal{D}a_\mu \, e^{i \int dt d^2x \, \mathcal{L}_{U(1)}}, \quad (6.3.2)$$

$$\mathcal{L}_{U(1)} = \frac{1}{2g^2} (e^2 - b^2), \quad e_i = \partial_0 a_i - \partial_i a_0, \quad b = \partial_1 a_2 - \partial_2 a_1,$$

where e is the ‘electric’ field and b is the ‘magnetic’ field of a_μ . The two path integrals are so different that it is hard to see any relationship between the two theories. However, we will show below that the two path integrals describe the same physical system.

We note that both theories are quadratic and thus exactly soluble. One way to understand the relationship between the two theories is to quantize the two theories and find all of the low-energy excitations in the two theories.

Let us first try to quantize the $U(1)$ gauge theory, i.e. find the Hilbert space and the Hamiltonian of the $U(1)$ gauge theory. We note that $\mathcal{L}_{U(1)}$ is invariant under the transformation $a_\mu \rightarrow a_\mu + \partial_\mu f$:

$$\mathcal{L}_{U(1)}(a_\mu) = \mathcal{L}_{U(1)}(a_\mu + \partial_\mu f)$$

If we treat a_μ and $a_\mu + \partial_\mu f$ as two different paths, then the theory would have a symmetry. However, here we treat a_μ and $a_\mu + \partial_\mu f$ as two different labels of the *same* path. In this case, we get a gauge theory and $a_\mu \rightarrow a_\mu + \partial_\mu f$ defines the gauge structure.

To obtain a Hamiltonian description, we first try to find a one-to-one labeling of the paths. Consider a path labeled by a_μ . The gauge transformation tells us that $a'_\mu = a_\mu + \partial_\mu f$ labels the same path. Among so many different labels of the same

path, we can always choose the one that satisfies

$$\partial \cdot \mathbf{a} = 0 \quad (6.3.3)$$

by adjusting f . So a_μ that satisfies eqn (6.3.3) provides a one-to-one labeling of the different paths. Equation (6.3.3) is called a gauge-fixing condition. There are many different gauge-fixing conditions that can lead to a one-to-one labeling of the paths. The particular gauge-fixing condition (6.3.3) is called the Coulomb gauge.

In the Coulomb gauge, the part of the action that contains a_0 has the form $S = \int dt d^2\mathbf{x} e^2 = \int dt d^2\mathbf{x} (\dot{a}_i^2 + \partial_i a_0 \partial_i a_0)$. The cross-term $\int dt d^2\mathbf{x} \partial_i a_0 \partial_0 a_i$ vanishes because $\partial_i a_i = 0$. We see that a_0 decouples from $\mathbf{a}$ and has no dynamics (i.e. no $\dot{a}_0$ terms). Thus, we can drop a_0 (by effectively setting $a_0 = 0$). So, in the Coulomb gauge, the path integral calculates the amplitude of the evolution from $a_i(\mathbf{x}, t_1)$ to $a_i(\mathbf{x}, t_2)$ that satisfies $\partial_i a_i(\mathbf{x}, t) = 0$, $t_1 \leq t \leq t_2$. Thus, the quantum state of the $U(1)$ gauge theory is described by a wave functional $\Psi[a_i(\mathbf{x})]$, where a_i satisfies $\partial_i a_i = 0$. This defines the Hilbert space for the physical states.

Next, we are going to find the eigenstates and their energies, by finding the gauge-fixed action. We first expand a_i as follows:

$$a_i = \frac{X_i(t)}{L_i} + b_0 x^1 \delta_{i,2} + \sum_{\mathbf{k}} i c_{\mathbf{k}}(t) \epsilon_{ij} k_j e^{i\mathbf{k} \cdot \mathbf{x}} \quad (6.3.4)$$

where $c_{\mathbf{k}} = c_{-\mathbf{k}}^*$. The above a_i always satisfies $\partial_i a_i = 0$. Thus, the wave functions of the physical states are functions of b_0 , X_i , and $c_{\mathbf{k}}$. The gauge-fixed Lagrangian takes the form

$$L = \frac{1}{2g^2} \left(-b_0^2 L_1 L_2 + \frac{L_2}{L_1} \dot{X}_1^2 + \frac{L_1}{L_2} \dot{X}_2^2 + \sum_{\mathbf{k}} (|\dot{c}_{\mathbf{k}}|^2 - \mathbf{k}^2 |c_{\mathbf{k}}|^2) L_1 L_2 \mathbf{k}^2 \right)$$

We see that the $U(1)$ gauge theory on a torus is described by a collection of oscillators $c_{\mathbf{k}}$, plus a free particle in two dimensions described by (X_1, X_2) . The energies of the eigenstates are

$$E = \frac{1}{2g^2} b_0^2 L_1 L_2 + \frac{g^2 L_1}{2L_2} P_1^2 + \frac{g^2 L_2}{2L_1} P_2^2 + \sum_{\mathbf{k}} |\mathbf{k}| n_{\mathbf{k}}$$

where $n_{\mathbf{k}}$ are the occupation numbers of the oscillators and P_i are the momenta that conjugate to X_i .

There are two kinds of $U(1)$ gauge theory, named compact and non-compact $U(1)$ theories. The gauge transformations in the two $U(1)$ gauge theories are defined differently. As a result, the two theories have different Hilbert spaces and should be distinguished. For the non-compact $U(1)$ theory, $a_\mu \rightarrow a_\mu + \partial_\mu f$ is the only allowed gauge transformation. We will see later that the gauge charge is not quantized in the non-compact $U(1)$ theory.

The compact $U(1)$ gauge theory has quantized gauge charge. Only the Wegner–Wilson loop amplitude

$$O_C = e^{-iq \oint_C dx^\mu a_\mu}$$

can be observed. Here q is the charge quantum. By choosing small loops with different orientations, one can show that O_C contains all of the components of the electric field e and the magnetic field b . As O_C is the only observable, any transformations of a_μ that make O_C invariant are gauge transformations. Such transformations have the form $a_\mu \rightarrow a_\mu + \partial_\mu f$, but now f may not be a single-valued function of space–time. In particular, the following f on the torus generates a valid gauge transformation:

$$f = \frac{2\pi N_1 x^1}{qL_1} + \frac{2\pi N_2 x^2}{qL_2} \quad (6.3.5)$$

For example, for a loop C that wraps around the torus in the x^1 direction, one can check directly that the Wegner–Wilson loop amplitude transforms as $O_C \rightarrow O_C e^{-iq \oint_C dx^\mu \partial_\mu f} = O_C$ under the above gauge transformation.

In general, any multi-valued f that makes e^{iqf} single valued will leave O_C invariant. These f will generate valid gauge transformations. We also note that a charge field with charge q transforms as

$$\phi \rightarrow e^{iqf} \phi.$$

A gauge transformation with a single-valued e^{iqf} will leave the charge field single valued.

The gauge transformations in eqn (6.3.5) are called large gauge transformations. These transformations, namely $e^{iqf} = \exp(iq(\frac{2\pi N_1 x^1}{qL_1} + \frac{2\pi N_2 x^2}{qL_2}))$, cannot be continuously deformed into the identity transformation $e^{iqf} = 1$. As x^1 moves from 0 to L_1 , e^{iqf} goes around the unit circle in the complex plane N_1 times. The winding number N_1 cannot be changed continuously.

We note that, under the gauge transformation (6.3.5),

$$X_i \rightarrow X_i + 2\pi q^{-1} N_i$$

Therefore, for the compact $U(1)$ theory, (X_1, X_2) , $(X_1 + 2\pi q^{-1}, X_2)$, and $(X_1, X_2 + 2\pi q^{-1})$ label the same physical state (because they are related by gauge transformations). The free particle described by X_i lives on a torus of size $\frac{2\pi}{q} \times \frac{2\pi}{q}$. The momenta P_i are quantized as $P_i = n_i q$.

For the non-compact $U(1)$ theory, the large gauge transformations are not allowed. As a result, X_1 and X_2 have no periodic conditions. The free particle described by X_i lives on the plane.

We see that, although their Hamiltonians have the same form, the compact and non-compact $U(1)$ theories have different physical Hilbert spaces. We also see that

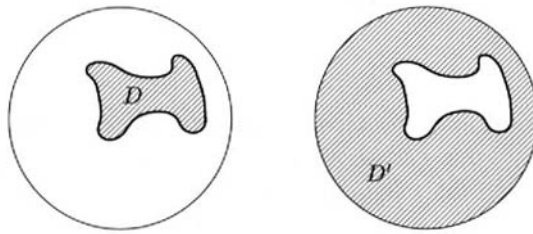


FIG. 6.5. A loop on a unit sphere can be the boundary of two different disks D and D' .

the $q \rightarrow 0$ limit of the compact $U(1)$ theory gives rise to the non-compact $U(1)$ theory. This is why we say that the charge is not quantized in the non-compact $U(1)$ theory.

As our system is defined on a torus, every physical quantity must be periodic in both the x^1 and x^2 directions. One may naively apply this requirement to a_i and require a_i to be periodic:

$$a_i(x^1, x^2) = a_i(x^1 + L_1, x^2) = a_i(x^1, x^2 + L_2).$$

However, we note that a_i itself is not physical. Different, but gauge-equivalent, a_i describe the same physical state. Thus, for a_i to describe a physical state on the torus, we only require that $a_i(x^1, x^2)$, $a_i(x^1 + L_1, x^2)$, and $a_i(x^1, x^2 + L_2)$ be gauge equivalent. This can happen only when the total magnetic flux is quantized (see Problem 6.3.1):

$$\int d^2x b = 2\pi N/q, \quad N = \text{integer} \quad (6.3.6)$$

The above quantization of the total flux is necessary for the loop amplitude $e^{-iq \int_C dx^\mu a_\mu}$ to make sense on a compact surface. This is because, for a loop C in a two-dimensional space S , $e^{-iq \int_C dx^\mu a_\mu} = e^{iq \int_D d^2x b}$, where D is a two-dimensional disk with the loop C as its boundary. For a compact two-dimensional space, the disk is not unique (see Fig. 6.5). Two different disks D and D' with the same boundary differ by the total area S of the compact space. Hence $q \int_D d^2x b - q \int_{D'} d^2x b = q \int_S d^2x b$. Therefore, $q \int_S d^2x b / 2\pi$ must be quantized as an integer in order for $e^{-iq \int_C dx^\mu a_\mu}$ to be well defined. We also note that, for the non-compact $U(1)$ theory, $q = 0$ and $\int_S d^2x b$ must vanish.

Now the total energy of the compact $U(1)$ gauge theory can be rewritten as

$$E = \frac{(2\pi)^2}{2g^2q^2L_1L_2}N + \frac{g^2q^2L_1}{2L_2}n_1^2 + \frac{g^2q^2L_2}{2L_1}n_2^2 + \sum_{\mathbf{k}} |\mathbf{k}| n_{\mathbf{k}} \quad (6.3.7)$$

All of the eigenstates of the $U(1)$ gauge theory on a torus are labeled by the set of integers $(N, n_i, n_{\mathbf{k}})$. In this way, we have solved the compact $U(1)$ gauge theory on a torus.

Next let us turn to the XY-model and its eigenstates. We expand

$$\theta(t, \mathbf{x}) = \theta_0(t) + \frac{2\pi}{L_i} m_i x^i + \sum_{\mathbf{k}} \lambda_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{x}}.$$

Here m_1 and m_2 are the winding number of θ in the x^1 and x^2 directions, respectively. The XY-model Lagrangian now takes the form

$$L = \frac{\chi}{2} \left(L_1 L_2 \dot{\theta}_0^2 - \frac{(2\pi)^2 L_2}{L_1} m_1^2 - \frac{(2\pi)^2 L_1}{L_2} m_2^2 + L_1 L_2 \sum_{\mathbf{k}} (|\dot{\lambda}_{\mathbf{k}}|^2 - \mathbf{k}^2 |\lambda_{\mathbf{k}}|^2) \right)$$

We see that the XY-model is described by a collection of oscillators (described by $\lambda_{\mathbf{k}}$) and a free particle on a circle (described by θ_0). The eigenstates of the XY-model are labeled by $(N, m_i, n_{\mathbf{k}})$ with energy

$$E = \frac{1}{2\chi L_1 L_2} N^2 + \frac{\chi(2\pi)^2 L_2}{2L_1} m_1^2 + \frac{\chi(2\pi)^2 L_1}{2L_2} m_2^2 + \sum_{\mathbf{k}} |\mathbf{k}| n_{\mathbf{k}} \quad (6.3.8)$$

We see that the XY-model and the $U(1)$ gauge theory have identical spectra if

$$\chi = \left(\frac{qg}{2\pi} \right)^2$$

and if we identify n_i as $\epsilon_{ij} m_j$ in the labeling of the eigenstates in the $U(1)$ theory and the XY-model. The eigenstates also carry definite total momenta $\mathbf{K}$, where

$$K_i = N m_i \frac{2\pi}{L_i} + \sum_{\mathbf{k}} k_i n_{\mathbf{k}} = N \epsilon_{ij} n_j \frac{2\pi}{L_i} + \sum_{\mathbf{k}} k_i n_{\mathbf{k}}$$

for the XY-model states $|N, m_i, n_{\mathbf{k}}\rangle$ or the $U(1)$ theory states $|N, n_i, n_{\mathbf{k}}\rangle$.

From the XY-model, we see that the label N has a physical meaning: it is the total number of bosons minus the number of bosons at equilibrium, namely $N = N_{tot} - N_0$. Therefore, $\frac{1}{\chi} \dot{\theta}$ and $\frac{q}{2\pi} b$ can be regarded as the boson-number density minus the equilibrium density:

$$\rho - \rho_0 = \frac{q}{2\pi} b = \frac{1}{\chi} \dot{\theta}$$

The constraint $\partial_0 b - \epsilon_{ij} \partial_i e_j = 0$ in the $U(1)$ theory can be regarded as a continuous equation for the conserved current, namely $\partial_0 \rho + \partial_i j_i = 0$. Thus, if $\frac{q}{2\pi} b$ is identified as the boson-number density, then the boson-number current density is given by

$$j_i = -\frac{q}{2\pi} \epsilon_{ij} e_j$$

in the $U(1)$ theory.

The equation of motion for the XY-model, namely $\partial_0^2 \theta - \partial_x^2 \theta = 0$, can also be regarded as a continuous equation for the conserved current. We find that the same current is given by

$$j_i = -\frac{1}{\chi} \partial_i \theta$$

in the XY-model. We discover a simple relationship between fields in the $U(1)$ theory and the XY-model:

$$\frac{q}{2\pi} b = \frac{1}{\chi} \dot{\theta}, \quad \frac{q}{2\pi} \epsilon_{ij} e_j = \frac{1}{\chi} \partial_i \theta. \quad (6.3.9)$$

In fact, substituting eqn (6.3.9) into eqn (6.3.2) will convert the $U(1)$ gauge Lagrangian into the XY-model Lagrangian (6.3.1). This is the simplest way to see the duality relation between the $U(1)$ gauge theory and the XY-model in $2+1$ dimensions.

The $U(1)$ gauge theory can be coupled to charges. Let J_μ be the density and the current density of the $U(1)$ charge. Then the coupled Lagrangian takes the form

$$\mathcal{L} = \frac{1}{2g^2} (e^2 - b^2) - J_\mu a_\mu$$

For a point charge at $x = 0$ we have $J_0 = q\delta(x)$ and we find, from the equation of motion, that

$$\frac{1}{g^2} \partial \cdot e = q\delta(x)$$

or

$$e = \frac{g^2 q x}{2\pi x^2}$$

The above e corresponds to a circular flow in the XY-model:

$$\partial_i \theta = \frac{\chi q}{2\pi} \frac{g^2 q}{2\pi x^2} \epsilon_{ij} x^j = \frac{\epsilon_{ij} x^j}{x^2}$$

We see that a quantized charge in the $U(1)$ gauge theory is just the quantized vortex in the XY-model.

Problem 6.3.1.

Show eqn (6.3.6). (Hint: You may find the expansion (6.3.4) useful.)

Problem 6.3.2.

Quantize the compact $U(1)$ gauge theory in $1+1$ dimensions, assuming that the space is a ring of length L and the charge quantum is q . Find the set of integers that label all of the energy eigenstates. Find the energies for these eigenstates.

6.3.2 Confinement of the compact $U(1)$ gauge theory in $1+2$ dimensions

- The instanton effect in the $(1+2)$ -dimensional $U(1)$ gauge theory gives the gauge boson a finite energy gap and causes a confinement between the $U(1)$ charges.
- The instanton effect can be described by a $\cos(\theta)$ term in the dual XY-model.

Here we consider the $U(1)$ gauge theory in imaginary time:

$$\mathcal{L}_{U(1)} = \frac{1}{2g^2} (e^2 + b^2) \quad (6.3.10)$$

Formally, the $U(1)$ gauge theory appears to be a free theory with gapless excitations. However, it is not that simple. For the compact $U(1)$ gauge theory with a finite cut-off scale, the theory contains instantons. The instanton effect will make the $U(1)$ gauge theory an interacting theory. We will see that the interaction will affect the low-energy properties of the $U(1)$ gauge theory in a drastic way.

What is the instanton? The instanton at $x^\mu = 0$ is described by the following configuration:

$$b(x^\mu) = \frac{1}{2q} \frac{x^0}{|x|^3}, \quad e_i(x^\mu) = \frac{1}{2q} \frac{x^i}{|x|^3}$$

The above instanton is designed to change the flux by $2\pi/q$, i.e.

$$\int_{x^0>0} d^2\mathbf{x} \, b - \int_{x^0<0} d^2\mathbf{x} \, b = \frac{2\pi}{q}$$

This is the smallest allowed instanton that is consistent with the charge quantum q . In the presence of a finite cut-off, the path integral should not only include smooth fluctuations of the gauge field, but it should also include instanton fluctuations.

When instanton effects are included, the $U(1)$ gauge theory is no longer a free theory. The problem now is how to understand the low-energy properties of the compact $U(1)$ theory with instantons. One approach is to use the duality between the XY-model and the $U(1)$ gauge theory.

For imaginary time, θ and a_μ are related by

$$\frac{q}{2\pi} b = \frac{1}{\chi} \dot{\theta}, \quad \frac{q}{2\pi} \epsilon_{ij} e_j = -\frac{1}{\chi} \partial_i \theta \quad (6.3.11)$$

which can be written as

$$\frac{q}{2\pi} \epsilon_{\mu\nu\lambda} \partial_\nu a_\lambda = \frac{1}{\chi} \partial_\mu \theta \quad (6.3.12)$$

After substituting eqn (6.3.12) into eqn (6.3.10), we obtain

$$\mathcal{L} = \frac{\chi}{2} (\partial_\mu \theta)^2$$

with $\chi = \left(\frac{qg}{2\pi}\right)^2$. We note that the instanton creates $2\pi/q$ amount of flux. As we have seen, such an amount of flux corresponds to a single particle in the XY-model. Thus, an instanton creates or annihilates a single particle. In the XY-model, it is the operator $e^{\pm i\theta}$ that creates or annihilates a single particle. We find that an instanton at x^μ corresponds to $e^{i\theta(x^\mu)}$. As both the instanton and the anti-instanton appear with equal weight in the path integral, we can include the instanton effect in the XY-model by adding a term $e^{i\theta} + e^{-i\theta} = 2\cos(\theta)$ as follows:

$$\mathcal{L} = \frac{\chi}{2}(\partial_\mu\theta)^2 - K\cos(\theta)$$

When χ is large, the fluctuations of θ around $\theta = 0$ are small. We may approximate $\mathcal{L}$ as

$$\mathcal{L} = \frac{\chi}{2}(\partial_\mu\theta)^2 + \frac{1}{2}K\theta^2$$

The correlations between $\partial_\mu\theta$ or (b, e_i) become short-ranged after including the instanton effect. We see that the instanton effect opens up an energy gap of the $U(1)$ gauge field in $1 + 2$ dimensions. Here we would like to stress that the $K\cos(\theta)$ term is a relevant perturbation and the $U(1)$ gauge boson will gain an energy gap, no matter how small K is.

If the gauge boson is gapless, then the charged particles will interact through a logarithmic potential in $1 + 2$ dimensions. The potential between a particle of charge q and a particle of charge $-q$ is given by

$$\pi \frac{g^2 q^2}{(2\pi)^2} \ln r = \pi\chi \ln r$$

After including the instanton effect, the gauge field gains an energy gap. But how does the instanton effect affect the long-range interaction between charged particles? We note that a particle of charge q is described by a vortex in the XY-model. In the presence of the instanton effect, that is, in the presence of the $K\cos(\theta)$ term, the potential between a vortex and an anti-vortex grows linearly with the separation of the two vortices (see Fig. 6.6). Thus, the instanton effect changes the logarithmic potential to a linear potential between charges.

We have shown that, in $1 + 2$ dimensions, the XY-model can be described by the compact $U(1)$ gauge theory. We have also shown that the instanton effect in the $U(1)$ gauge theory opens up an energy gap. These two results seem to contradict each other. How can the XY-model with its gapless Nambu–Goldstone mode be equivalent to the $U(1)$ gauge theory, which has an energy gap? However, we realize that an instanton creates flux. As the flux corresponds to particle numbers, the existence of the instanton implies that the particle number in the corresponding XY-model is not conserved. So everything is consistent. In the presence of the instanton, the particle number is not conserved. The compact $U(1)$ gauge theory

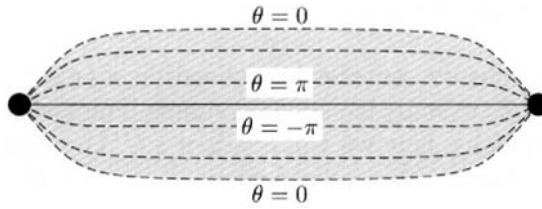


FIG. 6.6. For a vortex–anti-vortex pair in the XY-model separated by a distance l , the θ field must jump by 2π across the line connecting the two vortices. In the presence of the $-K \cos(\theta)$ term, the usual vortex ansatz $\theta(x, y) = \arctan(x/y)$ costs a huge energy which grows like l^2 . To minimize the energy, the nonzero θ is confined in the shaded region. (The dashed lines are the equal-value lines for the θ field.) This leads to a linear confining potential between the two vortices.

describes an XY-model that does not have the $U(1)$ symmetry. This explains why the instanton effect corresponds to a potential term $K \cos(\theta)$ in the XY-model. On the other hand, if the XY-model has the $U(1)$ symmetry and conserves particle number, then the instanton is not allowed in the corresponding $U(1)$ gauge theory.

Problem 6.3.3.

Duality between the $(1 + 2)$ -dimensional $U(1)$ gauge theory with instantons and the XY-model with the $\cos(\theta)$ term

1. Calculate the partition function $Z(x^\mu, y^\mu)$ of the $U(1)$ gauge theory with an instanton at x^μ and an anti-instanton at y^μ , up to an overall constant factor.
2. Use the above result to write down an expression for the partition function for the $U(1)$ gauge theory, including contributions from a multi-instanton gas.
3. Calculate the following partition function $Z(x^\mu, y^\mu)$ of the XY-model with an insertion $e^{i\theta}$ at x^μ and another insertion $e^{-i\theta}$ at y^μ , up to an overall constant factor:

$$Z(x^\mu, y^\mu) = \int \mathcal{D}\theta \, e^{i\theta(x^\mu)} e^{-i\theta(y^\mu)} e^{-\int d^3x^\mu \mathcal{L}_{XY}(\theta)}$$

4. Show that the $U(1)$ partition function with instantons agrees with the XY-model partition function with an added $K \cos(\theta)$ term (after expanding in powers of K).

6.4 The quantum $U(1)$ gauge theory on a lattice

6.4.1 The Lagrangian of a lattice $U(1)$ gauge theory

- A lattice $U(1)$ gauge theory is described by variables on the links a_{ij} and on the sites $a_0(i)$.
- The compact and non-compact lattice $U(1)$ gauge theories.

• The flux of the electric field e_{ij} .

The $U(1)$ gauge theory can also be defined on a lattice. The $U(1)$ lattice gauge theory allows us to study the strong-coupling limit and confinement in any dimension. For simplicity, here we will consider a $U(1)$ gauge theory on a two-dimensional square lattice. The scalar potential $a_0(\mathbf{x})$ is defined at each lattice site $\mathbf{i}$ and becomes $a_0(\mathbf{i})$. The vector potential $\mathbf{a}(\mathbf{x})$ is defined on each link and becomes a_{ij} . We note that a_{ij} and a_{ji} are not independent; they are related through $a_{ij} = -a_{ji}$. The continuous fields are related to the lattice fields as follows:

$$a_0(\mathbf{j}) = a_0(\mathbf{x})|_{\mathbf{x}=l\mathbf{j}}, \quad a_{j_1 j_2} = \int_{l\mathbf{j}_1}^{l\mathbf{j}_2} d\mathbf{x}^i a_i(\mathbf{x}) \quad (6.4.1)$$

where l is the lattice constant. The Lagrangian that describes the $U(1)$ lattice gauge theory is the following function of $a_0(\mathbf{i})$ and a_{ij} :

$$L = \frac{1}{4J} \sum_{\mathbf{i}, \mu=\mathbf{x}, \mathbf{y}} [\dot{a}_{\mathbf{i}, \mathbf{i}+\mu} + a_0(\mathbf{i}) - a_0(\mathbf{i} + \mu)]^2 - \frac{g}{2} \sum_{\mathbf{p}} \Phi_{\mathbf{p}}^2 \quad (6.4.2)$$

where $\Phi_{\mathbf{p}}$ is the $U(1)$ flux through the square labeled by $\mathbf{p}$:

$$\Phi_{\mathbf{p}} = a_{\mathbf{i}, \mathbf{i}+\mathbf{x}} + a_{\mathbf{i}+\mathbf{x}, \mathbf{i}+\mathbf{x}+\mathbf{y}} + a_{\mathbf{i}+\mathbf{x}+\mathbf{y}, \mathbf{i}+\mathbf{y}} + a_{\mathbf{i}+\mathbf{y}, \mathbf{i}}. \quad (6.4.3)$$

One can check directly that the above Lagrangian is invariant under the lattice $U(1)$ gauge transformation

$$a_{ij}(t) \rightarrow a_{ij}(t) + \phi_j(t) - \phi_i(t), \quad a_0(\mathbf{i}, t) \rightarrow a_0(\mathbf{i}, t) + \dot{\phi}_{\mathbf{i}}(t) \quad (6.4.4)$$

Equation (6.4.2) defines a non-compact lattice $U(1)$ gauge theory. A compact lattice $U(1)$ gauge theory is defined by regarding $u_{ij} = e^{ia_{ij}}$ as link variables. In the compact lattice $U(1)$ gauge theory, a_{ij} and $a'_{ij} = a_{ij} + 2\pi$ are regarded as being equivalent. The Lagrangian for the compact lattice $U(1)$ gauge theory has the form

$$L = \frac{1}{4J} \sum_{\mathbf{i}, \mu=\mathbf{x}, \mathbf{y}} [\dot{a}_{\mathbf{i}, \mathbf{i}+\mu} + a_0(\mathbf{i}) - a_0(\mathbf{i} + \mu)]^2 + g \sum_{\mathbf{p}} \cos(\Phi_{\mathbf{p}}) \quad (6.4.5)$$

which is modified from eqn (6.4.2) to be consistent with the periodic condition $a_{ij} \sim a_{ij} + 2\pi$. In the rest of this section, we will consider only the compact $U(1)$ lattice gauge theory.

The classical equation of motion is obtained from $\delta L/\delta a_{ij} = 0$ and $\delta L/\delta a_0(i) = 0$. We find that

$$\frac{1}{2J} \frac{de_{i,i+\mu}}{dt} + g[\sin(\Phi_{p_1}) - \sin(\Phi_{p_2})] = 0, \quad (6.4.6)$$

$$\sum_{\mu=\pm x, \pm y} e_{i,i+\mu} = 0, \quad e_{ij} = \dot{a}_{ij} + a_0(i) - a_0(j) \quad (6.4.7)$$

where p_1 and p_2 label the two squares on the two sides of the link $(i, i + \mu)$.

By comparing e_{ij} with the e in the continuum model, we see that e_{ij} can be interpreted as the flux of the electric field flowing through the link (i, j) . Equation (6.4.7) states that the electric flux is conserved. It corresponds to Gauss's law $\partial_x \cdot e = 0$ in the continuum model. Also, being the flux through the square S_p , Φ_p corresponds to $\Phi_p = \int_{S_p} d^2x b$ in the continuum model.

The simplest way to obtain the dynamical properties of the lattice $U(1)$ gauge theory described by eqn (6.4.2) or eqn (6.4.5) is to take the continuum limit. Assuming that a_μ is small and is a smooth function of t and x , we substitute the relation (6.4.1) between the continuous variables and the lattice variables into the lattice Lagrangians. We obtain the following standard $U(1)$ Lagrangian:

$$\mathcal{L} = \frac{1}{8\pi\alpha_{2D}} \left(\frac{1}{c} e^2 - cb^2 \right) \quad (6.4.8)$$

From the resulting Maxwell equation, we find that the lattice $U(1)$ gauge theory contains a gapless mode with velocity c . The coupling constant α_{2D} has dimension 1/length in 1 + 2 dimensions.

In Section 3.7.1, we used the gauge invariance to construct a coupled theory of charged bosons and the $U(1)$ gauge field in a continuum. The same construction applies to the lattice model. We find that a lattice $U(1)$ gauge theory that couples to a charged boson is described by the following Lagrangian:

$$\begin{aligned} L = & + \sum_{\langle ij \rangle} t_{ij} (\varphi_i^\dagger \varphi_j e^{-i a_{ij}} + h.c) + \sum_i \varphi_i^\dagger i [\partial_0 + i a_0(i)] \varphi_i \\ & + \frac{1}{4J} \sum_{i, \mu=x, y} [\dot{a}_{i, i+\mu} + a_0(i) - a_0(i + \mu)]^2 + g \sum_p \cos(\Phi_p) \end{aligned} \quad (6.4.9)$$

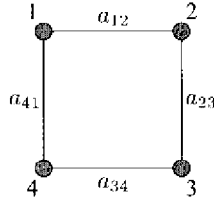
The Lagrangian is invariant under the following lattice gauge transformation:

$$c_i \rightarrow e^{i\phi(i,t)} c_i, \quad a_0(i) \rightarrow a_0(i) - \partial_0 \phi(i, t), \quad a_{ij} \rightarrow a_{ij} + \phi(i, t) - \phi(j, t).$$

Problem 6.4.1.

(a) Generalize eqn (6.4.1) to a three-dimensional cubic lattice.

(b) Use eqn (6.4.1) to show that the Lagrangian for the three-dimensional lattice gauge

FIG. 6.7. A $U(1)$ gauge theory on a square.

theory becomes $\frac{1}{8\pi\alpha}(c^{-1}e^2 - cb^2)$ in the continuum limit, where $e_i = \partial_0 a_i - \partial_i a_0$ and $b_i = \epsilon_{ijk}\partial_j a_k$. Find the values of α and c in terms of J , g , and the lattice constant l . Here the dimensionless constant α is the fine-structure constant and c is the speed of light.

6.4.2 The Hamiltonian of the lattice $U(1)$ gauge theory

- The Hamiltonian of a lattice $U(1)$ gauge theory can be obtained via gauge fixing or a phase-space path integral.
- The electric flux e_{ij} through a link is the canonical ‘momentum’ of the vector potential a_{ij} on the same link.

The Lagrangian (6.4.5) describes a classical $U(1)$ lattice gauge theory. In this section, we are going to find the quantum description. For simplicity, we will consider the lattice gauge theory on a single square (see Fig 6.7). The quantum theory is described by the path integral

$$Z = \int \mathcal{D}a \mathcal{D}a_0 \, e^{i \int dt \left(\frac{1}{4J} \sum_i (\dot{a}_{i,i+1} + a_0(i) - a_0(i+1))^2 + g \cos \Phi \right)}, \quad (6.4.10)$$

where $i = 1, 2, 3, 4$. Here $i = 1$ and $i = 5$ are regarded as the same point. As a gauge theory, $(a_{ij}(t), a_0(i, t))$ is a many-to-one labeling of the path. We can obtain a one-to-one labeling by ‘fixing a gauge’. We note that $\sum_{j=i\pm 1} a_{ij}$ transforms as $\sum_{j=i\pm 1} a_{ij} \rightarrow \sum_{j=i\pm 1} \tilde{a}_{ij} = \sum_{j=i\pm 1} (a_{ij} + \phi_i - \phi_j)$ under the gauge transformation. By tuning ϕ_i , we can always make $\sum_{j=i\pm 1} \tilde{a}_{ij} = 0$. Thus, for any path $(a_{ij}(t), a_0(i, t))$, we can always construct a gauge transformation to make $\sum_{j=i\pm 1} a_{ij} = 0$. Therefore, we can fix a gauge by choosing a gauge-fixing condition

$$\sum_{j=i\pm 1} a_{ij} = 0$$

Such a gauge is called the Coulomb gauge, which has the form $\partial \cdot \mathbf{a} = 0$ for a continuum theory. In the Coulomb gauge, our path integral becomes

$$Z = \int \mathcal{D}a \mathcal{D}a_0 e^{-i \int dt (\frac{1}{4J} \sum_i (\dot{a}_{i,i+1} + a_0(i) - a_0(i+1))^2 + g \cos \Phi)} \prod_{i,t} \delta(\sum_{j=i\pm 1} a_{ij})$$

We note that a coupling between $a_0(i)$ and a_{ij} has the form $a_0(i) \sum_{j=i\pm 1} \dot{a}_{ij}$. Thus, for a_{ij} satisfying the constraint $\sum_{j=i\pm 1} a_{ij} = 0$, $a_0(i)$ and a_{ij} decouple. As $a_0(i)$ has no dynamics (i.e. no $\dot{a}_0(i)$ terms), we can integrate out $a_0(i)$, which corresponds to simply dropping a_0 . The resulting path integral becomes

$$Z = \int \mathcal{D}a e^{-i \int dt (\frac{1}{4J} \sum_i \dot{a}_{i,i+1}^2 + g \cos \Phi)} \prod_{i,t} \delta(\sum_{j=i\pm 1} a_{ij}).$$

In general, a path integral in the Coulomb gauge can be obtained by the two simple steps of, firstly, inserting the gauge-fixing condition $\prod_{i,t} \delta(\sum_{j=i\pm 1} a_{ij})$ and, secondly, dropping the $a_0(i)$ field.

For our problem, the constraint $\prod_{i,t} \delta(\sum_{j=i\pm 1} a_{ij})$ requires that $a_{12} = a_{23} = a_{34} = a_{41} \equiv \Phi/4$. Here Φ describes the $U(1)$ flux through the square. The path integral takes the simple form

$$Z = \int \mathcal{D}\theta e^{-i \int dt (\frac{1}{16J} \dot{\Phi}^2 + g \cos \Phi)} \quad (6.4.11)$$

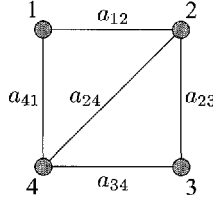
We note that the configuration $(a_{12}, a_{23}, a_{34}, a_{41}) = (\pi/2, \pi/2, \pi/2, \pi/2)$ is gauge equivalent to $(a_{12}, a_{23}, a_{34}, a_{41}) = (2\pi, 0, 0, 0)$ (i.e. there is a gauge transformation that transforms $(\pi/2, \pi/2, \pi/2, \pi/2)$ to $(2\pi, 0, 0, 0)$). Also, $a_{12} = 2\pi$ is equivalent to $a_{12} = 0$ because the $a_{i,i+1}$ live on a circle. Thus, $\Phi = 2\pi$ and $\Phi = 0$ correspond to the same physical point. The path integral (6.4.11) describes a particle of mass $(8J)^{-1}$ on a unit circle. The flux energy $-g \cos \Phi$ is the potential experienced by the particle. When $g = 0$, the energy levels are given by $E_n = 4Jn^2$.

There is another way to solve eqn (6.4.10). We first introduce the canonical momentum of a_{ij} , namely

$$\frac{\partial L}{\partial \dot{a}_{ij}} = \frac{1}{2J} e_{ij}$$

and write the path integral (6.4.10) as a phase-space path integral (see Problem 6.4.4)

$$Z = \int \mathcal{D}a \mathcal{D}a_0 \mathcal{D}e_{ij} e^{i \int dt (\sum_i \frac{e_{i,i+1}}{2J} (\dot{a}_{i,i+1} + a_0(i) - a_0(i+1)) - \frac{1}{4J} \sum_i e_{i,i+1}^2 + g \cos \Phi)}. \quad (6.4.12)$$

FIG. 6.8. A $U(1)$ gauge theory on a square with a diagonal link.

Then we integrate out a_0 and obtain⁴¹

$$Z = \int \mathcal{D}a \mathcal{D}e_{ij} e^{i \int dt \left(\sum_i \frac{e_{i,i+1}}{2J} \dot{a}_{i,i+1} - \frac{1}{4J} \sum_i e_{i,i+1}^2 + g \cos \Phi \right)} \prod_{i,t} \delta(e_{i,i+1} + e_{i,i-1}) \quad (6.4.13)$$

The quantum Hamiltonian of the above system is given by

$$H = \frac{1}{4J} \sum_i e_{i,i+1}^2 - g \cos \Phi, \quad [a_{i,i+1}, (2J)^{-1} e_{i,i+1}] = i\hbar.$$

The physical Hilbert space is formed by states that satisfy

$$(e_{i,i+1} + e_{i,i-1})|\text{phy}\rangle = 0$$

As $[e_{i,i+1} + e_{i,i-1}, H] = 0$, H acts within the physical Hilbert space.

To find the energy levels of the system, we note that a_{ij} is periodic with a period 2π and the momentum of a_{ij} is $e_{ij}/2J$. Thus, $e_{ij}/2J$ is quantized as integers, n_{ij} . When $g = 0$, the energy levels of the system are given by $J \sum_i n_{i,i+1}^2$, where the n_{ij} satisfy the constraint $\sum_{j=i\pm 1} n_{ij} = 0$. For our four-site system, the constraint requires that $n_{12} = n_{23} = n_{34} = n_{41} \equiv n$. The energy eigenstates are labeled by n and have energy $E_n = 4Jn^2$, which agrees with the previous calculation.

Problem 6.4.2.

- Generalize eqn (6.4.10) to a square with a diagonal link (see Fig. 6.8).
- Find the energies of the ground state and the excited states, assuming that $g = 0$.

Problem 6.4.3.

- Show that, for a non-compact lattice $U(1)$ gauge theory on a two-dimensional square

⁴¹ Note that the coupling between $a_0(i)$ and e_{ij} has the form $\sum_i a_0(i)(e_{i,i+1} + e_{i,i-1})$. The path integral $\int \mathcal{D}a_0 e^{i \int dt \sum_i a_0(i)(e_{i,i+1} + e_{i,i-1})}$ is proportional to $\prod_{i,t} \delta(e_{i,i+1}(t) + e_{i,i-1}(t))$.

lattice, the gauge-fixed path integral has the form

$$Z = \int \mathcal{D}a \, e^{-i \int dt \left(\frac{1}{4J} \sum_{i, \mu=x, y} \dot{a}_{i, i+\mu}^2 - \frac{1}{2g} \sum_p \Phi_p^2 \right)} \prod_{i, t} \delta \left(\sum_{\mu=\pm x, \pm y} a_{i, i+\mu} \right)$$

(b) Show that the constraint $\sum_{\mu=\pm x, \pm y} a_{i, i+\mu} = 0$ can be solved by introducing a real field ϕ_p defined on the squares, and write the gauge potential on the link as $a_{i, i+\mu} = \phi_{p_1} - \phi_{p_2}$, where p_1 and p_2 are the two squares on the two sides of the link $(i, i+\mu)$. Write the path integral in terms of the ϕ_p field.

(c) Find the dispersion relation of the ϕ_p fluctuations.

Problem 6.4.4.

Prove eqn (6.4.12). (Hint: The phase-space path integral is given by $\int \mathcal{D}p \mathcal{D}q \, e^{i \int dt [p\dot{q} - H(p, q)]}$ where $H(p, q) = p\dot{q} - L$ is the Hamiltonian.)

6.4.3 The Coulomb phase and the confined phase of the lattice $U(1)$ gauge theory

- In $1+3$ dimensions, the compact lattice $U(1)$ gauge theory is in the Coulomb phase when $g/J \gg 1$ and in the confined phase if $g/J \ll 1$.

We know that, in $1+2$ dimensions, the compact $U(1)$ gauge theory is always in the confined phase due to the instanton effect. In this section, we are going to study a compact $U(1)$ gauge theory on a cubic lattice. We will argue that the $U(1)$ gauge theory in $1+3$ dimensions has both the confined phase and the Coulomb phase.

The Lagrangian of the $U(1)$ gauge theory is given by

$$L = \frac{1}{4J} \sum_{i, \mu=x, y, z} [\dot{a}_{i, i+\mu} + a_0(i) - a_0(i+\mu)]^2 + g \sum_p \cos(\Phi_p)$$

We can express the action of the above Lagrangian in the following dimensionless form:

$$S = \sqrt{gJ} \int d\tilde{t} \left(\frac{1}{4} \sum_{i, \mu=x, y} [\partial_{\tilde{t}} a_{i, i+\mu} + \tilde{a}_0(i) - \tilde{a}_0(i+\mu)]^2 + \sum_p \cos(\Phi_p) \right)$$

where $\tilde{t} = \sqrt{gJ} t$ is the dimensionless time and $\tilde{a}_0 = a_0/\sqrt{gJ}$ is the dimensionless potential. We see that, when $g/J \gg 1$, the fluctuations of Φ_p are weak and we can assume that $a_{ij} \sim 0$. In this limit, L becomes a non-compact $U(1)$ gauge theory as follows:

$$L = \frac{1}{4J} \sum_{i, \mu=x, y, z} [\dot{a}_{i, i+\mu} + a_0(i) - a_0(i+\mu)]^2 - \frac{g}{2} \sum_p \Phi_p^2$$

The quadratic theory can be solved exactly. In the continuum limit, the above becomes the standard Maxwell Lagrangian of the $U(1)$ gauge theory. We find that,

at low energies, there are two linearly-dispersing modes $\omega = c_a |\mathbf{k}|$ representing the gapless photons.

When $g/J \ll 1$, we can drop the $\cos(\Phi_p)$ term. The resulting quadratic theory can also be solved exactly, as we did in the last section. The Lagrangian for the phase-space path integral has the form

$$L = \sum_{i, \mu=x, y, z} \frac{e_{i, i+\mu}}{2J} [\dot{a}_{i, i+\mu} + a_0(i) - a_0(i+\mu)] - \sum_{i, \mu=x, y, z} \frac{e_{i, i+\mu}^2}{4J} + g \sum_p \cos(\Phi_p)$$

After integrating out a_0 , we obtain the constraint

$$\sum_{\mu=\pm x, \pm y, \pm z} e_{i, i+\mu} = 0$$

The Hamiltonian is given by

$$H = \sum_{i, \mu=x, y, z} \frac{e_{i, i+\mu}^2}{4J} - g \sum_p \cos(\Phi_p)$$

Since $e_{ij}/2J$ is quantized as an integer n_{ij} , when $g = 0$, the energy levels are given by $E = J \sum_{i, \mu=x, y, z} n_{i, i+\mu}^2$, where $n_{i, i+\mu}$ satisfy $\sum_{\mu=\pm x, \pm y, \pm z} n_{i, i+\mu} = 0$. We see that all of the excitations in the $g = 0$ limit are gapped. The gapped excitations are loops formed by nonzero n_{ij} s. These loops represent lines of electric flux. In the presence of a pair of positive and negative charges, the two charges are connected by a line of electric flux which causes a linear confining potential between the two charges. Thus, the gapped phase is the confined phase of the $U(1)$ gauge theory.

Problem 6.4.5.

To show that charges interact with a linear potential in the confined phase, we consider the lattice $U(1)$ gauge theory in the presence of charges as follows:

$$L = \sum_{i, \mu=x, y, z} \frac{[\dot{a}_{i, i+\mu} + a_0(i) - a_0(i+\mu)]^2}{4J} + g \sum_p \cos(\Phi_p) - \sum_i a_0(i) q_i$$

where q_i is the $U(1)$ charge at site i .

- Find the Lagrangian in the phase-space path integral.
- Show that integrating out a_0 leads to the following constraint:

$$\sum_{\mu=\pm x, \pm y, \pm z} e_{i, i+\mu} = 2Jq_i$$

which is Gauss's law on the lattice.

- Find the ground-state energy of the gauge theory with a charge $q = 1$ at $i = 0$ and a charge $q = -1$ at $i = lx$, assuming that $g = 0$. Show that the interaction energy between the two charges grows linearly with l .

THEORY OF QUANTUM HALL STATES

- Fractional quantum Hall liquids represent new states of matter which contain extremely rich internal structures.
- Different fractional quantum Hall liquids cannot be characterized by order parameters and long-range orders of local operators. We need a completely new theory to describe fractional quantum Hall liquids.
- Fractional quantum Hall liquids contain gapless edge excitations which form the so-called chiral Luttinger liquids. The chiral Luttinger liquids at the edges also have very rich structures, as a reflection of the rich bulk internal structures.

The fractional quantum Hall (FQH) effect appears in two-dimensional electron systems in a strong magnetic field. Since its discovery in 1982 (Tsui *et al.*, 1982; Laughlin, 1983), experiments on FQH systems have continued to reveal many new phenomena and surprises. The observed rich hierarchical structures (Haldane, 1983; Halperin, 1984) indicate that electron states that demonstrate a fractional quantum Hall effect (these states are called FQH liquids) contain extremely rich internal structures. In fact, FQH liquids represent a whole new state of matter, which is very different from the symmetry-breaking states discussed in the last few chapters. One needs to develop new concepts and new techniques to understand this new kind of state. In this chapter, we will introduce a general theory of FQH states. A detailed discussion of the new orders (the topological orders) in FQH states will be given in Chapter 8.

7.1 The Aharonov–Bohm effect and fractional statistics

7.1.1 The Aharonov–Bohm effect—deflect a particle without touching

- The local phase for contractible loops and the global phase for non-contractible loops.
- The nonzero local phase represents a force, while a nonzero global phase does not correspond to any classical force.

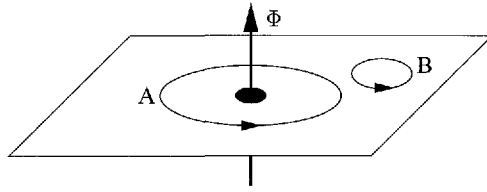


FIG. 7.1. A plane with the $r = 0$ point removed and a flux tube at $r = 0$. The loop A is non-contractible and has a winding number $n_w = 1$. The loop B is contractible.

To prepare for later discussions on the quantum interference in the quantum Hall (QH) state, let us first give a brief discussion of the Aharonov–Bohm (A–B) effects. First, we would like to introduce two concepts, namely local phase and global phase. The local phase is associated with contractible loops. Moving a charged particle slowly around a contractible loop will generate a local phase as follows:

$$\text{local phase} = \exp \left(ie \oint \mathbf{A} \, d\mathbf{x} \right) = \exp \left(ie \int \mathbf{B} \cdot d\mathbf{S} \right).$$

The magnetic field $\mathbf{B} = 0$ if and only if the local phases for all contractible loops vanish. Thus, a charged particle that feels no magnetic force has no local phases.

The global phase is related to non-contractible loops. To understand the global phase in a simple setting, let us consider a charged particle moving on a two-dimensional plane. We remove the $r = 0$ point, which changes the topology of the space. We then place an infinite thin flux tube of flux Φ at $r = 0$ (see Fig. 7.1). As the particle goes around $r = 0$, a phase $e^{ie\Phi}$ is generated. Such a phase is called the global phase. More generally,

$$\text{global phase} = e^{ie\Phi n_w}$$

$$n_w = \text{winding number}.$$

Clearly, despite the nonzero global phase, the particle feels no magnetic force because there is no magnetic field away from $r = 0$. We see that a free particle (i.e. a particle that feels no force) can have a nonzero global phase.

Global phase exists only when non-contractible loops exist (i.e. when the space is not simply connected). Another example of a non-simply-connected space is a torus (see Fig. 7.2). A free particle on a torus can also have global phases.

The above free particle (with the flux tube at $r = 0$) can be described by the following Hamiltonian:

$$H = -\frac{1}{2m}(\partial - i\mathbf{a})^2, \quad (a_x, a_y) = \frac{1}{r^2}(-y, -x)\frac{\phi}{2\pi}$$

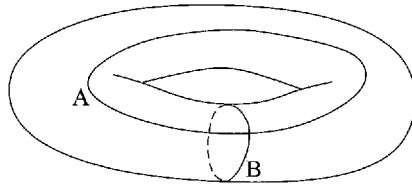


FIG. 7.2. A torus with two non-contractible loops A and B.

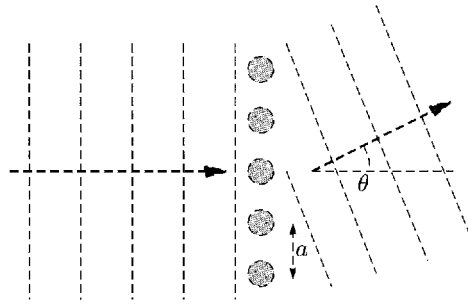


FIG. 7.3. A beam of particles passes through a grid of tubes, with a flux through each tube.

The field strength vanishes, i.e.

$$b = \partial_x a_y - \partial_y a_x = 0 \implies \text{local phase} = 0$$

For a loop around $x = 0$, we have

$$\oint \mathbf{a} \cdot d\mathbf{x} = \phi \implies \text{global phase} \neq 0.$$

A non-vanishing local phase will affect the classical equation of motion. The Berry phase for the spin is an example of local phase (see Section 2.3). In contrast, a global phase will not affect the classical equation of motion. However, a global phase will affect the quantum properties of the particle.

Problem 7.1.1.

Deflecting without touching Although a global phase does not produce any classical force, it can still deflect a moving particle. Consider the set-up in Fig. 7.3, where a beam of particles with charge e and momentum $\mathbf{k}$ passes through a grid of impenetrable tubes. Calculate the deflecting angle θ if there is a flux Φ through each tube.

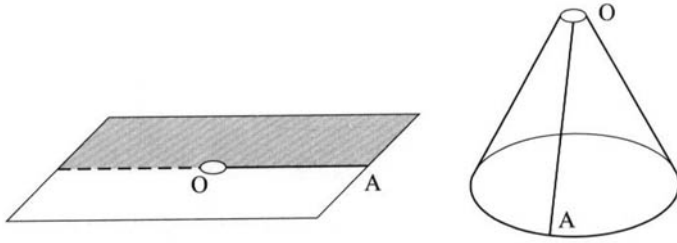


FIG. 7.4. The V_- space, with the $\mathbf{r} = 0$ point removed, and the points $\mathbf{r}$ and $-\mathbf{r}$ identified as the same point.

7.1.2 Particles with a hard-core condition and fractional statistics

- The configuration space for particles with a hard-core condition is not simply connected. The non-trivial topology of the configuration space allows non-trivial statistics.
- Fractional statistics exist and only exist in two spatial dimensions.

One realization of the A-B effect is the fractional statistics in two dimensions (Leinaas and Myrheim, 1977; Wilczek, 1982). Consider two free hard-core identical particles in two dimensions.⁴² The configuration space is given by

$$\begin{aligned} \text{configuration space} &= \{(\mathbf{r}_1, \mathbf{r}_2) | \mathbf{r}_1 \neq \mathbf{r}_2, (\mathbf{r}_1, \mathbf{r}_2) \sim (\mathbf{r}_2, \mathbf{r}_1)\} \\ &= \{(\mathbf{r}_+, \mathbf{r}_-) | \mathbf{r}_- \neq 0, (\mathbf{r}_+, \mathbf{r}_-) \sim (\mathbf{r}_-, -\mathbf{r}_+)\} = V_+ \otimes V_- \end{aligned}$$

where $\mathbf{r}_+ = \frac{1}{2}(\mathbf{r}_1 + \mathbf{r}_2)$, $\mathbf{r}_- = \mathbf{r}_1 - \mathbf{r}_2$, $V_+ = \{\mathbf{r}_+\}$, and $V_- = \{\mathbf{r}_- | \mathbf{r}_- \neq 0, \mathbf{r}_- \sim -\mathbf{r}_-\}$ (see Fig. 7.4). Here V_+ is the usual two-dimensional plane, while V_- is only half of the two-dimensional plane because $\mathbf{r}_-$ and $-\mathbf{r}_-$ are regarded as the same point. Also, the $\mathbf{r}_- = 0$ point is removed from V_- .

As pointed out in the last section, ‘free’ means that the local phase is zero. As the two-particle configuration space is not simply connected (V_- is not simply connected), there exist global phases and we have the freedom to choose the global phases. Note that the non-contractible loops in V_- are characterized by winding numbers n_w (Fig. 7.5). Thus, we can assign the following global phases for non-contractible loops:

$$\text{global phases} = e^{i\theta n_w}$$

Now it is clear that, in quantum mechanics, there are different kinds of free identical particles, labeled by a parameter θ .

⁴² Here, by free particles, we mean particles that do not experience any force when separated. However, the particles may be subject to a hard-core condition.

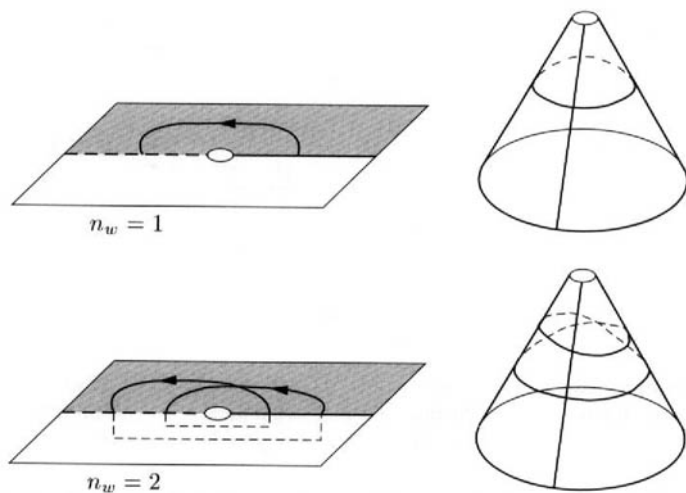


FIG. 7.5. Two loops with winding numbers $n_w = 1$ and $n_w = 2$ in the V_- space.

- The parameter θ describes the flux (the global phase) in many-particle configuration space.
- The parameter θ determines the statistics.

Note that the loop with $n_w = 1$ connects $\mathbf{r}_-$ to $-\mathbf{r}_-$, or $(\mathbf{r}_1, \mathbf{r}_2)$ to $(\mathbf{r}_2, \mathbf{r}_1)$, which corresponds to exchanging the two particles. Thus, the statistics of the particles are as follows:

$$\begin{cases} \theta = 0 \rightarrow & \text{boson} \\ \theta = \pi \rightarrow & \text{fermion} \\ \theta = \text{other} \rightarrow & \text{anyon} \end{cases}$$

The Hamiltonian for the above two free identical particles (i.e. for two anyons with statistics θ) is given by

$$H = -\frac{1}{2m}(\partial_{\mathbf{r}_+})^2 - \frac{1}{2} \frac{1}{m}(\partial_{\mathbf{r}_-} - i\mathbf{a})^2$$

$$(a_x, a_y) = \frac{\theta}{\pi r^2}(-y, x) \quad (7.1.1)$$

$$\text{Boundary Conditions : } \phi(\mathbf{r}_+, \mathbf{r}_-) = \phi(\mathbf{r}_+, -\mathbf{r}_-)$$

When $\theta = 0$ and $\mathbf{a} = 0$, H obviously describes two bosons. For $\theta = \pi$, we can make a gauge transformation

$$\phi(\mathbf{r}_+, \mathbf{r}_-) \rightarrow e^{if(\mathbf{r}_-)} \phi(\mathbf{r}_+, \mathbf{r}_-), \quad f(\mathbf{r}) \equiv \arctan\left(\frac{x}{y}\right)$$

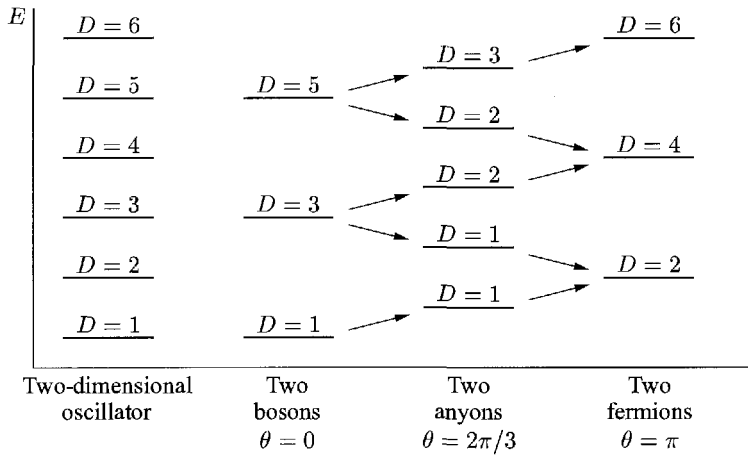


FIG. 7.6. The energy levels E and the degeneracies D (for the relative motion) of two anyons in a two-dimensional harmonic potential well.

As $\nabla f(\mathbf{r}) = \frac{1}{r^2}(-y, x)$, we find that $e^{-if}(\partial_{\mathbf{r}_-} - i\mathbf{a})^2 e^{if} = (\partial_{\mathbf{r}_-})^2$ and the Hamiltonian becomes

$$H = -\frac{1}{2m}(\partial_{\mathbf{r}_+})^2 - \frac{1}{2m}(\partial_{\mathbf{r}_-})^2$$

$$\phi(\mathbf{r}_+, \mathbf{r}_-) = -\phi(\mathbf{r}_+, \mathbf{r}_-)$$

$$\text{or } \phi(\mathbf{r}_1, \mathbf{r}_2) = -\phi(\mathbf{r}_2, \mathbf{r}_1)$$

which describes two free fermions. In general, we can consistently gauge away $\mathbf{a}$, even for N fermions, and describe the Fermi statistics through the 'boundary condition', i.e. the anti-symmetric condition. However, for an N -anyon system, we cannot gauge away $\mathbf{a}$ and represent the fractional statistics through the 'boundary condition'.

One can solve the system of two anyons in a two-dimensional harmonic potential well, and find the spectrum shown in Fig. 7.6. One can clearly see how the spectrum changes continuously from the bosonic one to the fermionic one.

An interesting question is do we have an anyon in three dimensions? Two identical free particles in three dimensions have a configuration space $V_+^{(3D)} \otimes V_-^{(3D)}$, with

$$V_-^{(3D)} = \{\mathbf{r}_- | \mathbf{r}_- \neq 0, \mathbf{r}_- \sim -\mathbf{r}_-\}$$

Here $V_-^{(3D)}$ is not simply connected, and a global phase exists. However, the non-contractible loop in $V_-^{(3D)}$ is characterized by Z_2 . This is because an exchange in one direction (described by the loop C_1) and the exchange in the opposite direction

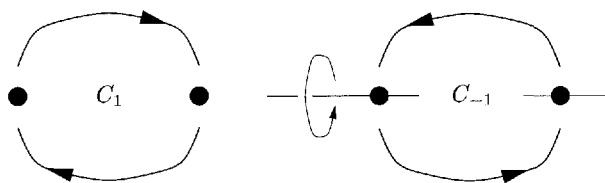


FIG. 7.7. The two exchange loops C_1 and C_{-1} can be continuously changed into each other in three dimensions, by rotating about the axis connecting the two particles.

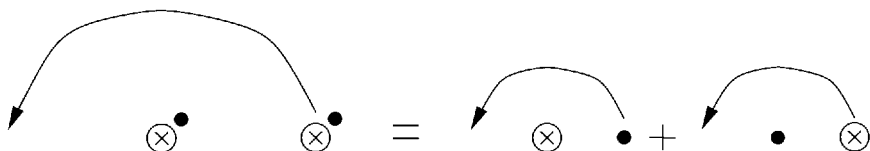


FIG. 7.8. The phase induced by moving a flux-charge bound state around another flux-charge bound state.

(described by the loop C_{-1}) belong to the same class, as the two loops $C_{\pm 1}$ can be continuously changed into each other (see Fig. 7.7). If we assign a global phase $e^{i\theta}$ to the loop C_1 , then the loop C_{-1} will have a global phase $e^{-i\theta}$. As the two loops $C_{\pm 1}$ differ only by a contractible loop which has a vanishing local phase, we also have $e^{i\theta} = e^{-i\theta}$. Therefore, θ can only take two possible values as follows:

$$\theta = \begin{cases} 0, & \text{boson} \\ \pi, & \text{fermion} \end{cases}$$

The above discussion demonstrates the anyon as a mathematical possibility in quantum mechanics. The next question is how an anyon may appear in some physical models which originally contained no anyons. In fact, an anyon can appear as excitations in boson/fermion systems. In the following, we will consider a model in which an anyon appears as a bound state of charge and flux. Consider a system of charge one boson ϕ . Assume that bosons form a q -boson bound state $\Phi_q = (\phi)^q$, and Φ_q undergoes a boson condensation. The effective Lagrangian is

$$\mathcal{L}_{\text{eff}} = \frac{1}{2} |(\partial_\mu - iqa_\mu) \Phi_q|^2 - \frac{1}{2} a |\Phi_q|^2 - \lambda |\Phi_q|^4 + \frac{1}{2g} (f_{\mu\nu})^2 \quad (7.1.2)$$

$$\langle \Phi_q \rangle \neq 0$$

where a_μ is the $U(1)$ gauge field that couples to the charge boson. In the condensed state, the vortex carries a flux $\frac{2\pi}{q}$ (see Problem 7.1.2). Consider a bound state of the vortex $\frac{2\pi}{q}$ and the original boson ϕ . Moving one bound state half-way around

the other, we obtain a phase

$$\theta = \frac{1}{2} \frac{2\pi}{q} + \frac{1}{2} \frac{2\pi}{q} = \frac{2\pi}{q}$$

The first term is from the charge going around the flux tube, and the second term is from the flux tube going around the charge (see Fig. 7.8). We see that the bound state is a boson if $q = 1$, a fermion if $q = 2$, and an anyon if $q > 2$.

For BCS superconductors, $q = 2$. However, ϕ is a fermion, and hence the bound state is a boson. The only known condensed matter system that supports anyon excitations is the FQH system.

Problem 7.1.2.

In polar coordinates (r, ϕ) , a vortex in eqn (7.1.2) is described by

$$\Phi_q = f(r) e^{i\phi}$$

where $f(\infty) = \langle \Phi_q \rangle$.

(a) Show that, if $a_\mu = 0$, then the energy of the vortex diverges as $\ln(L)$, where L is the size of the system.

(b) Find $a(r, \phi)$ that makes the vortex have a finite energy.

(c) Show that the total flux of the above $a(r, \phi)$ is $2\pi/q$.

7.2 The quantum Hall effect

7.2.1 The integral quantum Hall effects

- Integral quantum Hall (IQH) states are described by filled Landau levels.
- The many-body wave function of the $\nu = 1$ IQH state.
- The density profile of IQH wave functions.

Let us first discuss classical Hall effects. Consider a 2D gas of charge $q_e = -e$ electrons moving under the influence of an electric field in the plane and a magnetic field normal to the plane. To maintain a static current, the force from the electric field must balance the Lorentz force from the magnetic field, i.e.

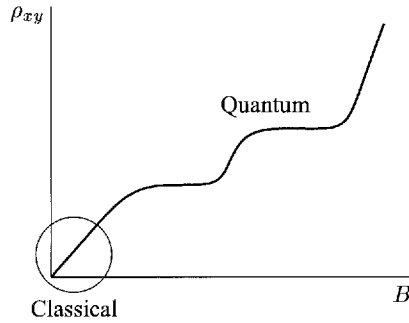
$$q_e \mathbf{E} = q_e \mathbf{v} \times \mathbf{B}$$

In this chapter, we will choose a unit such that the speed of light is $c = 1$. As the current $\mathbf{j}$ is given by $\mathbf{v}n$, where n is the density, $\mathbf{E}$ and $\mathbf{j}$ are related through

$$\mathbf{E} \perp \mathbf{j}, \quad |\mathbf{E}| = |\mathbf{j}| \frac{B}{n q_e c} = |\mathbf{j}| \frac{\hbar B / (\hbar c / q_e)}{q_e^2 n} = |\mathbf{j}| \left(\frac{\hbar}{q_e^2} \right) \frac{1}{\nu}$$

where

$$\nu \equiv \frac{n \hbar c}{q_e B} = \frac{\text{number of particles}}{\text{number of flux quanta}}$$

FIG. 7.9. The Hall resistance ρ_{xy} as a function of B .

is the filling fraction. The Hall resistance $\rho_{xy} = \left(\frac{h}{q_e^2}\right) \frac{1}{\nu}$ can be measured directly in experiments. According to the above classical theory, $\rho_{xy} \propto B$ if n is fixed.

Experimentally, one indeed finds that, for a two-dimensional electron gas, $\rho_{xy} \propto B$ at weak fields (see Fig. 7.9). In the early 1980s physicists put a two-dimensional electron gas under strong magnetic fields (~ 10 Tesla) and cooled it to very low temperatures ($\sim 1K^\circ$). They found that, for strong B , ρ_{xy} develops plateau structures, as shown in Fig. 7.9 (von Klitzing *et al.*, 1980; Tsui *et al.*, 1982). Such a phenomenon is called the QH effect. Physicists soon discovered many amazing properties of QH states. What is most striking is that the plateaus appear exactly at those ρ_{xy} corresponding to

$$\nu = \underbrace{1, 2, 3, \dots}_{\text{IQH}}, \underbrace{\frac{1}{3}, \frac{2}{3}, \frac{2}{5}, \frac{3}{7}, \dots}_{\text{FQH}}$$

The value of ρ_{xy} is so accurate and stable that it has become the standard for resistance.

The physics of the plateaus is a little complicated because it involves impurities. In short, we can understand the plateaus at these filling fractions, if the two-dimensional electron gas forms an incompressible liquid⁴³ at these ν s. As we will see below, electrons do form an incompressible state at $\nu = 1, 2, 3, \dots$, due to the Landau level structure. This allows us to understand the plateaus at integer ν and IQH effects.

Let us ignore the Coulomb interaction and consider a two-dimensional free electron gas in a strong magnetic field B . We will ignore the electron spins or assume that the electron spins are polarized. A single electron in a uniform

⁴³ That is, all charged excitations have a finite energy gap.

magnetic field is described by

$$H = -\frac{1}{2m}(\partial_i - iq_e A_i)^2, \quad \hbar = c = 1 \quad (7.2.1)$$

$$(A_x, A_y) = \frac{B}{2}(-y, x), \quad (\text{symmetric gauge})$$

$$B = \partial_x A_y - \partial_y A_x$$

The above Hamiltonian has an interesting symmetry property. It is not invariant under the direct translation $T_a = e^{-a \cdot \partial_a}$, i.e. $T_a H T_a^\dagger \neq H$, despite the uniform magnetic field. However, it is invariant under the translation T_a followed by the gauge transformation

$$G_a = e^{i\frac{1}{2}Bq_e(xa^y - ya^x)}. \quad (7.2.2)$$

or, more precisely,

$$G_a T_a H (G_a T_a)^\dagger = H$$

The combined transformation $G_a T_a$ is a symmetry of H and is called the magnetic translation. The magnetic translations in different directions do not commute:

$$(G_a T_a)(G_b T_b) = e^{-iq_e B(a^x b^y - a^y b^x)}(G_b T_b)(G_a T_a) \quad (7.2.3)$$

Thus, the eigenstates of H cannot be labeled by the momentum vector (k_x, k_y) , despite the Hamiltonian having magnetic translation symmetries in both the x and y directions. Note that the phase $q_e B(a^x b^y - a^y b^x)$ is just the phase obtained by moving the charge q_e around the parallelogram P_{ab} spanned by a and b , namely $e^{iq_e \oint_{P_{ab}} d\mathbf{x} \cdot \mathbf{A}}$.

To find the energy levels of eqn (7.2.1), we introduce the complex coordinate $z = x + iy$, so that

$$\begin{cases} \partial_z = \frac{1}{2}(\partial_x - i\partial_y), & x = \frac{1}{2}(z + \bar{z}), \\ \partial_{\bar{z}} = \frac{1}{2}(\partial_x + i\partial_y), & y = \frac{1}{2i}(z - \bar{z}). \end{cases}$$

In terms of z and $\bar{z}$, we have

$$H = -\frac{1}{m}(D_z D_{\bar{z}} + D_{\bar{z}} D_z)$$

where

$$\begin{cases} D_z = \partial_z - iq_e A_z, & A_z = \frac{1}{2}(A_x - iA_y) = \frac{B}{4i}\bar{z} \\ D_{\bar{z}} = \partial_{\bar{z}} - iq_e A_{\bar{z}}, & A_{\bar{z}} = \frac{1}{2}(A_x + iA_y) = -\frac{B}{4i}z \end{cases}$$

The Hamiltonian H can be simplified through a non-unitary transformation $e^{+|z|^2/4l_B^2}$, where l_B is the magnetic length defined by

$$l_B^2 = \left| \frac{c\hbar}{q_e B} \right|$$

(Note that $2\pi l_B^2 B = \frac{\hbar c}{q_e} = \Phi_0$ gives us a unit flux quantum.) We find that

$$H = e^{-|z|^2/4l_B^2} \left[-\frac{1}{m} \left(\tilde{D}_z \tilde{D}_{\bar{z}} + \tilde{D}_{\bar{z}} \tilde{D}_z \right) \right] e^{+|z|^2/4l_B^2}$$

where

$$\begin{aligned} \tilde{D}_z &= e^{+|z|^2/4l_B^2} D_z e^{-|z|^2/4l_B^2} = \partial_z - \frac{q_e B}{4c} \bar{z} - \frac{1}{4l_B^2} \bar{z} = \partial_z - \frac{1}{2l_B^2} \bar{z} \\ \tilde{D}_{\bar{z}} &= \partial_{\bar{z}} + \frac{q_e B}{4c} z - \frac{1}{4l_B^2} z = \partial_{\bar{z}} \end{aligned}$$

Here we have assumed that $q_e B > 0$. We see that the non-unitary transformation $e^{+|z|^2/4l_B^2}$ can be regarded as a 'gauge transformation' that makes $A_{\bar{z}} = 0$. We can further simplify H as follows:

$$H = e^{-|z|^2/4l_B^2} \left[-\frac{2}{m} \left(\partial_z - \frac{1}{2l_B^2} \bar{z} \right) \partial_{\bar{z}} \right] e^{+|z|^2/4l_B^2} + \frac{1}{2} \hbar \omega_c$$

where $\omega_c = q_e B / mc$ is the cyclotron frequency.

Now we can easily write down the eigenstates of an electron in a uniform magnetic field as follows:

$$\begin{aligned} \Psi_0 &= e^{-|z|^2/4l_B^2} f(z) & E &= \frac{1}{2} \hbar \omega_c \\ \Psi_1 &= e^{-|z|^2/4l_B^2} \left(\partial_z - \frac{1}{2l_B^2} \bar{z} \right) f(z) & E &= \left(\frac{1}{2} + 1 \right) \hbar \omega_c \\ \Psi_n &= e^{-|z|^2/4l_B^2} \left(\tilde{D}_z \right)^n f(z) & E &= \left(\frac{1}{2} + n \right) \hbar \omega_c \end{aligned}$$

which gives us the Landau level structure.

In the above, we have assumed that $q_e B > 0$. The electron wave function in the zeroth Landau level is an analytic function $f(z)$ times $e^{-|z|^2/4l_B^2}$. If $q_e B < 0$, then the non-unitary transformation $e^{+|z|^2/4l_B^2}$ will transform A^z to zero. The electron wave function in the zeroth Landau level will be an anti-analytic function $f(z^*)$ times $e^{-|z|^2/4l_B^2}$.

The wave functions Ψ_0 in the zeroth Landau level can be expanded by the basis states

$$\psi_m = \sqrt{N_m} z^m e^{-|z|^2/4l_B^2}$$

which carry angular momentum m . The wave function ψ_m has a ring shape. The position of the maximum of $|\psi_m| = e^{m \ln |z| - |z|^2/4l_B^2}$ is at

$$|z| = r_m = \sqrt{2ml_B}$$

We find that the ring of the m th state has an area of $\pi r_m^2 = 2\pi l_B^2 m$, which encloses m flux quanta (see Fig. 7.10). Therefore, there is one state for every flux quantum,

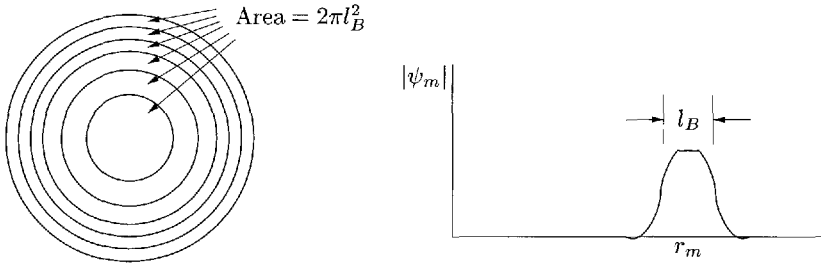
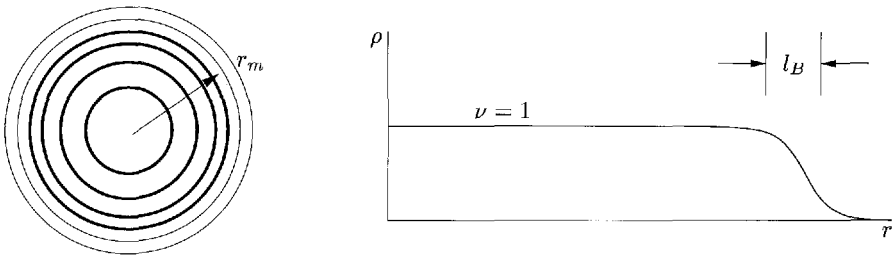


FIG. 7.10. The circular orbits in the zeroth Landau level.

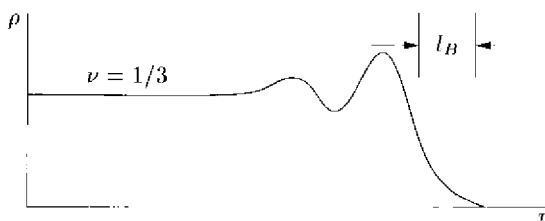
FIG. 7.11. The density profile of the $\nu = 1$ droplet, where the first m levels (represented by the thick lines) are filled.

and the number of states in the zeroth Landau level is equal to the number of flux quanta. So, when $\nu = 1$, every state in the zeroth Landau level is filled by an electron. This gives us a finite gap $\Delta = \hbar\omega_c$ for excitations, and explains the $\nu = 1$ plateau observed in experiments. Incompressibility is due to the Pauli exclusion principle (Fermi statistics).

The many-body wave function of the $\nu = 1$ state has the form

$$\begin{aligned} \Psi(z_1, \dots, z_N) &= \mathcal{A}[(z_1)^0(z_2)^1(z_3)^2 \dots] e^{-\sum |z_i|^2/4l_B^2} \\ &= \det \begin{vmatrix} 1 & 1 & \dots \\ z_1 & z_2 & \dots \\ (z_1)^2 & (z_2)^2 & \dots \\ \vdots & \vdots & \vdots \end{vmatrix} e^{-\sum |z_i|^2/4l_B^2} = \prod_{i < j} (z_i - z_j) e^{-\sum |z_i|^2/4l_B^2} \end{aligned}$$

where $\mathcal{A}$ is the anti-symmetrization operator. The above wave function describes a circular droplet with uniform density (Fig. 7.11), because every electron occupies the same area $\pi r_{m+1}^2 - \pi r_m^2 = 2\pi l_B^2$. This property is important. A generic wave function of N variables may not have a uniform density in the $N \rightarrow \infty$ limit. In this case, the wave function does not have a sensible thermodynamical limit.

FIG. 7.12. The density profile of the $\nu = 1/3$ droplet.**Problem 7.2.1.**

Find the energy levels of an electron in a potential $V = \frac{1}{2}m\omega_0^2 r^2$ and a uniform magnetic field B .

Problem 7.2.2.

- (a) Show that the magnetic translation $G_a T_a$ is a symmetry of the Hamiltonian in the uniform magnetic field (7.2.1) if the gauge transformation G_a is given by eqn (7.2.2).
 (b) Show that the magnetic translations satisfy the algebra (7.2.3).

7.2.2 Fractional quantum Hall effect

- The Laughlin wave functions of FQH states.
- The plasma analogue and the density profile of Laughlin states.

When $0 < \nu < 1$, the zeroth Landau level is only partially filled. This gives us a huge ground-state degeneracy $\sim \frac{N_\phi!}{N!(N_\phi - N)!}$ for free electrons (where N_ϕ is the number of states in the zeroth Landau level, which is also the number of flux quanta, and N is the number of electrons). Therefore, the experimentally observed $\nu = 1/3$ state must be due to the interaction. Two questions arise. Firstly, how can the interaction produce a gap (incompressibility)? Secondly, why is $\nu = 1/3$ special?

Laughlin answered the above two questions with the trial wave function

$$\Psi_3 = \prod_{i < j} (z_i - z_j)^3 e^{-\sum |z_i|^2 / 4l_B^2}$$

The wave function Ψ_3 is good due to the following reasons.

- Ψ_3 has a third-order zero $(z_i - z_j)^3$ for any pair of electrons. This is good for repulsive interactions.
- Ψ_3 describes a circular droplet of uniform density (Fig. 7.12), with $\nu = 1/3$.
- Ψ_3 is an incompressible state.

Due to the rotational invariance of $|\Psi_3|$, it is clear that $|\Psi_3|$ describes a circular droplet. If we assume that the droplet also has a uniform density, then it is easy to

understand why it has a filling fraction $\nu = 1/3$. This is because the highest power of z_i (with i fixed) is $3(N-1)$. The orbit at angular momentum $3(N-1)$ has a radius of $r_{\max} = \sqrt{2 \times 3(N-1)} l_B$. Here $r_{\max}$ is also the radius of the droplet, which is $\sqrt{3}$ times the radius $\sqrt{2 \times (N-1)} l_B$ of the $\nu = 1$ droplet. Thus, $\nu = \frac{1}{3}$ for Ψ_3 .

To understand why Ψ_3 has a uniform density, let us consider the following joint probability distribution of electron positions:

$$P(z_1 \cdots z_N) \propto |\Psi_m(z_1 \cdots z_N)|^2 = e^{-\beta V(z_1 \cdots z_N)} \quad (7.2.4)$$

where we have generalized the Ψ_3 state to the Ψ_m state, given by

$$\Psi_m = \prod_{i < j} (z_i - z_j)^m e^{-\sum |z_i|^2 / 4l_B^2}$$

We see that m must be an odd integer for Ψ_m to be totally anti-symmetric. Choosing $T = \frac{1}{\beta} = \frac{m}{2}$ in eqn (7.2.4), we can view

$$V = -m^2 \sum_{i < j} \ln |z_i - z_j| + \frac{m}{4l_B^2} \sum_i |z_i|^2$$

as the potential for a two-dimensional plasma of ‘charge’ m particles and $P(z_1 \cdots z_N)$ as the probability distribution of the plasma. The potential between two particles with ‘charges’ m_1 and m_2 is $-m_1 m_2 \ln(r)$, and the force is $m_1 m_2 / r$. Thus, to understand the density distribution of Ψ_m , we just need to calculate the ‘charge’ distribution of the plasma.

In V , we see that each particle in the plasma sees a potential $m |z|^2 / 4l_B^2$, which can be viewed as the potential produced by a background ‘charge’ distribution. In fact, such a potential is produced by a uniform background ‘charge’ with ‘charge’ density $\rho_\phi = 1/2\pi l_B^2$, which is just the density of the flux quanta. To see this, we note that the force of such a background ‘charge’ acting on a ‘charge’ m at radius r is a force between ‘charge’ m and the total ‘charge’ within a radius r . Such a force is given by $F = m \rho_\phi \pi r^2 / r = m \rho_\phi \pi r$. This gives us a potential $\frac{m}{2} \pi \rho_\phi r^2$, which is just the above potential. Due to the complete screening property of plasma, the plasma wants to be ‘charge’ neutral. The plasma ‘charge’ density is equal to the background ‘charge’ density $m n_e = \rho_\phi$, where n_e is the electron density. We find the filling fraction to be $\nu = n_e / \rho_\phi = \frac{1}{m}$.

Why is Ψ_m an incompressible state? Note that the total angular momentum of Ψ_m (the total power of z_i) is $M_0 = m \frac{N(N-1)}{2}$. Here Ψ_m is the state with the minimum total angular momentum, which has m th-order zeros between every pair of electrons. Compressing the droplet reduces the total angular momentum and creates lower-order zeros, which costs energy. We expect that the energy gap is of the order of the Coulomb energy e^2 / l_B . There is no mathematical proof of the above statement, but we have plenty of numerical evidence.

We would like to mention that, although Ψ_3 is only an approximate ground state for electrons with a Coulomb interaction, there exists an ideal Hamiltonian for which Ψ_3 is the exact ground state. The ideal Hamiltonian has the form

$$H = \frac{1}{2m} \sum_i (\partial_{x_i} - i q_e \mathbf{A})^2 - \sum_{i < j} UV(\mathbf{r}_i - \mathbf{r}_j), \quad V(\mathbf{r}) = \partial_{\mathbf{r}}^2 \delta(\mathbf{r}). \quad (7.2.5)$$

One can show that, for any state ψ , we have $\langle \psi | H | \psi \rangle \geq \frac{1}{2} N \hbar \omega_c$, and for Ψ_3 we have $\langle \Psi_3 | H | \Psi_3 \rangle = \frac{1}{2} N \hbar \omega_c$. Thus, Ψ_3 is indeed the exact ground state of H . Numerical calculations also indicate that Ψ_3 is incompressible for the ideal Hamiltonian.

Problem 7.2.3.

Show that $\langle \Psi_3 | H | \Psi_3 \rangle = \frac{1}{2} N \hbar \omega_c$ for the ideal Hamiltonian (7.2.5). Find an ideal Hamiltonian for the $\nu = 1/m$ Laughlin state Ψ_m .

Problem 7.2.4.

A double-layer FQH state is described by the wave function

$$\Psi_{lmn} = \prod_{i < j} (z_i - z_j)^l \prod_{i < j} (w_i - w_j)^m \prod_{i, j} (z_i - w_j)^n e^{-\sum |z_i|^2 / 4l_B^2} e^{-\sum |w_i|^2 / 4l_B^2} \quad (7.2.6)$$

where z_i are the coordinates of the electrons in the first layer and w_i in the second layer. Such a state is called the (lmn) state. Let N_1 and N_2 be the numbers of electrons in the two layers. Find the ratio N_1/N_2 such that the sizes of the electron droplets in the two layers are equal. Find the total filling fraction of the electrons in the two layers. (Hint: You may first assume that $n = 0$ to simplify the problem.)

7.2.3 Quasiparticles with fractional charge and fractional statistics

• Fractional charge and fractional statistics from the plasma analogue.

A quasi-hole excitation above the incompressible ground state Ψ_m is described by the many-body wave function

$$\Psi^h(\xi, \xi^*) = \sqrt{C(\xi, \xi^*)} \prod_i (\xi - z_i) \Psi_m,$$

where ξ is the position of the quasi-hole and $C(\xi, \xi^*)$ is the normalization coefficient. The simplest way to calculate the charge of the quasi-hole is to first remove an electron from the Ψ_m state and obtain the wave function $\prod_i (\xi - z_i)^m \Psi_m$, which has a charge e hole at ξ . However, $\prod_i (\xi - z_i)^m \Psi_m$ can also be viewed as the wave function with m quasi-holes at ξ . Thus, the quasi-hole charge is e/m .

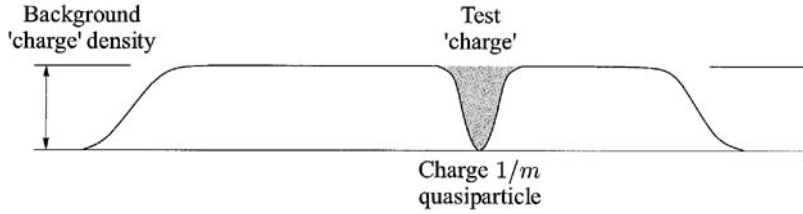


FIG. 7.13. The charge from the electrons and the test charge must neutralize the background charge.

A more direct calculation of the quasihole charge is through the plasma analogue. The electron position distribution in the quasihole state Ψ_h is given by

$$P_\xi^h(\{z_i\}) = |\Psi^h(\xi, \xi^*)|^2 = e^{-\beta[V(z_i) - m \ln|\xi - z_i|]}$$

Therefore, the electrons see an addition background potential $-m \ln|\xi - z_i|$. This potential is produced by a unit test charge. The background charge contains an extra unit test charge at ξ . There must be $1/m$ less electrons near ξ to maintain the charge neutrality (see Fig. 7.13). Thus, the quasihole carries a charge e/m .

To understand the fractional statistics of the quasiholes (Arovas *et al.*, 1984), let us start with the following effective Lagrangian for several quasiparticles:

$$Z = \int \mathcal{D}[\mathbf{x}_i(t)] e^{i \int dt \sum_i [-E_0 + \frac{1}{2} m \dot{\mathbf{x}}_i^2 + \mathbf{a}(\mathbf{x}_1, \mathbf{x}_2, \dots) \cdot \dot{\mathbf{x}}_i]}$$

The effective gauge potential $\mathbf{a}$ determines the statistics. Just like the spin systems discussed in Section 2.3, $\mathbf{a}$ is determined from the Berry phase of the coherent state $|\Psi^h(\xi(t), \xi^*(t))\rangle$.

Let us first consider only a single quasiparticle. For an adiabatic motion of the quasiparticle $\xi(t)$, the phase change of $\Psi^h(\xi(t), \xi^*(t))$ gives us the Berry phase. For a small Δt , the Berry phase is

$$\langle \Psi^h(\xi(t + \Delta t), \xi^*(t + \Delta t)) | \Psi^h(\xi(t), \xi^*(t)) \rangle = e^{i \Delta t \mathbf{a} \cdot \dot{\mathbf{x}}}.$$

Thus, $\mathbf{a}$ is given by

$$i \langle \Psi^h(\xi(t), \xi^*(t)) | \frac{d}{dt} | \Psi^h(\xi(t), \xi^*(t)) \rangle = \mathbf{a} \cdot \dot{\mathbf{x}}$$

As Ψ^h depends on t only through $\xi(t)$, we have

$$\begin{aligned} \mathbf{a} \cdot \dot{\mathbf{x}} &= i \langle \Psi^h(\xi(t), \xi^*(t)) | \frac{d}{dt} | \Psi^h(\xi(t), \xi^*(t)) \rangle \\ &= \frac{d\xi}{dt} i \langle \Psi^h(\xi, \xi^*) | \frac{\partial}{\partial \xi} | \Psi^h(\xi, \xi^*) \rangle + \frac{d\xi^*}{dt} i \langle \Psi^h(\xi, \xi^*) | \frac{\partial}{\partial \xi^*} | \Psi^h(\xi, \xi^*) \rangle \end{aligned}$$

Thus, if we write $\mathbf{a} \cdot d\mathbf{x} = a_\xi d\xi + a_{\xi^*} d\xi^*$, then⁴⁴

$$a_\xi = i \langle \Psi^h(\xi, \xi^*) | \frac{\partial}{\partial \xi} | \Psi^h(\xi, \xi^*) \rangle = -i\sqrt{C} \frac{\partial}{\partial \xi} \frac{1}{\sqrt{C}}$$

Here $C(\xi, \xi^*) = \langle \Psi(\xi) | \Psi(\xi) \rangle$ and $\Psi(\xi) = \prod (\xi - z_i) \Psi_m$ which is an analytic function of ξ . The key point of the above result is that the normalization of Ψ (an analytic function of ξ) gives us the Berry phase:

$$a_\xi = + \frac{i}{2} \frac{\partial}{\partial \xi} \ln C, \quad a_{\xi^*} = - \frac{i}{2} \frac{\partial}{\partial \xi^*} \ln C$$

The normalization $C(\xi, \xi^*)$ can be calculated from the plasma analogue. We note that, after including a term $e^{-|\xi|^2/4ml_B^2}$ representing the interaction between the test ‘charge’ and the background ‘charge’,⁴⁵

$$\int \prod_i d^2 z_i \left| e^{-\frac{1}{4ml_B^2}|\xi|^2} \prod (\xi - z_i) \prod (z_i - z_j)^m e^{-\frac{1}{4l_B^2}|z_i|^2} \right|^2 \propto e^{-\beta V(\xi, \xi^*)}$$

gives the total energy of the plasma with a test ‘charge’ at ξ . Again, due to the complete screening of the plasma, the total force acting on the inserted test ‘charge’ vanishes; hence the energy of plasma does not depend on ξ , i.e. $V(\xi, \xi^*) = \text{constant}$. We see that

$$C(\xi, \xi^*) \propto e^{\frac{1}{2l_B^2} \frac{1}{m} |\xi|^2}$$

This gives us

$$a_\xi = i \frac{1}{m} \frac{1}{4l_B^2} \xi^*, \quad a_{\xi^*} = -i \frac{1}{m} \frac{1}{4l_B^2} \xi$$

⁴⁴ We have used

$$\begin{aligned} i \langle \Psi^h(\xi, \xi^*) | \frac{\partial}{\partial \xi} | \Psi^h(\xi, \xi^*) \rangle &= i \langle \Psi(\xi) | \frac{1}{\sqrt{C(\xi, \xi^*)}} \frac{\partial}{\partial \xi} \frac{1}{\sqrt{C(\xi, \xi^*)}} | \Psi(\xi) \rangle \\ &= i \int \prod_i d^2 z_i \Psi^*(\xi^*) \frac{1}{\sqrt{C}} \frac{\partial}{\partial \xi} \left(\frac{1}{\sqrt{C}} \Psi(\xi) \right) \\ &= i \frac{\partial}{\partial \xi} \int \prod_i d^2 z_i \Psi^*(\xi^*) \frac{1}{C} \Psi(\xi) - i \int \prod_i d^2 z_i \Psi^*(\xi^*) \left(\frac{\partial}{\partial \xi} \frac{1}{\sqrt{C}} \right) \frac{1}{\sqrt{C}} \Psi(\xi) \\ &= -i\sqrt{C} \frac{\partial}{\partial \xi} \frac{1}{\sqrt{C}} \end{aligned}$$

⁴⁵ Note that $e^{-|\xi|^2/4l_B^2}$ corresponds to the interaction between an electron and the background ‘charge’. An electron corresponds to a bound state of m test ‘charges’.

The above vector potential $\mathbf{a}$ is proportional to the vector potential of the magnetic field:

$$a_\xi = -\frac{1}{m}q_e A_z, \quad a_{\xi^*} = -\frac{1}{m}q_e A_z^*$$

Around a loop, the quasihole picks up an A-B phase $\oint d\mathbf{x} \cdot \mathbf{a} = \left(-\frac{q_e}{m}\right) \oint d\mathbf{x} \cdot \mathbf{A}$, a phase for a charge $-q_e/m = e/m$ particle.

For two quasiholes, the total energy V of the plasma with two test 'charges' is given by

$$e^{-\beta V(\xi_1, \xi_1^*, \xi_2, \xi_2^*)} = \left| e^{-\frac{1}{4l_B^2} \frac{1}{m} (|\xi_1|^2 + |\xi_2|^2)} |\xi_1 - \xi_2|^{\frac{1}{m}} \prod_{a,i} (\xi_a - z_i) \prod (z_i - z_j)^m e^{-\frac{1}{4l_B^2} \sum |z_i|^2} \right|^2$$

where the extra term $|\xi_1 - \xi_2|^{2/m} = e^{-\beta(-)\ln|\xi_1 - \xi_2|}$ represents the direct interaction between the two test 'charges', each carries 'charge' 1. Again, $V(\xi_1, \xi_1^*, \xi_2, \xi_2^*)$ is independent of ξ_1 and ξ_2 . Thus,

$$|\Psi(\xi_1, \xi_2)|^2 \propto e^{\frac{1}{2l_B^2} \frac{1}{m} (|\xi_1|^2 + |\xi_2|^2)} |\xi_1 - \xi_2|^{-\frac{2}{m}}$$

Setting $\xi = \xi_1$ and $\xi_2 = 0$, we find that

$$a_\xi = i \frac{1}{m} \frac{1}{4l_B^2} \xi^* - \frac{i}{2m} \frac{\partial}{\partial \xi} \ln |\xi|^2 = i \frac{1}{m} \frac{1}{4l_B^2} \xi^* - \frac{i}{2m} \frac{1}{\xi}$$

$$a_{\xi^*} = -i \frac{1}{m} \frac{1}{4l_B^2} \xi + \frac{i}{2m} \frac{1}{\xi^*}$$

which gives us $(a_x, a_y) = (-y, x) \frac{1}{r^2} \frac{1}{m} + \dots$. As we move the quasihole ξ_1 around ξ_2 , the quasihole obtains a phase given by $\frac{e}{m} B \times \text{area} + \frac{2\pi}{m}$. The first term is due to the uniform magnetic field. The additional term $\frac{2\pi}{m}$ comes from $(-y, x) \frac{1}{r} \frac{1}{m}$. Such a term gives the quasiholes fractional statistics with a statistical angle $\theta = \frac{\pi}{m}$ (see eqn (7.1.1)).

Problem 7.2.5.

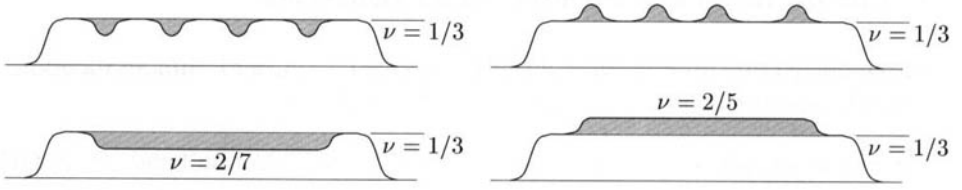
Consider the double-layer ($l m n$) state eqn (7.2.6). A quasihole in the first layer is described by the wave function

$$\Psi_1^h(\xi_1) = \prod_i (\xi_1 - z_i) \Psi_{l m n}(\{z_i, w_j\})$$

and a quasihole in the second layer by

$$\Psi_2^h(\xi_2) = \prod_i (\xi_2 - w_i) \Psi_{l m n}(\{z_i, w_j\})$$

Find the fractional charge and the fractional statistics of the quasiholes in the first and second layers. Find the mutual statistics between a quasihole in the first layer and a quasihole

FIG. 7.14. The hierarchical $\nu = 2/7$ and $\nu = 2/5$ states.

in the second layer. The mutual statistics between two non-identical particles are defined by the Berry phase generated by moving one particle all the way around the other.

7.2.4 Hierarchical fractional quantum Hall states—generalization of Laughlin theory

- Hierarchical FQH states are equivalent to Laughlin states for quasiparticles.

To construct more general FQH states at filling fractions other than $1/m$, we may start with a $\nu = 1/m$ Laughlin state. We add quasiholes or quasiparticles to change the average electron density. When its density reaches a certain value, the gas of quasiholes/quasiparticles can also form a Laughlin state by itself. The resulting state is a hierarchical FQH state (see Fig. 7.14).

When the charge $e/3$ quasiholes on top of the $\nu = 1/3$ state condense into a Laughlin state, we obtain the $\nu = 2/7$ FQH state. The wave function has the form

$$\begin{aligned} \Psi(z_1 \cdots z_N) \\ = \int \prod_i d^2 \xi_i \prod (z_i - z_j)^3 e^{-\frac{1}{4B} \sum |z_i|^2} \prod (\xi_i - z_j)(\xi_i^* - \xi_j^*)^2 e^{-\frac{1}{4B} \sum |\xi_i|^2} \end{aligned}$$

When the charge $e/3$ quasiparticles on top of the $\nu = 1/3$ state condense into a Laughlin state, we get the $\nu = 2/5$ FQH state. The wave function is given by

$$\Psi_c(z_i - z_N) = \int \prod_i d^2 \xi_i \prod_{i < j} (\xi_i - \xi_j)^2 (\xi_i^* - 2\partial_{z_i}) \prod (z_i - z_j)^3 e^{-\frac{1}{4B} \sum |z_i|^2}$$

We can calculate the properties of the above two states using the plasma analogue. However, the calculation is complicated. In the next section, we are going to derive the effective field theory for FQH states. We will use the effective theory to calculate the properties of the above hierarchical FQH states. The effective field theory greatly simplifies the calculation. We can easily obtain the filling fractions of the FQH states and the quantum numbers of the quasiparticles.

7.3 Effective theory of fractional quantum Hall liquids

- Low-energy effective theories of FQH states are $U(1)$ Chern–Simons theories, which capture the universal properties of the FQH states.

We have already seen several different FQH states. On one hand, these QH states have different fractional charges and different fractional statistics. This suggests that the different FQH states belong to different quantum phases. On the other hand, these FQH states all have the same symmetry. We cannot use broken symmetry to distinguish them. In fact, FQH states contain a new kind of order—topological order (see Chapter 8). We need to use new tools to characterize topological orders.

One way to systematically study topological orders is to construct a low-energy effective theory for FQH liquids. The effective theory can capture the universal properties of FQH liquids and provide hints on how to characterize and label different topological orders in FQH liquids. In the following, we will introduce a way to construct effective theories that is closely related to the hierarchical construction proposed by Haldane and Halperin (Haldane, 1983; Halperin, 1984).

The first attempt to characterize the internal structures of FQH liquids was proposed by Girvin and MacDonald (1987), where it was shown that a Laughlin state contains an off-diagonal long-range order in a *non-local* operator. Such an observation leads to a description of FQH liquids in terms of Ginzburg–Landau–Chern–Simons effective theory (Read, 1989; Zhang *et al.*, 1989). These developments have led to many interesting and important results, and a deeper understanding of FQH liquids. However, here I will try to describe the internal structures of FQH liquids from a more general point of view. It appears that some internal structures of FQH liquids (especially those in the so-called non-abelian FQH liquids) cannot be described by the Ginzburg–Landau–Chern–Simons effective theory and the associated off-diagonal long-range orders. Thus, we need to develop more general concepts and formulations, such as topological orders and topological field theory, to describe the internal structures in FQH liquids (see Chapter 8). The concept of off-diagonal long-range order, in my opinion, does not capture the essence of the internal structure in FQH liquids. Here we will develop an effective theory without using off-diagonal long-range order (Blok and Wen, 1990a,b). The resulting effective theory contains only pure Chern–Simons terms. The pure Chern–Simons form is more compact and more clearly reveals the internal structure of the hierarchical FQH states.

7.3.1 Effective theory of the Laughlin states

- The hydrodynamics of FQH liquids and effective Chern–Simons theory.

In this section, we will consider only single-layer spin-polarized QH systems. To construct the effective theory for the hierarchical states, we will first try to obtain the effective theory of the Laughlin state. Then we will use the hierarchical construction to obtain the effective theory for the hierarchical states in the next section.

Consider an electron system in a magnetic field. To include more general situations, we will assume that the electrons can have bosonic or fermionic statistics. The Lagrangian of the system has the following form (the first quantized version):

$$\begin{aligned}\mathcal{L} &= -eA^i J^i + \text{kinetic/potential energy} \\ &= eA_i J^i + \text{kinetic/potential energy}\end{aligned}\quad (7.3.1)$$

where

$$J(x) = \sum_i v_i \delta(x - x_i), \quad J^0(x) = \sum_i \delta(x - x_i) \quad (7.3.2)$$

are the current and the density of the electrons, respectively, and (x_i, v_i) are the position and the velocity of the i th electron. The kinetic/potential energy is given by $\sum_i \frac{1}{2} m v_i^2 + \sum_{i < j} V(x_i - x_j)$, whose precise form is not important to us. We have also assumed that the charge of each electron is $-e$ and the speed of light is $c = 1$.

In a hydrodynamical approach, we assume that the low-energy collective modes can be described by the density and the current J^μ , and the low-energy effective theory has the form $\mathcal{L}(A_\mu, J^\nu) = eA_i J^i + \mathcal{L}'(J^\mu)$. From our discussion in Sections 5.1.3 and 5.3.3, we see that sometimes a state may have too many low-lying excitations to be described by single density mode. As the gapped FQH states have even fewer low-energy excitations than a superfluid, it is reasonable to assume that a single density mode is enough to describe the low-energy fluctuations in the FQH states.

At a filling fraction of $\nu = 1/m$, where m is an even integer for a bosonic electron and an odd integer for a fermionic electron, the ground state of the electron system is given by the Laughlin wave function (Laughlin, 1983)

$$\left[\prod (z_i - z_j)^m \right] e^{-\frac{1}{4l_B^2} \sum |z_i|^2} \quad (7.3.3)$$

(Here we have assumed that $B < 0$, so that $(-e)B > 0$.) To construct the effective theory, we note that the state (7.3.3) is an incompressible fluid, so that its density is tied to the magnetic field, i.e. $J^0 = (-e)\nu B/2\pi$. When combined with the finite Hall conductance $\sigma_{xy} = \frac{\nu e^2}{h} = \frac{\nu e^2}{2\pi\hbar} = \frac{\nu e^2}{2\pi}$, we find that the electron number current J^μ has the following response to a change of electromagnetic field:

$$-e\delta J^\mu = -\sigma_{xy}\varepsilon^{\mu\nu\lambda}\partial_\nu\delta A_\lambda = -\frac{\nu e^2}{2\pi}\varepsilon^{\mu\nu\lambda}\partial_\nu\delta A_\lambda \quad (7.3.4)$$

We choose the effective Lagrangian $\mathcal{L}(A_\mu, J^\nu)$ in such a way that it produces the equation of motion (7.3.4). It is convenient to introduce a $U(1)$ gauge field a_μ to

describe the electron number current:

$$J^\mu = \frac{1}{2\pi} \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda}$$

The current defined in this way automatically satisfies the conservation law. Then the effective Lagrangian that produces eqn (7.3.4) takes the following form:

$$\mathcal{L} = \left[-m \frac{1}{4\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} + \frac{e}{2\pi} A_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} \right] \quad (7.3.5)$$

Equation (7.3.5) describes only the linear response of the ground state to the external electromagnetic fields. To have a more complete description of the topological fluid, such as the FQH liquid, we need to introduce the electron excitations into our effective theory. In particular, we want to make sure that the effective theory contains an excitation that carries the same quantum numbers as the electron.

7.3.2 Electron and quasiparticle excitations in the effective theory

- Effective Chern–Simons theory alone is not a complete description of the FQH state. We need to specify the electron operator to have a complete effective theory.
- The required trivial mutual statistics between the electrons and the quasiparticle/quasihole determine the quantum numbers of the quasiparticles and quasiholes.

Let us introduce a particle that carries an a_μ charge of l . In the effective theory (7.3.5), such a particle corresponds to the following source term:

$$l a_0 \delta(\mathbf{x} - \mathbf{x}_0) \quad (7.3.6)$$

The source term will create an excitation of charge

$$Q = -le/m. \quad (7.3.7)$$

This can be seen from the equation of motion $\delta\mathcal{L}/\delta a_0 = 0$, with

$$J_0 = \frac{1}{2\pi} \varepsilon_{ij} \partial_i a_j = -\frac{e}{2\pi m} B + \frac{l}{m} \delta(\mathbf{x} - \mathbf{x}_0)$$

The first term on the right-hand side indicates that the filling fraction $\nu \equiv 2\pi \frac{J_0}{-eB}$ is indeed $\nu = 1/m$, and the second term corresponds to the increase in the electron density associated with the excitation.

We also see that the excitation created by the source term (7.3.6) is associated with l/m units of the a_μ flux. Thus, if we have two excitations carrying a_μ charges

of l_1 and l_2 , then moving one excitation around the other will induce a phase $2\pi \times (\text{number of } a_\mu\text{-flux quanta}) \times (a_\mu \text{ charge})$,⁴⁶ namely

$$2\pi \times \frac{l_1}{m} \times l_2 \quad (7.3.8)$$

If $l_1 = l_2 \equiv l$, then the two excitations will be identical. Interchanging them will induce half of the phase in eqn (7.3.8), namely

$$\theta = \pi \frac{l^2}{m} \quad (7.3.9)$$

Here θ is simply the statistical angle of the excitation that carries l units of the a_μ charge.

Our electrons carry a charge of $-e$. From the above discussion, we see that a charge $-e$ excitation corresponds to a particle that carries m units of the a_μ charge. Such an excitation has a statistical angle $\theta = \pi m$ (see eqn (7.3.9)); thus, it is a boson if m is even and a fermion if m is odd. Therefore, for the case of a bosonic electron with even m and the case of a fermionic electron with odd m , we can identify the excitations of m units of the a_μ charge as the electrons that form the FQH liquid. The existence of the excitations with the electric charge and the statistics of the electron support the correctness of our Chern–Simons effective theory.

We would like to stress that the identification of the fundamental electrons in the effective theory is very important. It is this identification, together with the effective Lagrangian, that provides a complete description of the topological properties of the FQH liquid. We will see below that this identification allows us to determine the fractional charge and the fractional statistics of the quasiparticle excitations.

A quasihole excitation at a position described by a complex number $\xi = x + iy$ is created by multiplying $\prod_i (\xi - z_i)$ with the ground-state wave function (7.3.3). Note that the phase of the wave function changes by 2π as an electron goes around the quasihole. The 2π phase implies that the electron and the quasihole have trivial mutual statistics. In general, the electron and an allowed excitation must have trivial mutual statistics in order for the excitation to have a single-valued electron wave function.

Now let us try to create an excitation by inserting a source term of l units of the a_μ charge. Moving an electron around such an excitation will induce a phase $2\pi l$

⁴⁶ The careful reader may note that the two excitations carry both an a_μ charge and an a_μ flux. So the phase induced by moving one excitation around the other should receive two contributions, one from moving the charge around the flux and the other from moving the flux around the charge, as described in Fig. 7.8. This will lead to a phase $2\pi \times \frac{l_1}{m} \times l_2 + 2\pi \times \frac{l_2}{m} \times l_1$, which is twice as large as the one in eqn (7.3.8). However, a more careful calculation indicates that the picture of Fig. 7.8 does not apply to the charge–flux bound state induced by the Chern–Simons term. The naive result in eqn (7.3.8) happens to be the correct one (see Problem 7.3.1).

(see eqn (7.3.8)). The single-value property of the electron wave function requires such a phase to be a multiple of 2π . So the a_μ charges must be quantized as integers, and only those charges correspond to allowed excitations.

From the charge of the excitations given in eqn (7.3.7), we find that $l = -1$ corresponds to the fundamental quasihole excitation, while $l = 1$ corresponds to the fundamental quasiparticle excitation. The quasiparticle excitation carries an electric charge $-e/m$ and the quasihole carries an electric charge e/m . Both of them have statistics $\theta = \pi/m$, as one can see from eqn (7.3.9). We see that the effective theory reproduces the well-known results for quasiparticles in Laughlin states (Arovas *et al.*, 1984). The full effective theory with quasiparticle excitations is given by

$$\mathcal{L} = \left[-m \frac{1}{4\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} + \frac{e}{2\pi} A_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} \right] + l a_\mu j^\mu + \text{kinetic/potential energy} \quad (7.3.10)$$

where j^μ is the current of the quasiparticles, which has the form given in eqn (7.3.2). Equation (7.3.10), together with the quantization condition on l , is a complete low-energy effective theory which captures the topological properties of the $1/m$ Laughlin state.

Problem 7.3.1.

Prove eqn (7.3.8). (Hint: You may start with the following effective theory with two excitations:

$$\mathcal{L} = \left[-\frac{1}{2} \frac{m}{2\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} + j^\mu a_\mu \right]$$

where $j^\mu = j_1^\mu + j_2^\mu$ and

$$\begin{aligned} j_1^0(\mathbf{x}, t) &= l_1 \delta(\mathbf{x} - \mathbf{x}_1(t)), & j_2^0(\mathbf{x}, t) &= l_2 \delta(\mathbf{x} - \mathbf{x}_2(t)), \\ j_1^i(\mathbf{x}, t) &= l_1 \dot{x}_1^i \delta(\mathbf{x} - \mathbf{x}_1(t)), & j_2^i(\mathbf{x}, t) &= l_2 \dot{x}_2^i \delta(\mathbf{x} - \mathbf{x}_2(t)). \end{aligned}$$

Here j^μ is the total current of the two excitations and $\mathbf{x}_{1,2}(t)$ are the locations of the two excitations. You can then integrate out a_μ and obtain

$$\int d^3x \frac{1}{2} j^\mu \frac{1}{\frac{m}{2\pi} \partial_\lambda \varepsilon^{\mu\lambda\nu}} j_\nu$$

The cross-term can be written as

$$\int d^3x j_1^\mu \frac{1}{\frac{m}{2\pi} \partial_\lambda \varepsilon^{\mu\lambda\nu}} j_2^\nu = \int d^3x j_1^\mu f_\mu$$

where f_μ satisfies

$$\frac{m}{2\pi} \partial_\lambda \varepsilon^{\mu\lambda\nu} f_\nu = j_2^\mu$$

You can find f_i by assuming that $\mathbf{x}_2 = \dot{\mathbf{x}}_2 = 0$.)

Problem 7.3.2.

We have been concentrating on the topological properties of the Laughlin states. To have some understanding of the dynamical properties of the Laughlin states, we need to include the Maxwell term in the effective Chern–Simons theory as follows:

$$\mathcal{L} = -m \frac{1}{4\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} + \frac{1}{2g_1} e^2 - \frac{1}{2g_2} b^2 \quad (7.3.11)$$

where e and b are the electric field and the magnetic field, respectively, of a_μ , and $e_i = \dot{a}_i - \partial_i a^0$ and $b = \partial_1 a_2 - \partial_2 a_1$. Find the equation of motion for the collective fluctuations described by e and b . Find the energy gap of the collective excitations. Assume that the energy gap in the $1/m$ Laughlin state is of order $e^2/m^2 l_B$ and the size of the quasihole is of order l_B . Estimate the values of g_1 and g_2 .

7.3.3 Effective theory of the hierarchical fractional quantum Hall states

- Different hierarchical FQH states can be characterized by an integer K -matrix and a charge vector q .
- The topological properties of the FQH state, such as filling fractions and quasiparticle quantum numbers, can be calculated easily from the K -matrix and the charge vector.
- The equivalence relations between different pairs of K -matrix and charge vector.

To obtain an effective theory of the hierarchical FQH states, let us start with a $1/m$ Laughlin state formed by fermionic electrons. Let us increase the filling fraction by creating the fundamental quasiparticles, which are labeled by $l = 1$. Equation (7.3.10) with $l = 1$ describes the $1/m$ state in the presence of these quasiparticles. Now, two equivalent pictures emerge.

(a) In a mean-field theory approach, we may view the gauge field a_μ in eqn (7.3.10) as a fixed background and we do not allow a_μ to respond to the inserted source term j^μ . In this case, the quasiparticle gas behaves like bosons in the ‘magnetic’ field $b = \partial_i a_j \varepsilon_{ij}$, as one can see from the second term in eqn (7.3.10). These bosons do not carry any electric charge because the quasiparticle number current j^μ does not directly couple with the electromagnetic gauge potential A_μ . When the boson density satisfies

$$j^0 = \frac{1}{p_2} \frac{b}{2\pi}$$

where p_2 is even, the bosons have a filling fraction $\frac{1}{p_2}$. The ground state of the bosons can again be described by a Laughlin state. The final electronic state that we obtained is just the second-level hierarchical FQH state constructed by Haldane (1983).

(b) If we let a_μ respond to the insertion of j^μ , then quasiparticles will be dressed

by the a_μ flux. The dressed quasiparticles carry an electric charge of $-e/m$ and statistics of $\theta = \pi/m$. When the quasiparticles have the density

$$j^0 = \frac{1}{(p_2 - \frac{\theta}{\pi})} \frac{(-e)B}{2\pi m}$$

where p_2 is even, the quasiparticle will have a filling fraction $\frac{1}{(p_2 - \frac{\theta}{\pi})}$. In this case, the quasiparticle system can form a Laughlin state described by the wave function

$$\prod_{i < j} (z_i - z_j)^{p_2 - \frac{\theta}{\pi}}$$

The final electronic state obtained this way is again a second-level hierarchical FQH state. This construction was first proposed by Halperin (1984). The two constructions in (a) and (b) lead to the same hierarchical state and are equivalent.

In the following, we will follow Haldane's hierarchical construction to derive the Chern–Simons effective theory of hierarchical FQH states (Blok and Wen, 1990a,b; Read, 1990; Fröhlich and Kerler, 1991; Wen and Zee, 1992a). Notice that, under the assumption (a), the boson Lagrangian (the second term in eqn (7.3.10) with $l = 1$) is just eqn (7.3.1) with an external electromagnetic field eA_μ replaced by a_μ . Thus, we can follow the same steps as from eqn (7.3.3) to eqn (7.3.10) to construct the effective theory of the boson Laughlin state. Introducing a new $U(1)$ gauge field $\tilde{a}_\mu$ to describe the boson current, we find that the boson effective theory takes the form

$$\mathcal{L} = -\frac{p_2}{4\pi} \tilde{a}_\mu \partial_\nu \tilde{a}_\lambda \varepsilon^{\mu\nu\lambda} + \frac{1}{2\pi} a_\mu \partial_\nu \tilde{a}_\lambda \varepsilon^{\mu\nu\lambda} \quad (7.3.12)$$

In eqn (7.3.12), the new gauge field $\tilde{a}_\mu$ describes the density j^0 and the current j^i of the bosons, and is given by

$$j^\mu = \frac{1}{2\pi} \partial_\nu \tilde{a}_\lambda \varepsilon^{\mu\nu\lambda}$$

This reduces the coupling between a_μ and the boson current, $a_\mu j^\mu$, to a Chern–Simons term between a_μ and $\tilde{a}_\mu$ (which becomes the second term in eqn (7.3.12)). The total effective theory (including the original electron condensate) has the form

$$\begin{aligned} \mathcal{L} = & \left[-\frac{p_1}{4\pi} a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} + \frac{e}{2\pi} A_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} \right] \\ & + \left[-\frac{p_2}{4\pi} \tilde{a}_\mu \partial_\nu \tilde{a}_\lambda \varepsilon^{\mu\nu\lambda} + \frac{1}{2\pi} a_\mu \partial_\nu \tilde{a}_\lambda \varepsilon^{\mu\nu\lambda} \right] \end{aligned} \quad (7.3.13)$$

where $p_1 = m$ is an odd integer. Equation (7.3.13) is the effective theory of a second-level hierarchical FQH state.

The effective theory can be used to determine the physical properties of the hierarchical FQH state. The total filling fraction is determined from the equation of motion $\delta\mathcal{L}/\delta a_0 = \delta\mathcal{L}/\delta\tilde{a}_0 = 0$, with

$$-eB = p_1 b - \tilde{b}, \quad b = p_2 \tilde{b}$$

We find that

$$\nu = \frac{b}{-eB} = \frac{1}{p_1 - \frac{1}{p_2}}. \quad (7.3.14)$$

Equation (7.3.13) can be written in the following more compact form by introducing $(a_{1\mu}, a_{2\mu}) = (a_\mu, \tilde{a}_\mu)$:

$$\mathcal{L} = -\frac{1}{4\pi} K_{IJ} a_{I\mu} \partial_\nu a_{J\lambda} \epsilon^{\mu\nu\lambda} + \frac{e}{2\pi} q_I A_\mu \partial_\nu a_{I\lambda} \epsilon^{\mu\nu\lambda} \quad (7.3.15)$$

where K is the integer matrix

$$K = \begin{pmatrix} p_1 & -1 \\ -1 & p_2 \end{pmatrix}$$

and $\mathbf{q}$ is the integer vector $\mathbf{q}^\top = (q_1, q_2) = (1, 0)$. Here $\mathbf{q}$ will be called the charge vector. The filling fraction (7.3.14) can be rewritten as $\nu = \mathbf{q}^\top K^{-1} \mathbf{q}$. When $(p_1, p_2) = (3, 2)$, the hierarchical state corresponds to the $\nu = 2/5$ FQH state observed in experiments.

The second-level hierarchical FQH state contains two kinds of quasiparticle. One is the quasihole (or vortex) in the original electron condensate, and the other is the quasihole (or vortex) in the new boson condensate. The two kinds of quasiholes are created by inserting the source terms $-j^\mu a_\mu$ and $-\tilde{j}^\mu \tilde{a}_\mu$, respectively, where $\tilde{j}^\mu$ and j^μ have a similar form to eqn (7.3.2). The first kind of quasihole is created by multiplying $\prod_i (\xi - z_i)$ with the electron wave function, while the second kind is created by multiplying $\prod_i (\eta - \xi_i)$ with the boson Laughlin wave function (here ξ_i are the complex coordinates of the bosons and η is the position of the quasihole).

A generic quasiparticle consists of a number l_1 of quasiparticles of the first kind and a number l_2 of quasiparticles of the second kind, and is labeled by two integers. Such a quasiparticle carries l_1 units of the $a_{1\mu}$ charge and l_2 units of the $a_{2\mu}$ charge, and is described by

$$(l_1 a_{1\mu} + l_2 a_{2\mu}) j^\mu \quad (7.3.16)$$

After integrating out the gauge fields, we find that such a quasiparticle carries $\sum_J K_{IJ} l_J$ units of the $a_{I\mu}$ flux. Hence the statistics of such a quasiparticle are

given by

$$\theta = \pi \mathbf{l}^\top K^{-1} \mathbf{l} = \frac{1}{p_2 p_1 - 1} (p_2 l_1^2 + p_1 l_2^2 + 2 l_1 l_2)$$

and the electric charge of the quasiparticle is

$$Q_q = -e \mathbf{q}^\top K^{-1} \mathbf{l} = -e \frac{p_2 l_1 + l_2}{p_2 p_1 - 1}$$

For the $\nu = 2/5$ state (i.e. $(p_1, p_2) = (3, 2)$), the quasiparticle with minimal electric charge is labeled by $(l_1, l_2) = (0, 1)$. The minimal electric charge is $-e/5$. Such a quasiparticle has statistics $\theta = \frac{3}{5}\pi$.

We can also construct more general FQH states. The effective theory for these FQH states still has the form given in eqn (7.3.15), but now I runs from 1 to an integer n (n will be called the level of the FQH state). To obtain the form of the matrix K , let us assume that at the level $n - 1$ the effective theory is given by eqn (7.3.15) with $a_{I\mu}$, $I = 1, \dots, n - 1$, and $K = K^{(n-1)}$. The quasiparticles carry integer charges of the $a_{I\mu}$ gauge fields. Now consider an n th-level hierarchical state which is obtained by the ‘condensation’ of quasiparticles with the $a_{I\mu}$ charge $l_I|_{I=1, \dots, n-1}$. The effective theory of this n th-level hierarchical state will be given by eqn (7.3.15) with n gauge fields. The n th gauge field $a_{n\mu}$ comes from the new condensate. The matrix K is given by

$$K^{(n)} = \begin{pmatrix} K^{(n-1)} & -\mathbf{l} \\ -\mathbf{l}^\top & p_n \end{pmatrix}$$

with $p_n = \text{even}$. The charge vector $\mathbf{q}$ is still given by $(1, 0, 0, \dots)$. By iteration, we see that the generalized hierarchical states are always described by integer symmetric matrices, with $K_{II} = \text{even}$, except for $K_{11} = \text{odd}$. The new condensate gives rise to a new kind of quasiparticle, which again carries an integer charge of the new gauge field $a_{n\mu}$. Hence, a generic quasiparticle always carries integral charges of the $a_{I\mu}$ field.

Let us summarize the above results in more general terms. We know that a hierarchical (or generalized hierarchical) FQH state contains many different condensates. The different condensates are not independent. The particles in one condensate behave like a flux tube to the particles in other condensates. To describe such a coupling, it is convenient to use $U(1)$ gauge fields to describe the density and the current $J_{I\mu}$ of the I th condensate

$$J_I^\mu = \frac{1}{2\pi} \varepsilon^{\mu\alpha\beta} \partial_\alpha a_{I\beta} \quad (7.3.17)$$

In this case, the couplings between different condensates are described by the Chern–Simons term of the gauge fields as follows:

$$\mathcal{L} = -\frac{1}{4\pi} K_{IJ} a_{I\mu} \partial_\nu a_{J\lambda} \varepsilon^{\mu\nu\lambda} + \frac{e}{2\pi} q_I A_\mu \partial_\nu a_{I\lambda} \varepsilon^{\mu\nu\lambda} \quad (7.3.18)$$

When K is taken to be a general integral $n \times n$ matrix with $K_{II}|_{I=1} = \text{odd}$, $K_{II}|_{I>1} = \text{even}$, and $\mathbf{q}^\top = (1, 0, \dots, 0)$. Equation (7.3.18) describes the most general (abelian) FQH states (Wen and Zee, 1992a). The filling fraction is given by

$$\nu = \mathbf{q}^\top K^{-1} \mathbf{q} \quad (7.3.19)$$

The quasiparticle excitations can be viewed as vortices in different condensates. A generic quasiparticle is labeled by n integers $l_I|_I = 1, \dots, n$, and can be generated by the following source term:

$$\mathcal{L} = l_I a_{I\mu} j^\mu \quad (7.3.20)$$

Such a quasiparticle will be denoted by ψ_l . The j^μ in eqn (7.3.20) has the form

$$\begin{aligned} j(x) &= \dot{x}_0 \delta(x - x_0) \\ j^0(x) &= \delta(x - x_0) \end{aligned} \quad (7.3.21)$$

which creates a quasiparticle at x_0 .

As we create a quasiparticle ψ_l , it will induce a change in the densities of all of the condensates, namely δJ_I^0 . From the equation of motion of eqn (7.3.18) and eqn (7.3.20), we find that δJ_I^0 satisfies

$$\int d^2x \delta J_I^0 = l_J (K^{-1})_{JI} \int d^2x j^0 = (l^\top K^{-1})_I \quad (7.3.22)$$

The charge and the statistics of the quasiparticle ψ_l are given by

$$\theta_l = \pi l^\top K^{-1} l, \quad Q_l = -e q_I \int d^2x \delta J_I^0 = -e l^\top K^{-1} \mathbf{q} \quad (7.3.23)$$

The result of quasiparticle statistics is obtained by noting that $2\pi\delta J_I^0$ is the flux of the $a_{I\mu}$ and the quasiparticle carries l_I units of $a_{I\mu}$ charge.

A generic electron excitation can be viewed as a special kind of (generic) quasiparticle, $\psi_e = \psi_{l_e}$, where the integer vector l_e is given by

$$l_{eI} = K_{IJ} L_J, \quad L_I = \text{integers}, \quad q_I L_I = 1 \quad (7.3.24)$$

We can show that these electron excitations satisfy the following properties: they carry a unit charge (see eqn (7.3.23)); they have fermionic statistics; moving an electron excitation $\psi_e = \psi_{l_e}$ around any quasiparticle excitations ψ_l always induces a phase of a multiple of 2π ; and the excitations defined in eqn (7.3.24) are all of the excitations satisfying the above three conditions.

From the generic effective theory in eqn (7.3.15), we find that the generalized hierarchical states can be labeled by an integer-valued K -matrix and a charge vector $\mathbf{q}$. Now we would like to ask the following question: do different $(K, \mathbf{q})$ s

describe different FQH states? Notice that, through a redefinition of the gauge fields $a_{I\mu}$, one can always diagonalize the K -matrix into one with ± 1 as the diagonal elements. Thus, it seems that all K -matrices with the same signature describe the same FQH states, because they lead to the same effective theory after a proper redefinition of the gauge fields. Certainly this conclusion is incorrect. We would like to stress that the effective Lagrangian (7.3.15) alone does not provide a proper description of the internal orders (or topological orders) in the hierarchical states. It is the effective Lagrangian (7.3.15) together with the quantization condition of the $a_{I\mu}$ charges that characterize the topological order. A $U(1)$ gauge theory equipped with a quantization condition on the allowed $U(1)$ charges is called a compact $U(1)$ theory. Our effective theory (7.3.15) is actually a compact $U(1)$ theory with all $U(1)$ charges quantized as integers. Thus, the allowed $U(1)$ charges form an n -dimensional cubic lattice, which will be called the charge lattice. Therefore, when one considers the equivalence of two different K -matrices, one can only use the field redefinitions that keep the charge quantization condition unchanged (i.e. keep the charge lattice unchanged). The transformations that map the charge lattice onto itself belong to the group $SL(n, Z)$ (a group of integer matrices with unit determinant):

$$a_{I\mu} \rightarrow W_{IJ} a_{J\mu}, \quad W \in SL(n, Z) \quad (7.3.25)$$

From the above discussion, we see that the two FQH states described by $(K_1, \mathbf{q}_1)$ and $(K_2, \mathbf{q}_2)$ are equivalent (i.e. they belong to the same universality class) if there exists a $W \in SL(n, Z)$ such that

$$\mathbf{q}_2 = W \mathbf{q}_1, \quad K_2 = W K_1 W^T \quad (7.3.26)$$

This is because, under the transformation (7.3.25), an effective theory described by $(K_1, \mathbf{q}_1)$ simply changes into another effective theory described by $(K_2, \mathbf{q}_2)$.

We would like to remark that in the above discussion we have ignored another topological quantum number—the spin vector. Because of this, the equivalence condition in eqn (7.3.26) does not apply to rotationally-invariant systems. However, eqn (7.3.26) does apply to disordered FQH systems because the angular momentum is not conserved in disordered systems and the spin vector is not well defined. A discussion of the spin vectors can be found in Wen and Zee (1992c,d). Table 7.1 lists the K -matrix, the charge vector $\mathbf{q}$, and the spin vector $\mathbf{s}$ for some common single-layer spin-polarized FQH states.

Problem 7.3.3.

Prove the four properties listed below eqn (7.3.24).

7.3.4 Effective theory of simple multi-layer fractional quantum Hall states

- The K -matrix and multi-layer FQH wave functions.

TABLE 7.1. Expressions for $(K, \mathbf{q}, \mathbf{s})$ for some simple single-layer spin-polarized FQH states.

ν	$\mathbf{q}$	K	$\mathbf{s}$
$1/m$	(1)	(m)	($m/2$)
$1 - 1/m$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 1 & 1 \\ 1 & -(m-1) \end{pmatrix}$	$\begin{pmatrix} 1/2 \\ (1-m)/2 \end{pmatrix}$
$2/5$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix}$	$\begin{pmatrix} 1/2 \\ 1 \end{pmatrix}$
$3/7$	$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$	$\begin{pmatrix} 1/2 \\ 1 \\ 1 \end{pmatrix}$

So far, we only considered the so called single-layer FQH states where the 2D electron gas lives on an interface of two different semiconductors. We can also make samples with multiple layers of interfaces. The 2D electron gases on different layers can form a multi-layer FQH state when placed under a strong magnetic field.

The same approach used to construct the effective theory of the hierarchical states can also be used to construct the effective theory for the multi-layer FQH states. The connection between the FQH wave function and the K -matrix becomes very transparent for the multi-layer FQH states. In this section, we will concentrate on double-layer FQH states. However, the generalization to the n -layer FQH states is straightforward.

We would like to construct an effective theory for the following simple double-layer FQH state:

$$\prod_{i < j} (z_{1i} - z_{1j})^l \prod_{i < j} (z_{2i} - z_{2j})^m \prod_{i, j} (z_{1i} - z_{2j})^n e^{-\frac{1}{4l_B^2} (\sum_i |z_{1i}|^2 + \sum_j |z_{2j}|^2)}, \quad (7.3.27)$$

where z_{Ii} is the complex coordinate of the i th electron in the I th layer. Here l and m are odd integers, so that the wave function is consistent with the Fermi statistics of the electrons, while n can be any non-negative integer. The above wave function was first suggested by Halperin as a generalization of the Laughlin wave function (Halperin, 1983). It appears that these wave functions can explain some of the dominant FQH filling fractions observed in double-layer FQH systems.

We start with a single-layer FQH state in the first layer, namely $\prod_{i < j} (z_{1i} - z_{1j})^l e^{-\sum_i |z_{1i}|^2/4}$, which is a $1/l$ Laughlin state and is described by the effective theory

$$\mathcal{L} = \left[-l \frac{1}{4\pi} a_{1\mu} \partial_\nu a_{1\lambda} \varepsilon^{\mu\nu\lambda} + \frac{e}{2\pi} A_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} \right] \quad (7.3.28)$$

where $a_{1\mu}$ is the gauge field that describes the electron density and current in the first layer.

Examining the wave function in eqn (7.3.27), we see that an electron in the second layer is bounded to a quasi-hole excitation in the first layer. Such a quasi-hole excitation is formed by n fundamental quasi-hole excitations and carries an $a_{1\mu}$ charge of $-n$. A gas of the quasi-holes is described by the following effective theory:

$$\mathcal{L} = -na_{1\mu}j^\mu + \text{kinetic/potential energy} \quad (7.3.29)$$

where j^μ has the form given in eqn (7.3.2). As we mentioned before, in the mean-field theory, if we ignore the response of the $a_{1\mu}$ field, then eqn (7.3.29) simply describes a gas of bosons in a magnetic field nb_1 , where $b_1 = -\varepsilon_{ij}\partial_i a_{1j}$. Now we would like to attach an electron (in the second layer) to each quasi-hole in eqn (7.3.29). Such an operation has the following two effects: the bound state of the quasi-holes and the electrons can directly couple with the electromagnetic field A_μ because the electron carries the charge $-e$; and the bound state behaves like a fermion. The effective theory for the bound states has the following form:

$$\mathcal{L} = (eA_\mu - na_{1\mu})j^\mu + \text{kinetic/potential energy} \quad (7.3.30)$$

which now describes a gas of fermions (in mean-field theory). These fermions see an effective magnetic field $-eB + nb_1$.

When the electrons (i.e. the fermions in eqn (7.3.30)) in the second layer have a density of $\frac{1}{m}(-eB + nb_1)/2\pi$ (i.e. they have an effective filling fraction $1/m$), they can form a $1/m$ Laughlin state, which corresponds to the $\prod_{i<j}(z_{2i} - z_{2j})^m$ part of the wave function. (Note that here $eB < 0$.) Introducing a new gauge field $j^\mu = \frac{1}{2\pi}\partial_\nu a_{2\lambda}\varepsilon^{\mu\nu\lambda}$ to describe the fermion current j^μ in eqn (7.3.30), the effective theory of the $1/m$ state in the second layer has the form

$$\mathcal{L} = -m\frac{1}{4\pi}a_{2\mu}\partial_\nu a_{2\lambda}\varepsilon^{\mu\nu\lambda} \quad (7.3.31)$$

Putting eqns (7.3.28), (7.3.30), and (7.3.31) together, we obtain the total effective theory of the double-layer state, which has the form given in eqn (7.3.15) with the K -matrix and the charge vector $\mathbf{q}$ given by

$$K = \begin{pmatrix} l & n \\ n & m \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (7.3.32)$$

We see that the elements of the K -matrix are simply the exponents in the wave function. The filling fraction of the FQH state is still given by eqn (7.3.19).

There are two kinds of (fundamental) quasi-hole excitation in the double-layer state. The first kind is created by multiplying $\prod_i(\xi - z_{1i})$ with the ground-state wave function and the second kind is created by $\prod_i(\xi - z_{2i})$. As the vortices in the two electron condensates in the first and the second layers, a first kind of quasi-hole is created by the source term $-a_{1\mu}j^\mu$, and the second kind of quasi-hole is created by $-a_{2\mu}j^\mu$. Thus, a generic quasiparticle in the double-layer state is a

TABLE 7.2. Expressions for $(K, \mathbf{q}, \mathbf{s})$ for some double-layer FQH states.

ν	$\mathbf{q}$	K	$\mathbf{s}$
$1/m$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} m & m \\ m & m \end{pmatrix}$	$\begin{pmatrix} m/2 \\ m/2 \end{pmatrix}$
$2/5$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$	$\begin{pmatrix} 3/2 \\ 3/2 \end{pmatrix}$
$1/2$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$	$\begin{pmatrix} 3/2 \\ 3/2 \end{pmatrix}$
$2/3$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$	$\begin{pmatrix} 3/2 \\ 3/2 \end{pmatrix}$
$2/3$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$	$\begin{pmatrix} 1/2 \\ 1/2 \end{pmatrix}$

bound state of several quasiholes of the first and the second kind, and is described by eqn (7.3.16). The quantum numbers of such quasiparticles are still given by eqn (7.3.23).

In general, a multi-layer FQH state of the type (7.3.27) is described by a K -matrix whose elements are integers and whose diagonal elements are odd integers. The charge vector has the form $\mathbf{q}^\top = (1, 1, \dots, 1)$. People usually label the double-layer FQH state (7.3.27) by (l, m, n) . In Table 7.2, we list the K -matrix, the charge vector $\mathbf{q}$, and the spin vector $\mathbf{s}$ for some simple double-layer states.

From the above, we see that the (332) double-layer state has the filling fraction $2/5$ —a filling fraction that also appears in single-layer hierarchical states. Now the following question arises: do the double-layer $2/5$ state and the single-layer $2/5$ state belong to the same universality class? This question has experimental consequences. We can imagine the following experiment. We start with a (332) double-layer state in a system with very weak inter-layer tunneling. As we make the inter-layer tunneling stronger and stronger, while keeping the filling fraction fixed, the double-layer state will eventually become a single-layer $2/5$ state. The question is whether the transition between the double-layer (332) state and the single-layer $2/5$ state is a smooth crossover or a phase transition. If we ignore the spin vector, then we see that the K -matrices and the charge vectors of the two $2/5$ states are equivalent because they are related by an $SL(2, \mathbb{Z})$ transformation. Therefore, in the absence of rotational symmetry (in which case the spin vector is not well defined), the two $2/5$ states can change into each other smoothly. When we include the spin vector, the two $2/5$ states are not equivalent and, for rotationally-invariant systems, they are separated by a first-order phase transition.

From Table 7.2, we also see that there are two different $2/3$ double-layer states. When the intra-layer interaction is much stronger than the inter-layer interaction (a situation found in real samples), the ground-state wave function prefers to have higher-order zeros between electrons in the same layers. Thus, the (330) state should have lower energy than the (112) state. The K -matrix and the charge vector of the single-layer $2/3$ state are equivalent to those of the (112) state, and are not equivalent to those of the (330) state. Thus, to change

a double-layer $2/3$ state (i.e. the (330) state) into the single-layer $2/3$ state, one must go through a phase transition, regardless of the rotational symmetry.

The double-layer (mmm) state is an interesting state because $\det K = 0$. Fertig (1989), Brey (1990), and MacDonald *et al.* (1990) studied the (111) state and discovered a gapless collective mode. Wen and Zee (1992b) studied the more general (mmm) state and pointed out that the (mmm) state (in the absence of inter-layer tunneling) spontaneously breaks a $U(1)$ symmetry and is a neutral superfluid. More detailed discussions (including their experimental implications) can be found in Wen and Zee (1993), Murphy *et al.* (1994), and Yang *et al.* (1994).

Problem 7.3.4.

Let us first ignore the spin vector s . Show that (K, q) for the single-layer $2/5$ state in Table 7.1 and the double-layer $2/5$ state in Table 7.2 are equivalent. Show that, however, (K, q, s) for the two states are not equivalent. (In the presence of the spin vector s , the equivalence relation becomes $s_2 = Ws_1$, $q_2 = Wq_1$, and $K_2 = WK_1W^\dagger$.)

Problem 7.3.5.

Consider the double-layer (lmn) state (7.2.6). Use the effective theory (7.3.15) and eqn (7.3.32) to find the filling fraction. Also find the fractional charge and the fractional statistics of the quasiholes in the first and the second layers. (You may compare the results with those of Problem 7.2.5.)

Problem 7.3.6.

Assume that the effective theory for a double-layer (mmm) state has the form

$$\mathcal{L} = -\frac{1}{4\pi} K_{IJ} a_{I\mu} \partial_\nu a_{J\lambda} \epsilon^{\mu\nu\lambda} + \frac{1}{2g_1} \mathbf{e}_I \cdot \mathbf{e}_I - \frac{1}{2g_2} \mathbf{b}_I \mathbf{b}_I$$

where $\mathbf{e}_I$ and $\mathbf{b}_I$ are the ‘electric’ field and the ‘magnetic’ field, respectively, of $a_{I\mu}$.

1. Show that the (mmm) state, just like a superfluid, has gapless excitations. Find the velocity of the gapless excitations.
2. Also like the superfluid, the (mmm) state contains vortex excitations with energy $\gamma \ln L$, where L is the linear size of the system. Find the value of γ for the vortex with minimal energy. (Hint: The $a_{I\mu}$ charges of $\mathbf{b}_I$ are still quantized as integers in the (mmm) state.)
3. If we do regard the (mmm) state as a superfluid, can you identify the broken symmetry and the order parameter?

7.4 Edge excitations in fractional quantum Hall liquids

- FQH states *always* have gapless edge excitations.
- Edge excitations of an FQH state form a chiral Luttinger liquid.
- The structure of edge excitations is determined by bulk topological orders.

Due to the repulsive interaction and strong correlation between the electrons, a QH liquid is an incompressible state, despite the fact that the first Landau level is only partially filled. All of the bulk excitations in QH states have finite energy gaps. QH states and insulators are very similar in the sense that both states have finite energy gaps and short-ranged electron propagators. Due to this similarity, people were puzzled by the fact that QH systems apparently have very different transport properties to ordinary insulators. Halperin first pointed out that the IQH states contain gapless edge excitations (Halperin, 1982). The non-trivial transport properties of the IQH states come from the gapless edge excitations (Halperin, 1982; Trugman, 1983; MacDonald and Streda, 1984; Streda *et al.*, 1987; Buttiker, 1988; Jain and Kivelson, 1988a,b). For example, a two-probe measurement of an IQH sample can result in a finite resistance only when the source and the drain are connected by the edges. If the source and the drain are not connected by any edge, then the two-probe measurement will yield an infinite resistance at zero temperature, a result very similar to the insulators. Halperin also studied the dynamical properties of the edge excitations of the IQH states and found that the edge excitations are described by a chiral one-dimensional Fermi liquid theory.

Due to the similar transport properties of the FQH and IQH states, it is natural to conjecture that the transport in the FQH states is also governed by the edge excitations (Beenakker, 1990; MacDonald, 1990). However, because the FQH states are intrinsically many-body states, the edge excitations in the FQH states cannot be constructed by filling single-particle energy levels. In other words, the edge excitations of the FQH states should not be described by a Fermi liquid. Thus, we need a completely new approach to understand the dynamical properties of the edge states of FQH liquids (Wen, 1992, 1995).

In the next section, we will use Fermi liquid theory to study the edge excitations of the $\nu = 1$ IQH state. Then we will use the current algebra (or bosonization, see Tomonaga, 1950; Luttinger, 1963; Coleman, 1975) to develop a theory for edge excitations of the FQH states.

7.4.1 Fermi liquid theory of integral quantum Hall edge states

- The ground states of IQH liquids are obtained by filling single-particle energy levels. As a result, the low-energy edge excitations can be obtained by filling these single-particle levels in slightly different ways.

Consider a non-interacting electron gas in a uniform magnetic field B . In the symmetric gauge, the angular momentum is a good quantum number. The common eigenstates of energy and momentum form a ring-like shape (see Fig. 7.10). Therefore, in the presence of a smooth circular potential $V(r)$, the single-particle energy levels are as illustrated in Fig. 7.15. The ground state is obtained by filling

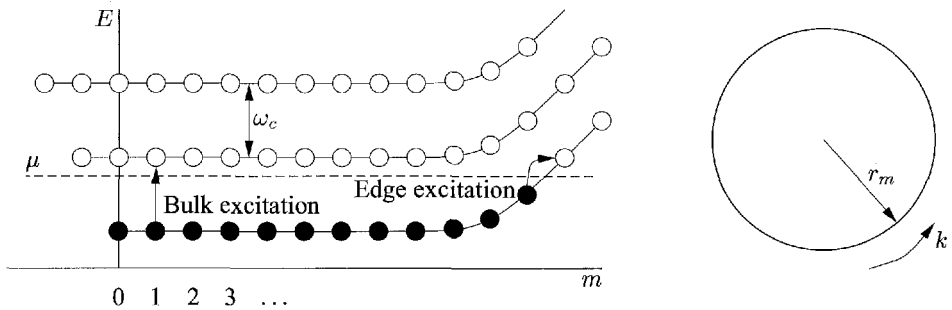


FIG. 7.15. Energy levels in the first three Landau levels in the presence of a smooth potential $V(r)$. Here m is the angular momentum of the levels. The levels below the chemical potential μ are filled by one electron, which gives rise to an IQH state. Modifying the occupation numbers near the edge produces gapless edge excitations of the IQH states.

N lowest energy levels. Such a state corresponds to a circular droplet of uniform density (see Fig. 7.11).

In the zeroth Landau level, the energy of the angular momentum m state is given by

$$E_m = \frac{1}{2}\hbar\omega_c + V(r_m)$$

where $r_m = \sqrt{2ml_B}$ is the radius of the m state. From Fig. 7.15, we see that bulk excitations have a finite energy gap $\hbar\omega_c$. An edge excitation can be created by moving an electron from the m state to the $m+1$ state near the edge (see Fig. 7.15). As $r_{m+1} - r_m \rightarrow 0$ in the thermodynamical limit $r_m \rightarrow \infty$, the edge excitations are gapless.

As the m state has radius r_m , we can also view the angular momentum m as the momentum along the edge $k = m/r_m$. This allows us to regard E_m as an energy-momentum relation

$$E(k) = \frac{1}{2}\hbar\omega_c + V(2l_B^2 k)$$

In terms of $E(k)$, our electron system can be described by a non-interacting Hamiltonian

$$H = \sum_k E(k) c_k^\dagger c_k \quad (7.4.1)$$

The above Hamiltonian describes a one-dimensional chiral Fermi liquid. We call it a chiral Fermi liquid because it has only one Fermi point and all of the low-energy excitations propagate in the same direction.⁴⁷ We conclude that the edge

⁴⁷ In contrast, the usual one-dimensional Fermi liquid has two Fermi points and contains both left and right movers.

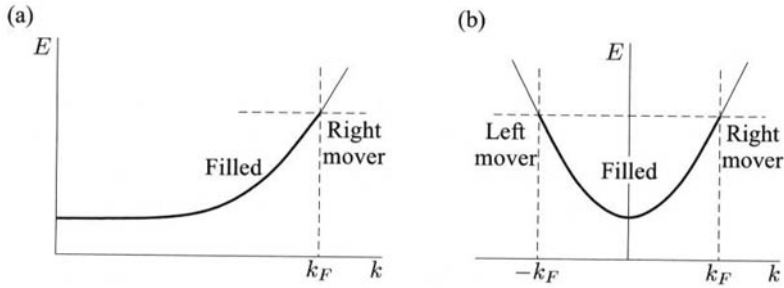


FIG. 7.16. (a) The energy–angular momentum relation E_m can be regarded as an energy–momentum relation $E(k)$. The edge excitations can be regarded as excitations of a one-dimensional chiral Fermi liquid, which contains only right movers. (b) The dispersion relation of a usual one-dimensional Fermi liquid contains both right and left movers.

excitations of the $\nu = 1$ IQH state are described by the one-dimensional chiral Fermi liquid (7.4.1).

Problem 7.4.1.

Show that the velocity of the IQH edge excitations $v = \partial E(k)/\partial k$ is given by cE/B , where c is the speed of light and $E = -\partial V(r)/\partial r$ is the electric field on the edge produced by the confining potential $V(r)$. Find the Fermi momentum k_F for the chiral Fermi liquid, assuming that the $\nu = 1$ IQH droplet contains N electrons.

Problem 7.4.2.

A $\nu = 1$ IQH state is confined by a circular smooth potential $V(r)$ and contains N electrons. Find the total ground-state angular momentum M_0 . Find the number of low-energy particle–hole excitations with total angular momentum $M_0 + m$ for $m = -1$ and $m = 1, 2, 3, 4, 5$. Show that the excitations with the same total angular momentum have the same energy in the thermodynamical limit.

Problem 7.4.3.

Consider a one-dimensional non-interacting fermion system described by $H = \sum_k (v_1 k c_{1,k}^\dagger c_{1,k} + v_2 k c_{2,k}^\dagger c_{2,k})$. The mixing term $\sum_k (\gamma c_{1,k}^\dagger c_{2,k} + h.c.)$ is a relevant perturbation. Show that, when the system has both right and left movers (i.e. $v_1 v_2 < 0$), the relevant mixing term opens up a finite energy gap and drastically changes the low-energy properties of the system. However, for a chiral Fermi liquid with all excitations moving in one direction (i.e. $v_1 v_2 > 0$), the relevant mixing term does not induce any energy gap. In fact, we believe that no perturbations can induce an energy gap for one-dimensional chiral Fermi liquids. The gapless excitations in a one-dimensional chiral Fermi liquid are robust against all perturbations and are topologically stable.

7.4.2 The hydrodynamical approach—the $1/m$ Laughlin state

- Edge excitations from the current algebra (the Kac–Moody algebra).

- There is no way to construct FQH edge excitations by filling single-particle energy levels.

As the $1/m$ Laughlin FQH state cannot be obtained by filling single-particle energy levels, we cannot use the picture in the last section to construct edge excitations of the $1/m$ Laughlin state. Not knowing how to derive the FQH edge theory from the Hamiltonian of interacting electrons, in the following we instead try to guess a low-energy effective theory.

The simplest way to guess the dynamical theory of edge excitations is to use the hydrodynamical approach. In this approach, one uses the fact that FQH states are incompressible, irrotational liquids that contain no low-energy bulk excitations. Therefore, the only low-lying excitations (below the bulk energy gap) are surface waves from the deformations of the FQH droplet. These surface waves are identified as edge excitations of the FQH state.

In the hydrodynamical approach (Wen, 1992), we first study the classical theory of the surface wave on the FQH droplet. Then we quantize the classical theory to obtain the quantum description of the edge excitations. It is amazing that the simple quantum description obtained from the classical theory provides a complete description of the edge excitations at low energies and allows us to calculate the electron and the quasiparticle propagators along the edges.

Consider an FQH droplet with a filling fraction ν confined by a potential well. Due to the nonzero conductance, the electric field of the potential well generates a persistent current flowing along the edge, given by

$$\mathbf{j} = \sigma_{xy} \hat{z} \times \mathbf{E}, \quad \sigma_{xy} = \nu \frac{e^2}{h}$$

This implies that the electrons near the edge drift along the edge with a velocity

$$v = \frac{E}{B} c$$

where c is the velocity of light. Thus, the edge wave (the deformation of the edge) also propagates with velocity v . Let us use the one-dimensional density $\rho(x) = nh(x)$ to describe the edge wave, where $h(x)$ is the displacement of the edge, x is the coordinate along the edge, and $n = \nu/2\pi l_B^2$ is the two-dimensional electron density in the bulk. (Here $l_B = \sqrt{c/eB}$ is the magnetic length.) We see that the propagation of the edge waves is described by the following wave equation:

$$\partial_t \rho + v \partial_x \rho = 0 \quad (7.4.2)$$

Notice that the edge waves always propagate in one direction; there are no waves that propagate in the opposite direction.

The Hamiltonian (i.e. the energy) of the edge waves is given by

$$H = \int dx \frac{1}{2} e \rho E h = \int dx \pi \frac{v}{\nu} \rho^2 \quad (7.4.3)$$

where we have used $h = \rho/n = \rho \frac{2\pi c}{eB}$. In momentum space, eqns (7.4.2) and (7.4.3) can be rewritten as

$$\dot{\rho}_k = -i\nu k \rho_k, \quad H = 2\pi \frac{v}{\nu} \sum_{k>0} \rho_{-k} \rho_k \quad (7.4.4)$$

where $\rho_k = \int dx \frac{1}{\sqrt{L}} e^{ikx} \rho(x)$ and L is the length of the edge. We find that, if we identify $\rho_k|_{k>0}$ as the ‘coordinates’ and $\pi_k = i2\pi\rho_{-k}/\nu k$ as the corresponding canonical ‘momenta’, then the standard Hamiltonian equation

$$\dot{q} = \frac{\partial H}{\partial p}, \quad \dot{p} = -\frac{\partial H}{\partial q}$$

will reproduce the equation of motion $\dot{\rho}_k = i\nu k \rho_k$. This allows us to identify the canonical ‘coordinates’ and ‘momenta’. It is interesting to see that the displacement $h(x)$ contains both the ‘coordinates’ and the ‘momenta’. This is due to the chiral property of the edge wave.

Knowing the canonical coordinates and momenta, it is easy to quantize the classical theory. We simply view ρ_k and π_k as operators that satisfy $[\rho_k, \pi_{k'}] = i\delta_{kk'}$. Thus, after quantization we have

$$[\rho_k, \rho_{k'}] = \frac{\nu}{2\pi} k \delta_{k+k'}, \quad k, k' = \text{integer} \times \frac{2\pi}{L} \quad (7.4.5)$$

$$H = 2\pi \frac{v}{\nu} \sum_{k>0} \rho_{-k} \rho_k$$

The above algebra is called the $(U(1))$ Kac–Moody (K–M) algebra (Kac, 1983; Goddard and Olive, 1985, 1986). A similar algebra has also appeared in the Tomonaga model (Tomonaga, 1950; Luttinger, 1963; Coleman, 1975). Notice that eqn (7.4.5) simply describes a collection of decoupled harmonic oscillators (generated by (ρ_k, ρ_{-k})). Thus, eqn (7.4.5) is a one-dimensional free phonon theory (with only a single branch). For $k > 0$, $\rho_k^\dagger = \rho_{-k}$ creates a phonon with momentum k and energy νk , while ρ_k annihilates a phonon. Equation (7.4.5) provides a complete description of the low-lying edge excitations of the Laughlin state.

The edge excitations considered here do not change the total charge of the system and hence are neutral. In the following, we will discuss the charged excitations and calculate the electron propagator from the K–M algebra (7.4.5).

The low-lying charge excitations obviously correspond to adding (removing) electrons to (from) the edge. These charged excitations carry integer charges and

are created by the electron operators $\Psi^\dagger$. The above theory of edge excitations is formulated in terms of a one-dimensional density operator $\rho(x)$. So the central objective is to write the electron operator in terms of the density operator. The electron operator on the edge creates a localized charge and should satisfy

$$[\rho(x), \Psi^\dagger(x')] = \delta(x - x') \Psi^\dagger(x') \quad (7.4.6)$$

As ρ satisfies the K-M algebra (7.4.5), one can show that

$$[\rho(x_1), \rho(x_2)] = i \frac{\nu}{2\pi} \delta'(x_1 - x_2)$$

or

$$[\rho(x_1), \phi(x_2)] = -i\nu \delta(x_1 - x_2)$$

where ϕ is given by $\rho = \frac{1}{2\pi} \partial_x \phi$. We see that $\rho(x)$ can be regarded as a functional derivative of ϕ , i.e. $\rho(x) = -i\nu \frac{\delta}{\delta\phi(x)}$. Using this relation, one can show that the operators that satisfy eqn (7.4.6) are given by (Wen, 1992)

$$\Psi \propto e^{i\frac{1}{\nu}\phi} \quad (7.4.7)$$

Equation (7.4.6) only implies that the operator Ψ carries the charge e . In order to identify Ψ as an electron operator, we need to show that Ψ is a fermionic operator. Using the K-M algebra (7.4.5), we find that (see Problem 7.4.8)

$$\Psi(x)\Psi(x') = (-)^{1/\nu} \Psi(x')\Psi(x) \quad (7.4.8)$$

We see that the electron operator Ψ in eqn (7.4.7) is fermionic only when $1/\nu = m$ is an odd integer, in which case the FQH state is a Laughlin state.

In the above discussion, we have made an assumption that is not generally true. We have assumed that the incompressible FQH liquid contains only *one* component of incompressible fluid, which leads to one branch of edge excitations. The above result implies that, when $\nu \neq 1/m$, the edge theory with only one branch does not contain the electron operators and is not self-consistent. Our one-branch edge theory only applies to the $\nu = 1/m$ Laughlin state.

Now let us calculate the electron propagator along the edge of the $\nu = 1/m$ Laughlin state. As ϕ is a free phonon field with a propagator

$$\langle \phi(x, t) \phi(0) \rangle = -\nu \ln(x - vt) + \text{constant}$$

the electron propagator can be easily calculated as

$$G(x, t) = \langle T(\Psi^\dagger(x, t) \Psi(0)) \rangle = \exp\left[\frac{1}{\nu^2} \langle \phi(x, t) \phi(0) \rangle\right] \propto \frac{1}{(x - vt)^m} \quad (7.4.9)$$

The first thing we see is that the electron propagator on the edge of an FQH state acquires a non-trivial exponent $m = 1/\nu$ that is not equal to one. This implies that

the electrons on the edge of the FQH state are strongly correlated and cannot be described by Fermi liquid theory. We will call this type of electron state a chiral Luttinger liquid.

We would like to emphasize that the exponent m is quantized. The quantization of the exponent is directly related to the fact that the exponent is linked to the statistics of the electrons (see eqn (7.4.8)). Thus, the exponent is a topological number that is independent of electron interactions, edge potential, etc. Despite the exponent being a property of the edge states, the only way to change the exponent is through a phase transition in the bulk state. Therefore, the exponent can be regarded as a quantum number that characterizes the topological orders in the *bulk* FQH states.

In momentum space, the electron propagator has the form

$$G(k, \omega) \propto \frac{(vk + \omega)^{m-1}}{\omega - vk + i0^+ \text{sgn}(\omega)}$$

The anomalous exponent m can be measured in tunneling experiments. From $\text{Im}G$, we find that the tunneling density of states of electrons is given by

$$N(\omega) \propto |\omega|^{m-1}$$

This implies that the differential conductance has the form $\frac{dI}{dV} \propto V^{m-1}$ for a metal–insulator–FQH junction.

Problem 7.4.4.

Prove eqn (7.4.8). (Hint: First show that $[\phi(x), \phi(y)] = \frac{\pi}{q} \text{sgn}(x - y)$.)

Problem 7.4.5.

Bosonization and fermionization When $m = 1$ the discussion in this section implies that the $\nu = 1$ IQH edge state can be described by the free phonon theory

$$H_b = 2\pi v \sum_{k>0} \rho_k^\dagger \rho_k$$

In Section 7.4.1, we found that the $\nu = 1$ IQH edge state can be described by the free chiral fermion model

$$H_f = \sum_k vk : c_k^\dagger c_k := \sum_{k>0} vk c_k^\dagger c_k - \sum_{k<0} vk c_k c_k^\dagger$$

The normal order is defined as $: c_k^\dagger c_k := c_k^\dagger c_k$ if $k > 0$ and $: c_k^\dagger c_k := -c_k c_k^\dagger$ if $k < 0$. In this problem, we are going to study the direct relationship between the two descriptions. Switching from the fermion description to the boson description is called bosonization, and switching from the boson description to the fermion description is called fermionization.

1. Find the energies of the first five energy levels and their degeneracy for the bosonic model. Compare your result with that from Problem 7.4.2.

2. Let $\rho_f(x) =: c^\dagger(x)c(x)$. In k space, we have $\rho_{f,k} = \sum_q' : c_q^\dagger c_{q+k} :$, where the summation $\sum_q'$ is limited to the range for which both the momenta of c_{q+k} and c_q are within the range $[-\Lambda, +\Lambda]$. The normal ordered $\rho_f(x)$ has a zero average in the ground state, which agrees with ρ in boson theory. Show that $\rho_{f,k}$ satisfies the same K-M algebra as ρ_k (see eqn (7.4.5) with $\nu = 1$). (Hint: We have $c_k^\dagger c_k = 0$ for large positive k and $c_k^\dagger c_k = 0$ for large negative k .)
3. Show that $H_f = 2\pi\nu \sum_{k>0} \rho_{f,k}^\dagger \rho_{f,k}$.

7.4.3 A microscopic theory for the edge excitations

- Relationship between the K-M algebra and the Laughlin wave functions.
- Electron equal-time correlation from the plasma analogue.

In this section, we will present a microscopic theory for the edge excitations in the Laughlin states. To be specific, let us consider an electron gas in the first Landau level. We assume that the electrons are described by the ideal Hamiltonian (7.2.5). The $\nu = 1/3$ Laughlin wave function

$$\Psi_3(z_i) = Z^{-1/2} \prod_{i<j} (z_i - z_j)^3 \prod_k e^{-|z_k|^2/4l_B} \quad (7.4.10)$$

has zero energy⁴⁸ and is an exact ground state of the ideal Hamiltonian. In this section, we will assume that the magnetic length $l_B = 1$. In eqn (7.4.10), Z is the normalization factor. However, the Laughlin state (7.4.10) is not the only state with zero energy. One can easily check that the following type of states all have zero energy:

$$\Psi(z_i) = P(z_i)\Psi_3(z_i) \quad (7.4.11)$$

where $P(z_i)$ is a symmetric polynomial of z_i . In fact, the reverse is also true; all of the zero-energy states are of the form (7.4.11). This is because, in order for a fermion state to have zero energy, Ψ must vanish at least as fast as $(z_i - z_j)^3$ when any two electrons i and j are brought together (the possibility $(z_i - z_j)^2$ is excluded by fermion statistics). Because the Laughlin wave function is zero only when $z_i = z_j$, then $P = \Psi/\Psi_3$ is a finite function. As Ψ and Ψ_3 are both anti-symmetric functions in the first Landau level, P is a symmetric holomorphic function that can only be a symmetric polynomial.

Among all of the states in eqn (7.4.11), the Laughlin state describes a circular droplet with the smallest radius. All of the other states are a deformation and/or an inflation of the droplet of the Laughlin state. Thus, the states generated by P correspond to the edge excitations of the Laughlin state.

⁴⁸ Here, for convenience, we shift the zero of energy to the zeroth Landau level.

First, let us consider the zero-energy space (i.e. the space of symmetric polynomials). We know that the space of symmetric polynomials is generated by the polynomials $s_n = \sum_i z_i^n$ (through multiplication and addition). Let $M_0 = 3\frac{N(N-1)}{2}$ be the total angular momentum of the Laughlin state (7.4.10). Then the state Ψ will have an angular momentum $M = \Delta M + M_0$, where ΔM is the order of the symmetric polynomial P . As we have only one order-zero and one order-one symmetric polynomial, namely $s_0 = 1$ and $s_1 = \sum_i z_i$, respectively, the zero-energy states for $\Delta M = 0, 1$ are non-degenerate. However, when $\Delta M = 2$, we have two zero-energy states generated by $P = s_2$ and $P = s_1^2$. For general ΔM , the degeneracy of the zero-energy states is given by

$$\begin{array}{lcl} \Delta M : & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ \text{degeneracy :} & 1 & 1 & 2 & 3 & 5 & 7 & 11 \end{array} \quad (7.4.12)$$

Here we would like to point out that the degeneracy in eqn (7.4.12) is exactly what we expected from the macroscopic theory. We know that, for a circular droplet, the angular momentum ΔM can be regarded as the momentum along the edge $k = 2\pi\Delta M/L$, where L is the perimeter of the FQH droplet. According to the macroscopic theory, the (neutral) edge excitations are generated by the density operators ρ_k . One can easily check that the edge states generated by the density operators have the same degeneracies as those in eqn (7.4.12) for every ΔM . For example, the two states at $\Delta M = 2$ are generated by $\rho_{\kappa_0}^2$ and $\rho_{2\kappa_0}$, where $\kappa_0 = 2\pi/L$. Therefore, the space generated by the K–M algebra (7.4.5) and the space of symmetric polynomials are identical.

Now let us ask the following physical question: do the symmetric polynomials generate all of the low-energy states? If this is true, then from the above discussion we see that all of the low-energy excitations of the HQ droplet are generated by the K–M algebra, and we can say that eqn (7.4.5) is a complete theory of the low-lying excitations. Unfortunately, up to now we do not have an analytic proof of the above statement. This is because, although states which are orthogonal to the states generated by the symmetric polynomials have nonzero energies, it is not clear that these energies remain finite in the thermodynamical limit. It is possible that the energy gap approaches zero in the thermodynamical limit. To resolve this problem, we currently have to rely on numerical calculations. In Fig. 7.17 we present the energy spectrum of a system of six electrons in the first twenty-two orbits for the Hamiltonian introduced at the beginning of this section. The degeneracies of the zero-energy states at $M = 45, \dots, 51$ (or $\Delta M = 0, \dots, 6$) are found to be 1, 1, 2, 3, 5, 7, 11, respectively, which agrees with eqn (7.4.12). More importantly, we clearly see that a finite energy gap separate all of the other states from the zero-energy states. Thus, the numerical results imply that all of the low-lying edge excitations of the Laughlin state are generated by the symmetric polynomials or the K–M algebra (7.4.5).

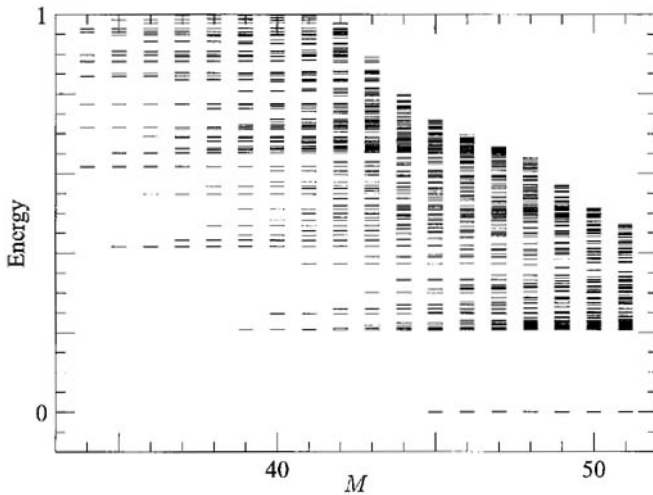


FIG. 7.17. The energy spectrum of a system of six electrons in the first twenty-two orbits for the Hamiltonian H_V . The degeneracies of the zero-energy states at $M = 45, \dots, 51$ are found to be 1, 1, 2, 3, 5, 7, 11, respectively.

In the following, we are going to calculate the equal-time correlation of the quasiparticle/electron from the Laughlin wave function Ψ_m . First, we calculate the norm of $|\xi\rangle = \prod_i (1 - \frac{z_i}{\xi})^n \Psi_m(\{z_i\})$ as follows:

$$\langle \xi | \xi \rangle = |\xi|^{-2Nn} \frac{Z_1}{Z} \quad (7.4.13)$$

$$Z = \int \prod d^2 z_i \exp \left(-\beta \sum_{ij} (-m^2) \ln |z_i - z_j| - \beta \sum_k \frac{m}{4} |z_k|^2 \right)$$

$$Z_1 = \int \prod d^2 z_i \exp \left[\beta \sum_{ij} m^2 \ln |z_i - z_j| - \beta \sum_k \left(\frac{m}{4} |z_k|^2 - nm \ln |\xi - z_k| \right) \right]$$

where $\beta = 2/m$. Notice that Z is the partition function of a plasma formed by N 'charge' m particles, and Z_1 is the partition function of the plasma interacting with a particle of 'charge' n at ξ . We may write

$$Z = e^{-\beta E}, \quad Z_1 = e^{-\beta E_1}$$

where E and E_1 are the total energies of the plasma. The change in energy $E_1 - E$ is given by the interaction between the added 'charge' n particle at ξ and N 'charge' m particles at z_i . When $|\xi|$ is large, the N 'charge' m particles form a circular droplet and we may treat them as a point 'charge' mN at $z = 0$. We

obtain $E_1 - E = -nmN \ln |\xi|$. For smaller $|\xi|$, the ‘charge’ n changes the shape of the droplet and

$$E_1 - E = -nmN \ln |\xi| - \frac{n^2}{2} \left[-\ln \left(|\xi| - \frac{R^2}{|\xi|} \right) + \ln(|\xi|) \right] \quad (7.4.14)$$

where $R = \sqrt{mN}l_B$ is the radius of the droplet. The first term in eqn (7.4.14) is the interaction between the ‘charge’ n and the *undeformed* droplet. The second term is the correction due to the deformation of the droplet. Since the plasma behaves like a metal, this correction can be represented by the interaction of the ‘charge’ n with its mirror images at $|z| = R^2/|\xi|$ and $z = 0$.

In the above discussion, we ignore the discreteness of the charges and treat the plasma as a continuous medium. We expect this approximation to give rise to the correct ratio Z_1/Z if ξ is not too close to the droplet, i.e. $|\xi| - R \gg l_B$.

From eqn (7.4.14), we find the norm of $|\psi^n(\xi)\rangle$ to be

$$\langle \xi | \xi \rangle = \left(\frac{\xi \xi^*}{\xi \xi^* - R^2} \right)^{n^2/m} \quad (7.4.15)$$

As the inner product $\langle \tilde{\xi} | \xi \rangle$ is a holomorphic function of ξ and an anti-holomorphic function of $\tilde{\xi}$, eqn (7.4.15) implies that

$$\langle \tilde{\xi} | \xi \rangle = \left(\frac{\xi \tilde{\xi}^*}{\xi \tilde{\xi}^* - R^2} \right)^{n^2/m}$$

Notice that, when $n = m$,

$$\xi^{Nm} \tilde{\xi}^{*Nm} \langle \tilde{\xi} | \xi \rangle = \xi^{Nm} \tilde{\xi}^{*Nm} \left(\frac{\xi \tilde{\xi}^*}{\xi \tilde{\xi}^* - R^2} \right)^m$$

is proportional to the electron propagator G_e along the edge of an $N_e = N + 1$ electron system at equal time. Choosing $\xi = R e^{i\frac{x}{R}}$ and $\tilde{\xi} = R$, we obtain

$$G_e(x) = L^{-m} a^{m-1} e^{im(N_e - \frac{1}{2})\frac{2\pi x}{L}} \sin^{-m}(\pi x/L) \quad (7.4.16)$$

which reduces to eqn (7.4.9) when x is much less than $L \equiv 2l_B$. Here a is a length scale of order l_B . Equation (7.4.16) can be expanded as follows:

$$\begin{aligned} G_e(x) &= L^{-m} a^{m-1} e^{im(N_e - 1)\frac{2\pi x}{L}} \left[\sum_{n=0}^{\infty} e^{-i\frac{2\pi x}{L}n} \right]^m \\ &= L^{-m} a^{m-1} e^{im(N_e - 1)\frac{2\pi x}{L}} \sum_{n=0}^{\infty} C_n^{m+n-1} e^{-i\frac{2\pi x}{L}n} \\ C_n^{m+n-1} &= \frac{(n+m-1)!}{(m-1)!n!} \end{aligned} \quad (7.4.17)$$

From this expansion, we obtain the electron occupation number n_M at the angular momentum M state as follows:

$$\begin{aligned} n_M &= 0, & \delta M &= m(N_e - 1) - M < 0 \\ n_M &= \frac{a^{m-1}}{L^{m-1}} C_{\delta M}^{\delta M+m-1}, & \delta M &\geq 0 \end{aligned} \quad (7.4.18)$$

We see that the exact position of the Fermi edge is at the last partially occupied single-particle orbit, i.e. at the angular momentum $m(N_e - 1)$ (or $k_F = \sqrt{m(N_e - 1)}/l_B$). Note that, when $m(N_e - 1) - M \gg m$, we have $n_M \propto (m(N_e - 1) - M)^{m-1}$. In terms of momentum along the edge, we have $n_k \propto (k_F - k)^{m-1}$. We find that, unlike a Fermi liquid, the occupation number n_k does not have a jump at the Fermi momentum.

We would like to remark that eqn (7.4.16) is correct only when x is much larger than the magnetic length l_B . Therefore, eqn (7.4.18) is valid only when $m(N_e - 1) - M \ll \sqrt{N_e}$.

If the dispersion of the edge excitations is linear, then G_e can only depend on $x - vt$ at low energies. We immediately see that the dynamical electron Green's function is

$$G_e(x, t) = L^{-m} a^{m-1} e^{im(N-\frac{1}{2})\frac{2\pi(x-vt)}{L}} \sin^{-m}[\pi(x-vt)/L]$$

When $m = 1$, the above becomes the electron Green's function of one-dimensional free fermions (for k near a Fermi point).

7.4.4 The hydrodynamical approach—the 2/5 and 2/3 states

- The edge states of hierarchical FQH liquids.
- The structure of electron and quasiparticle operators.
- The edge interaction can modify the exponents of electron/quasiparticle propagators if and only if the edge excitations can propagate in opposite directions.
- When $\nu_1 = -\nu_2 = 1$, the discussion here describes the one-dimensional Tomonaga–Luttinger liquid.

In this section, we will use the hydrodynamical approach to study the edge structures of second-level hierarchical states. We will concentrate on the 2/5 and 2/3 states as examples. In particular, we will study the structures of the electron and the quasiparticle operators on the edges of the hierarchical states. We will also see that the 2/3 state contains two edge modes that propagate in opposite directions, which is quite counter-intuitive.

First, let us consider the $\nu = \frac{2}{5}$ FQH state. According to the hierarchical theory, the $\nu = \frac{2}{5}$ FQH state is generated by the condensation of quasiparticles on top of

the $\nu = \frac{1}{3}$ FQH state. Thus, the $2/5$ state contains two components of incompressible fluids. To be definite, let us consider a special edge potential such that the FQH state consists of two droplets (see Fig. 7.14); one is the electron condensate with a filling fraction $\nu_1 = \frac{1}{3}$ and radius r_1 , and the other is the quasiparticle condensate (on top of the $1/3$ state) with a filling fraction $\nu_2 = \frac{1}{15}$ (note that $\frac{1}{3} + \frac{1}{15} = \frac{2}{5}$) and radius $r_2 < r_1$.

When $r_1 - r_2 \gg l_B$, the two edges are independent. Generalizing the hydrodynamical approach in Section 7.4.2, we can show that there are two branches of the edge excitations whose low-energy dynamics are described by

$$\begin{aligned} [\rho_{Ik}, \rho_{Jk'}] &= \frac{\nu_I}{2\pi} k \delta_{IJ} \delta_{k+k'} \\ H &= 2\pi \sum_{I,k>0} \frac{v_I}{\nu_I} \rho_{I,-k} \rho_{I,k} \end{aligned} \quad (7.4.19)$$

where $I = 1, 2$ labels the two branches and v_I are the velocities of the edge excitations. In eqn (7.4.19), ρ_I are the one-dimensional electron densities. In order for the Hamiltonian to be bounded from below, we require that $\nu_I v_I > 0$. We find that the stability of the $\nu = \frac{2}{5}$ FQH state requires both of the v_I to be positive.

Generalizing the discussion in Section 7.4.2, the electron operators on the two edges are found to be

$$\Psi_I = e^{i \frac{1}{\nu_I} \phi_I(x)}, \quad I = 1, 2 \quad (7.4.20)$$

with $\partial_x \phi_I = \frac{1}{2\pi} \rho_I$. The electron propagators have the form

$$\langle T(\Psi_I(x, t) \Psi_I^\dagger(0)) \rangle = \frac{e^{ik_I x}}{(x - v_I t)^{-1/|\nu_I|}}, \quad I = 1, 2$$

where $k_I = r_I/2l_B^2$.

According to the hierarchical picture, the $\nu = \frac{2}{3}$ FQH state is also formed by two condensates, an electron condensate with a filling fraction 1 and a hole condensate with a filling fraction $-\frac{1}{3}$. Thus, the above discussion can also be applied to the $\nu = \frac{2}{3}$ FQH state by choosing $(\nu_1, \nu_2) = (1, -\frac{1}{3})$. Again, there are two branches of the edge excitations, but now with *opposite* velocities if the Hamiltonian is positive definite.

As we bring the two edges together ($r_1 - r_2 \sim l_B$), the interaction between the two branches of the edge excitations can no longer be ignored. In this case, the Hamiltonian has the form

$$H = 2\pi \sum_{k>0} V_{IJ} \rho_{I,-k} \rho_{J,k} \quad (7.4.21)$$

The Hamiltonian (7.4.21) can be diagonalized. For $\nu_1 \nu_2 > 0$, we may choose

$$\begin{aligned}\tilde{\rho}_{1k} &= \cos(\theta) \frac{1}{\sqrt{|\nu_1|}} \rho_{1k} + \sin(\theta) \frac{1}{\sqrt{|\nu_2|}} \rho_{2k} \\ \tilde{\rho}_{2k} &= \cos(\theta) \frac{1}{\sqrt{|\nu_2|}} \rho_{2k} - \sin(\theta) \frac{1}{\sqrt{|\nu_1|}} \rho_{1k} \\ \tan(2\theta) &= 2 \frac{\sqrt{|\nu_1 \nu_2|} V_{12}}{|\nu_1| V_{11} - |\nu_2| V_{22}}\end{aligned}\quad (7.4.22)$$

One can check that the $\tilde{\rho}$ s satisfy

$$\begin{aligned}[\tilde{\rho}_{Ik}, \tilde{\rho}_{Jk'}] &= \frac{\text{sgn}(\nu_I)}{2\pi} k \delta_{IJ} \delta_{k+k'} \\ H &= 2\pi \sum_{I, k > 0} \text{sgn}(\nu_I) \tilde{v}_I \tilde{\rho}_I \tilde{\rho}_{I, -k} \tilde{\rho}_{I, k}\end{aligned}\quad (7.4.23)$$

where the new velocities $\tilde{v}_I$ of the edge excitations are given by

$$\begin{aligned}\text{sgn}(\nu_1) \tilde{v}_1 &= \frac{\cos^2(\theta)}{\cos(2\theta)} |\nu_1| V_{11} - \frac{\sin^2(\theta)}{\cos(2\theta)} |\nu_2| V_{22} \\ \text{sgn}(\nu_2) \tilde{v}_2 &= \frac{\cos^2(\theta)}{\cos(2\theta)} |\nu_2| V_{22} - \frac{\sin^2(\theta)}{\cos(2\theta)} |\nu_1| V_{11}\end{aligned}\quad (7.4.24)$$

We see that there are still two branches of the edge excitations. However, in this case, the edge excitations with a definite velocity are mixtures of those on the inner edge and the outer edge. One can also show that, as long as the Hamiltonian (7.4.21) is bounded from below, the velocities $\tilde{v}_I$ of the two branches are always positive.

After rewriting the electron operator Ψ_I in eqn (7.4.20) in terms of $\tilde{\rho}_I$ by inverting eqn (7.4.22), we can calculate their propagators using eqn (7.4.24) as follows:

$$\langle T(\Psi_I(x, t) \Psi_I^\dagger(0)) \rangle = e^{ik_I x} \frac{1}{(x - \tilde{v}_1 t)^{\alpha_I}} \frac{1}{(x - \tilde{v}_2 t)^{\beta_I}}$$

where

$$(\alpha_1, \alpha_2) = \left(\frac{\cos^2 \theta}{|\nu_1|}, \frac{\sin^2 \theta}{|\nu_2|} \right), \quad (\beta_1, \beta_2) = \left(\frac{\sin^2 \theta}{|\nu_1|}, \frac{\cos^2 \theta}{|\nu_2|} \right)$$

However, when the two edges are close to each other within the magnetic length, the Ψ_I are no longer the most general electron operators on the edge. The generic electron operator may contain charge transfers between the two edges. For

the $\nu = 2/5$ FQH state, the inner edge and the outer edge are separated by the $\nu = \frac{1}{3}$ Laughlin state. Thus, the elementary charge transfer operator is given by

$$\eta(x) = e^{i(\phi_1 - \frac{\nu_1}{\nu_2}\phi_2)} = (\Psi_1\Psi_2^\dagger)^{\nu_1}$$

which transfers a $\nu_1 e = e/3$ charge from the outer edge to the inner edge. The generic electron operator then takes the form

$$\begin{aligned}\Psi(x) &= \sum_{n=-\infty}^{+\infty} c_n \psi_n(x) \\ \psi_n(x) &= \Psi_1(x) \eta^n(x)\end{aligned}\tag{7.4.25}$$

To understand this result, we notice that each operator ψ_n always creates a unit localized charge and is a fermionic operator, regardless of the value of the integer n . Therefore, each ψ_n is a candidate for the electron operator on the edge. For a generic interacting system, the electron operator on the edge is expected to be a superposition of different ψ_n s, as represented in eqn (7.4.25). Note that $\Psi_2 = \psi_{-1/\nu_1}$. The propagator of ψ_n can be calculated in a similar way to that outlined above, and is given by

$$\langle T(\psi_n(x, t) \psi_m^\dagger(0)) \rangle \propto \delta_{n,m} e^{i[k_1 + n\nu_1(k_2 - k_1)]x} \prod_I (x - \tilde{v}_I t)^{-\gamma_{In}}$$

where the γ_{In} are

$$\begin{aligned}\gamma_{1n} &= \left[\left(n + \frac{1}{|\nu_1|} \right) \sqrt{|\nu_1|} \cos \theta - \frac{n\nu_1}{\nu_2} \sqrt{|\nu_2|} \sin \theta \right]^2 \\ \gamma_{2n} &= \left[\left(n + \frac{1}{|\nu_1|} \right) \sqrt{|\nu_1|} \sin \theta + \frac{n\nu_1}{\nu_2} \sqrt{|\nu_2|} \cos \theta \right]^2\end{aligned}\tag{7.4.26}$$

From eqns (7.4.25) and (7.4.4), we see that the electron propagator has singularities at the discrete momenta $k = k_1 + n\nu_1(k_2 - k_1)$. They are analogous to the $k_F, 3k_F, \dots$ singularities of the electron propagator in the interacting one-dimensional electron systems.

For the $\nu = \frac{2}{3}$ FQH state, we have $\nu_1 \nu_2 < 0$. In this case, we need to choose

$$\begin{aligned}\tilde{\rho}_{1k} &= \cosh(\theta) \frac{1}{\sqrt{|\nu_1|}} \rho_{1k} + \sinh(\theta) \frac{1}{\sqrt{|\nu_2|}} \rho_{2k} \\ \tilde{\rho}_{2k} &= \cosh(\theta) \frac{1}{\sqrt{|\nu_2|}} \rho_{2k} + \sinh(\theta) \frac{1}{\sqrt{|\nu_1|}} \rho_{1k} \\ \tanh(2\theta) &= 2 \frac{\sqrt{|\nu_1 \nu_2|} V_{12}}{|\nu_1| V_{11} + |\nu_2| V_{22}}\end{aligned}\tag{7.4.27}$$

to diagonalize the Hamiltonian. One can check that $\tilde{\rho}_I$ also satisfies the K-M algebra (7.4.23), but now

$$\begin{aligned}\operatorname{sgn}(\nu_1)\tilde{v}_1 &= \frac{\cosh^2(\theta)}{\cosh(2\theta)}|\nu_1|V_{11} - \frac{\sinh^2(\theta)}{\cosh(2\theta)}|\nu_2|V_{22} \\ \operatorname{sgn}(\nu_2)\tilde{v}_2 &= \frac{\cosh^2(\theta)}{\cosh(2\theta)}|\nu_2|V_{22} - \frac{\sinh^2(\theta)}{\cosh(2\theta)}|\nu_1|V_{11}\end{aligned}\quad (7.4.28)$$

Again, as long as the Hamiltonian H is positive definite, the velocities $\tilde{v}_I$ of the edge excitations always have opposite signs. The electron operator still has the form (7.4.25), with $\eta = (\Psi_1\Psi_2^\dagger)^{\nu_1}$. The propagator of ψ_n is still given by eqn (7.4.4), with

$$\begin{aligned}\gamma_{1n} &= \left[\left(n + \frac{1}{|\nu_1|} \right) \sqrt{|\nu_1|} \cosh\theta + \frac{n\nu_1}{\nu_2} \sqrt{|\nu_2|} \sinh\theta \right]^2 \\ \gamma_{2n} &= \left[\left(n + \frac{1}{|\nu_1|} \right) \sqrt{|\nu_1|} \sinh\theta + \frac{n\nu_1}{\nu_2} \sqrt{|\nu_2|} \cosh\theta \right]^2\end{aligned}\quad (7.4.29)$$

At equal space, the long-time correlation of the electron operator ψ_n is controlled by the total exponent $g_e^{(n)}$ as follows:

$$\langle \psi_n^\dagger(0, t) \psi_n(0, 0) \rangle \sim t^{-g_e^{(n)}}, \quad g_e^{(n)} = \sum_I \gamma_{In}$$

The equal-space electron correlation determines the I - V curve in the edge tunneling experiments (see Sections 3.7.4 and 4.2.2). The minimum value of the exponents $g_e \equiv \operatorname{Min}(g_e^{(n)})$ controls the scaling properties of the tunneling of electrons between two edges. For example, the tunneling conductance scales at finite temperatures as

$$\sigma \propto T^{2g_e-2}$$

Here $g_e = 3$ for the $\nu = 2/5$ state. For the $\nu = 2/3$ state, g_e depends on the interactions between the two edges.

Problem 7.4.6.

Bosonization of a one-dimensional interacting fermion system Consider a free one-dimensional fermion model

$$H_0 = \sum_k \epsilon_k : c_k^\dagger c_k :$$

The normal order is defined as $: c_k^\dagger c_k := c_k^\dagger c_k$ if $\epsilon_k > 0$ and $: c_k^\dagger c_k := -c_k c_k^\dagger$ if $\epsilon_k < 0$. Here $\epsilon_k = 0$ when $k = \pm k_F$.

1. Let $\rho_{1,k} = \sum'_{q \sim k_F} c_q^\dagger c_{q+k} :$, where the summation $\sum'_{q \sim k_F}$ is limited to the range for which both the momenta of c_{q+k} and c_q are within the range $[-\Lambda + k_F, \Lambda + k_F]$.

Let $\rho_{2,k} = \sum'_{q \sim -k_F} : c_q^\dagger c_{q+k} :$, where the summation $\sum'_{q \sim -k_F}$ is such that c_{q+k} and c_q are within the range $[-\Lambda - k_F, +\Lambda - k_F]$. Here $k < \Lambda < k_F$. We note that $\rho_{1,k}$ creates right-moving excitations near k_F , while $\rho_{2,k}$ creates left-moving ones near $-k_F$. Show that $\rho_{I,k}$ satisfies the K-M algebra (7.4.19), with $\nu_1 = 1$ and $\nu_2 = -1$. Show that $H_0 = 2\pi v_F \sum_{k>0} (\rho_{1,k} \rho_{1,-k} - \rho_{2,-k} \rho_{2,k})$. So the one-dimensional free fermion system can be bosonized.

2. We can write the electron operator as $c(x) = \psi_1(x) + \psi_2(x)$, where $\psi_1(x) = \sum_{-\Lambda+k_F}^{\Lambda+k_F} e^{ikx} c_k$ is the electron operator near k_F , and $\psi_2(x) = \sum_{-\Lambda-k_F}^{\Lambda-k_F} e^{ikx} c_k$ is the electron operator near $-k_F$. Express $c(x)$ in terms of ϕ_I , where $2\pi^{-1} \partial_x \phi_I(x) = \rho_I(x)$.
3. To verify that a one-dimensional interacting fermion system can also be bosonized, show that an interacting term $H_V = \sum_{q,k_1,k_2} V_q (c_{k_1}^\dagger c_{k_1+q})(c_{k_2}^\dagger c_{k_2+q})$, when limited near to the two Fermi points, becomes $2\pi \sum V_{IJ} \rho_{I,-k} \rho_{J,k} + \text{constant}$. Find V_{IJ} .
4. Calculate the electron Green's function $\langle c^\dagger(x,t)c(0) \rangle$ for the interacting model $H_0 + H_V$.

7.4.5 Bulk effective theory and the edge states

- The structure of the edge states can be directly determined from the bulk topological order characterized by the K -matrix.

In this section, we will directly derive the macroscopic theory of the edge excitations from the Chern–Simons effective theory of the bulk FQH states. In this approach, we do not rely on a specific construction of the FQH states. The relationship between the bulk topological orders and edge states becomes quite clear. I have to warn the reader that the calculations presented in this section are very formal. The correctness of the results is not guaranteed by the formal calculation, but by the comparison with other independent calculations, such as the one in Section 7.4.3.

To understand the relationship between the effective theory and the edge states, let us first consider the simplest FQH state of the filling fraction $\nu = 1/m$ and try to rederive the results of Section 7.4.2 from the bulk effective theory. Such an FQH state is described by the $U(1)$ Chern–Simons theory with the action

$$S = -\frac{m}{4\pi} \int a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} d^3x \quad (7.4.30)$$

Suppose that our sample has a boundary. For simplicity, we shall assume that the boundary is the x axis and that the sample covers the lower half-plane.

There is one problem with the effective action (7.4.30) for FQH liquids with boundaries. It is not invariant under gauge transformations $a_\mu \rightarrow a_\mu + \partial_\mu f$ due to the presence of the boundary, i.e. $\Delta S = -\frac{m}{4\pi} \int_{y=0} dx dt f(\partial_0 a_1 - \partial_1 a_0)$. To solve

this problem, we will restrict the gauge transformations to be zero on the boundary $f(x, y = 0, t) = 0$. Due to this restriction, some degrees of freedom of a_μ on the boundary become dynamical.

We know that the effective theory (7.4.30) is derived only for a bulk FQH state without a boundary. Here we will take eqn (7.4.30) with the restricted gauge transformation as the definition of the effective theory for an FQH state with a boundary. Such a definition is definitely self-consistent. In the following, we will show that such a definition reproduces the results obtained in Section 7.4.2.

One way to study the dynamics of gauge theory is to choose the gauge condition $a_0 = 0$ and regard the equation of motion for a_0 as a constraint. For the Chern–Simons theory, such a constraint becomes $f_{ij} = 0$. Under this constraint, we may write a_i as $a_i = \partial_i \phi$. Substituting this into eqn (7.4.30), one obtains an effective one-dimensional theory on the edge with the action (Elitzur *et al.*, 1989)

$$S_{\text{edge}} = -\frac{m}{4\pi} \int \partial_t \phi \partial_x \phi dx dt \quad (7.4.31)$$

This approach, however, has a problem. It is easy to see that a Hamiltonian associated with the action (7.4.31) is zero and the boundary excitations described by eqn (7.4.31) have zero velocity. Therefore, this action cannot be used to describe any physical edge excitations in real FQH samples. The edge excitations in FQH states always have finite velocities.

The appearance of finite velocities for edge excitations is a boundary effect. The bulk effective theory defined by eqn (7.3.18) does not contain the information about the velocities of the edge excitations. To determine the dynamics of the edge excitations from the effective theory, we must find a way to input the information about the edge velocity. The edge velocities must be treated as the external parameters that are not contained in the bulk effective theory. The problem is how to put these parameters into the theory.

Let us now note that the condition $a_0 = 0$ is not the only choice for the gauge-fixing condition. A more general gauge-fixing condition has the form

$$a_\tau = a_0 + v a_x = 0 \quad (7.4.32)$$

Here a_x is the component of the vector potential parallel to the boundary of the sample and v is a parameter that has the dimension of velocity.

It is convenient to choose new coordinates that satisfy

$$\tilde{x} = x - vt, \quad \tilde{t} = t, \quad \tilde{y} = y. \quad (7.4.33)$$

Note that the gauge potential a_μ transforms as $\partial/\partial x^\mu$ under the coordinate transformation. We find that in the new coordinates the components of the gauge field

are given by

$$\tilde{a}_{\tilde{t}} = a_t + v a_x, \quad \tilde{a}_{\tilde{x}} = a_x, \quad \tilde{a}_{\tilde{y}} = a_y. \quad (7.4.34)$$

The gauge-fixing condition becomes the one discussed before in the new coordinates. It is easy to see that the form of the Chern–Simons action is preserved under the transformation given in eqns (7.4.33) and (7.4.34):

$$S = -\frac{m}{4\pi} \int d^3x \, a_\mu \partial_\nu a_\lambda \varepsilon^{\mu\nu\lambda} = -\frac{m}{4\pi} \int d^3x \, \tilde{a}_{\tilde{\mu}} \partial_{\tilde{\nu}} \tilde{a}_{\tilde{\lambda}} \varepsilon^{\tilde{\mu}\tilde{\nu}\tilde{\lambda}}$$

Repeating the previous derivation, we find that the edge action is given by

$$S = -\frac{m}{4\pi} \int d\tilde{t} d\tilde{x} \, \partial_{\tilde{t}} \phi \partial_{\tilde{x}} \phi$$

In terms of the original physical coordinates, the above action acquires the form

$$S = -\frac{m}{4\pi} \int dt dx \, (\partial_t + v \partial_x) \phi \partial_x \phi \quad (7.4.35)$$

which is a chiral boson theory (Gross *et al.*, 1985; Floreanini and Jackiw, 1988). From the equation of motion $\partial_t \phi + v \partial_x \phi = 0$, we can see that the edge excitations described by eqn (7.4.35) have a nonzero velocity v and move only in one direction.

To obtain the quantum theory for eqn (7.4.35), we need to quantize the chiral boson theory. The quantization can be done in the momentum space:

$$S = \frac{m}{2\pi} \int dt \sum_{k>0} (ik \dot{\phi}_k \phi_{-k} - vk^2 \phi_k \phi_{-k}).$$

One can show that, if we regard ϕ_k with $k > 0$ as ‘coordinates’, then ϕ_{-k} will be proportional to the corresponding momentum $\pi_k = \partial S / \partial \dot{\phi}_k$. After identifying the ‘coordinates’ and the ‘momentum’, we can obtain the Hamiltonian and the commutation relation $[\phi_k, \pi_{k'}] = i\delta_{kk'}$, which completely define the quantum system. If we introduce $\rho = \frac{1}{2\pi} \partial_x \phi$, then we find that the quantum system is described by

$$[\rho_k, \rho_{k'}] = \frac{k\delta_{k+k'}}{2\pi m}, \quad H = 2\pi m v \sum_{k>0} \rho_k^\dagger \rho_k \quad (7.4.36)$$

The velocity v of the edge excitations enters our theory through the gauge-fixing condition. Notice that, under the restricted gauge transformations, the gauge-fixing conditions (7.4.32) with different v cannot be transformed into each other. They are physically inequivalent. This agrees with our assumption that v in

the gauge-fixing condition is physical and actually determines the velocity of the edge excitations.

The Hamiltonian (7.4.36) is bounded from below only when $vm < 0$. The consistency of our theory requires v and m to have opposite signs. Therefore, the sign of the velocity (the chirality) of the edge excitations is determined by the sign of the coefficient in front of the Chern–Simons terms.

The above results can be easily generalized to the generic FQH states described by eqn (7.3.18) because the matrix K can be diagonalized. The resulting effective edge theory has the form

$$S_{\text{edge}} = \frac{1}{4\pi} \int dt dx [K_{IJ} \partial_t \phi_I \partial_x \phi_J - V_{IJ} \partial_x \phi_I \partial_x \phi_J] \quad (7.4.37)$$

The Hamiltonian is given by

$$H_{\text{edge}} = \frac{1}{4\pi} \int dt dx V_{IJ} \partial_x \phi_I \partial_x \phi_J$$

Therefore, V must be a positive-definite matrix. Using this result, one can show that a positive eigenvalue of K corresponds to a left-moving branch and a negative eigenvalue corresponds to a right-moving one.

The effective theory of the $\nu = 2/5$ FQH state is given by $K = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$. As K has two positive eigenvalues, the edge excitations of the $\nu = 2/5$ FQH state have two branches moving in the same direction. The $\nu = 1 - \frac{1}{n}$ FQH state is described by the effective theory with $K = \begin{pmatrix} 1 & 0 \\ 0 & -n \end{pmatrix}$. The two eigenvalues of K now have opposite signs; hence, the two branches of the edge excitations move in opposite directions.

Problem 7.4.7.

Show that, after quantization, the system (7.4.35) is described by eqn (7.4.36).

7.4.6 Charged excitations and the electron propagator

• Generic electron/quasiparticle operators on a generic FQH edge.

We have studied the dynamics of edge excitations in a few simple FQH liquids. We found that the low-lying edge excitations are described by a free phonon theory. In this section, we will concentrate on the generic charge excitations. In particular, we will calculate the propagators of the electrons and the quasiparticles for the most general (abelian) FQH states. The key point again is to write the electron or the quasiparticle operators in terms of the phonon operator ρ_I . Once we have done so, the propagators can be easily calculated because the phonons are free (at low energies and long wavelength).

We know that, for the FQH state described by eqn (7.3.18), the edge states are described by the action (7.4.37). The Hilbert space of the edge excitations forms a representation of the K–M algebra

$$[\rho_{Ik}, \rho_{Jk'}] = (K^{-1})_{IJ} \frac{1}{2\pi} k \delta_{k+k'} \\ k, k' = \text{integer} \times \frac{2\pi}{L} \quad (7.4.38)$$

where $\rho_I = \frac{1}{2\pi} \partial_x \phi_I$ is the edge density of the I th condensate in the FQH state, $I, J = 1, \dots, \kappa$, and κ is the dimension of K . The electron density on the edge is given by

$$\rho_e = -eq_I \rho_I$$

The dynamics of the edge excitations are described by the Hamiltonian

$$H = 2\pi \sum_{IJ} V_{IJ} \rho_{I,-k} \rho_{J,k} \quad (7.4.39)$$

where V_{IJ} is a positive-definite matrix.

Let us first try to write down the quasiparticle operator Ψ_l on the edge which creates a quasiparticle labeled by l_I . We know that inserting the quasiparticle on the edge will cause a change $\delta\rho_I$ in the edge density of the I th condensate (see eqn (7.3.22)). Here $\delta\rho_I$ satisfies

$$\int dx \delta\rho_I = (l^\top K^{-1})_I.$$

Because Ψ_l is a local operator that only causes a local change in the density, we have

$$[\rho_I(x), \Psi_l(x')] = l_J (K^{-1})_{JI} \delta(x - x') \Psi_l(x') \quad (7.4.40)$$

Using the Kac–Moody algebra (7.4.38), one can show that the quasiparticle operators that satisfy eqn (7.4.40) are given by

$$\Psi_l \propto e^{i\phi_I l_I} \quad (7.4.41)$$

The charge of the quasiparticle Ψ_l is determined from the commutator $[\rho_e, \Psi_l]$, and is found to be

$$Q_l = -eq^\top K^{-1} l \quad (7.4.42)$$

From eqns (7.3.24) and (7.4.41), we see that the electron operator can be written as

$$\Psi_{e,L} \propto e^{i \sum_I l_I \phi_I}, \quad l_I = K_{IJ} L_J, \quad q_I L_I = 1 \quad (7.4.43)$$

The above operators carry a unit charge, as one can see from eqn (7.4.42). The commutation of $\Psi_{e,L}$ can be calculated as follows:

$$\begin{aligned}\Psi_{e,L}(x)\Psi_{e,L}(x') &= (-)^\lambda \Psi_{e,L}(x')\Psi_{e,L}(x) \\ \lambda &= l_I(K^{-1})_{IJ}l_J = L_I K_{IJ} L_J\end{aligned}\quad (7.4.44)$$

In the hierarchical basis, $K_{II}|_{I=1} = \text{odd}$, $K_{II}|_{I>1} = \text{even}$, $\mathbf{q}^\top = (1, 0, \dots, 0)$, and $L_1 = 1$. Using these conditions, we can show that $(-)^\lambda = -1$. The electron operators defined in eqn (7.4.43) are indeed fermion operators.

As all of the operators $\Psi_{e,L}$ for different choices of L_I carry a unit charge and are fermionic, each $\Psi_{e,L}$ can be a candidate for the electron operators. In general, the true electron operator is the following superposition of the $\Psi_{e,L}$ s:

$$\Psi_e = \sum_L C_L \Psi_{e,L}$$

Here, when we say that there are many different electron operators on the edge, we really mean that the true physical electron operator is a superposition of these operators.

Using the K–M algebra (7.4.38) and the Hamiltonian (7.4.39), we can calculate the propagators of a generic quasiparticle operator

$$\Psi_I \propto e^{il_I \phi_I}$$

(which includes the electron operators for suitable choices of I). First, we note that the transformation

$$\rho_I \rightarrow U_{IJ} \tilde{\rho}_J$$

changes (V, K) to $(\tilde{V}, \tilde{K}) = (U^\top V U, U^\top K U)$. By choosing a suitable U , K and V can be simultaneously diagonalized;⁴⁹ in terms of $\tilde{\rho}_I$, eqns (7.4.38) and (7.4.39) become

$$\begin{aligned}[\tilde{\rho}_{Ik}, \tilde{\rho}_{Jk'}] &= \sigma_I \delta_{IJ} \frac{1}{2\pi} k \delta_{k+k'} \\ H &= 2\pi \sum_{I, k>0} |v_I| \tilde{\rho}_{I,-k} \tilde{\rho}_{I,k}\end{aligned}\quad (7.4.45)$$

where $\sigma_I = \pm 1$ is the sign of the eigenvalues of K . The velocity of the edge excitations created by $\tilde{\rho}_I$ is given by $v_I = \sigma_I |v_I|$.

⁴⁹ As V is a positive-definite symmetric matrix, we can find U_1 which transforms $V \rightarrow V_1 = U_1^\top V U_1 = 1$ and $K \rightarrow K_1 = U_1^\top K U_1$. Note that K_1 is a symmetric matrix whose eigenvalues have the same sign as the eigenvalues of K (although their absolute values may differ). Then we make an orthogonal transformation to diagonalize K_1 , namely $(K_1)_{IJ} \rightarrow (K_2)_{IJ} = \sigma_I |v_I|^{-1} \delta_{IJ}$. Under the orthogonal transformation, $V_1 \rightarrow V_2 = 1$ is not changed and remains diagonal. Finally, we use a diagonal matrix to scale K_2 to $(K_3)_{IJ} = \sigma_I \delta_{IJ}$. This changes V_2 to $V_3 = |v_I| \delta_{IJ}$. The new (V_3, K_3) lead to eqn (7.4.45).

In terms of $\tilde{\rho}_I$, the operator Ψ_I has the form

$$\Psi_I \propto e^{i \sum_I \tilde{l}_I \tilde{\phi}_I}, \quad \tilde{l}_J = l_I U_{IJ} \quad (7.4.46)$$

From eqns (7.4.45) and (7.4.46), we see that the propagator of Ψ_I has the following general form:

$$\langle \Psi_I^\dagger(x, t) \Psi_I(0) \rangle \propto e^{i l_I k_I x} \prod_I (x - v_I t + i \sigma_I \delta)^{-\gamma_I}, \quad \gamma_I = \tilde{l}_I^2 \quad (7.4.47)$$

where $v_I = \sigma_I |v_I|$ are the velocities of the edge excitations. The total exponent from the right movers minus the total exponent from the left movers satisfies the sum rule⁵⁰

$$\sum_I \sigma_I \gamma_I \equiv \lambda_I = l^\top K^{-1} l \quad (7.4.48)$$

We see that the difference λ_I is independent of the edge interaction V and is a topological quantum number. From eqns (7.4.44) and (7.4.48), we see that the difference is directly related to the statistics of Ψ_I . If Ψ_I represents the electron operator (or other fermionic operators), then λ_I will be an odd integer. From eqn (7.4.47), we also see that the operator Ψ_I creates an excitation with momentum near to $\sum_I l_I k_I$.

Problem 7.4.8.

Prove that Ψ_I in eqn (7.4.41) satisfies eqn (7.4.40).

7.4.7 Phenomenological consequences of chiral Luttinger liquids

- Three classes of edge states.
- The effects of long-range interactions and impurity scattering.

In the last four sections, we have shown that the electrons at the edges of an FQH liquid form a chiral Luttinger liquid. As one of the characteristic properties of chiral Luttinger liquids, the electron and the quasiparticle propagators obtain anomalous exponents as follows:

$$\langle \Psi_c^\dagger(t, x=0) \Psi_e(0) \rangle \sim t^{-g_e}, \quad \langle \Psi_q^\dagger(t, x=0) \Psi_q(0) \rangle \sim t^{-g_q}$$

These anomalous exponents can be directly measured by tunneling experiments between the FQH edges.

⁵⁰ We have used

$$\lambda_I = \sum_{I, J, I'} l_I U_{I'I} \sigma_I (U^\top)_{IJ} l_J, \quad (K^{-1})_{IJ} = U_{I'I'} \sigma_{I'} \delta_{I'J'} (U^\top)_{J'I}.$$

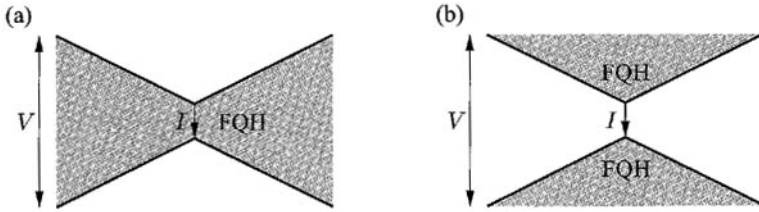


FIG. 7.18. (a) Quasiparticle tunneling between two edges of the same FQH state. (b) Electron tunneling between two edges of the different FQH states.

The following two situations need to be considered separately: (i) two edges of FQH liquids are separated by an insulator; and (ii) two edges are separated by an FQH liquid. In case (i), only an electron can tunnel between the edges (see Fig. 7.18(b)); in case (ii), the quasiparticles supported by the FQH liquid that separates the two edges can tunnel (see Fig. 7.18(a)). The tunneling operator A is given by $A \propto \Psi_{e1} \Psi_{e2}^\dagger$ for case (i), where Ψ_{e1} and Ψ_{e2} are the electron operators on the two edges. The tunneling operator A has the form $A \propto \Psi_{q1} \Psi_{q2}^\dagger$ for case (ii), where Ψ_{q1} and Ψ_{q2} are the quasiparticle operators on the two edges. The physical properties of the tunneling can be calculated from the correlation of the tunneling operator, which in turn can be expressed as a product of the electron or quasiparticle propagators on the two edges.

Let $g = g_e$ for case (i) and $g = g_q$ for case (ii). Then, at zero temperature, the anomalous exponents of the electron and quasiparticle operators lead to $\langle A(t)A(0) \rangle \propto |t|^{-2g}$, which results in a nonlinear tunneling I - V curve (see Section 3.7.4)

$$I \propto V^{2g-1}$$

The noise spectrum of the tunneling current also contains a singularity at the frequency $f = QV/h$, i.e.

$$S(f) \sim \left| f - \frac{QV}{h} \right|^{2g-1} \quad (7.4.49)$$

where V is the voltage difference between the two edges and Q is the electric charge of the tunneling electron or the tunneling quasiparticle. At a finite temperature T , the zero-bias conductance also has the following power law dependence:

$$\sigma \propto T^{2g-2}$$

We see that the anomalous exponents can be easily measured by the tunneling experiments. The noise spectrum further reveals the charges of the tunneling particles.

The exponents g_e and g_q are calculated in Section 7.4.6. To summarize the results in a simple way, it is convenient to divide the FQH edges into the following two classes: (A) all edge excitations move in one direction; and (B) edge excitations propagate in both directions.

For the class A edges, the exponents g_e and g_q are directly related to the statistics of the electrons and quasiparticles. In this case, g_e and g_q are universal and independent of the details of the electron interactions and the edge potentials. Table 7.3 lists some FQH states which support the class A edge states, as well as the corresponding values of the exponents $g_{e,q}$ and the electric charge of the associated particles (note that only the minimum values of g_e and g_q are listed).

TABLE 7.3. (g_e, g_q, Q_q) for some FQH states with the class A edge states.

FQH states	$\nu = 1/m$	$\nu = 2/5$	$\nu = p/(pq+1)$	$(331)_{\nu=1/2}$	$(332)_{\nu=2/5}$
g_e	m	3	$q+1$	3	3
g_q	$1/m$	$2/5$	$p/(pq+1)$	$3/8$	$2/5$
Charge Q_q/e	$1/m$	$2/5$	$p/(pq+1)$	$1/4$	$2/5$

Let us discuss how we obtain the above results for the hierarchical states with the filling fractions $\nu = \frac{p}{pq+1}$ ($q = \text{even}$). These states include the $\nu = 2/5, 3/7, 2/9, \dots$ FQH states. The hierarchical states with $\nu = \frac{p}{pq+1}$ are described by the $p \times p$ matrices $K = 1 + qC$ in the basis where $\mathbf{q}^T = (1, 1, \dots, 1)$. Here C is the pseudo-identity matrix, namely $C_{IJ} = 1$, $I, J = 1, \dots, p$, and $K^{-1} = 1 - \frac{q}{pq+1}C$. Because all of the edge excitations move in the same direction, we have

$$\langle \Psi_l^\dagger(x=0, t) \Psi_l(0) \rangle \propto t^{-\lambda_l}$$

where λ_l is given by eqn (7.4.48). The fundamental quasiparticle is given by $\mathbf{l}^T = (1, 0, \dots, 0)$ and carries the charge $\frac{1}{pq+1}$. The exponent in its propagator is $\lambda_l = 1 - \frac{q}{pq+1}$. The quasiparticle with the smallest exponent is given by $\mathbf{l}^T = (1, \dots, 1)$ and carries the charge $\frac{p}{pq+1}$. The exponent is $\lambda_l = \frac{p}{pq+1}$, which is less than $1 - \frac{q}{pq+1}$ (note that we have $q \geq 2$ and $p \geq 1$). Such a quasiparticle (with the charge $\frac{p}{pq+1}$) dominates the tunneling between two edges of the *same* FQH fluid at low energies (see Section 7.4.7).

The electron operators are given by $\Psi_{e,L} = \psi_l$, with $\mathbf{l}$ satisfying $\sum_I l_I = pq+1$. The exponent in the propagator is given by $\lambda_l = \sum_I l_I^2 - q(pq+1)$. The electron operator with a minimum exponent in its propagator is given by $\mathbf{l}^T = (q, \dots, q, q+1)$. The value of the minimum exponent is $\lambda_l = q+1$. Such an electron operator dominates the tunneling between edges of two *different* FQH fluids at low energies.

For the class B edges, g_e and g_q are not universal. Their values depend on the electron interactions and the edge potentials.

Problem 7.4.9.

Derive eqn (7.4.49) using the correlation of the tunneling operator $\langle A(t)A(0) \rangle$ at a finite voltage V .

TOPOLOGICAL AND QUANTUM ORDER—BEYOND LANDAU'S THEORIES

In Chapters 3, 4, and 5, we discussed several interacting boson/fermion systems in detail. These simple models illustrate Landau's symmetry-breaking theory (plus RG theory) and Landau's Fermi liquid theory (plus perturbation theory). The two Landau theories explain the behavior of many condensed matter systems and form the foundation of traditional many-body theory. The discussions in the three chapters only scratch the surface of Landau's theories and their rich applications. Readers who want to learn more about Landau's theories (and RG theory and perturbation theory) may find the books by Abrikosov *et al.* (1975), Ma (1976), Mahan (1990), Negele and Orland (1998), and Chaikin and Lubensky (2000) useful.

Landau's theories are very successful and for a long time people could not find any condensed matter systems that could not be described by Landau's theories. For fifty years, Landau's theories dominated many-body physics and essentially defined the paradigm of many-body physics. After so many years, it has become a common belief that we have figured out all of the important concepts and understood the essential properties of all forms of matter. Many-body theory has reached its end and is a more or less complete theory. The only thing to be done is to apply Landau's theories (plus the renormalization group picture) to all different kinds of systems.

From this perspective, we can understand the importance of the fractional quantum Hall (FQH) effect discovered by Tsui *et al.* (1982). The FQH effect opened up a new chapter in condensed matter physics. As we have seen in the last chapter, FQH liquids cannot be described by Fermi liquid theory. Different FQH states have the same symmetry and cannot be described by Landau's symmetry-breaking theory. Thus, FQH states are completely beyond the two Landau theories. The existence of FQH liquids indicates that there is a new world beyond the paradigm of Landau's theories. Recent studies suggest that the new paradigm is much richer than the paradigm of Landau's theories. Chapters 7 to 10 of this book are devoted to the new paradigm beyond Landau's theories.

To take a glimpse at the new paradigm of condensed matter physics, we are going to study quantum rotor systems, hard-core boson systems, and quantum spin systems, in addition to the FQH states. These systems are all strongly-correlated

many-body systems. Usually, the strong correlation in these systems will lead to long-range orders and symmetry breaking. However, we have already discussed long-range order and symmetry breaking, using a boson superfluid and a fermion SDW state as examples. So, to study the world beyond Landau's theories, we concentrate on quantum liquid states (such as the FQH states) that cannot be described by long-range order and symmetry breaking.

These quantum liquid states represent new states of matter that contain a completely new kind of order—topological/quantum order. I will show that the new order may have a deep impact on our understanding of the quantum phase and the quantum phase transition, as well as gapless excitations in the quantum phase. In particular, topological/quantum order might provide an origin for light and electrons (as well as other gauge bosons and fermions) in nature.

In this chapter, we will give a general discussion of topological/quantum order to paint a larger picture. We will use FQH states and Fermi liquid states as examples to discuss some basic issues in topological/quantum order. The problems and issues in topological/quantum order are very different from those in traditional many-body physics. It is very important to understand what the problems are before going into any detailed calculations.

8.1 States of matter and the concept of order

- Matter can have many different states (or different phases). The concept of order is introduced to characterize different internal structures in different states of matter.
- We used to believe that different orders are characterized by their different symmetries.

At sufficiently high temperatures, all matter is in the form of a gas. Gas is one of the simplest states. The motion of an atom in a gas hardly depends on the positions and motion of other molecules. Thus, gases are weakly-correlated systems which contain no internal structure. However, as the temperature is lowered the motion of the atoms becomes more and more correlated. Eventually, the atoms form a very regular pattern and a crystal order is developed. In a crystal, an individual atom can hardly move by itself. Excitations in a crystal always correspond to the collective motion of many atoms (which are called phonons). A crystal is an example of a strongly-correlated state.

With the development of low-temperature technology in around 1900, physicists discovered many new states of matter (such as superconductors and superfluids). These different states have different internal structures, which are called different kinds of orders. The precise definition of order involves phase transition. Two states of a many-body system have the same order if we can smoothly change

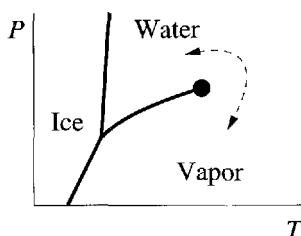


FIG. 8.1. The phase diagram of water.

one state into the other (by smoothly changing the Hamiltonian) without encountering a phase transition (i.e. without encountering a singularity in the free energy). If there is no way to change one state into the other without a phase transition, then the two states will have different orders. We note that our definition of order is a definition of an equivalent class. Two states that can be connected without a phase transition are defined to be equivalent. The equivalent class defined in this way is called the universality class. Two states with different orders can also be said to be two states belonging to different universality classes. According to our definition, water and ice have different orders, while water and vapor have the same order (see Fig. 8.1).

After discovering so many different kinds of order, a general theory is needed to gain a deeper understanding of the states of matter. In particular, we like to understand what makes two orders really different, so that we cannot change one order into the other without encountering a phase transition. The key step in developing the general theory for order and the associated phase and phase transition is the realization that orders are associated with symmetries (or rather, the breaking of symmetries). We find that, when two states have different symmetries, then we cannot change one into the other without encountering a singularity in the free energy (i.e. without encountering a phase transition). Based on the relationship between orders and symmetries, Landau developed a general theory of orders and transitions between different orders (Ginzburg and Landau, 1950; Landau and Lifschitz, 1958). Landau's theory is very successful. Using Landau's theory and the related group theory for symmetries, we can classify all of the 230 different kinds of crystals that can exist in three dimensions. By determining how symmetry changes across a continuous phase transition, we can obtain the critical properties of the phase transition. The symmetry breaking also provides the origin of many gapless excitations, such as phonons, spin waves, etc., which determine the low-energy properties of many systems (Nambu, 1960; Goldstone, 1961). Many of the properties of those excitations, including their gaplessness, are directly determined by the symmetry. Introducing order parameters associated with symmetries,

Ginzburg and Landau developed Ginzburg–Landau theory, which became the standard theory for phase and phase transition. As Landau’s symmetry-breaking theory has such a broad and fundamental impact on our understanding of matter, it became a corner-stone of condensed matter theory. The picture painted by Landau’s theory is so satisfactory that one starts to have a feeling that we understand, at least in principle, all kinds of orders that matter can have.

8.2 Topological order in fractional quantum Hall states

- The FQH state opened up a new chapter in condensed matter physics, because it was the second state to be discovered by experimentalists that could not be characterized by symmetry breaking and (local) order parameters.
- The FQH states contain a new kind of order—topological order.

However, nature never ceases to surprise us. With advances in semiconductor technology, physicists learnt how to confine electrons on an interface between two different semiconductors, and hence made a two-dimensional electron gas (2DEG). In 1982, Tsui *et al.* (1982) put a 2DEG under strong magnetic fields and discovered a new state of matter—the FQH liquid (Laughlin, 1983). As the temperatures are low and the interaction between the electrons is strong, the FQH state is a strongly-correlated state. However, such a strongly-correlated state is not a crystal, as people had originally expected. It turns out that the strong quantum fluctuations of electrons, due to their very small mass, prevent the formation of a crystal. Thus, the FQH state is a quantum liquid. (A crystal can melt in two ways, namely by thermal fluctuations as we raise temperatures, which leads to an ordinary liquid, or by quantum fluctuations as we reduce the mass of the particles, which leads to a quantum liquid.)

As we have seen in the last chapter, quantum Hall liquids have many amazing properties. A quantum Hall liquid is more ‘rigid’ than a solid (a crystal), in the sense that a quantum Hall liquid cannot be compressed. Thus, a quantum Hall liquid has a fixed and well-defined density. When we measure the electron density in terms of the filling fraction, defined by

$$\nu = \frac{\text{density of electron}}{\text{density of magnetic flux quanta}},$$

we find that all of the discovered quantum Hall states have densities such that the filling fractions are given exactly by some rational numbers, such as $\nu = 1, 1/3, 2/3, 2/5, \dots$. Knowing that FQH liquids exist only at certain magical filling fractions, one cannot help but guess that FQH liquids should have some internal orders or ‘patterns’. Different magical filling fractions should be due to these different internal ‘patterns’. However, the hypothesis of internal ‘patterns’ appears

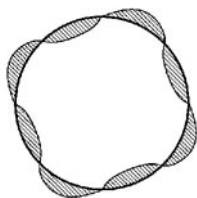


FIG. 8.2. A particle wave on a circle has a quantized wavelength.

to have one difficulty—the FQH states are liquids, and how can liquids have any internal ‘patterns’?

To gain some intuitive understanding of the internal order in FQH states, let us try to visualize the quantum motion of electrons in an FQH state. We know that a particle also behaves like a wave, according to quantum physics. Let us first consider a particle moving on a circle with momentum p . Such a particle corresponds to a wave with wavelength $\lambda = h/p$, where h is the Planck constant. Only waves that can fit into the circle are allowed (i.e. the circle must contain an integer number of wavelengths) (see Fig. 8.2). Thus, due to quantum physics, the motion of a particle on a circle is highly restricted (or quantized), and only certain discrete values of momentum are allowed. Such a quantization condition can be viewed in a more pictorial way. We may say that the particle dances around the circle in steps, with a step length given by the wavelength. The quantization condition requires that the particle always takes an integer number of steps to go around the circle.

Now let us consider a single electron in a magnetic field. Under the influence of the magnetic field, the electron always moves along circles (which are called cyclotron motions). In quantum physics, only certain discrete cyclotron motions are allowed due to the wave property of the particle. The quantization condition is such that the circular orbit of an allowed cyclotron motion contains an integer number of wavelengths. We may say that an electron always takes an integer number of steps to go around the circle. If the electron takes n steps around the circle, then we say that the electron is in the n th Landau level. The electrons in the first Landau level have the lowest energy, and the electron will stay in the first Landau level at low temperatures.

When we have many electrons to form a 2DEG, electrons not only do their own cyclotron motion in the first Landau level, but they also go around each other and exchange places. These additional motions are also subject to the quantization condition. For example, an electron must take integer steps to go around another electron. As electrons are fermions, exchanging two electrons introduces a minus sign into the wave function. Also, exchanging two electrons is equivalent to moving one electron half-way around the other electron. Thus, an electron must take

half-integer steps to go half-way around another electron. (The half-integer steps introduce a minus sign into the electron wave function.) In other words, an electron must take an odd number of steps to go around another electron. Electrons in an FQH state not only move in a way that satisfies the quantization condition, but they also try to stay away from each other as much as possible, due to the strong Coulomb repulsion and the Fermi statistics between electrons. This means that an electron tries to take more steps to go around another electron, if possible.

Now we see that, despite the absence of crystal order, the quantum motions of electrons in an FQH state are highly organized. All of the electrons in an FQH state dance collectively, following strict dancing rules.

1. All of the electrons do their own cyclotron motion in the first Landau level, i.e. they take one step to go around the circle.
2. An electron always takes an odd number of steps to go around another electron.
3. Electrons try to stay away from each other, i.e. they try to take as many steps as possible to go around another electron.

If every electron follows these strict dancing rules, then only one unique global dancing pattern is allowed. Such a dancing pattern describes the internal quantum motion in the FQH state. It is this global dancing pattern that corresponds to the internal order in the FQH state. Different FQH states are distinguished by their different dancing patterns.

A more precise mathematical description of the quantum motion of electrons outlined above is given by the famous Laughlin wave function (Laughlin, 1983)

$$\Psi_m = \left[\prod (z_i - z_j)^m \right] e^{-\frac{1}{4l_B^2} \sum |z_i|^2}$$

where m is an odd integer and $z_j = x_j + iy_j$ is the coordinate of the j th electron. Such a wave function describes a filling fraction $\nu = 1/m$ FQH state. We see that the wave function vanishes as $z_i \rightarrow z_j$, so that the electrons do not like to stay close to each other. Also, the wave function changes its phase by $2\pi m$ as we move one electron around another. Thus, an electron always takes m steps to go around another electron in the Laughlin state.

We would like to stress that the internal orders (i.e. the dancing patterns) of FQH liquids are very different from the internal orders in other correlated systems, such as crystals, superfluids, etc. The internal orders in the latter systems can be described by order parameters associated with broken symmetries. As a result, the ordered states can be described by the Ginzburg–Landau effective theory. The internal order in FQH liquids is a new kind of ordering which cannot be described

by long-range orders associated with broken symmetries.⁵¹ In 1989, the concept of ‘topological order’ was introduced to describe this new kind of ordering in FQH liquids (Wen, 1990, 1995).

We would like to point out that topological orders are general properties of any states at zero temperature with a finite energy gap. Non-trivial topological orders not only appear in FQH liquids, but they also appear in spin liquids at zero temperature. In fact, the concept of topological order was first introduced (Wen, 1990) in a study of chiral spin liquids (Kalmeyer and Laughlin, 1987; Khveshchenko and Wiegmann, 1989; Wen *et al.*, 1989). In addition to chiral spin liquids, non-trivial topological orders were also found in anyon superfluids (Chen *et al.*, 1989; Fetter *et al.*, 1989; Wen and Zee, 1991) and short-ranged resonating valence bound states for spin systems (Kivelson *et al.*, 1987; Rokhsar and Kivelson, 1988; Read and Chakraborty, 1989; Read and Sachdev, 1991; Wen, 1991a). The FQH liquid is not even the first experimentally observed state with non-trivial topological orders. That honor goes to the superconducting state discovered in 1911 (Onnes, 1911; Bardeen *et al.*, 1957). In contrast to a common point of view, a superconducting state, with dynamical electromagnetic interactions, cannot be characterized by broken symmetries. It has neither long-range orders nor local order parameters. A superconducting state contains non-trivial topological orders. It is fundamentally different from a superfluid state (Coleman and Weinberg, 1973; Halperin *et al.*, 1974; Fradkin and Shenker, 1979).

It is instructive to compare FQH liquids with crystals. FQH liquids are similar to crystals in the sense that they both contain rich internal patterns (or internal orders). The main difference is that the patterns in the crystals are static, related to the positions of atoms, while the patterns in QH liquids are ‘dynamic’, associated with the ways that electrons ‘dance’ around each other. However, many of the same questions for crystal orders can also be asked and should be addressed for topological orders. We know that crystal orders can be characterized and classified by symmetries. Thus, one important question is how do we characterize and classify the topological orders? We also know that crystal orders can be measured by X-ray diffraction. The second important question is how do we experimentally measure the topological orders?

In the following, we are going to discuss topological orders in FQH states in more detail. It turns out that FQH states are quite typical topologically-ordered states. Many other topologically-ordered states share many similar properties with FQH states.

8.2.1 Characterization of topological orders

- Any new concepts in physics must be introduced (or defined) via quantities that can be measured by experiments. To define a physical quantity or concept is to design an experiment.

⁵¹ Although it was suggested that the internal structures of Laughlin states can be characterized by an ‘off-diagonal long-range order’ (Girvin and MacDonald, 1987), the operator that has long-range order itself is not a local operator. For local operators, there is no long-range order and there are no symmetry-breaking Laughlin states.

- The concept of topological order is (partially) defined by ground-state degeneracy, which is robust against *any* perturbations that can break all of the symmetries.

In the above, the concept of topological order (the dancing pattern) is introduced through the ground-state wave function. This is not quite correct because the ground-state wave function is not universal. To establish a new concept, such as topological order, one needs to find physical characterizations or measurements of topological orders. In other words, one needs to find universal quantum numbers that are robust against any perturbation, such as changes in interactions, effective mass, etc., but which can take different values for different classes of FQH liquids. The existence of such quantum numbers implies the existence of topological orders.

One way to show the existence of topological orders in FQH liquids is to study their ground-state degeneracies (in the thermodynamical limit). FQH liquids have a very special property. Their ground-state degeneracy depends on the topology of space (Haldane, 1983; Haldane and Rezayi, 1985). For example, the $\nu = \frac{1}{q}$ Laughlin state has q^g degenerate ground states on a Riemann surface of genus g . The ground-state degeneracy in FQH liquids is *not* a consequence of the symmetry of the Hamiltonian. The ground-state degeneracy is robust against arbitrary perturbations (even impurities that break all of the symmetries in the Hamiltonian) (Wen and Niu, 1990). The robustness of the ground-state degeneracy indicates that the internal structures that give rise to ground-state degeneracy are universal and robust, hence demonstrating the existence of universal internal structures—topological orders.

To understand the topological degeneracy of FQH ground states, we consider a $\nu = 1/m$ Laughlin state on a torus. We will use two methods to calculate the ground-state degeneracy. In the first method, we consider the following tunneling process. We first create a quasiparticle–quasihole pair. Then we bring the quasiparticle all the way around the torus. Finally, we annihilate the quasiparticle–quasihole pair and go back to the ground state. Such a tunneling process produces an operator that maps ground states to ground states. Such an operator is denoted by U_x if the quasiparticle goes around the torus in the x direction, and U_y if it goes around the torus in the y direction (see Fig. 8.3(a,b)). Then, the four tunnelings in the x , y , $-x$, and $-y$ directions generate $U_y^{-1}U_x^{-1}U_yU_x$ (see Fig. 8.3(c)). We note that the path of the above four tunnelings can be deformed into two linked loops (see Fig. 8.3(d)). The two linked loops correspond to moving one quasiparticle around the other. It gives rise to a phase $e^{2i\theta}$, where θ is the statistical angle of the quasiparticle. For the $1/m$ Laughlin state, we have $\theta = \pi/m$. Therefore, we have

$$U_y^{-1}U_x^{-1}U_yU_x = e^{i2\pi/m}$$

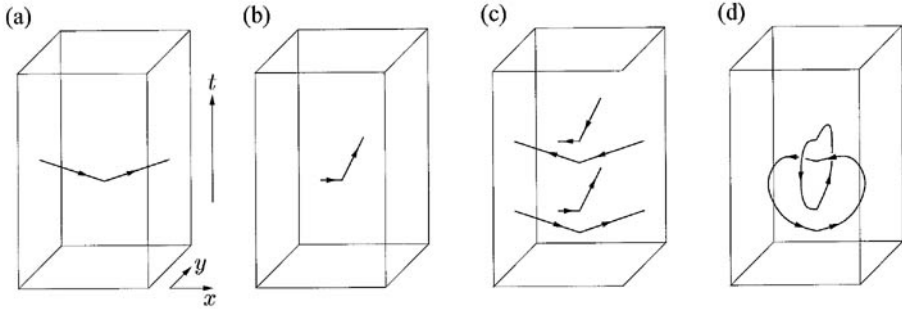


FIG. 8.3. (a) Tunneling in the x direction generates U_x . (b) Tunneling in the y direction generates U_y . (c) The four tunnelings in the x , y , $-x$, and $-y$ directions generate $U_y^{-1}U_x^{-1}U_yU_x$. (d) The above four tunnelings can be deformed into two linked loops.

As $U_{x,y}$ acts within the ground states, the ground states form the representation of the above algebra. The algebra has only one m -dimensional irreducible representation. Thus, the $1/m$ Laughlin state has $m \times$ integer number of degenerate ground states. This approach allows us to see the direct connection between the quasiparticle statistics and the ground-state degeneracy.

The second way to calculate the ground-state degeneracy is to use the effective theory (7.3.11). The degenerate ground states arise from the following collective fluctuations:

$$a_i(t, x, y) = \theta_i(t)/L, \quad i = x, y \quad (8.2.1)$$

where L is the size of the torus in the x and y directions. All of the other fluctuations generate a nonzero ‘magnetic’ field $b = f_{xy}$ and have a finite energy gap, as one can see from the classical equation of motion. The following Lagrangian describing the dynamics of the collective excitations in eqn (8.2.1) can be obtained by substituting eqn (8.2.1) into eqn (7.3.11):

$$L = -\frac{m}{4\pi}(\dot{\theta}_x\theta_y - \dot{\theta}_y\theta_x) + \frac{1}{2g_1}\dot{\theta}_i^2. \quad (8.2.2)$$

Since the charge of a_μ is quantized as an integer, the gauge transformation $U(x, y)$ that acts on the quasiparticle field, $\psi_q \rightarrow U\psi_q$, must be a periodic function on the torus. Thus, the gauge transformation must have the form $U(x, y) = \exp(2\pi i(\frac{nx}{L} + \frac{my}{L}))$, where n and m are integers. As the a_μ charge of ψ_q is 1, such a gauge transformation changes the gauge field a_i to $a'_i = a_i - iU^{-1}\partial_i U$ as follows:

$$(a'_x, a'_y) = (a_x + \frac{2\pi n}{L}, a_y + \frac{2\pi m}{L}) \quad (8.2.3)$$

Equation (8.2.3) implies that (θ_x, θ_y) and $(\theta_x + 2\pi n, \theta_y + 2\pi m)$ are gauge equivalent and should be identified. The gauge-inequivalent configurations are given by

a point on a torus $0 < \theta_i < 2\pi$. As a result, the Lagrangian (8.2.2) describes a particle with unit charge moving on a torus parametrized by (θ_x, θ_y) .

The first term in eqn (8.2.2) indicates that there is a uniform ‘magnetic’ field $B = m/2\pi$ on the torus. The total flux passing through the torus is equal to $2\pi \times m$. The Hamiltonian of eqn (8.2.2) is given by

$$H = \frac{g_1}{2} [-(\partial_{\theta_x} - iA_{\theta_x})^2 - (\partial_{\theta_y} - iA_{\theta_y})^2]. \quad (8.2.4)$$

The energy eigenstates of eqn (8.2.4) form Landau levels. The gap between the Landau levels is of order g_1 , which is independent of the size of the system. The number of states in the first Landau level is equal to the number of flux quanta passing through the torus, which is m in our case. Thus, the ground-state degeneracy of the $1/m$ Laughlin state is m .

To understand the robustness of the ground-state degeneracy, let us add an arbitrary perturbation to the electron Hamiltonian. Such a perturbation will cause a change in the effective Lagrangian, $\delta\mathcal{L}(a_\mu)$. The key here is that $\delta\mathcal{L}(a_\mu)$ only depends on a_μ through its field strength. In other words, $\delta\mathcal{L}$ is a function of e and b . As the collective fluctuation in eqn (8.2.1) is a pure gauge locally, e and b , and hence $\delta\mathcal{L}$, do not depend on θ_i . The $\delta\mathcal{L}(a_\mu)$ cannot generate any potential terms $V(\theta_i)$ in the effective theory of θ_i . It can only generate terms that only depend on $\dot{\theta}_i$. Such a correction renormalizes the value of g_1 . It cannot lift the degeneracy.

More general FQH states are described by eqn (7.3.18). When K is diagonal, the above result implies that the ground-state degeneracy is $\det(K)$. It turns out that, for a generic K , the ground-state degeneracy is also given by $\det(K)$. On a Riemann surface of genus g , the ground-state degeneracy becomes $(\det(K))^g$.

We see that in a compact space the low-energy physics of FQH liquids are very unique. There are only a finite number of low-energy excitations (i.e. the degenerate ground states), yet the low-energy dynamics are non-trivial because the ground-state degeneracy depends on the topology of the space. Such special low-energy dynamics, which depend only on the topology of the space, are described by the so-called topological field theory, which was studied intensively in the high-energy physics community (Witten, 1989; Elitzur *et al.*, 1989; Fröhlich and King, 1989). Topological theories are effective theories for FQH liquids, just as the Ginzburg–Landau theory is for superfluids (or other symmetry-broken phases).

The dependence of the ground-state degeneracy on the topology of the space indicates the existence of some kind of long-range order (the global dancing pattern mentioned above) in FQH liquids, despite the absence of long-range correlations for all local physical operators. In some sense, we may say that FQH liquids contain hidden long-range orders.

8.2.2 Classification of topological orders

- It is important to understand the mathematical framework behind topological order. (Just like it is important to understand group theory—the mathematical framework behind the symmetry-breaking order.)
- The understanding of the mathematical framework will allow us to classify all of the possible topological orders.

A long-standing problem has been how to label and classify the rich topological orders in FQH liquids. We are able to classify all crystal orders because we know that the crystal orders are described by a symmetry group. However, our understanding of topological orders in FQH liquids is very poor, and the mathematical structure behind topological orders is unclear.

Nevertheless, we have been able to find a simple and unified treatment for a class of FQH liquids—abelian FQH liquids (Blok and Wen, 1990a,b; Read, 1990; Fröhlich and Kerler, 1991; Fröhlich and Studer, 1993). Laughlin states represent the simplest abelian FQH states that contain only one component of incompressible fluid. More general abelian FQH states with a filling fraction such as $\nu = 2/5, 3/7, \dots$ contain several components of incompressible fluids, and have more complicated topological orders. The topological orders (or the dancing patterns) in the abelian FQH state can also be described by the dancing steps. The dancing patterns can be characterized by an integer symmetric matrix K and an integer charge vector $\mathbf{q}$. An entry of $\mathbf{q}$, q_i , is the charge (in units of e) carried by the particles in the i th component of the incompressible fluid. An entry of K , K_{ij} , is the number of steps taken by a particle in the i th component to go around a particle in the j th component. In the $(K, \mathbf{q})$ characterization of FQH states, the $\nu = 1/m$ Laughlin state is described by $K = m$ and $\mathbf{q} = 1$, while the $\nu = 2/5$ abelian state is described by $K = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

All of the physical properties associated with the topological orders can be determined in terms of K and $\mathbf{q}$. For example, the filling fraction is simply given by $\nu = \mathbf{q}^T K^{-1} \mathbf{q}$ and the ground-state degeneracy on the genus g Riemann surface is $\det(K)^g$. All of the quasiparticle excitations in this class of FQH liquids have abelian statistics, which leads to the name abelian FQH liquids.

The above classification of FQH liquids is not complete. Not every FQH state is described by K -matrices. In 1991, a new class of FQH states—non-abelian FQH states—was proposed (Moore and Read, 1991; Wen, 1991b). A non-abelian FQH state contains quasiparticles with non-abelian statistics. The observed filling fraction $\nu = 5/2$ FQH state (Willett *et al.*, 1987) is very likely to be one such state (Haldane and Rezayi, 1988a,b; Greiter *et al.*, 1991; Read and Green, 2000). Many studies (Moore and Read, 1991; Blok and Wen, 1992; Iso *et al.*, 1992; Cappelli *et al.*, 1993; Wen *et al.*, 1994) have revealed a connection between the topological

orders in FQH states and conformal field theories. However, we are still quite far from a complete classification of all possible topological orders in non-abelian states.

8.2.3 Edge excitations—a practical way to measure topological orders

- The edge excitations for FQH states play a similar role to X-rays for crystals. We can use edge excitations to experimentally probe the topological orders in FQH states. In other words, compared to ground-state degeneracy, edge excitations provide a more complete definition of topological orders.

Topological degeneracy of the ground states only provides a partial characterization of topological orders. Different topological orders can sometimes lead to the same ground-state degeneracy. The issue here is whether we have a more complete characterization/measurement of topological orders. Realizing that the topological orders cannot be characterized by local order parameters and long-range correlations of local operators, it seems difficult to find any methods to characterize topological order. Amazingly, FQH states find a way out in an unexpected fashion. The bulk topological orders in FQH states can be characterized/measured by edge excitations (Wen, 1992). This phenomenon of two-dimensional topological orders being encoded in one-dimensional edge states shares some similarities with the holomorphic principle in superstring theory and quantum gravity ('t Hooft, 1993; Susskind, 1995).

FQH liquids as incompressible liquids have a finite energy gap for all of their bulk excitations. However, FQH liquids of finite size always contain one-dimensional gapless edge excitations, which is another unique property of FQH fluids. The structures of edge excitations are extremely rich, which reflects the rich bulk topological orders. Different bulk topological orders lead to different structures of edge excitations. Thus, we can study and measure the bulk topological orders by studying the structures of edge excitations.

As we have seen in the last chapter that, due to the non-trivial bulk topological order, the electrons at the edges of (abelian) FQH liquids form a new kind of correlated state—chiral Luttinger liquids (Wen, 1992). The electron propagator in chiral Luttinger liquids develops an anomalous exponent: $\langle c^\dagger(t, x)c(0) \rangle \propto (x - vt)^{-g}$, $g \neq 1$. (For Fermi liquids, we have $g = 1$.) The exponent g , in many cases, is a topological quantum number which does not depend on detailed properties of the edges. Thus, g is a new quantum number that can be used to characterize the topological orders in FQH liquids. Many experimental groups have successfully measured the exponent g through the temperature dependence of tunneling conductance between two edges (Milliken *et al.*, 1995; Chang *et al.*, 1996), which was predicted to have the form $\sigma \propto T^{2g-2}$ (Wen, 1992). This experiment demonstrates

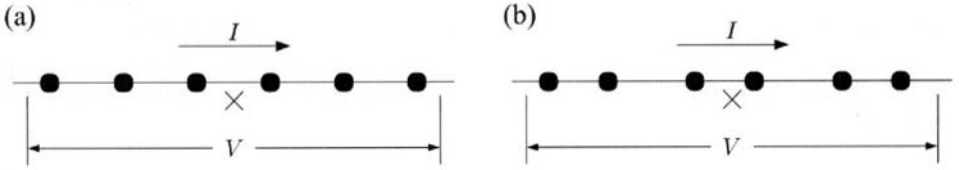


FIG. 8.4. A one-dimensional crystal passing an impurity will generate narrow-band noise in the voltage drop.

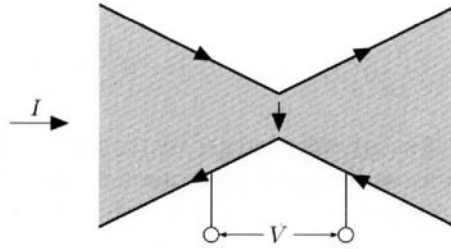


FIG. 8.5. An FQH fluid passing through a constriction will generate narrow-band noises due to the back-scattering of the quasiparticles.

the existence of new chiral Luttinger liquids and opens the door to the experimental study of the rich internal and edge structures of FQH liquids.

The edge states of non-abelian FQH liquids form more exotic one-dimensional correlated systems which have not yet been named. These edge states were found to be closely related to conformal field theories in $1 + 1$ dimensions (Wen *et al.*, 1994).

We know that crystal orders can be measured by X-ray diffraction experiments. In the following, we would like to suggest that the topological orders in FQH liquids can be measured (in principle) through a noise spectrum in an edge transport experiment. Let us first consider a one-dimensional crystal driven through an impurity (see Fig. 8.4(a)). Due to the crystal order, the voltage across the impurity has a narrow-band noise at a frequency of $f = I/e$ if each unit cell has only one charged particle. More precisely, the noise spectrum has a singularity, i.e. $S(f) \sim A\delta(f - \frac{I}{e})$. If each unit cell contains two charged particles (see Fig. 8.4(b)), then we will see an additional narrow-band noise at $f = I/2e$, so that $S(f) \sim B\delta(f - \frac{I}{2e}) + A\delta(f - \frac{I}{e})$. In this example, we see that the noise spectrum allows us to measure crystal orders in one-dimension. A similar experiment can also be used to measure topological orders in FQH liquids. Let us consider an FQH sample with a narrow constriction (see Fig. 8.5). The constriction induces a back-scattering through quasiparticle tunneling between the two

edges. The back-scattering causes a noise in the voltage across the constriction. In the weak-back-scattering limit, the noise spectrum contains singularities at certain frequencies, which allows us to measure the topological orders in the FQH liquids.⁵² To be more specific, the singularities in the noise spectrum have the form (see eqn (7.4.49))

$$S(f) \sim \sum_a C_a |f - f_a|^{\gamma_a} \quad (8.2.5)$$

The frequencies and the exponents of the singularities (f_a, γ_a) are determined by the topological orders. For the abelian state characterized by the matrix K and the charge vector $\mathbf{q}$, the allowed values of the pair (f_a, γ_a) are given by

$$f_a = \frac{I}{e\nu} \mathbf{q}^\top K^{-1} \mathbf{l}, \quad \gamma_a = 2\mathbf{l}^\top K^{-1} \mathbf{l} - 1 \quad (8.2.6)$$

where $\mathbf{l}^\top = (l_1, l_2, \dots)$ is an arbitrary integer vector and $\nu = \mathbf{q}^\top K^{-1} \mathbf{q}$ is the filling fraction. The singularities in the noise spectrum are caused by quasiparticle tunneling between the two edges. The frequency of the singularity f_a is determined by the electric charge of the tunneling quasiparticle Q_q , i.e. $f_a = \frac{I Q_q}{e}$. The exponent γ_a is determined by the statistics of the tunneling quasiparticle θ_q , namely $\gamma = 2\frac{[\theta_q]}{\pi} - 1$. Thus, the noise spectrum measures the charge and the statistics of the allowed quasiparticles, which in turn determines the topological orders in FQH states.

8.3 Quantum orders

- Quantum states generally contain a new kind of order—quantum order. Quantum orders cannot be completely characterized by broken symmetries and the associated order parameters.
- Quantum order describes the pattern of quantum entanglements in many-body ground states.
- The fluctuations of quantum order can give rise to gapless gauge bosons and gapless fermions. Quantum order protects the gaplessness of these excitations, just like symmetry protects gapless Nambu–Goldstone bosons in symmetry-breaking states.
- Topological order is a special kind of quantum order in which all excitations have finite energy gaps.

⁵² The discussion presented here applies only to the FQH states whose edge excitations all propagate in the same direction. This requires, for abelian states, all of the eigenvalues of K to have the same sign.

The topological order, by definition, only describes the internal order of gapped quantum states. Here we make a leap of faith. We will assume that the gap is not important and that gapless quantum states can also contain orders that cannot be described by symmetry and long-range correlations. We will call the non-symmetry-breaking order in quantum ground states the quantum order.

If you believe in this line of thinking, then the only things that need to be done are to show that quantum orders do exist, and to find mathematical descriptions (or symbols) that characterize the quantum orders. We will show that quantum orders do exist in Section 8.3.2 and in Chapter 10. As quantum orders cannot be characterized by broken symmetries and order parameters, we need to develop a new theory to describe quantum orders. At present, we do not have a complete theory that can describe all possible quantum orders. However, in Chapter 9 we manage to find a mathematical object—the projective symmetry group (PSG)—that can describe a large class of quantum orders.

One may ask, why do we need to introduce the new concept of quantum order? What use can it have? To answer such a question, we would like to ask, why do we need the concept of symmetry breaking? Is the symmetry-breaking description useful? Symmetry breaking is useful because it leads to a classification of crystal orders (such as the 230 different crystals in three dimensions), and it determines the structure of low-energy excitations without the need to know the details of a system (such as three branches of phonons from three broken translational symmetries in a solid) (Nambu, 1960; Goldstone, 1961). The quantum order and its PSG description are useful in the same sense; a PSG can classify different quantum states that have the same symmetry (Wen, 2002c), and quantum orders determine the structure of low-energy excitations without the need to know the details of a system (Wen, 2002a,c; Wen and Zee, 2002). The main difference between symmetry-breaking orders and quantum orders is that symmetry-breaking orders generate and protect gapless Nambu–Goldstone modes (Nambu, 1960; Goldstone, 1961), which are scalar bosonic excitations, while quantum orders can generate and protect gapless gauge bosons and gapless fermions. Fermion excitations can even emerge in pure local bosonic models, as long as the boson ground state has a proper quantum order.

One way to visualize quantum order is to view quantum order as a description of the pattern of the quantum entanglement in a many-body ground state. Different patterns of entanglement give rise to different quantum orders. The fluctuations of entanglement correspond to collective excitations above a quantum-ordered state. We will see that these collective excitations can be gauge bosons and fermions.

The concept of topological/quantum order is also useful in the field of quantum computation. People have been designing different kinds of quantum-entangled states to perform different computing tasks. When the number of qubits becomes larger and larger, it is more and more difficult to understand the pattern of quantum

entanglements. One needs a theory to characterize different quantum entanglements in many-qubit systems. The theory of topological/quantum order (Wen, 1995, 2002c) is just such a theory. Also, the robust topological degeneracy in topologically-ordered states discovered by Wen and Niu (1990) can be used in fault-tolerant quantum computation (Kitaev, 2003).

In the following, we will discuss the connection between quantum phase transitions and quantum orders. Then we will use the quantum phase transitions in free fermion systems to study the quantum orders there.

8.3.1 Quantum phase transitions and quantum orders

- Quantum phase transitions are defined by the singularities of the ground-state energy as a function of the parameters in the Hamiltonian.

Classical orders can be studied through classical phase transitions. Classical phase transitions are marked by singularities in the free-energy density f . The free-energy density can be calculated through the partition function as follows:

$$f = -\frac{T \ln Z}{V_{\text{space}}}, \quad Z = \int \mathcal{D}\phi \, e^{-\beta \int dx \, h(\phi)} \quad (8.3.1)$$

where $h(\phi)$ is the energy density of the classical system and V_{space} is the volume of space.

Similarly, to study quantum orders we need to study quantum phase transitions at zero temperature $T = 0$. Here the energy density of the ground state plays the role of the free-energy density. A singularity in the ground-state-energy density marks a quantum transition. The similarity between the ground-state-energy density and the free-energy density can be clearly seen in the following expression for the energy density of the ground state:

$$\rho_E = i \frac{\ln Z}{V_{\text{space-time}}}, \quad Z = \int \mathcal{D}\phi \, e^{i \int dx dt \, \mathcal{L}(\phi)} \quad (8.3.2)$$

where $\mathcal{L}(\phi)$ is the Lagrangian density of the quantum system and $V_{\text{space-time}}$ is the volume of space-time. Comparing eqns (8.3.1) and (8.3.2), we see that a classical system is described by a path integral of a positive functional, while a quantum system is described by a path integral of a complex functional. In general, a quantum phase transition, marked by a singularity of the path integral of a complex functional, can be more general than classical phase transitions that are marked by a singularity of the path integral of a positive functional.

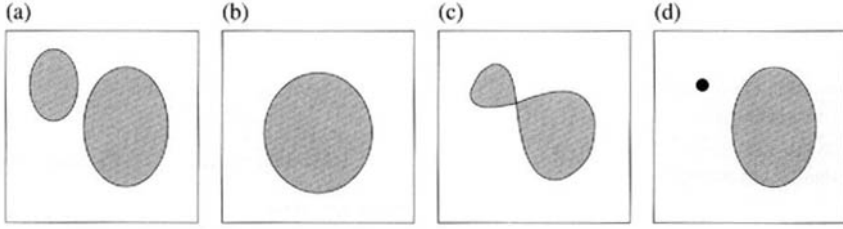


FIG. 8.6. The two sets of oriented Fermi surfaces in (a) and (b) represent two different quantum orders. The two possible transition points between the two quantum orders in (a) and (b) are described by the Fermi surfaces in (c) and (d).

8.3.2 Quantum orders and quantum transitions in free fermion systems

- Free fermion systems contain quantum phase transitions that do not change any symmetry, indicating that free fermion systems contain non-trivial quantum order.
- Different quantum orders in free fermion systems are classified by the topologies of a Fermi surface.

Let us consider a free fermion system with only the translational symmetry and the $U(1)$ symmetry from the fermion number conservation. The Hamiltonian has the form

$$H = \sum_{\langle ij \rangle} \left(c_i^\dagger t_{ij} c_j + h.c. \right)$$

with $t_{ij}^* = t_{ji}$. The ground state is obtained by filling every negative energy state with one fermion. In general, the system contains several pieces of Fermi surfaces.

To understand the quantum order in the free fermion ground state, we note that the topology of the Fermi surfaces can change in two ways as we continuously change t_{ij} : a Fermi surface can shrink to zero (Fig. 8.6(d)); and two Fermi surfaces can join (Fig. 8.6(c)). When a Fermi surface is about to disappear in a d -dimensional system, the ground-state-energy density has the form

$$\rho_E = \int \frac{d^d \mathbf{k}}{(2\pi)^d} (\mathbf{k} \cdot \mathbf{M} \cdot \mathbf{k} - \mu) \Theta(-\mathbf{k} \cdot \mathbf{M} \cdot \mathbf{k} + \mu) + \dots$$

where ‘...’ represents the non-singular contribution and the symmetric matrix $\mathbf{M}$ is positive (or negative) definite. We find that the ground-state-energy density has a singularity at $\mu = 0$, i.e. $\rho_E = c\mu^{(2+d)/2}\Theta(\mu) + \dots$, where $\Theta(x > 0) = 1$ and $\Theta(x < 0) = 0$. When two Fermi surfaces are about to join, the singularity is still determined by the above equation, but now $\mathbf{M}$ has both negative and positive eigenvalues. The ground-state-energy density has a singularity of the form $\rho_E = c\mu^{(2+d)/2}\Theta(\mu) + \dots$ when d is odd and $\rho_E = c\mu^{(2+d)/2} \log |\mu| + \dots$ when d is even.

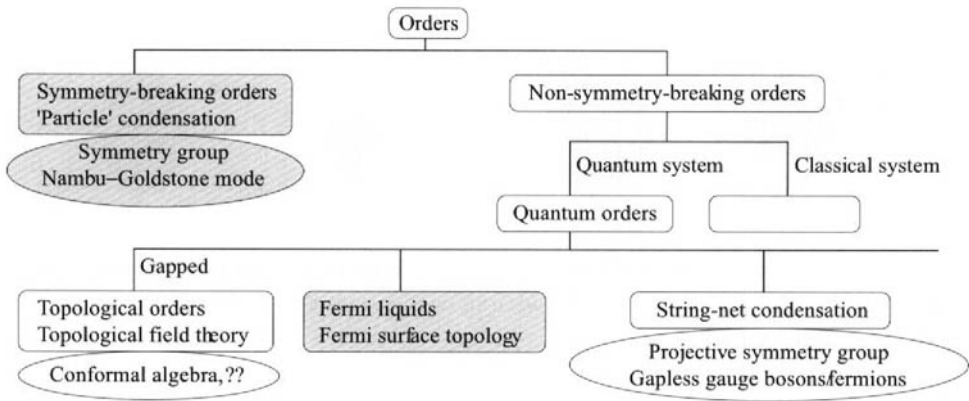


FIG. 8.7. A new classification of orders. The phases in the shaded boxes can be described Landau’s theories. Other phases are beyond Landau’s theories.

The singularity of the ground-state-energy density at $\mu = 0$ indicates a quantum phase transition. This kind of transition was first studied by Lifshitz (1960). Clearly, there is no change of symmetry across the transition and there is no local order parameters to characterize the phases on the two sides of the transition. This suggests that the two states can only be distinguished by their quantum orders. As the $\mu = 0$ point is exactly the place where the topology of the Fermi surface changes, we find that the topology of the Fermi surface is a ‘quantum number’ that characterizes the quantum order in a free fermion system (see Fig. 8.6). A change in the topology signals a *continuous* quantum phase transition that changes the quantum order.

Problem 8.3.1.

Consider a two-dimensional spin-1/2 free electron system. As we change the chemical potential μ , the system undergoes a quantum phase transition, as illustrated in Fig. 8.6(d). Find the singular behavior of the spin susceptibility near the transition point μ_c .

8.4 A new classification of orders

- Quantum orders have many classes. FQH states and free fermion systems represent only two of the many classes of quantum orders.

The concept of topological/quantum order allows us to have a new classification of orders, as illustrated in Fig. 8.7. According to this classification, a quantum order is simply a non-symmetry-breaking order in a quantum system, and a topological order is simply a quantum order with a finite energy gap.

From the FQH states and the Fermi liquid states discussed above, we see that quantum order can be divided into several different classes. The FQH states and the free fermion systems only provide two examples of quantum orders. In the next few chapters, we will study quantum orders in some strongly-correlated systems. We will show that these quantum orders belong to a different class, which is closely related to a condensation of nets of strings in the correlated ground state. Chapter 9 studies and classifies this class of quantum order using a projective construction (i.e. the slave-boson approach). Chapter 10 relates the quantum-ordered state studied in Chapter 9 to string-net-condensed states.

MEAN-FIELD THEORY OF SPIN LIQUIDS AND QUANTUM ORDER

- The mean-field theory of the spin-liquid state using the slave-boson approach (or projective construction).
- The importance of the gauge interaction and how to apply the mean-field theory (with the gauge interaction) to study the physical properties of the spin liquids.
- The characterization of the spin liquids and the concept of topological order and quantum order.

In this chapter, we are going to develop a mean-field theory for the spin-liquid state. Here by ‘spin-liquid state’ we mean an insulator with spin-rotation symmetry and with an *odd* number of electrons per unit cell. Usually, a state with an odd number of electrons per unit cell has a half-filled band and is a conductor. Thus, the spin liquids, if they exist, are very unusual states. The spin liquids are so strange that many people do not believe that they can ever exist. Indeed, even now, we do not know of any spin Hamiltonian that can be shown to reliably give rise to a spin-liquid ground state.

For those who believe in the existence of spin liquids, they have to accept or believe the following strange (or fascinating) properties of spin liquids. (i) The excitations in spin liquids always carry fractional quantum numbers, such as neutral spin-1/2, which are impossible to obtain from any collection of electrons. Sometimes excitations can even carry fractional statistics. (ii) Different spin liquids cannot be distinguished by their symmetry properties. Spin liquids are examples of quantum-ordered states. (iii) The ground states of gapped spin liquids always have topological degeneracy, which cannot be related to any symmetry. (iv) Spin liquids always contain certain kinds of gauge fluctuations.

In this chapter, we will assume that spin liquids do exist and develop a mean-field theory for spin liquids. The mean-field theory allows us to understand some physical properties of spin liquids. After including important mean-field fluctuations, we show that some of the mean-field states are stable against these fluctuations and represent real spin-liquid states. As spin liquids are typical states with non-trivial topological/quantum orders, we will use the spin liquids to develop a theory of quantum orders.

9.1 Projective construction of quantum spin-liquid states

In this section, we are going to use the slave-boson approach (or the projective construction) (Baskaran *et al.*, 1987; Affleck *et al.*, 1988; Baskaran and Anderson, 1988; Dagotto *et al.*, 1988; Wen and Lee, 1996; Senthil and Fisher, 2000) to construct two-dimensional spin liquids. The gauge structure discovered by Baskaran and Anderson (1988) in the slave-boson approach plays a crucial role in our understanding of strongly-correlated spin liquids. In addition to the slave-boson approach, one can also use another type of projective construction—the slave-fermion or Schwinger boson approach—to construct spin-liquid states (Arovas and Auerbach, 1988; Read and Sachdev, 1991; Sachdev and Park, 2002). As the slave-fermion approach can only lead to spin liquids with finite energy gaps, here we will concentrate on the slave-boson approach.

9.1.1 Mean-field theory of spin-liquid states

- A mean-field theory of spin liquids can be obtained from a projective construction.
- Mean-field theory and the mean-field ansatz of the π -flux phase.
- Gauge fluctuations of $a_0(i)$ impose the constraint.

Let us consider a Hubbard model (5.5.1) on a two-dimensional square lattice. At half-filling, the Hubbard model reduces to the Heisenberg model as follows:

$$H = \sum_{\langle ij \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j \quad (9.1.1)$$

We would like to point out that the above spin-1/2 model can also be viewed as a hard-core boson model, where $|\downarrow\rangle$ corresponds to an empty site and $|\uparrow\rangle$ corresponds to a site occupied by one boson. The above Hamiltonian is hard to solve. So we use a mean-field approximation to understand its physical properties. In the mean-field approximation, we replace one of the $\mathbf{S}_i$ by its quantum average $\langle \mathbf{S}_i \rangle$ and obtain the following mean-field Hamiltonian:

$$H_{\text{mean}} = \sum_{\langle ij \rangle} J_{ij} (\langle \mathbf{S}_i \rangle \cdot \mathbf{S}_j + \mathbf{S}_i \cdot \langle \mathbf{S}_j \rangle - \langle \mathbf{S}_i \rangle \cdot \langle \mathbf{S}_j \rangle)$$

The mean-field Hamiltonian H_{mean} is easy to solve and we can obtain the ground state of H_{mean} , namely $|\Phi_{\text{mean}}\rangle$. The only thing we need to do is to choose the values of $\langle \mathbf{S}_i \rangle$ carefully, so that they satisfy the so-called self-consistency equation

$$\langle \mathbf{S}_i \rangle = \langle \Phi_{\text{mean}} | \mathbf{S}_i | \Phi_{\text{mean}} \rangle$$

This is the standard mean-field approach for spin systems.

The problem with the above standard mean-field approach is that it can only be used to study the ordered spin states, because we assume that at the beginning $\langle S_i \rangle \neq 0$. For a long time, it seemed impossible to have a mean-field theory for spin liquids. In 1987, we finally found a strange trick—the slave-boson approach⁵³—to do so (Baskaran *et al.*, 1987). To obtain the mean-field ground state of *spin liquids*, we introduce the spinon operators $f_{i\alpha}$, $\alpha = 1, 2$, which are spin-1/2 charge-neutral operators. The spin operator S_i is represented by

$$S_i = \frac{1}{2} f_{i\alpha}^\dagger \sigma_{\alpha\beta} f_{i\beta} \quad (9.1.2)$$

In terms of the spinon operators, the Hamiltonian (9.1.1) can be rewritten as

$$H = \sum_{\langle ij \rangle} -\frac{1}{2} J_{ij} f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{i\beta} + \sum_{\langle ij \rangle} J_{ij} \left(\frac{1}{2} n_i - \frac{1}{4} n_i n_j \right) \quad (9.1.3)$$

Here we have used $\sigma_{\alpha\beta} \cdot \sigma_{\alpha'\beta'} = 2\delta_{\alpha\beta'}\delta_{\alpha'\beta} - \delta_{\alpha\beta}\delta_{\alpha'\beta'}$ and n_i is the number of fermions at site i . The second term in eqn (9.1.3) is a constant and will be dropped in the following discussions. Notice that the Hilbert space of eqn (9.1.3) with four states per site is larger than that of eqn (9.1.1), which has two states per site. The equivalence between eqn (9.1.1) and eqn (9.1.3) is valid only in the subspace where there is exactly one fermion per site. Therefore, to use eqn (9.1.3) to describe the spin state, we need to impose the constraint (Baskaran *et al.*, 1987; Baskaran and Anderson, 1988)

$$f_{i\alpha}^\dagger f_{i\alpha} = 1, \quad f_{i\alpha} f_{i\beta} \epsilon_{\alpha\beta} = 0 \quad (9.1.4)$$

The second constraint is actually a consequence of the first one.

A mean-field ground state at zeroth-order is obtained by making the following approximations. First we replace the constraint (9.1.4) by its ground-state average

$$\langle f_{i\alpha}^\dagger f_{i\alpha} \rangle = 1 \quad (9.1.5)$$

Such a constraint can be enforced by including a *site-dependent* and time-independent Lagrangian multiplier $a_0(i)(f_{i\alpha}^\dagger f_{i\alpha} - 1)$ in the Hamiltonian. Second, we replace the operator $f_{i\alpha}^\dagger f_{j\alpha}$ by its ground-state expectation value χ_{ij} , again ignoring their fluctuations. In this way, we obtain the zeroth-order mean-field

⁵³ The slave-boson and the slave-fermion are two very strange and confusing terms. The slave-boson approach has no bosons and the slave-fermion approach has no fermions. This is why I prefer to use projective construction to describe the two approaches.

Hamiltonian

$$H_{\text{mean}} = \sum_{\langle ij \rangle} -\frac{1}{2} J_{ij} \left[(f_{i\alpha}^\dagger f_{j\alpha} \chi_{ji} + h.c.) - |\chi_{ij}|^2 \right] + \sum_i a_0(i) (f_{i\alpha}^\dagger f_{i\alpha} - 1) \quad (9.1.6)$$

The χ_{ij} in eqn (9.1.6) must satisfy the self-consistency condition

$$\chi_{ij} = \langle f_{i\alpha}^\dagger f_{j\alpha} \rangle \quad (9.1.7)$$

and the site-dependent chemical potential $a_0(i)$ is chosen such that eqn (9.1.5) is satisfied by the mean-field ground state. For convenience, we will call χ_{ij} the ‘ansatz’ of the spin-liquid state.

Let $\bar{\chi}_{ij}$ be a solution of eqn (9.1.7) and let $\bar{a}(i)$ be a solution of eqn (9.1.5). Such an ansatz corresponds to the mean-field ground state. As χ_{ij} and a_0 do not change under the spin-rotation transformation, the mean-field ground state is invariant under the spin-rotation transformation. If we ignore the fluctuations of $\bar{\chi}_{ij}$ and $\bar{a}(i)$, then the excitations around the mean-field state are described by

$$H_{\text{mean}} = \sum_{\langle ij \rangle} -\frac{J_{ij}}{2} \left[(f_{i\alpha}^\dagger f_{j\alpha} \bar{\chi}_{ji} + h.c.) - |\bar{\chi}_{ij}|^2 \right] + \sum_i \bar{a}_0(i) (f_{i\alpha}^\dagger f_{i\alpha} - 1)$$

We see that the excitations in the zeroth-order mean-field theory are free spinons described by $f_{i\alpha}$. The spinons are spin-1/2 neutral fermions. It is amazing to see that fermionic excitations can emerge from a purely bosonic model eqn (9.1.1).

Now the question is whether we should trust the mean-field result. Should we believe in the existence of spin liquids with spin-1/2 neutral fermionic excitations? One way to check this is to include the fluctuations around the mean-field ansatz, and to see if the fluctuations alter the mean-field result. So, in the following, we will consider the effects of fluctuations.

First, we would like to point out that, if we had included the fluctuations (i.e. the time dependence) of the a_0 , then the constraint (9.1.5) would have become the original constraint (9.1.4). To see this, let us consider the path integral formulation of H_{mean} :

$$Z = \int \mathcal{D}f \mathcal{D}[a_0(i)] \mathcal{D}\chi_{ij} e^{i \int dt (\mathcal{L} - \sum_i a_0(i, t) (f_i^\dagger f_i - 1))} \quad (9.1.8)$$

$$\mathcal{L} = \sum_i f_i^\dagger i \partial_t f_i - \sum_{\langle ij \rangle} -\frac{1}{2} J_{ij} \left[(f_{i\alpha}^\dagger f_{j\alpha} \chi_{ji} + h.c.) - |\chi_{ij}|^2 \right]$$

We see that the integration of a time-dependent $a_0(i, t)$ produces a constraint

$$\prod_{i,t} \delta(f_i^\dagger(t) f_i(t) - 1)$$

which is enforced at every site i and at every time t . We also note that the integration of $\chi_{ij}(t)$ will reproduce the original Hamiltonian (9.1.3). Thus, eqn (9.1.8) is

an exact representation of the spin model (9.1.3). The fluctuations in χ_{ij} and $a_0(i)$ describe the collective excitations above the mean-field ground state.

Here χ_{ij} has two kinds of fluctuation, the amplitude fluctuations and the phase fluctuations. The amplitude fluctuations have a finite energy gap and are not essential in our discussion. So here we will only consider the phase fluctuations a_{ij} around the mean-field ansatz $\bar{\chi}_{ij}$ as follows:

$$\chi_{ij} = \bar{\chi}_{ij} e^{-i a_{ij}} \quad (9.1.9)$$

Including these phase fluctuations and fluctuations of a_0 , the mean-field Hamiltonian becomes

$$H = \sum_{\langle ij \rangle} -J_{ij} (f_{i\alpha}^\dagger f_{j\alpha} \bar{\chi}_{ji} e^{-i a_{ji}} + h.c.) - \sum_i a_0(i) (f_{i\alpha}^\dagger f_{i\alpha} - 1) \quad (9.1.10)$$

We would like to call eqn (9.1.10) the first-order mean-field Hamiltonian. From eqn (9.1.10), we see that the fluctuations described by a_0 and a_{ij} are simply a $U(1)$ lattice gauge field (see eqn (6.4.9)) (Baskaran and Anderson, 1988; Lee and Nagaosa, 1992). The Hamiltonian is invariant under the gauge transformation

$$a_{ij} \rightarrow a_{ij} + \theta_i - \theta_j, \quad f_i \rightarrow f_i e^{i\theta_i}. \quad (9.1.11)$$

So, in first-order mean-field theory, the excitations are described by spinons coupled to the $U(1)$ gauge field (instead of free spinons in the zeroth-order mean-field theory).

In the usual mean-field theory, we make an approximation to the interacting Hamiltonian to simplify the problem. The Hilbert space is not changed by the mean-field approximation. The projective construction (or slave-boson approach) is very different in this aspect. Not only is the Hamiltonian changed, but the Hilbert space is also changed. This makes the mean-field results from the projective construction not only quantitatively incorrect, but also qualitatively incorrect. For example, the mean-field ground state $|\Psi_{\text{mean}}\rangle$ (the ground state of H_{mean} in eqn (9.1.6)) is not even a valid spin wave function because some sites can have zero or two fermions. However, do not give up. In the usual mean-field theory, we can improve the approximation quantitatively by including fluctuations around the mean-field state. In the above, we see that, in the projective construction, we can improve the approximation qualitatively and recover the original Hilbert space by including gauge fluctuations (a_0, a_{ij}) around the mean-field state. Thus, to obtain even qualitative results from the projective construction, it is important to include, at least, the gauge fluctuations around the mean-field state. In other words, we cut a spin into two halves in the projective construction of spin liquids. It is important to glue them back together to obtain the correct physical results.

According to zeroth-order mean-field theory, the low-energy excitations in the spin liquids are free spinons. From the above discussion, we see that such a result

is incorrect. According to first-order mean-field theory, spinons interact via gauge fluctuations. The gauge interaction may drastically change the properties of the spinons. We will return to this problem later.

Problem 9.1.1.

Prove eqn (9.1.3).

9.1.2 To believe or not to believe

- The deconfined phase of first-order mean-field theory leads to new states of matter—quantum-ordered states.
- The projection to the physical spin wave function and the meaning of the $U(1)$ gauge structure.
- Emergent gauge bosons and fermions as fluctuations of entanglements.

According to first-order mean-field theory, the fluctuations around the mean-field ground state are described by gauge fields and fermion fields. Remember that our original model is just an interacting spin model which is a purely bosonic model. How can a purely bosonic model have an effective theory described by gauge fields and fermion fields? This is incredible. Let us examine how we get here. We first split the bosonic spin operator into a product of two fermionic operators (the spinon operators). We then introduce a gauge field to glue the spinons back into a bosonic spin. From this point of view, it looks like the first-order mean-field theory is just a fake theory. It appears that the gauge bosons and the fermions are fake. In the end, all we have are the bosonic spin fluctuations.

However, we should not discard first-order mean-field theory too quickly. It can actually reproduce the above picture of bosonic spin fluctuations if the gauge field is in a confining phase (see Section 6.4.3). In the confining phase, the spinons interact with each other through a linear potential and can never appear as quasiparticles at low energies. The gauge bosons have a large energy gap in the confining phase, and are absent from the low-energy spectrum. The only low energy excitations are the spinon pairs which correspond to the bosonic spin fluctuations. So first-order mean-field theory may be useless, but it is not wrong. It is capable of producing pictures that agree with common sense (although through a long detour).

On the other hand, first-order mean-field theory is also capable of producing pictures that defy common sense when the gauge field is in a deconfined phase. In this case, the spinons and gauge bosons will appear as well-defined quasiparticles. The question is do we believe the picture of deconfined phase? Do we believe the possibility of emergent gauge bosons and fermions from a purely bosonic model? Clearly, the projective construction outlined above is far too formal to convince most people to believe such drastic results.

I have to say that this business of cutting spins into two halves and gluing them back together turns off a lot of people. It is hard to see that any new physical insights and results can possibly be obtained by such a formal manipulation. If any new results do appear from this approach, then many people will attribute them to artifacts of this strange construction rather than true physical properties of the spin liquids. Indeed, the projective construction is very formal, but it is also a gift beyond its time. We now know that what the projective construction really does is to produce a string-net-condensed state. String-net condensation gives rise to a new type of correlated state that has non-trivial quantum orders (see Chapter 10). So, believe me that the striking results from the projective construction can be trusted, once we properly include the effects of gauge fluctuations.

Here let us try to understand the physical picture behind the projective construction, without using the picture of string-net condensation. First, we need to understand how a mean-field ansatz χ_{ij} is connected to a physical spin wave function which has exactly one fermion per site. We know that the mean-field wave state $|\Psi_{\text{mean}}^{(\chi_{ij})}\rangle$ (the ground state of H_{mean}) is not a valid wave function for the spin system, because it may not have one fermion per site. To connect this to a physical spin wave function, we need to include fluctuations of a_0 to enforce the one-fermion-per-site constraint. With this understanding, we may obtain a valid wave function of the spin system, $\Psi_{\text{spin}}(\{\alpha_i\})$, by projecting the mean-field state to the subspace of one fermion per site:

$$\Psi_{\text{spin}}^{(\chi_{ij})}(\{\alpha_i\}) = \langle 0_f | \prod_i f_{i\alpha_i} |\Psi_{\text{mean}}^{(\chi_{ij})}\rangle. \quad (9.1.12)$$

where $|0_f\rangle$ is the state with no f -fermions, i.e. $f_{i\alpha}|0_f\rangle = 0$. Equation (9.1.12) connects the mean-field ansatz (including its fluctuations) to the physical spin wave function. It allows us to understand the physical meaning of the mean-field ansatz and mean-field fluctuations.

For example, the projection (9.1.12) gives the gauge transformation (9.1.11) a physical meaning. The two mean-field ansatz χ_{ij} and $\tilde{\chi}_{ij}$, related by a gauge transformation

$$\tilde{\chi}_{ij} = e^{i\theta_i} \chi_{ij} e^{-i\theta_j} \quad (9.1.13)$$

give rise to the same projected spin state

$$\Psi_{\text{spin}}^{(\tilde{\chi}_{ij})}(\{\alpha_i\}) = e^{i\sum_i \theta_i} \Psi_{\text{spin}}^{(\chi_{ij})}(\{\alpha_i\}) \quad (9.1.14)$$

For different choices of χ_{ij} , the ground states of the H_{mean} in eqn (9.1.6) correspond to different mean-field wave functions $|\Psi_{\text{mean}}^{(\chi_{ij})}\rangle$. After projection, they lead to different physical spin wave functions $\Psi_{\text{spin}}^{(\tilde{\chi}_{ij})}(\{\alpha_i\})$. Thus, we can regard χ_{ij} as labels that label different physical spin states. Equation (9.1.14) tells us that the

label is not a one-to-one label, but a many-to-one label. This property is important for us to understand the unusual dynamical properties of the χ_{ij} fluctuations. Using many labels to label the same physical state also makes our theory a gauge theory according to the definition in Section 6.1.1.

Let us consider how the many-to-one property or the gauge structure of χ_{ij} affect its dynamical properties. If χ_{ij} was a one-to-one label of physical states, then χ_{ij} would be like the condensed boson amplitude $\langle \phi(\mathbf{x}, t) \rangle$ in a boson superfluid or the condensed spin moment $\langle \mathbf{S}_i(t) \rangle$ in an SDW state. The fluctuations of χ_{ij} would correspond to a bosonic mode similar to a spin wave mode.⁵⁴ However, χ_{ij} does not behave like local order parameters, such as $\langle \phi(\mathbf{x}, t) \rangle$ and $\langle \mathbf{S}_i(t) \rangle$, which label physical states without redundancy. As a many-to-one label, some fluctuations of χ_{ij} do not change the physical state and are unphysical. These fluctuations are called pure gauge fluctuations. For a generic fluctuation $\delta\chi_{ij}$, part of it is physical and the other part is unphysical. The effective theory for χ_{ij} must be gauge invariant. This drastically changes the dynamical properties of the fluctuations. It is this property that makes fluctuations of χ_{ij} behave like gauge bosons, which are very different from the sound mode and the spin wave mode.

We have argued that the phase fluctuations of χ_{ij} correspond to gauge fluctuations. The projective construction (9.1.12) allows us to obtain the physical spin wave function that corresponds to a gauge fluctuation a_{ij} :

$$\Psi_{\text{spin}}^{(a_{ij})} = \langle 0 | \prod_i f_{i\alpha_i} | \Psi_{\text{mean}}^{(\tilde{\chi}_{ij} e^{i a_{ij}})} \rangle.$$

Similarly, the projective construction also allows us to obtain the physical spin wave function that corresponds to a pair of spinon excitations. We start with the mean-field ground state with a pair of particle-hole excitations. After the projection (9.1.12), we obtain the physical spin wave functions that contain a pair of spinons:

$$\Psi_{\text{spin}}^{\text{spinon}}(i_1, \alpha_1; i_2, \alpha_2) = \langle 0 | (\prod_i f_{i\alpha_i}) f_{i_1\lambda_1}^\dagger f_{i_2\lambda_2} | \Psi_{\text{mean}}^{(\tilde{\chi}_{ij})} \rangle.$$

If you are not satisfied with the physical picture presented and still do not believe that the projective construction can produce emergent gauge bosons and fermions, then you may go directly to Chapter 10. In Chapter 10, we construct several spin models which can be solved exactly (or quasi-exactly) by the projective construction. These models have emergent $Z_2/U(1)$ gauge structure and fermions. The exactly soluble models reveal the string-net origin of the emergent gauge bosons and fermions. In the rest of this chapter we assume that the projective construction makes sense and study its consequences.

⁵⁴ More precisely, the spin wave mode corresponds to scalar bosons. The fluctuations of the local order parameters always give rise to scalar bosons.

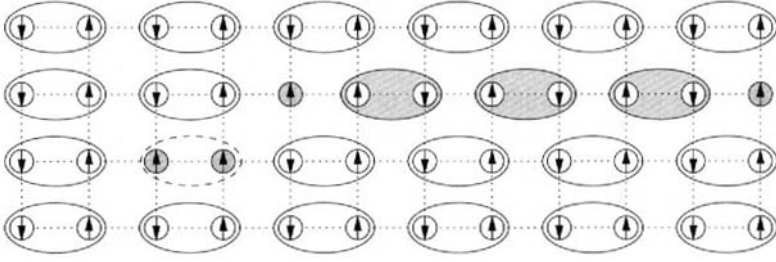


FIG. 9.1. A dimer state is formed by spin-singlet pairs. Changing a dimer into a triplet state creates a spin-1 excitation. Two separated spin-1/2 excitations are connected by a string of displaced dimers.

9.1.3 The dimer state

- The first-order mean-field theory for the dimer state has a confining $U(1)$ gauge interaction. Only spinon-pair bound states appear as physical excitations.

Let us first discuss a specific mean-field ground state, the dimer state (Majumdar and Ghosh, 1969; Affleck *et al.*, 1987; Read and Sachdev, 1989) in the Heisenberg model with nearest-neighbor coupling:

$$H = J_1 \sum_i (\mathbf{S}_i \mathbf{S}_{i+x} + \mathbf{S}_i \mathbf{S}_{i+y}). \quad (9.1.15)$$

The dimer state is described by the following ansatz:

$$\begin{aligned} \chi_{i,i+x} &= \frac{1}{2}(1 + (-)^{i_x}) \\ \text{others} &= 0 \end{aligned} \quad (9.1.16)$$

and $a_0(\mathbf{i}) = 0$. One can check that the self-consistency equation (9.1.7) and the constraint (9.1.5) are satisfied by the mean-field ground states of H_{mean} in eqn (9.1.6). The spinon spectrum has no dispersion in the mean-field dimer state:

$$E_k = \pm \frac{1}{2} J_1 \quad (9.1.17)$$

The valence band with $E_k = -\frac{1}{2} J_1$ is completely filled by the spinons. The spin excitations have a finite energy gap in the dimer state. It is also clear that the dimer state breaks the translational symmetry in the x direction.

In zeroth-order mean-field theory, the excitations above the dimer ground state are spin-1/2 spinons. Such a result is obviously wrong because, physically, the dimer state is formed by spin-singlet dimers filling the lattice (see Fig. 9.1). The elementary excitations correspond to spin-triplet dimers, which are bosonic spin-1

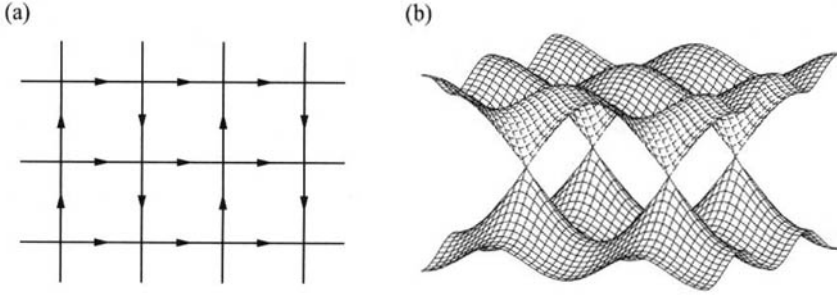


FIG. 9.2. (a) The mean-field ansatz of the π -flux state. Here $\chi_{ij} = i\chi_1$ in the direction of the arrow. (b) The fermion dispersion in the π -flux state. The valence band is filled. The low-energy excitations exist near the four Fermi points, where the valence band and the conduction band touch.

excitations. In first-order mean-field theory, the spinons are coupled to the $U(1)$ gauge field. In $2 + 1$ dimensions, $U(1)$ gauge theory is confining. Thus, individual spinons are not observable. Only bosonic bound states (which carry integral spin) can appear in the physical spectrum. If we do create two separated spin-1/2 excitations, then, from Fig. 9.1, we see that they are connected by a string of displaced dimers. The string leads to a linear confining interaction between the two spin-1/2 excitations, which agrees with the picture obtained from first-order mean-field theory. We see that first-order mean-field theory gives us qualitatively correct results.

9.1.4 The π -flux state

- After projection, the mean-field π -flux phase gives rise to a translation, a rotation, and a parity-symmetric spin-liquid wave function.
- The low-energy effective theory (the first-order mean-field theory) of the π -flux phase contains gapless Dirac fermions coupled to a $U(1)$ gauge field. The Dirac fermions carry spin-1/2 and are electrically neutral.
- The concepts of stable, marginal, and unstable mean-field states.
- First-order mean-field theory is reliable only for stable mean-field states or marginal mean-field states with weak fluctuations. The π -flux phase is not one of these mean-field states.

The second mean-field state that we are going to study is the π -flux state (Affleck and Marston, 1988; Kotliar, 1988) for the nearest-neighbor Heisenberg model. The π -flux state is given by the ansatz (see Fig. 9.2)

$$\chi_{i,i+x} = i\chi_1, \quad \chi_{i,i+y} = i\chi_1(-)^{i_x}, \quad a_0(i) = 0. \quad (9.1.18)$$

The mean-field Hamiltonian (9.1.6) is equivalent to an electron hopping problem with π flux per plaquette, because around a plaquette $\prod \chi_{ij} \propto e^{i\pi}$. The spinon spectrum in the π -flux phase is given by (see Fig. 9.2)

$$E_k = \pm J_1 \chi_1 \sqrt{\sin(k_x)^2 + \sin(k_y)^2} \quad (9.1.19)$$

where $-\frac{\pi}{2} < k_x < \frac{\pi}{2}$ and $-\pi < k_y < \pi$. The value of χ_1 can be obtained by minimizing the mean-field energy

$$-\sum_{\mathbf{k}}' J_1 \chi_1 \sqrt{\sin(k_x)^2 + \sin(k_y)^2} + J_1 |\chi_1|^2 N_{\text{site}}$$

We find that

$$\chi_1 = \frac{1}{4} \int \frac{d^2 \mathbf{k}}{(2\pi)^2} \sqrt{\sin(k_x)^2 + \sin(k_y)^2}$$

In addition to the π -flux phase, there are many other different mean-field ansatz which locally minimize the mean-field energy. However, for the Heisenberg model with only nearest-neighbor coupling J_1 , the π -flux phase has the lowest mean-field energy among translationally-invariant spin liquids.

Let us discuss some physical properties of the π -flux phase. First, let us consider the symmetry of the mean-field state. We know that mean-field theory is designed to describe a spin state whose wave function is given by $\Psi_{\text{spin}}(\{\alpha_i\})$, where $\alpha_i = \pm 1/2$ are the values of S^z at the site i . When we say ‘the symmetry of the mean-field state’, we really mean ‘the symmetry of the corresponding spin wave function $\Psi_{\text{spin}}(\{\alpha_i\})$ ’. The corresponding spin wave function is obtained by projecting the mean-field state to the subspace of one fermion per site (see eqn (9.1.12)).

As the mean-field ansatz χ_{ij} does not depend on the spin orientation, in the mean-field ground state every negative energy level is occupied by a spin-up and a spin-down fermion. Thus, the mean-field state $|\Psi_{\text{mean}}^{(\chi_{ij})}\rangle$ is spin-rotation invariant. As a result, the physical spin wave function obtained by the projection is also spin-rotation invariant.

However, the mean-field ansatz is not invariant under translation in the x direction by one lattice spacing. Therefore, the mean-field state $|\Psi_{\text{mean}}^{(\chi_{ij})}\rangle$ breaks the translational symmetry. This seems to suggest that the physical spin wave function also breaks the translational symmetry. In fact, the physical spin wave function does not break the translational symmetry. This is because the translated ansatz $(\chi_{i,i+x}, \chi_{i,i+y}) = (-i\chi_1(-)^{i_x}, i\chi_1)$ is related to the original ansatz $(\chi_{i,i+x}, \chi_{i,i+y}) = (i\chi_1(-)^{i_x}, i\chi_1)$ by the gauge transformation (9.1.11) with $e^{i\theta_i} = (-)^{i_x}$. Therefore, the two ansatz give rise to the same physical spin wave function. The projected spin wave function is invariant under translation. The π -flux phase is a translationally-symmetric spin liquid.

Second, we consider the properties of low-energy excitations. The Fermi surfaces at half-filling (i.e. one fermion per site on average) are points at $(k_x, k_y) = (0, 0)$ and $(0, \pi)$. At the mean-field level, the π -flux state contains gapless spin excitations which correspond to the particle-hole excitations across the Fermi points. In addition to these gapless spin excitations, the mean-field flux state also contains $U(1)$ gauge fluctuations (a_{ij}, a_0) . In first-order mean-field theory, the low-energy excitations are described by gapless spinons coupled to the $U(1)$ gauge field.

If we ignore the interaction between the spinons f_i and the gauge fluctuations (a_{ij}, a_0) , then the π -flux phase will have gapless neutral spin-1/2 excitations described by fermionic quasiparticles. However, one should not believe this striking result unless it is shown that this result remains valid after the inclusion of the gauge interaction.

Now let us consider the interactions between the spin excitations and the gauge fluctuations. For convenience, let us add the Maxwell term $\frac{1}{8\pi g^2}(v^2 f_{12}^2 - f_{0i}^2)$ to the Hamiltonian (9.1.10), where g is a coupling constant and $f_{\mu\nu}$ is the field strength of a_μ . The original theory corresponds to the $g \rightarrow \infty$ limit. The Maxwell term is generated in the process of integrating out the fermions. Let $g(\Lambda)$ be the coupling constant obtained by integrating out the fermions between the energy scales Λ and Λ_0 , where Λ_0 is the energy cut-off in the original theory. Using a dimensional analysis, we find that $dg^{-2}(\Lambda) \sim d\Lambda/\Lambda^2$ (note that $dg^{-2}(\Lambda) \propto d\Lambda$). Thus $g^{-2}(\Lambda) \sim \Lambda^{-1} - \Lambda_0^{-1}$.

Due to the coupling with the gauge field, a spinon creates an ‘electric’ field f_{0i} of the gauge field (a_0, a_{ij}) (notice that a spinon carries a unit charge of the gauge field). The potential energy between a particle-hole pair in a particle-hole excitation is

$$V(\mathbf{r}_1 - \mathbf{r}_2) = 2\pi g^2 \ln |\mathbf{r}_1 - \mathbf{r}_2| \quad (9.1.20)$$

We find that the particle and the hole interact at long distances. To estimate the strength and effect of the interaction, let us compare the kinetic energy $E_K \sim 1/|\mathbf{r}_1 - \mathbf{r}_2|$ and the potential energy $V(\mathbf{r}_1 - \mathbf{r}_2)$ of the particle-hole pair. As $g^2 \sim v/|\mathbf{r}_1 - \mathbf{r}_2|$ (assuming that $\Lambda \sim v/|\mathbf{r}_1 - \mathbf{r}_2|$), we see that $V/E_K \sim 1$. This implies that the interaction is marginal and we cannot ignore the interaction at low energies. This result suggests that the quantum fluctuations (e.g. the gauge fluctuations) in the π -flux state are important and will drastically change the low-energy properties of the zeroth-order mean-field theory. In this case, zeroth-order mean-field theory (with free spinon excitations) does not provide a reliable picture of the low-energy properties of spin liquids. To obtain reliable low-energy properties of the π -flux state, one has to deal with the coupled system of gapless fermions and

the gauge field. The striking properties obtained by ignoring the gauge interaction may not be valid.⁵⁵

We see that the unbelievers of the projective construction are right. The results from zeroth-order mean-field theory are misleading and cannot be trusted. However, the believers are also right. First-order mean-field theory does tell us that the results from zeroth-order mean-field theory are incorrect. So far, the projective construction has not misled us and can be trusted, once we include the proper fluctuations. It is the results from first-order mean-field theory (not zeroth-order mean-field theory) that correspond to the physical properties of spin liquids.

To characterize the importance of the fluctuations around the mean-field state, here we would like to introduce the three concepts of stable, marginal, and unstable mean-field states. In a stable mean-field state, the fluctuations are weak. The interaction induced by the fluctuations vanishes at low energies (i.e. the interactions are irrelevant perturbations). In a marginal mean-field state, the ratios of the interactions and the energy approach finite constants in the low-energy limit. In this case, the interactions are marginal perturbations. In an unstable mean-field state, the ratios of the interactions and the energy diverge in the low-energy limit. Then the interactions are relevant perturbations. The properties of zeroth-order mean-field theories can survive the fluctuations only for stable mean-field states. For unstable mean-field states, the fluctuations will drive a phase transition at low energies and we cannot deduce any physical properties of spin liquids from zeroth-order mean-field theory. In this case, first-order mean-field theory is not useful because it does not help us to deduce the physical properties of the spin system. The π -flux state discussed above is a marginal mean-field state. For marginal mean-field states, first-order mean-field theory can be useful if the ratios of the interaction and the energy are small. In this case, the physical properties of the corresponding spin state can be calculated perturbatively. For the π -flux state, the ratios of the interaction and the energy are of order 1. Thus, it is hard to obtain the low-energy physical properties of the π -flux state from first-order mean-field theory.

From the above discussion, we see that first-order mean-field theory is useful only for stable mean-field states and for marginal mean-field states with weak fluctuations. The key to using mean-field theory to study spin liquids is to find stable mean-field states (see Sections 9.1.6, 9.2.4 and 9.2.6), or marginal mean-field states with weak fluctuations (see Section 9.8).

Problem 9.1.2.

The rotational symmetry of the π -flux state

1. Prove eqn (9.1.14).
2. Show that the π -flux state does not break the 90° -rotational symmetry.

⁵⁵ When mean-field fluctuations are weak, a marginal mean-field state can lead to an algebraic spin liquid — an spin liquid with no free quasiparticles at low energies. See Sections 9.9.5 and 9.10.

Problem 9.1.3.

Let $|\Psi_{\text{mean}}\rangle$ be the mean-field ground state for the π -flux ansatz. Calculate $\langle\Psi_{\text{mean}}|\mathbf{S}_i\mathbf{S}_{i+\mathbf{y}}|\Psi_{\text{mean}}\rangle$. Use this result to find the variational mean-field energy $\langle\Psi_{\text{mean}}|H|\Psi_{\text{mean}}\rangle$, where H is the Heisenberg model with nearest-neighbor interactions (see eqn (9.1.15)). Compare the mean-field energy of the π -flux state to the energy of the dimer state.

Problem 9.1.4.**Dirac fermions**

1. Show that, in the continuum limit, the low-energy spinons in the π -flux state are described by

$$H = v \sum_{\mathbf{k} \sim 0} \lambda_{\alpha, \mathbf{k}}^\dagger (k_x \Gamma_x + k_y \Gamma_y) \lambda_{\alpha, \mathbf{k}}$$

where $\lambda_{\alpha, \mathbf{k}}^\dagger = (f_{\alpha, \mathbf{k}}, f_{\alpha, \mathbf{k}+\mathbf{Q}})$, $\mathbf{Q} = (\pi, \pi)$, and $\Gamma_x^2 = \Gamma_y^2 = 1$. Find v and $\Gamma_{x,y}$. In real space, H can be rewritten as $H = \int d^2x \, v \lambda_\alpha^\dagger (-i\Gamma_x \partial_x - i\Gamma_y \partial_y) \lambda_\alpha$.

2. The corresponding Lagrangian $\mathcal{L} = i\lambda_\alpha^\dagger \partial_t \lambda_\alpha - v \lambda_\alpha^\dagger (-i\Gamma_x \partial_x - i\Gamma_y \partial_y) \lambda_\alpha$ can be rewritten as

$$\mathcal{L} = i\bar{\lambda}_\alpha (\gamma^0 \partial_t + v\gamma^x \partial_x + v\gamma^y \partial_y) \lambda_\alpha$$

where $\bar{\lambda}_\alpha = \lambda_\alpha^\dagger \gamma^0$, $\gamma_0^2 = 1$, and $(\Gamma_x, \Gamma_y) = (-\gamma^0 \gamma^x, -\gamma^0 \gamma^y)$. Find $\gamma^{0,x,y}$. The fermion described by $\mathcal{L}$ is called the massless Dirac fermion.

3. Show that $(\gamma^0, \gamma^x, \gamma^y)$ satisfies the following Dirac algebra in 1 + 2 dimensions:

$$\{\gamma^\mu, \gamma^\nu\} = \eta^{\mu\nu}, \quad \mu, \nu = 0, x, y,$$

where $\eta^{\mu\nu}$ is a diagonal matrix with $\eta^{00} = 1$ and $\eta^{xx} = \eta^{yy} = -1$.

4. A massive Dirac fermion is described by the Lagrangian

$$\mathcal{L} = i\bar{\lambda}_\alpha (\gamma^0 \partial_t + v\gamma^x \partial_x + v\gamma^y \partial_y) \lambda_\alpha + m\bar{\lambda}_\alpha \lambda_\alpha.$$

Find the solutions of the corresponding equation of motion $i(\gamma^0 \partial_t + v\gamma^x \partial_x + v\gamma^y \partial_y + m)\lambda_\alpha = 0$. Show that the fermion has dispersion $\omega_{\mathbf{k}} = \sqrt{v^2 \mathbf{k}^2 + m^2}$.

5. *Coupling to the $U(1)$ gauge field*

Use the procedure described in Section 3.7.1 to find the Lagrangian of the Dirac fermion that is minimally coupled to a $U(1)$ gauge field. Make sure that the resulting Lagrangian has a $U(1)$ gauge invariance.

9.1.5 How to kill gapless $U(1)$ gauge bosons

- How to obtain a stable mean-field state: give gauge fluctuations a finite energy gap via the Chern–Simons term and/or the Anderson–Higgs mechanism.
- The stable spin liquids always contain electrically-neutral spin-1/2 spinons with only short-ranged interactions between them.

We have seen that the gapless $U(1)$ gauge bosons interact with gapless fermions strongly in the π -flux state, even down to zero energy. It is hard to obtain the properties of the first-order mean-field theory (i.e. the fermion–gauge coupled system) for the π -flux phase. Although the first-order mean-field theory does not mislead us, it seems too complicated to be useful. Now the question is whether it is possible to construct a stable mean-field state in which mean-field fluctuations interact weakly at low energies. In this case, the properties of the first-order mean-field theory can be obtained easily.

As the first-order mean-field theory must contain, at least, the gauge fluctuations to enforce the one-fermion-per-site constraint, one way to have a stable mean-field state is to give gauge bosons an energy gap. The gapped gauge boson can only mediate a short-range interaction between the fermions. We know how to handle fermions with a short-range interaction, based on our experience with Landau Fermi liquid theory.

First let us clarify what we mean by ‘give gauge bosons an energy gap’. We know that gauge bosons are simply fluctuations of the mean-field ansatz χ_{ij} . The dynamics of these fluctuations depends on the mean-field ansatz. So, by ‘give gauge bosons an energy gap’ we mean to find the mean-field ansatz such that the collective mode fluctuations of the ansatz are gapped.

To motivate our search for a stable mean-field ansatz, let us first consider in what ways a $U(1)$ gauge boson can gain an energy gap. In fact, in Sections 3.7.5 and 4.4, we have already encountered two ways in which a $U(1)$ gauge boson can be gapped. The first way is the Anderson–Higgs mechanism, where the gauge bosons couple to condensed charge bosons. The dynamics of the coupled system is described by

$$\mathcal{L} = \frac{1}{8\pi g^2}(\mathbf{e}^2 - b^2) + c_1 a_0^2 - c_2 \mathbf{a}^2$$

in $2 + 1$ dimensions and in the continuum limit (see eqn (3.7.17)), where c_1 and c_2 come from the condensed charge bosons. The second way is via the Chern–Simons term. If the mean-field ansatz is such that the filled band has a nonzero Hall conductance, then the dynamics of $U(1)$ gauge bosons will be described by

$$\mathcal{L} = \frac{1}{8\pi g^2}(\mathbf{e}^2 - b^2) + \frac{K}{4\pi} \epsilon^{\mu\nu\lambda} a_\mu \partial_\nu a_\lambda. \quad (9.1.21)$$

In Problem 9.1.5 we will find that the Chern–Simons term gives the gauge boson a nonzero energy gap. The key to finding a stable mean-field state is to find a mean-field ansatz that realizes one of the above two mechanisms.

In both cases, the gauge field can only mediate short-range interactions. As a consequence, the spinons are not confined. The quasiparticles above the spin liquid are described by free spinons which carry spin-1/2 and zero electric charge. In the presence of the Chern–Simons term, the spinons can even have fractional statistics.

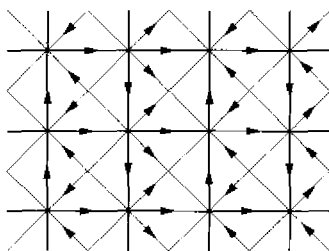


FIG. 9.3. The mean-field ansatz of a chiral spin state. Here χ_{ij} is $i\chi$ in the direction of the arrow.

Problem 9.1.5.

Show that the fluctuations described by eqn (9.1.21) have a finite energy gap. (You may do so by calculating the classical equation of motion.)

9.1.6 Chiral spin state

- The mean-field chiral spin ansatz gives rise to a translationally- and rotationally-symmetric spin-liquid state. However, the time-reversal and parity symmetries are broken.
- The mean-field chiral spin state is stable. The chiral spin liquid obtained from the mean-field theory does represent a stable quantum phase of real spin systems.
- The chiral spin liquid contains fractionalized excitations—spinons. The spinons carry spin-1/2 and no electric charge, and have fractional statistics.

In this section, we will discuss a mean-field state which realizes the second mechanism discussed above. The mean-field state is described by an ansatz χ_{ij} , where χ_{ij} is complex and generates flux. The spinons described by eqn (9.1.6) behave as if they are moving in a magnetic field. When the flux has a correct commensuration with the spinon density (which is one fermion per site), an integral number of Landau levels (or, more precisely, Landau bands, due to the lattice) are completely filled. In this case, the effective lattice gauge theory contains a Chern–Simons term due to the finite Hall conductance of the filled Landau level. We will call such a stable mean-field state a chiral spin state (Wen *et al.*, 1989; Khveshchenko and Wiegmann, 1989).

The simplest chiral spin state is given by the following ansatz (see Fig. 9.3):

$$\begin{aligned} \chi_{i,i+x} &= i\chi_1, & \chi_{i,i+y} &= i\chi_1(-)^{i_x}, & a_0(i) &= 0, \\ \chi_{i,i+x+y} &= -i\chi_2(-)^{i_x}, & \chi_{i,i+x-y} &= i\chi_2(-)^{i_x} \end{aligned} \quad (9.1.22)$$

The above χ_{ij} induce π flux for each square and $\frac{1}{2}\pi$ flux for each triangle. The π -flux phase discussed in the last section is somewhat special. As π flux is equivalent to $-\pi$ flux, the π -flux phase respects time-reversal symmetry (T) and parity (P). However, the $\frac{1}{2}\pi$ -flux phase described by eqn (9.1.22) is not equivalent to the $-\frac{1}{2}\pi$ -flux phase. Under T or P , the $\frac{1}{2}\pi$ flux changes into $-\frac{1}{2}\pi$ flux, and χ_2 in eqn (9.1.22) changes into $-\chi_2$. Therefore, the chiral spin state with nonzero χ_2 breaks T and P spontaneously. Using the identity

$$E_{123} \equiv \mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3) = 2i(\hat{\chi}_{12}\hat{\chi}_{23}\hat{\chi}_{31} - \hat{\chi}_{13}\hat{\chi}_{32}\hat{\chi}_{21}), \quad \hat{\chi}_{ij} = f_i^\dagger f_j \quad (9.1.23)$$

we find that $\langle E_{123} \rangle \sim \text{Im}(\chi_{12}\chi_{23}\chi_{31})$ is nonzero in the chiral spin state. Notice that E_{123} is odd under T or P ; thus, E_{123} can be regarded as the T - and P -breaking order parameter.

After specifying the chiral spin state, we would like to know which spin Hamiltonian supports the chiral spin state. Let us consider the following frustrated spin Hamiltonian:

$$H = J_1 \sum_i (\mathbf{S}_i \cdot \mathbf{S}_{i+x} + \mathbf{S}_i \cdot \mathbf{S}_{i+y}) + J_2 \sum_i (\mathbf{S}_i \cdot \mathbf{S}_{i+x+y} + \mathbf{S}_i \cdot \mathbf{S}_{i+x-y}) \quad (9.1.24)$$

We would like to know when the mean-field Hamiltonian (9.1.6) supports the mean-field chiral spin state. In the mean-field Hamiltonian (9.1.6), J_{ij} is equal to J_1 for the nearest neighbor and J_2 for the second-nearest neighbor.

When $\chi_2 = 0$, the spinon spectrum determined by the mean-field Hamiltonian is given by eqn (9.1.19). The conduction band and the valence band touch at the points $(k_x, k_y) = (0, 0)$ and $(0, \frac{\pi}{a})$. When $\chi_2 \neq 0$, an energy gap between the conduction band and the valence band is opened. The mean-field spinon spectrum is given by

$$E_{\mathbf{k}}^\pm = \pm \sqrt{J_1^2 \chi_1^2 (\sin(k_x)^2 + \sin(k_y)^2) + J_2^2 \chi_2^2 (\cos(k_x + k_y) + \cos(k_x - k_y))^2} \quad (9.1.25)$$

The mean-field ground state is obtained by filling the valence band. Due to the energy gap, the spin-spin correlation is short-ranged. Using the ground-state wave function, we can calculate $\langle f_{\alpha i}^\dagger f_{\alpha j} \rangle$ to check the self-consistency condition (9.1.7). We find that, when $J_2/J_1 < 0.49$, the self-consistency equation only supports one solution with $\chi_1 \neq 0$ and $\chi_2 = 0$, i.e. the π -flux phase. When $J_2/J_1 > 0.49$, eqn (9.1.7) also supports a second solution with $\chi_2 \neq 0$. The second solution is found to have a lower mean-field energy. In this case, T and P are spontaneously broken.

Notice that the mean Hamiltonian for the chiral state is equivalent to the problem of electron hopping in a magnetic field. The coupling between the slave fermions and the gauge field a_μ is identical to the coupling between the electrons and the electromagnetic field. Thus, one expects that the slave-fermion system

described by eqn (9.1.10) has a phenomenon similar to the Hall effect. The ‘Hall effect’ in this case implies that an ‘electric’ field of the a_μ gauge field induces a spinon current in the transverse direction:

$$j_x = \sigma_{xy} e_y, \quad e_y = \partial_t a_y - \partial_y a_0 \quad (9.1.26)$$

where σ_{xy} is the Hall conductance. There is a theorem stating that the Hall conductance of a filled band is always quantized as an integer times $1/2\pi$ (Thouless *et al.*, 1982; Avron *et al.*, 1983). In our case, this implies that $\sigma_{xy} = 2n/2\pi$, where the factor 2 comes from the spins. The value of the ‘Hall’ conductance can be obtained using the method discussed in Section 4.4. We find that

$$\sigma_{xy} = 2/2\pi. \quad (9.1.27)$$

The simplest way to understand this result is to note that there is one flux quantum for every spin-up spinon, and one flux quantum for every spin-down spinon. Thus, both spin-up and spin-down spinons have a fill fraction of $\nu = 1$. After turning off the lattice potential, the valence band in the mean-field chiral spin state becomes the first Landau level. Both the spin-up and spin-down slave fermions in the valence band contribute $1/2\pi$ to the ‘Hall’ conductance. Therefore, the total ‘Hall’ conductance is given by eqn (9.1.27).

The effective action for the gauge fluctuations is obtained by integrating out the spinons in eqn (9.1.10). Using the relationship between the ‘Hall’ conductance and the Chern–Simons term, we can easily write down the effective action in the continuum limit as follows:

$$S = \int d^3x \frac{1}{2} \sigma_{xy} a_\mu \partial_\nu a_\lambda \epsilon_{\mu\nu\lambda} + \frac{1}{8\pi g^2} (e^2 - v^2 b^2) + \dots \quad (9.1.28)$$

Here g^2 in eqn (9.1.28) is of the order of the spinon gap, and v is of order $1/aJ$, the typical spinon velocity. Using the effective Lagrangian, we can calculate the low-energy dynamical properties of the gauge fluctuations.

Due to the nonzero ‘Hall’ conductance σ_{xy} , we can change the spinon density *without* creating spinons in the conduction band or creating holes in the valence band. This property is important in understanding the properties of the quasiparticles. Let us slowly turn on the flux of the a_μ field, $\Phi = \int d^2x b$. If the flux is distributed in a large region, the ‘magnetic’ field b in the flux is small. In this case, the energy gap between the valence band and the conduction band remains finite and all of the energy levels in the valence band are filled by a spin-up and a spin-down spinon. Turning on the flux induces a circular ‘electric’ field e_θ , which in turn generates a spinon current in the radial direction $\hat{r}$ due to the nonzero σ_{xy} . Thus, some charges are accumulated near the origin. We find that the total number

of induced spinons is given by

$$N \equiv \sum_i (\langle f_{\alpha i}^\dagger f_{\alpha i} \rangle - 1) = -\sigma_{xy} \Phi = -\frac{\Phi}{\pi} \quad (9.1.29)$$

We would like to stress that the flux only changes the spinon density. It does not induce any spin quantum number. No matter how many spinons are induced by the flux, the flux tube always carries zero spin because every energy level in the valence band is filled by a spin-up and a spin-down spinon.

Now we are ready to discuss the quantum number of the quasiparticle excitations in the chiral spin state. The simplest excitation in the mean-field chiral spin state can be obtained by adding a spinon into the conduction band. However, this excitation is not physical because the additional spinon in the conduction band violates the constraint (9.1.5). To satisfy the constraint, we can add flux to change the spinon density in the valence band. From the above discussion, we see that the extra spinon density arising from the spinon in the conduction band can be cancelled by introducing π flux (see eqn (9.1.29)). Therefore, the physical quasiparticles in the chiral spin state are spinons *dressed* by π flux. The dressed spinons carry spin-1/2 because the flux cannot induce any spin quantum numbers. However, as a bound state of a charge and flux (note that the spinon carries a unit charge of the a_μ gauge field), the spinon has fractional statistics. Exchanging two dressed spinons is equivalent to moving one spinon half-way around the other, which induces a phase $\theta = \frac{1}{2}q\Phi$. Here $q = 1$ is the charge of the spinon and $\Phi = \pi$ is the flux bounded to the spinon. Hence the statistical angle of the dressed spinon is $\theta = \frac{\pi}{2}$. Particles with such statistics are half-way between bosons (with $\theta = 0$) and fermions (with $\theta = \pi$), and are called semions.

To understand the low-energy dynamics of the spinons, we would like to derive the effective Lagrangian for the spinons. First let us ignore the gauge fluctuations by setting $a_\mu = 0$. In this case, a spinon in the conduction band or the valence band is described by the dispersion given in eqn (9.1.25). Here E_k^+ (E_k^-) has two minima (maxima) at $(0, 0)$ and $(0, \pi/a)$. Therefore, in the continuum limit, we have four species of spinons described by the following effective Lagrangian:

$$\mathcal{L} = \sum_{I=1,2} \left[-i f_{I\alpha}^\dagger \partial_t f_{I\alpha} + \frac{1}{2m_s} f_{I\alpha}^\dagger \partial_i^2 f_{I\alpha} - i \bar{f}_{I\alpha}^\dagger \partial_t \bar{f}_{I\alpha} + \frac{1}{2m_s} \bar{f}_{I\alpha}^\dagger \partial_i^2 \bar{f}_{I\alpha} \right] \quad (9.1.30)$$

Here $f_{1\alpha}$ and $f_{2\alpha}$ correspond to the spinons near the two minima of the conduction band, and $\bar{f}_{1\alpha}$ and $\bar{f}_{2\alpha}$ correspond to the holes near the maxima of the valence band. When $a_\mu \neq 0$, the coupling between the spinons and the gauge field can be obtained by replacing ∂_μ in eqn (9.1.30) by $\partial_\mu \pm ia_\mu$. Such a form of coupling is determined by the requirement of gauge invariance. After including the coupling

to the gauge field, the total effective Lagrangian has the form

$$\begin{aligned}\mathcal{L} = & \sum_{I=1,2} \left[-i f_{I\alpha}^\dagger (\partial_t - i a_0) f_{I\alpha} + \frac{1}{2m_s} f_{I\alpha}^\dagger (\partial_i - i a_i)^2 f_{I\alpha} \right] \\ & + \sum_{I=1,2} \left[-i \bar{f}_{I\alpha}^\dagger (\partial_t + i a_0) \bar{f}_{I\alpha} + \frac{1}{2m_s} \bar{f}_{I\alpha}^\dagger (\partial_i + i a_i)^2 \bar{f}_{I\alpha} \right] \\ & - \frac{2}{4\pi} a_\mu \partial_\nu a_\lambda \epsilon_{\mu\nu\lambda} + \frac{1}{8\pi g^2} (\mathbf{c}^2 - v^2 b^2)\end{aligned}\quad (9.1.31)$$

From the equation of motion $\frac{\delta \mathcal{L}}{\delta a_i} = 0$, we find that

$$n_1 + n_2 = -\frac{1}{\pi} b \quad (9.1.32)$$

where $n_I = \sum_\alpha (f_{I\alpha}^\dagger f_{I\alpha} - \bar{f}_{I\alpha}^\dagger \bar{f}_{I\alpha})$, $I = 1, 2$, is the density of the spinons. Equation (9.1.32) tells us that a spinon is dressed by π flux, which agrees with the previous results. The statistics of the spinons can also be directly calculated from eqn (9.1.31).

We would like to point out that, according to eqn (9.1.31), there are two kinds of quasiparticles for each fixed I , namely the spin-1/2 spinons in the conduction band and the spin-1/2 holes in the valence band. This result is *incorrect*. For each fixed value of I there should only be one kind of quasiparticles, namely the spin-1/2 spinons. The spinons in the conduction band and hole in the valence band give rise to the same spinon after the projection (Affleck *et al.*, 1988; Dagotto *et al.*, 1988). This over-counting problem can be resolved after realizing that the chiral spin ansatz actually has an $SU(2)$ gauge structure (see Problem 9.2.4).

Problem 9.1.6.

Find the equations that determine χ_1 and χ_2 in the ansatz (9.1.22) of the chiral spin state by minimizing the mean-field energy (9.1.6).

Problem 9.1.7.

Prove eqn (9.1.25).

Problem 9.1.8.

Prove eqn (9.1.27) by using the result of Section 4.4.

9.2 The $SU(2)$ projective construction

9.2.1 The hidden $SU(2)$ gauge structure

- The projective construction has a hidden $SU(2)$ gauge structure.

- The mean-field ansatz $(U_{ij}, a_0^l(i))$ is a many-to-one label of physical spin liquids. The $SU(2)$ gauge-equivalent ansatz labels the same physical wave function.
- The invariance of a physical wave function under a symmetry transformation only requires the invariance of the corresponding mean-field ansatz up to a gauge transformation.

9.2.1.1 Spinon-pair condensation

In the last section, we constructed a stable mean-field state using a Chern–Simons term. In this section, we would like to consider another class of stable mean-field state due to the Anderson–Higgs mechanism. In order for the Anderson–Higgs mechanism to work, we first need a boson that carries a_μ charge. Secondly, the charged bosons should have proper dynamics, so that they condense. In the mean-field theory obtained from the projective construction, we do not have bosons that carry a_μ charge; but we have fermions that carry a_μ charge. So we can make charged bosons from pairs of charged fermions and let those fermion pairs condense. We see that the Anderson–Higgs mechanism can be achieved through fermion-pair condensations.⁵⁶ In the following, we will include the fermion-pair condensation in our mean-field ansatz. We hope that these mean-field ansatz will represent stable mean-field states.

Remember that, in terms of the spinon operators, the Hamiltonian (9.1.1) can be rewritten as

$$H = \sum_{\langle ij \rangle} -\frac{1}{2} J_{ij} \left(f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{i\beta} + \frac{1}{2} f_{i\alpha}^\dagger f_{i\alpha} f_{j\beta}^\dagger f_{j\beta} \right) \quad (9.2.1)$$

where we have added proper constant terms $\sum_i f_{i\alpha}^\dagger f_{i\alpha}$ to obtain the above result.

To obtain the mean-field Hamiltonian that contains fermion-pair condensation, we replace both the operators $f_{i\alpha}^\dagger f_{j\beta}$ and $f_{i\alpha} f_{i\beta}$ by their ground-state expectation value

$$\begin{aligned} \eta_{ij} \epsilon_{\alpha\beta} &= -2 \langle f_{i\alpha} f_{j\beta} \rangle, & \eta_{ij} &= \eta_{ji}, \\ \chi_{ij} \delta_{\alpha\beta} &= 2 \langle f_{i\alpha}^\dagger f_{j\beta} \rangle, & \chi_{ij} &= \chi_{ji}^\dagger. \end{aligned} \quad (9.2.2)$$

The χ_{ij} term was included in the previous mean-field ansatz. The η_{ij} term is new and describes fermion-pair condensation. We also replace the constraint (9.1.4) by

⁵⁶ If the fermions are electrons, then the fermion-pair condensed state is a BCS superconducting state.

its ground-state average

$$\langle f_{i\alpha}^\dagger f_{i\alpha} \rangle = 1, \quad \langle f_{i\alpha} f_{i\beta} \epsilon_{\alpha\beta} \rangle = 0 \quad (9.2.3)$$

Such a constraint can be enforced by introducing *site-dependent* and time-independent Lagrangian multipliers $a_0^l(i)$, $l = 1, 2, 3$. In this way, we obtain the following zeroth-order mean-field Hamiltonian that includes fermion-pair condensation:

$$\begin{aligned} H_{\text{mean}} = & \sum_{\langle ij \rangle} -\frac{3}{8} J_{ij} \left[(\chi_{ji} f_{i\alpha}^\dagger f_{j\alpha} + \eta_{ij} f_{i\alpha}^\dagger f_{j\beta}^\dagger \epsilon_{\alpha\beta} + h.c.) - |\chi_{ij}|^2 - |\eta_{ij}|^2 \right] \\ & + \sum_i \left[a_0^3 (f_{i\alpha}^\dagger f_{i\alpha} - 1) + [(a_0^1 + i a_0^2) f_{i\alpha} f_{i\beta} \epsilon_{\alpha\beta} + h.c.] \right] \end{aligned} \quad (9.2.4)$$

Here χ_{ij} and η_{ij} in eqn (9.1.6) must satisfy the self-consistency condition (9.2.2) and the site-dependent fields $a_0^l(i)$ are chosen such that eqn (9.2.3) is satisfied by the mean-field ground state. Such χ_{ij} , η_{ij} , and a_0^l give us a mean-field solution.

For the Heisenberg model with nearest-neighbor spin coupling, see eqn (9.1.15), the following mean-field ansatz with fermion-pair condensation is a solution of the mean-field equation (9.2.2):

$$\begin{aligned} \chi_{i,i+\mathbf{x}} &= \chi, & \chi_{i,i+\mathbf{y}} &= \chi, \\ \eta_{i,i+\mathbf{x}} &= \eta, & \eta_{i,i+\mathbf{y}} &= -\eta, \\ a_0^l &= 0. \end{aligned} \quad (9.2.5)$$

Such an ansatz actually corresponds to a d -wave BCS state. We will call it a d -wave state.⁵⁷ Due to the fermion-pair condensation, we expect that the mean-field state described by the above ansatz contains no gapless $U(1)$ gauge bosons due to the Anderson–Higgs mechanism.

Well, despite that everything seems to fit together very well and the physical picture seems to be very reasonable, the above result turns out to be incorrect. Where did we make a mistake? The physical picture and reasoning presented here are correct. The mistake turns out to be a mathematical one. We have claimed that the mean-field theory (9.1.10) (or, more generally, eqn (9.2.4)) has a $U(1)$ gauge structure, such that the two ansatz related by the $U(1)$ gauge transformation (9.1.13) correspond to the same physical spin wave function (see eqn (9.1.14)) after the projection. In fact, the mean-field theory has a larger $SU(2)$ gauge structure. Two ansatz related by the $SU(2)$ gauge transformation correspond to the same

⁵⁷ We note that the fermion-pairing order parameter η_{ij} changes sign under a 90° rotation. This is similar to a wave function that carries an angular momentum of 2. This is why we call the state the d -wave state.

physical spin wave function. The different gauge structure has a profound consequence on our understanding of the properties of mean-field states. In particular, the fluctuations in χ_{ij} , η_{ij} , and $a_0^l(i)$ describe $SU(2)$ gauge fluctuations.

9.2.1.2 The $SU(2)$ formulation

To understand the $SU(2)$ gauge structure in the mean-field Hamiltonian (9.2.4) and in the constraints (9.1.4) (Affleck *et al.*, 1988; Dagotto *et al.*, 1988), we introduce a doublet

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \begin{pmatrix} f_{\uparrow} \\ f_{\downarrow} \end{pmatrix} \quad (9.2.6)$$

and a matrix

$$U_{ij} = \begin{pmatrix} \chi_{ij}^{\dagger} & \eta_{ij} \\ \eta_{ij}^{\dagger} & -\chi_{ij} \end{pmatrix} = U_{ji}^{\dagger} \quad (9.2.7)$$

Using eqns (9.2.6) and (9.2.7), we can rewrite eqns (9.2.3) and (9.2.4) as follows:

$$\langle \psi_i^{\dagger} \tau^l \psi_i \rangle = 0, \quad l = 1, 2, 3 \quad (9.2.8)$$

$$H_{\text{mean}} = \sum_{\langle ij \rangle} \frac{3}{8} J_{ij} \left[\frac{1}{2} \text{Tr}(U_{ij}^{\dagger} U_{ij}) - (\psi_i^{\dagger} U_{ij} \psi_j + h.c.) \right] + \sum_i a_0^l \psi_i^{\dagger} \tau^l \psi_i \quad (9.2.9)$$

where τ^l , $l = 1, 2, 3$, are the Pauli matrices. From eqn (9.2.9), we can clearly see that the Hamiltonian is invariant under a local $SU(2)$ transformation W_i :

$$\begin{aligned} \psi_i &\rightarrow W_i \psi_i \\ U_{ij} &\rightarrow W_i U_{ij} W_j^{\dagger} \end{aligned} \quad (9.2.10)$$

The $SU(2)$ gauge structure actually originates from eqn (9.1.2). Here $SU(2)$ is the most general transformation between the spinons that leaves the physical spin operator unchanged. Those transformations become the gauge transformation, since the physical Hamiltonian is a function of the spin operators.

9.2.1.3 The meaning of the $SU(2)$ gauge transformation

Just like the $U(1)$ gauge transformation, the $SU(2)$ gauge transformation has the following meaning: two ansatz related by a $SU(2)$ gauge transformation correspond to the same physical spin wave function. To see this, we note that the

f -fermion states and the ψ -fermion states at each site have the following relation:

$$\begin{aligned} |0_f\rangle &= \psi_2^\dagger |0_\psi\rangle, & f_\uparrow^\dagger f_\downarrow^\dagger |0_f\rangle &= \psi_1^\dagger |0_\psi\rangle, \\ f_\downarrow^\dagger |0_f\rangle &= |0_\psi\rangle, & f_\uparrow^\dagger |0_f\rangle &= \psi_1^\dagger \psi_2^\dagger |0_\psi\rangle \end{aligned}$$

where $|0_\psi\rangle$ is the state with no ψ -fermion, i.e. $\psi_\alpha |0_\psi\rangle = 0$. Thus, the physical one- f -fermion-per-site states correspond to states with even numbers of ψ -fermions per site. These even- ψ -fermion-per-site states are local $SU(2)$ singlet states (i.e. they are an $SU(2)$ singlet on every site). The empty ψ -fermion state corresponds to a down-spin and the doubly-occupied ψ -fermion state corresponds to an up-spin. With this understanding, we may obtain a valid wave function of the spin system by projecting the mean-field state to the even- ψ -fermion-per-site subspace. Let $i_1, i_2, \dots$ be the locations of the up-spins. The physical spin wave function can be written as a function of $i_1, i_2, \dots$, i.e. $\Psi_{\text{spin}}(i_1, i_2, \dots)$. As $i_1, i_2, \dots$ are the only sites with two ψ -fermions, we find that

$$\Psi_{\text{spin}}(\{i_n\}) = \langle 0_\psi | \prod_n \psi_{1,i_n} \psi_{2,i_n} | \Psi_{\text{mean}}^{(U_{ij})} \rangle. \quad (9.2.11)$$

As $\langle 0_\psi |$ and $\psi_{1,i} \psi_{2,i}$ are invariant under the local $SU(2)$ transformation, the two ansatz U_{ij} and U'_{ij} related by an $SU(2)$ gauge transformation $U'_{ij} = W_i U_{ij} W_j^\dagger$ are just two different labels which label the *same physical state*:

$$\langle 0_\psi | \prod_n \psi_{1,i_n} \psi_{2,i_n} | \Psi_{\text{mean}}^{(U_{ij})} \rangle = \langle 0_\psi | \prod_n \psi_{1,i_n} \psi_{2,i_n} | \Psi_{\text{mean}}^{(W_i U_{ij} W_j^\dagger)} \rangle \quad (9.2.12)$$

The relationship between the mean-field state and the physical spin wave function (9.2.11) allows us to construct transformations of a physical spin wave function from those of the corresponding mean-field ansatz. For example, the mean-field state $|\Psi_{\text{mean}}^{(U_{ij})}\rangle$ with translated ansatz $U'_{ij} = U_{i-l,j-l}$ produces a translated physical spin wave function after the projection.

It is obvious that the translationally-invariant ansatz will lead to a translationally-invariant physical spin wave function. However, the translational symmetry of the physical wave function after projection does not require the translational invariance of the corresponding ansatz. The physical state is translationally symmetric if and only if the translated ansatz U'_{ij} is gauge equivalent to the original ansatz U_{ij} . We see that the gauge structure can complicate our analysis of symmetries, because the physical spin wave function $\Psi_{\text{spin}}(\{\alpha_i\})$ may have more symmetries than the mean-field state $|\Psi_{\text{mean}}^{(U_{ij})}\rangle$ before projection.

9.2.1.4 Spin rotation invariance

We note that both of the components of ψ carry spin-up. Thus, the spin-rotation symmetry is not explicit in our formalism and it is hard to tell whether eqn (9.2.9)

describes a spin-rotation-invariant state or not. In fact, for a general U_{ij} satisfying $U_{ij} = U_{ji}^\dagger$, eqn (9.2.9) may not describe a spin-rotation-invariant state. However, if U_{ij} has the form

$$\begin{aligned} U_{ij} &= \chi_{ij}^\mu \tau^\mu, \quad \mu = 0, 1, 2, 3, \\ \chi_{ij}^0 &= \text{imaginary}, \quad \chi_{ij}^l = \text{real}, \quad l = 1, 2, 3, \end{aligned} \quad (9.2.13)$$

then eqn (9.2.9) will describe a spin-rotation-invariant state. This is because the above U_{ij} can be rewritten in the form of eqn (9.2.7). In this case, eqn (9.2.9) can be rewritten as eqn (9.2.4), where the spin-rotation invariance is explicit. In eqn (9.2.13), τ^0 is the identity matrix and $\tau^{1,2,3}$ are the Pauli matrices.

9.2.1.5 Variational approach

We would like to remark that, in addition to the self-consistency equation (9.2.2), there is another way to obtain the mean-field solutions. We can view the mean-field ground state of H_{mean} (see eqn (9.2.9)), namely $|\Psi_{\text{mean}}^{(U_{ij})}\rangle$, as a trial wave function and U_{ij} as variational parameters. Introducing

$$(\tilde{U}_{ij})_{\alpha\beta} \equiv -2 \langle \Psi_{\text{mean}}^{(U_{ij})} | \psi_{i,\alpha} \psi_{j,\beta}^\dagger | \Psi_{\text{mean}}^{(U_{ij})} \rangle$$

we find that the mean-field energy for $|\Psi_{\text{mean}}^{(U_{ij})}\rangle$ is given by

$$E_{\text{mean}}(\{U_{ij}\}) = - \sum_{\langle ij \rangle} \frac{3}{16} J_{ij} \text{Tr}(\tilde{U}_{ij}^\dagger \tilde{U}_{ij}) \quad (9.2.14)$$

Here $E_{\text{mean}}(\{U_{ij}\})$ is a functional of U_{ij} which has $SU(2)$ gauge invariance:

$$E_{\text{mean}}(\{U_{ij}\}) = E_{\text{mean}}(\{W_i U_{ij} W_j^\dagger\})$$

The mean-field solution U_{ij} can now be obtained by minimizing E_{mean} .

9.2.1.6 First-order mean-field theory

To obtain the first-order mean-field theory, we start with the zeroth-order mean-field theory described by the mean field Hamiltonian

$$H_{\text{mean}} = \sum_{\langle ij \rangle} -\frac{3}{8} J_{ij} (\psi_i^\dagger \bar{U}_{ij} \psi_j + h.c.) + \sum_i a_0^l \psi_i^\dagger \tau^l \psi_i \quad (9.2.15)$$

where $\bar{U}_{ij}$ is the mean-field solution, which satisfies the self-consistency condition

$$\bar{\chi}_{ij} = \langle \psi_{i\alpha}^\dagger \psi_{j\alpha} \rangle, \quad \bar{\eta}_{ij} = -\langle \psi_{i\alpha} \psi_{i\beta} \epsilon_{\alpha\beta} \rangle \quad (9.2.16)$$

The a_0^l in eqn (9.2.15) are chosen such that eqn (9.2.3) is satisfied. The important fluctuations around the mean-field ground state are the following ‘phase’

fluctuations of U_{ij} :

$$U_{ij} = \bar{U}_{ij} e^{ia_{ij}} \quad (9.2.17)$$

where $a_{ij} = a_{ij}^l \tau^l$ is a 2×2 traceless hermitian matrix. Due to the $SU(2)$ gauge structure, these fluctuations are the $SU(2)$ gauge fluctuations. We must include these fluctuations in order to obtain qualitatively correct results for spin-liquid states at low energies. This leads us to the first-order mean-field theory

$$H_{\text{mean}} = \sum_{\langle ij \rangle} -\frac{3}{8} J_{ij} (\psi_i^\dagger \bar{U}_{ij} e^{ia_{ij}^l \tau^l} \psi_j + h.c.) + \sum_i a_0^l \psi_i^\dagger \tau^l \psi_i \quad (9.2.18)$$

which describes spinons coupled to $SU(2)$ lattice gauge fields.

We would like to point out that the mean-field ansatz of the spin liquids U_{ij} can be divided into two classes: the unfrustrated ansatz where U_{ij} only link an even lattice site to an odd lattice site; and the frustrated ansatz where U_{ij} are nonzero between two even sites and/or two odd sites. An unfrustrated ansatz has only pure $SU(2)$ flux through each plaquette, while a frustrated ansatz has $U(1)$ flux of a multiple of $\pi/2$ through some plaquettes in addition to the $SU(2)$ flux.

Problem 9.2.1.

Use eqn (9.2.2) to obtain eqn (9.2.4) from eqn (9.2.1).

Problem 9.2.2.

The equivalence of the π -flux state and the d -wave state

1. Show that the π -flux state (9.1.18) is described by the $SU(2)$ link variables

$$U_{i,i+x} = -i\chi_1 \tau^0, \quad U_{i,i+y} = -i\chi_1 (-)^{i_x} \tau^0$$

2. Show that, when $\eta = \chi$, the d -wave state (9.2.5) is described by the $SU(2)$ link variables

$$U_{i,i+x} = \chi \tau^3 + \chi \tau^1, \quad U_{i,i+y} = \chi \tau^3 - \chi \tau^1$$

3. Show that, when χ and χ_1 are related in a certain way, the above two ansatz are gauge equivalent. Find an $SU(2)$ gauge transformation W_i that changes one ansatz to the other. (Hint: Consider the $SU(2)$ gauge transformation $(i\tau^1)^{i_x} (i\tau^3)^{i_y}$.) Find the relation between χ and χ_1 .
4. Show that, when χ and χ_1 are related in this way, the π -flux state and the d -wave state have an identical spinon spectrum.

9.2.2 Dynamics of the $SU(2)$ gauge fluctuations

- The Anderson–Higgs mechanism can be realized through the condensation of $SU(2)$ gauge flux without using Higgs bosons.
- Collinear $SU(2)$ flux breaks the $SU(2)$ gauge structure down to a $U(1)$ gauge structure.

- Non-collinear $SU(2)$ flux breaks the $SU(2)$ gauge structure down to a Z_2 gauge structure. In this case, all of the $SU(2)$ gauge bosons gain a gap.

Just like the $U(1)$ projective construction, to obtain a stable mean-field state in the $SU(2)$ projective construction, we need to find ways to kill the gapless $SU(2)$ gauge bosons. The $SU(2)$ Chern–Simons term is one way to give $SU(2)$ gauge bosons a nonzero energy gap. In the following, we will discuss how to use the Anderson–Higgs mechanism to kill the gapless $SU(2)$ gauge bosons.

We know that $U(1)$ gauge bosons do not carry their own gauge charge. So we need additional charged bosons to implement the Anderson–Higgs mechanism. In contrast, $SU(2)$ gauge bosons themselves carry nonzero gauge charges. Thus, we do not need additional Higgs bosons to realize the Anderson–Higgs mechanism. The $SU(2)$ gauge bosons are capable of killing themselves.

To see how $SU(2)$ gauge bosons commit suicide, we consider a lattice $SU(2)$ gauge theory. The lattice $SU(2)$ gauge field is given by the link variables $U_{ij} \in SU(2)$. The first-order mean-field theory (9.2.18) is an $SU(2)$ lattice gauge theory. The energy of a configuration is a function of U_{ij} , i.e. $E(U_{ij})$. The energy is invariant under the $SU(2)$ gauge transformation

$$E(\tilde{U}_{ij}) = E(U_{ij}), \quad \tilde{U}_{ij} = W_i(U_{ij})W_j, \quad W_i \in SU(2) \quad (9.2.19)$$

To understand the dynamics of the lattice $SU(2)$ gauge fluctuations, we write $U_{ij} = \bar{U}_{ij} e^{ia_{ij}^l \tau^l}$, where the 2×2 matrices a_{ij} on the links describe the gauge fluctuations. The energy can now be written as $E(\bar{U}_{ij}, e^{ia_{ij}^l \tau^l})$. To see whether the $SU(2)$ gauge fluctuations gain an energy gap or not, we need to examine whether $E(\bar{U}_{ij}, e^{ia_{ij}^l \tau^l})$ contains a mass term $(a_{ij}^l)^2$, or not, in the small- a_{ij}^l limit.

To understand how the mean-field ansatz $\bar{U}_{ij}$ affects the dynamics of the gauge fluctuations, it is convenient to introduce the following loop variable of the mean-field solution:

$$P(C_i) = \bar{U}_{ij} \bar{U}_{jk} \dots \bar{U}_{li} \quad (9.2.20)$$

Here $P(C_i)$ is called the $SU(2)$ flux through the loop C_i given by $i \rightarrow j \rightarrow k \rightarrow \dots \rightarrow l \rightarrow i$, with base point i . We will also call it the $SU(2)$ -flux operator. The loop variable corresponds to gauge field strength in the continuum limit. Under the gauge transformations, $P(C_i)$ transforms as follows:

$$P(C_i) = W_i P(C_i) W_i \quad (9.2.21)$$

We note that the $SU(2)$ flux has the form $P(C) = \chi^0(C)\tau^0 + i\chi^l(C)\tau^l$. Thus, when $\chi^l \neq 0$, the $SU(2)$ flux has a sense of direction in the $SU(2)$ space which is indicated by χ^l . From eqn (9.2.21), we see that the local $SU(2)$ gauge transformations rotate the direction of the $SU(2)$ flux. As the direction of the $SU(2)$ flux for loops with different base points can be rotated independently by the local $SU(2)$

gauge transformations, it is meaningless to directly compare the directions of the $SU(2)$ flux for different base points. However, it is quite meaningful to compare the directions of the $SU(2)$ flux for loops with the same base point. We can divide different $SU(2)$ flux configurations into the following three classes, based on the $SU(2)$ fluxes through loops with the *same* base point: a trivial $SU(2)$ flux where all $P(C) \propto \tau^0$; a collinear $SU(2)$ flux where all of the $SU(2)$ fluxes point in the same direction; and a non-collinear $SU(2)$ flux where the $SU(2)$ fluxes for loops with the same base point are in different directions.

First, let us consider an ansatz $\bar{U}_{ij}$, with trivial $SU(2)$ flux for all of the loops. The $SU(2)$ flux is invariant under the $SU(2)$ gauge transformation. We can choose a mean-field ansatz (by performing gauge transformations) such that all $\bar{U}_{ij} \propto \tau^0$. In this case, the gauge invariance of the energy implies that

$$E(\bar{U}_{ij}, e^{ia_{ij}^l \tau^l}) = E(\bar{U}_{ij}, e^{i\theta_i^l \tau^l} e^{ia_{ij}^l \tau^l} e^{-i\theta_j^l \tau^l}). \quad (9.2.22)$$

As a result, none of the mass terms $(a_{ij}^1)^2$, $(a_{ij}^2)^2$, and $(a_{ij}^3)^2$ are allowed in the expansion of E .⁵⁸ Thus, the $SU(2)$ gauge fluctuations are gapless and appear at low energies. As the ansatz $\bar{U}_{ij} \propto \tau^0$ is invariant under the global $SU(2)$ gauge transformation, we say that the $SU(2)$ gauge structure is not broken.

Second, let us assume that the $SU(2)$ flux is collinear. This means that the $SU(2)$ fluxes for different loops with the same base point all point in the same direction. However, the $SU(2)$ fluxes for loops with different base points may still point in different directions (even for the collinear $SU(2)$ fluxes). Using the local $SU(2)$ gauge transformation, we can always rotate the $SU(2)$ fluxes for different base points into the same direction, and we can pick this direction to be the τ^3 direction. In this case, all of the $SU(2)$ fluxes have the form $P(C) \propto \chi^0(C) + i\chi^3(C)\tau^3$. We can choose a mean-field ansatz (by performing $SU(2)$ gauge transformations) such that all of the $\bar{U}_{ij}$ have the form $i e^{i\phi_{ij}\tau^3}$. As the ansatz is invariant under the global $U(1)$ gauge transformation $e^{i\theta\tau^3}$, but not $e^{i\theta\tau^{1,2}}$, we say that the $SU(2)$ gauge structure is broken down to a $U(1)$ gauge structure. The gauge invariance of the energy implies that

$$E(\bar{U}_{ij}, e^{ia_{ij}^l \tau^l}) = E(\bar{U}_{ij}, e^{i\theta_i \tau^3} e^{ia_{ij}^l \tau^l} e^{-i\theta_j \tau^3}). \quad (9.2.23)$$

When $a_{ij}^{1,2} = 0$, the above reduces to

$$E(\bar{U}_{ij}, e^{ia_{ij}^3 \tau^3}) = E(\bar{U}_{ij}, e^{i(a_{ij}^3 + \theta_i - \theta_j)\tau^3}). \quad (9.2.24)$$

We find that the mass term $(a^3)^2$ is incompatible with eqn (9.2.24). Therefore, at least the gauge boson a^3 is massless. How about the a^1 and a^2 gauge bosons? Let

⁵⁸ Under the gauge transformation $e^{i\theta_i^l \tau^l}$, a_{ij}^l transforms as $a_{ij}^l = a_{ij}^l + \theta_i^l - \theta_j^l$. The mass term $(a_{ij}^l)^2$ is not invariant under such a transformation and is thus not allowed.

$P_A(\mathbf{i})$ be the $SU(2)$ flux through a loop with base point $\mathbf{i}$. If we assume that all of the gauge-invariant terms that can appear in the energy function do appear in the energy function, then $E(U_{ij})$ will contain the following gauge-invariant term:

$$E = a \text{Tr}[P_A(\mathbf{i})U_{\mathbf{i},\mathbf{i}+\mathbf{x}}P_A(\mathbf{i}+\mathbf{x})U_{\mathbf{i}+\mathbf{x},\mathbf{i}}] + \dots \quad (9.2.25)$$

If we write $U_{\mathbf{i},\mathbf{i}+\mathbf{x}}$ as $\chi e^{i\phi_{ij}\tau^3} e^{ia_x^l \tau^l}$, use the fact that $U_{\mathbf{i},\mathbf{i}+\mathbf{x}} = -U_{\mathbf{i}+\mathbf{x},\mathbf{i}}^\dagger$ (see eqn (9.2.13)), and expand to order $(a_x^l)^2$, then eqn (9.2.25) becomes

$$E = -\frac{1}{2}a\chi^2 \text{Tr}([P_A, a_x^l \tau^l]^2) + \dots \quad (9.2.26)$$

We see from eqn (9.2.26) that the mass terms for a^1 and a^2 are generated if $P_A \propto \tau^3$.

To summarize, we find that, if the $SU(2)$ flux is collinear, then the ansatz is invariant under the $U(1)$ rotation $e^{i\theta \mathbf{n} \cdot \boldsymbol{\tau}}$, where $\mathbf{n}$ is the direction of the $SU(2)$ flux. As a result, the $SU(2)$ gauge structure is broken down to a $U(1)$ gauge structure. The corresponding mean-field state will have gapless $U(1)$ gauge fluctuations.

Third, we consider the situation where the $SU(2)$ flux is non-collinear. In the above, we have shown that an $SU(2)$ flux P_A can induce a mass term of the form $\text{Tr}([P_A, a_x^l \tau^l]^2)$. For a non-collinear $SU(2)$ flux configuration, we can have two $SU(2)$ fluxes, P_A and P_B , pointing in different directions. The mass term will also contain a term $\text{Tr}([P_B, a_x^l \tau^l]^2)$. In this case, the mass terms for all of the $SU(2)$ gauge fields, namely $(a_{ij}^1)^2$, $(a_{ij}^2)^2$, and $(a_{ij}^3)^2$, will be generated. All of the $SU(2)$ gauge bosons will gain an energy gap (see Section 3.7.5) and the mean-field state described by the ansatz will be a stable state. As the ansatz is invariant under the Z_2 transformation $W_i = -\tau^0$, but not other more general global $SU(2)$ gauge transformations, the non-collinear $SU(2)$ flux breaks the $SU(2)$ gauge structure down to a Z_2 gauge structure. So we may guess that the low-energy effective theory is a Z_2 gauge theory. In Sections 9.2.4 and 9.3, we will study the low-energy properties of states with non-collinear $SU(2)$ fluxes. We will show that the low-energy properties of such states, such as the existence of a Z_2 vortex and ground-state degeneracy, are indeed identical to those of a Z_2 gauge theory. So we will call the mean-field state with non-collinear $SU(2)$ fluxes a mean-field Z_2 state.

In a mean-field Z_2 state, all of the gauge fluctuations are gapped. These fluctuations can only mediate short-range interactions between spinons. The low-energy spinons interact weakly and behave like free spinons. Therefore, including mean-field fluctuations does not qualitatively change the properties of the mean-field state. The mean-field state is stable at low energies.

A stable mean-field spin-liquid state implies the existence of a real physical spin liquid. The physical properties of the stable mean-field state apply to the

physical spin liquid. If we believe these two statements, then we can study the properties of a physical spin liquid by studying its corresponding stable mean-field state. As the spinons are not confined in mean-field Z_2 states, the physical spin liquid derived from a mean-field Z_2 state contains neutral spin-1/2 fermions as its excitation. This is a very striking result, after realizing that the spin model is a purely bosonic model.

Problem 9.2.3.

(a) Show that the ansatz (9.2.5) is an $SU(2)$ -collinear state.

(b) Find the $SU(2)$ gauge transformation that transforms U_{ij} into the form $e^{i\phi_{ij}\tau^3}$. (Hint: Try $W_i = (i\tau^3)^{i_x+i_y}$.)

Problem 9.2.4.

(a) Show that the chiral spin ansatz U_{ij} obtained from eqn (9.1.22) is invariant under the global $SU(2)$ gauge transformation $U_{ij} = WU_{ij}W^\dagger$.

(b) Show that the chiral spin ansatz U_{ij} with $a_0^l = 0$ satisfies the constraint (9.2.8).

(c) Show that the ansatz described by eqn (9.1.22) and the following translationally-invariant ansatz:

$$\begin{aligned} u_{i,i+x} &= -\chi\tau^3 - \chi\tau^1, & u_{i,i+y} &= -\chi\tau^3 + \chi\tau^1, \\ u_{i,i+x+y} &= \eta\tau^2, & u_{i,i-x+y} &= -\eta\tau^2, \\ a_0^l &= 0, \end{aligned}$$

are gauge equivalent. In terms of the f -fermions (see eqns (9.2.4) and (9.2.7)), the above ansatz describes a $d_{x^2-y^2} + id_{xy}$ ‘superconducting’ state.

As the $SU(2)$ gauge structure is unbroken, the low-energy effective theory for the chiral spin state is actually an $SU(2)$ Chern–Simons theory (of level 1).

9.2.3 Spin liquids from translationally-invariant ansatz

- Uniform RVB state, π -flux state and staggered-flux state.
- The symmetry of projected physical wave functions.
- The low energy $SU(2)$ or $U(1)$ gauge fluctuations.

As an application of the $SU(2)$ projective construction, we will study a spin liquid obtained from a translationally-invariant ansatz

$$U_{i+l,j+l} = U_{ij}, \quad a_0^l(i) = a_0^l,$$

Such an ansatz will lead to a physical wave function with translational symmetry.⁵⁹ First, let us introduce

$$u_{ij} = \frac{3}{8} J_{ij} U_{ij} = u_{ij}^\mu \tau^\mu$$

where $u_i^{1,2,3}$ are real, and u_i^0 is imaginary. Within zeroth-order mean-field theory, the spinon spectrum is determined by the Hamiltonian (see eqn (9.2.9))

$$H = - \sum_{\langle ij \rangle} \left(\psi_i^\dagger u_{ij} \psi_j + h.c. \right) + \sum_i \psi_i^\dagger a_0^l \tau^l \psi_i \quad (9.2.27)$$

In $\mathbf{k}$ space, we have

$$H = - \sum_{\mathbf{k}} \psi_{\mathbf{k}}^\dagger (u^\mu(\mathbf{k}) - a_0^\mu) \tau^\mu \psi_{\mathbf{k}}$$

where $\mu = 0, 1, 2$,

$$u^\mu(\mathbf{k}) = \sum_l u_{i, i+l}^\mu e^{i\mathbf{l} \cdot \mathbf{k}},$$

$a_0^0 = 0$, and N is the total number of sites. The fermion spectrum has two branches and is given by

$$\begin{aligned} E_{\pm}(\mathbf{k}) &= u^0(\mathbf{k}) \pm E_0(\mathbf{k}) \\ E_0(\mathbf{k}) &= \sqrt{\sum_l (u^l(\mathbf{k}) - a_0^l)^2} \end{aligned} \quad (9.2.28)$$

The constraints can be obtained from $\partial E_{\text{ground}} / \partial a_0^l = 0$ and have the form

$$N \langle \psi_i^\dagger \tau^l \psi_i \rangle = \sum_{\mathbf{k}, E_-(\mathbf{k}) < 0} \frac{u^l(\mathbf{k}) - a_0^l}{E_0(\mathbf{k})} - \sum_{\mathbf{k}, E_+(\mathbf{k}) < 0} \frac{u^l(\mathbf{k}) - a_0^l}{E_0(\mathbf{k})} = 0 \quad (9.2.29)$$

which allow us to determine a_0^l , $l = 1, 2, 3$.

Let us try the following simple ansatz⁶⁰ (Baskaran *et al.*, 1987; Affleck and Marston, 1988; Kotliar and Liu, 1988):

$$a_0^l = 0, \quad u_{i, i+\mathbf{x}} = \chi \tau^3 + \eta \tau^1, \quad u_{i, i+\mathbf{y}} = \chi \tau^3 - \eta \tau^1. \quad (9.2.30)$$

First, we would like to show that the corresponding spin liquid has all of the symmetries.

⁵⁹ Here we will distinguish between the invariance of an ansatz and the symmetry of an ansatz. We say that an ansatz has translational invariance when the ansatz itself does not change under a translation. We say that an ansatz has translational symmetry when the physical spin wave function obtained from the ansatz has translational symmetry. Due to the gauge structure, a translationally-non-invariant ansatz can have translational symmetry.

⁶⁰ The ansatz is actually the d -wave ansatz (9.2.5) introduced earlier.

To study the symmetry of the corresponding physical spin state, we need to obtain its wave function $|\Psi_{\text{phy}}\rangle$. Using the ansatz, we can obtain the mean-field ground state $|\Psi_{\text{mean}}\rangle$ of the mean-field Hamiltonian (9.2.27). The physical wave function $|\Psi_{\text{phy}}\rangle$ is obtained by projecting into the subspace with even numbers of ψ -fermions per site, i.e. $|\Psi_{\text{phy}}\rangle = \mathcal{P}|\Psi_{\text{mean}}\rangle$.

The ansatz is already invariant under the two translations $T_x : \mathbf{i} \rightarrow \mathbf{i} + \mathbf{x}$ and $T_y : \mathbf{i} \rightarrow \mathbf{i} + \mathbf{y}$. The ansatz is also invariant under the two parity transformation $P_x : x \rightarrow -x$ and $P_y : y \rightarrow -y$. The $P_{xy} : (x, y) \rightarrow (y, x)$ parity transformation changes $u_{\mathbf{i}, \mathbf{i}+\mathbf{x}} \rightarrow u_{\mathbf{i}, \mathbf{i}+\mathbf{y}}$ and $u_{\mathbf{i}, \mathbf{i}+\mathbf{y}} \rightarrow u_{\mathbf{i}, \mathbf{i}+\mathbf{x}}$. The transformed ansatz

$$a_0^l = 0, \quad u_{\mathbf{i}, \mathbf{i}+\mathbf{x}} = \chi\tau^3 - \eta\tau^1, \quad u_{\mathbf{i}, \mathbf{i}+\mathbf{y}} = \chi\tau^3 + \eta\tau^1.$$

is different from the original ansatz. However, note that a uniform gauge transformation $G_{P_{xy}}(\mathbf{i}) = i\tau^3$ changes $u_{\mathbf{i}\mathbf{j}}$ to $G_{P_{xy}} u_{\mathbf{i}\mathbf{j}} G_{P_{xy}}^\dagger$. It induces the changes $(\tau^1, \tau^2, \tau^3) \rightarrow (-\tau^1, -\tau^2, \tau^3)$. Thus the transformed ansatz is gauge equivalent to the original ansatz through a gauge transformation $G_{P_{xy}}$. In other words, the P_{xy} parity transformation, when followed by a gauge transformation $G_{P_{xy}}$, leaves the ansatz unchanged. As a result, the physical wave function is invariant under the P_{xy} parity transformation. A 90° rotation R_{90} is generated by $P_x P_{xy}$. The physical wave function also has 90° -rotational symmetry. The above ansatz also has time-reversal symmetry, because the time-reversal transformation $u_{\mathbf{i}\mathbf{j}} \rightarrow -u_{\mathbf{i}\mathbf{j}}$ followed by a gauge transformation $G_T(\mathbf{i}) = i\tau^2$ leaves the ansatz unchanged.

To summarize, the ansatz is invariant under the following combined symmetry and gauge transformations: $G_x T_x$, $G_y T_y$, $G_{P_x} P_x$, $G_{P_y} P_y$, $G_{P_{xy}} P_{xy}$, and $G_T T$. The associated gauge transformations are given by

$$\begin{aligned} G_x &= \tau^0, & G_y &= \tau^0, & G_T &= i\tau^2, \\ G_{P_x} &= \tau^0, & G_{P_y} &= \tau^0, & G_{P_{xy}} &= i\tau^3. \end{aligned} \quad (9.2.31)$$

So the spin liquid described by the ansatz (9.2.30) has all of the symmetries. We will call such a state a symmetric spin-liquid state.

We note that the link variables $u_{\mathbf{i}\mathbf{j}}$ in eqn (9.2.30) point to different directions in the $\tau^{1,2,3}$ space, and thus are not collinear. As a result, the ansatz is not invariant under the uniform $SU(2)$ gauge transformations. One might expect that the $SU(2)$ gauge structure is broken down to a Z_2 gauge structure and that the ansatz (9.2.30) describes a Z_2 spin liquid. In fact, this reasoning is incorrect. As we pointed out in Section 9.2.2, to determine if the $SU(2)$ gauge structure is broken down to a Z_2 gauge structure, we need to check the collinearity of the $SU(2)$ flux, not the collinearity of the $SU(2)$ link variables.

When $\chi = \eta$ or when $\eta = 0$, the $SU(2)$ fluxes P_C for all of the loops are trivial, i.e. $P_C \propto \tau^0$. In this case, the $SU(2)$ gauge structure is not broken and the $SU(2)$ gauge fluctuations are gapless (in the weak-coupling limit). The spinon in the spin